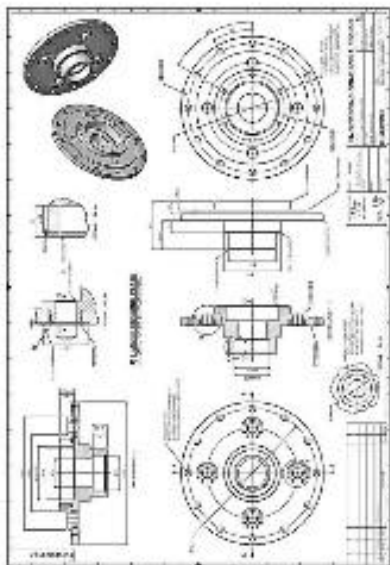


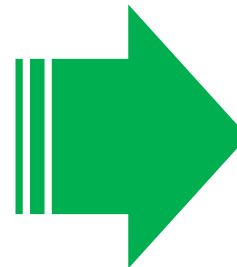
Fundamentos de desenho técnico mecânico

NORMAS DE DESENHO

Como um projeto se materializa?



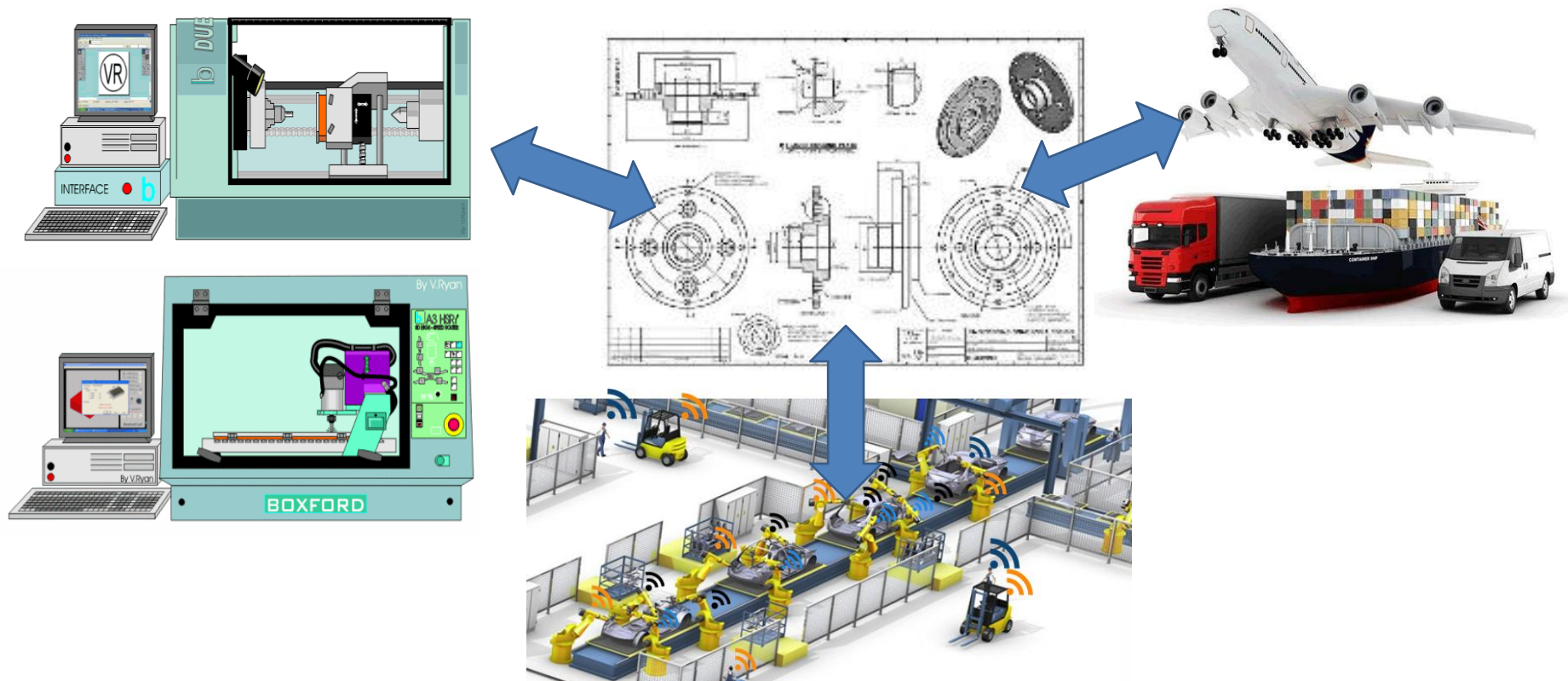
Projeto



Produto

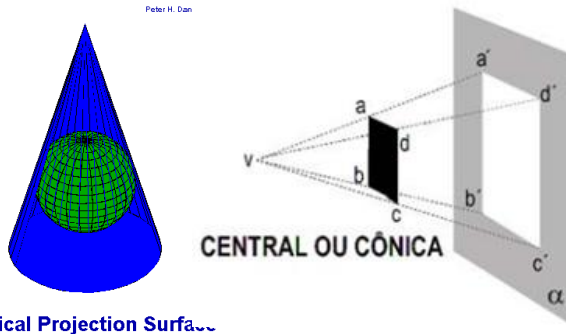
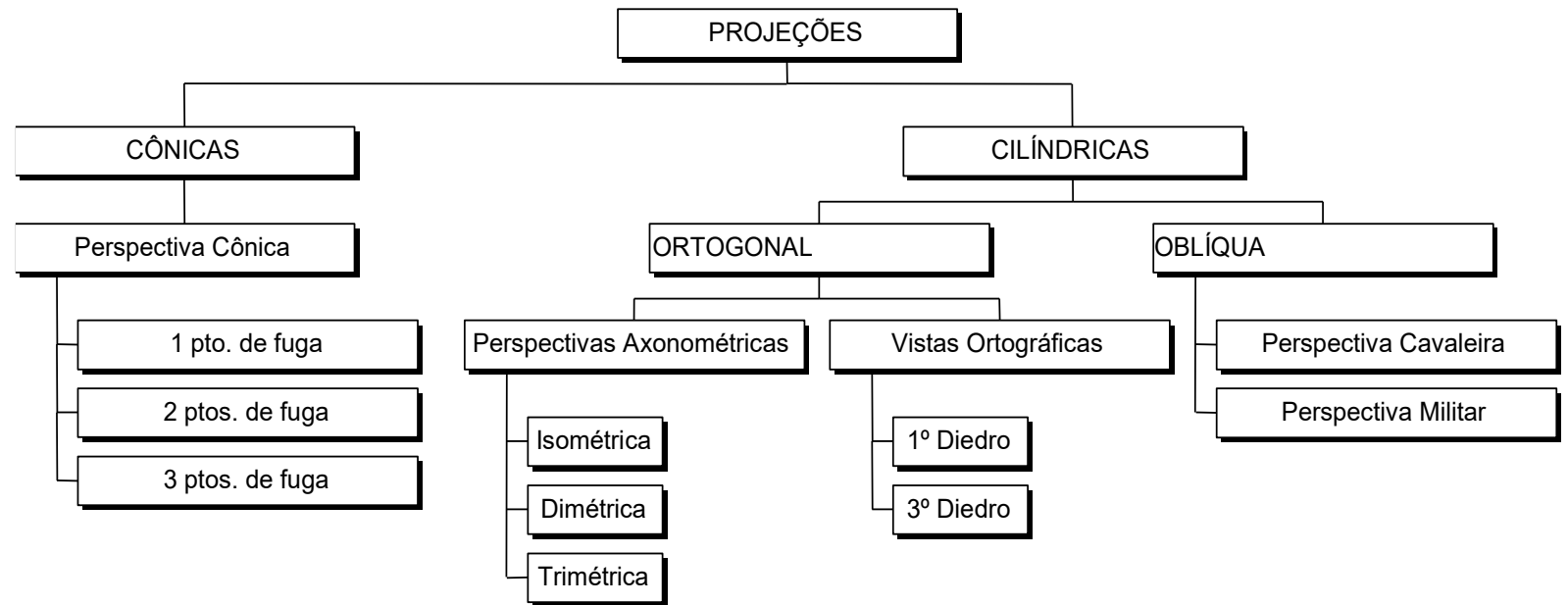
Fabricação

Como fornecedores, fabricação e montagem se harmonizam?



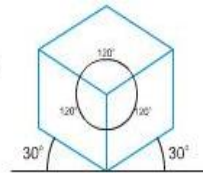
NORMALIZAÇÃO

➤ Classificação das Projeções e Perspectivas

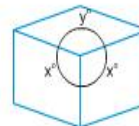


Conical Projection Surface

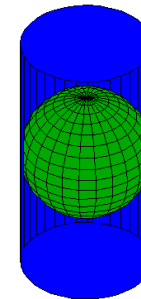
ISOMÉTRICA



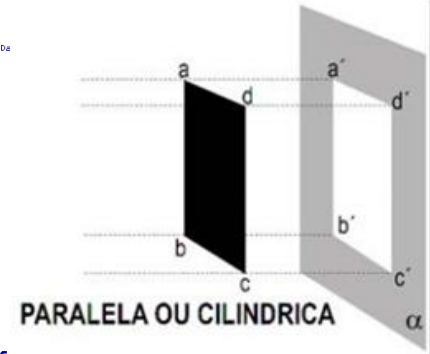
DIMÉTRICA



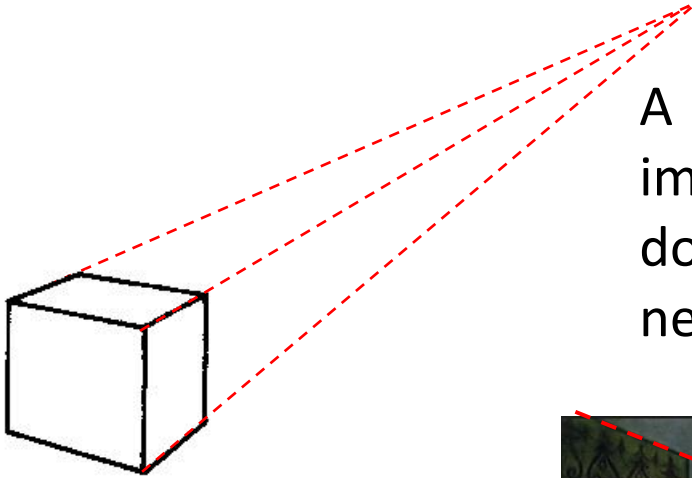
TRIMÉTRICA



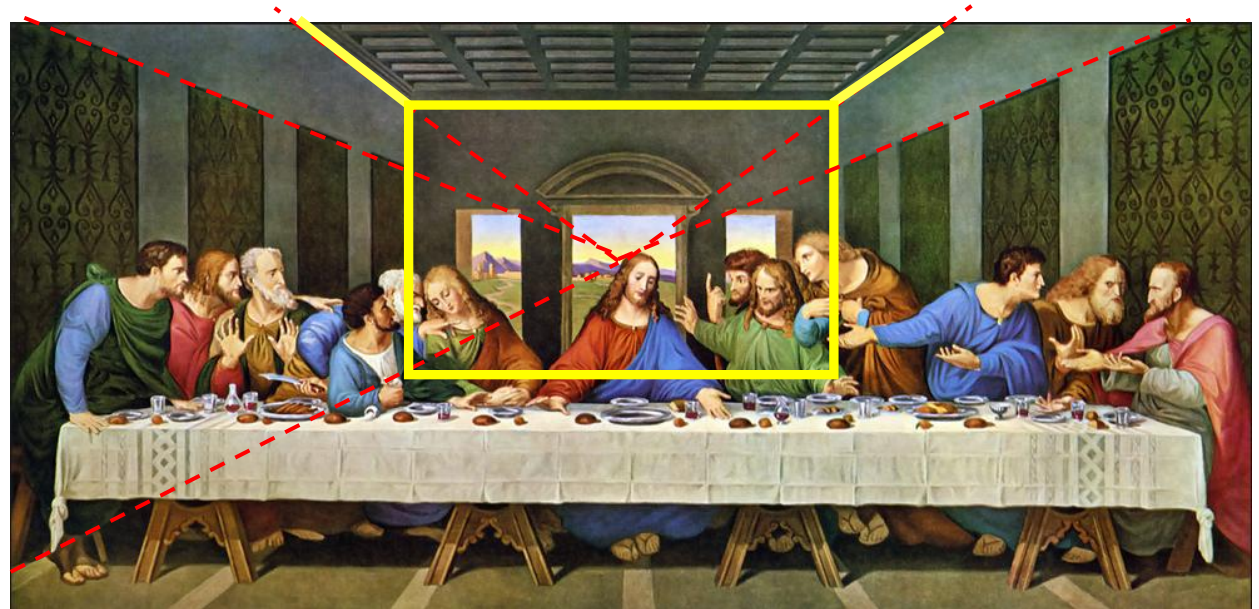
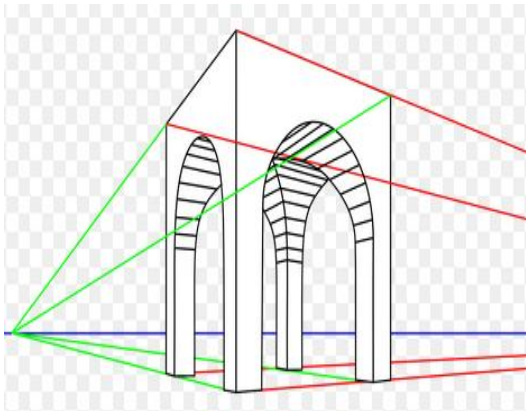
Cylindrical Projection Surface



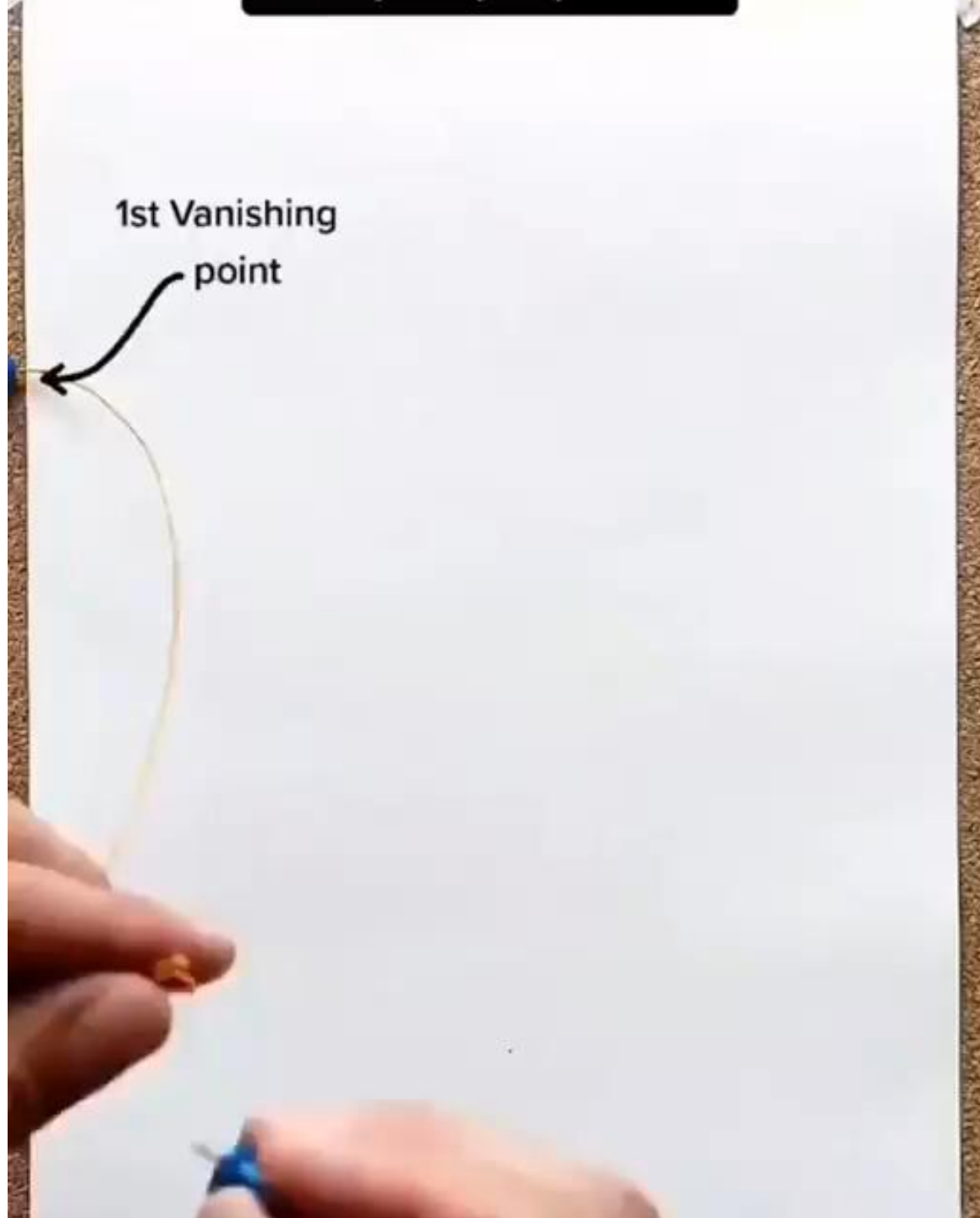
➤ Projeção cônica (perspectiva cônica)



A projeção cônica é utilizada por ter uma imagem mais real do objeto. Pode ter um, dois ou três pontos de fuga, com duas ou nenhuma das direções dominantes paralelas.



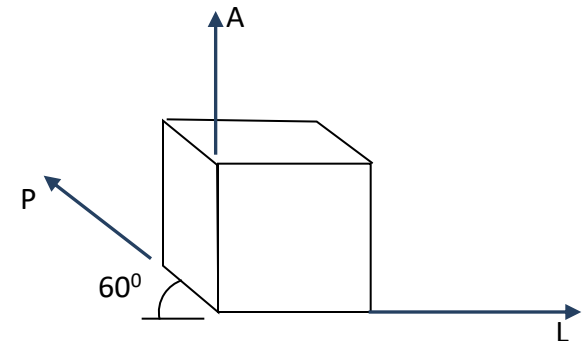
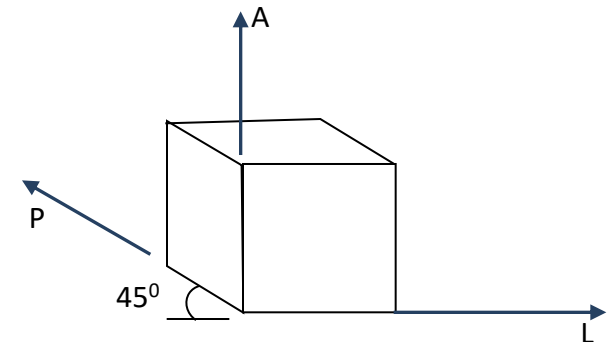
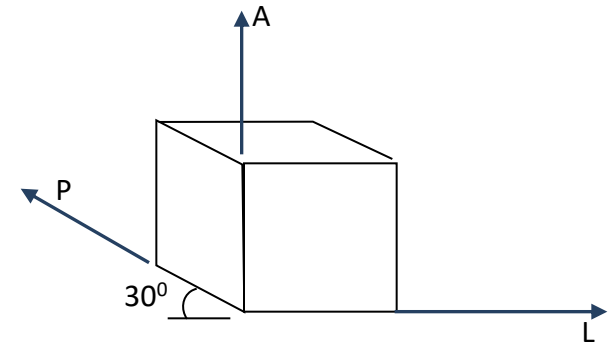
➤ Projeção cônica



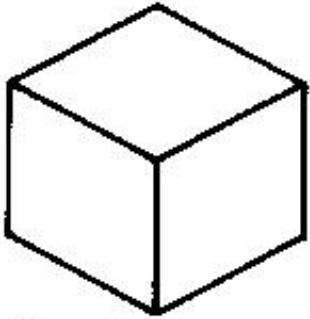
➤ Projeção cilíndrica oblíqua (perspectiva cavaleira)

Nessa perspectiva, os eixos da altura e largura formam entre si um ângulo reto, com as medidas mantendo suas verdadeiras grandezas. Já para a profundidade, o eixo está inclinado e suas medidas sofrem certa deformação.

TIPOS	COEFICIENTE DE REDUÇÃO DAS ESCALAS DOS EIXOS		
	L	A	P
CAVALEIRA 30°	1	1	2/3
CAVALEIRA 45°	1	1	1/2
CAVALEIRA 60°	1	1	1/3



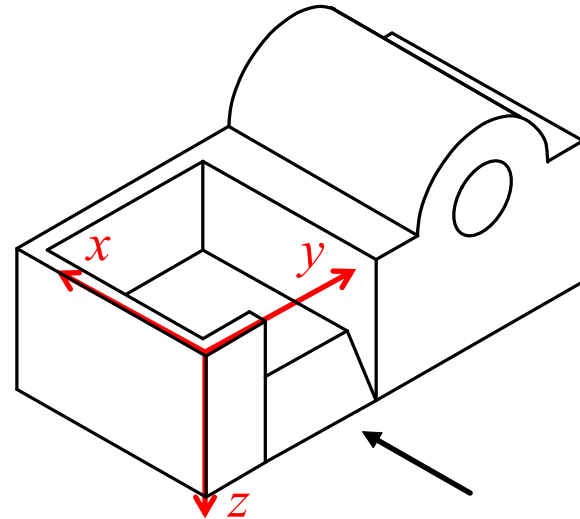
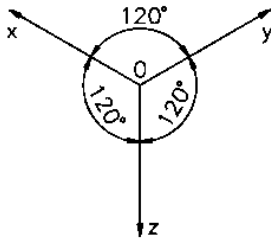
➤ Projeção cilíndrica ortogonal (Perspectiva isométrica)



Comparando as três formas de representação, a perspectiva isométrica é a que dá a ideia menos deformada do objeto.

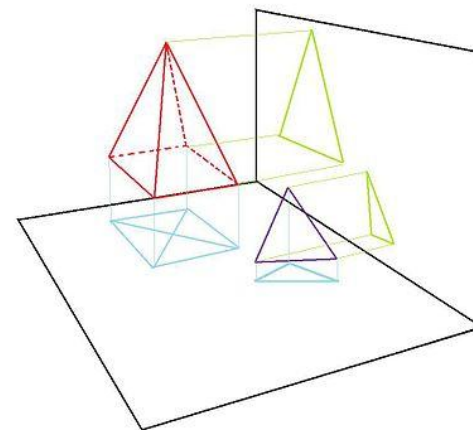
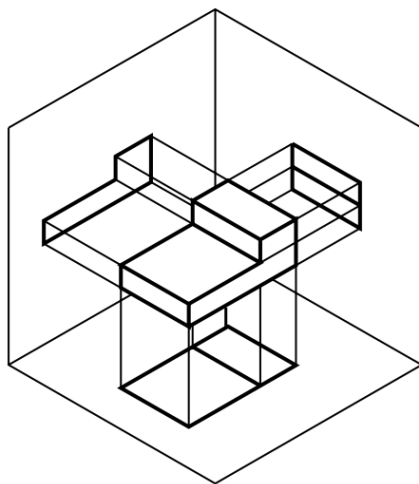
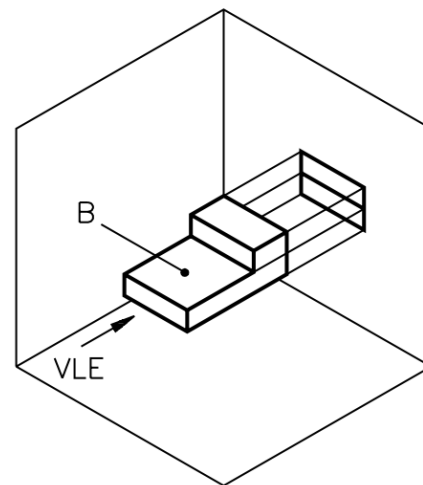
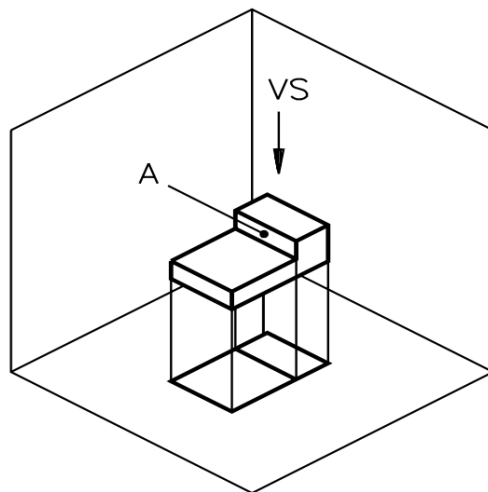
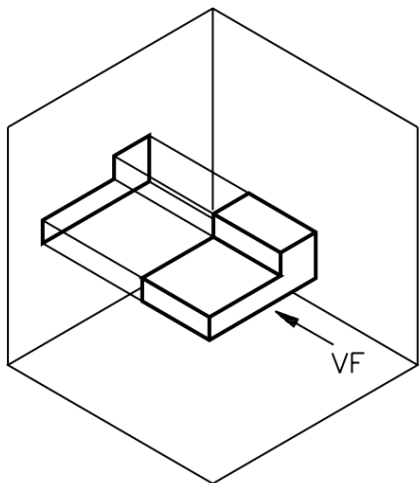
Iso quer dizer mesma; métrica quer dizer medida. A **perspectiva isométrica** mantém as **mesmas proporções do comprimento**, da largura e da altura do objeto representado.

Os três eixos axonométricos têm a mesma inclinação em relação ao quadro, com **ângulos de 120°** entre si, e **todas as cotas estão na mesma escala**. Os três coeficientes de redução das escalas são iguais.

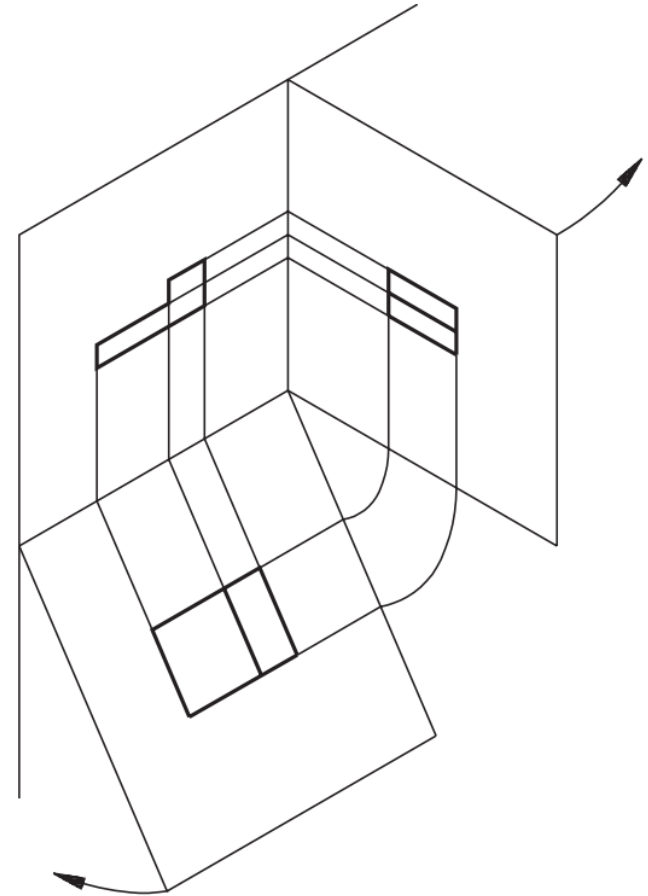
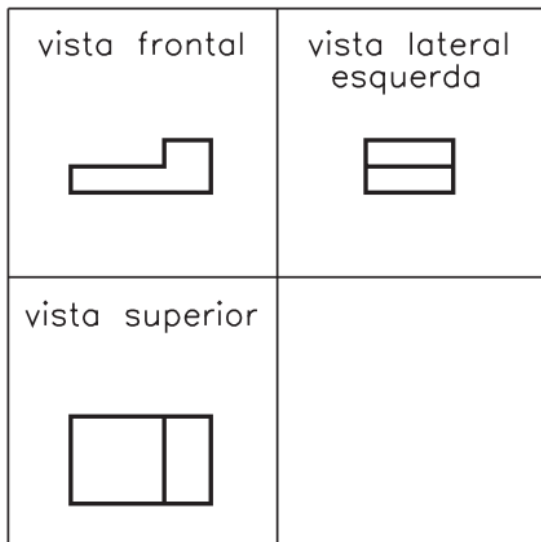
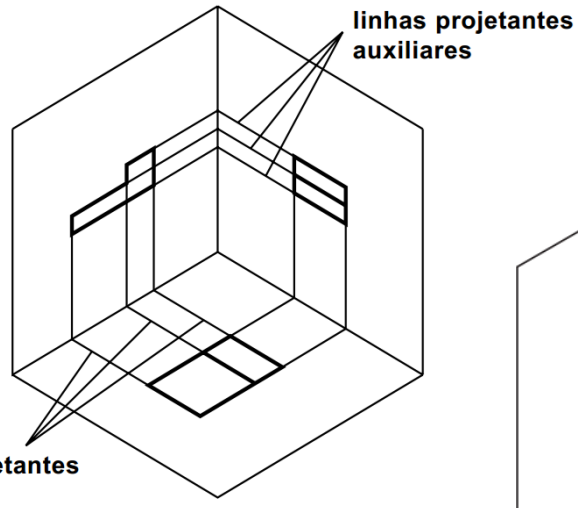
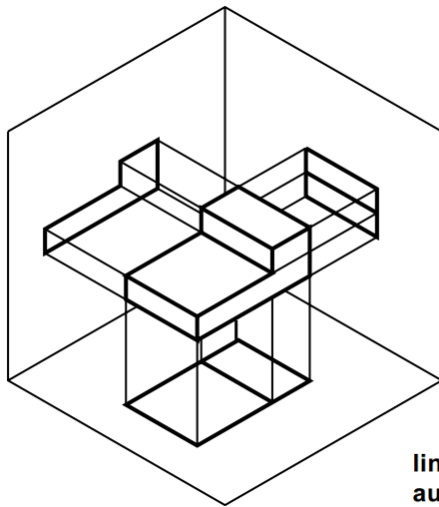


Isométrica: faces Frontal, Lateral Esquerda e Superior

➤ Vistas ortogonais

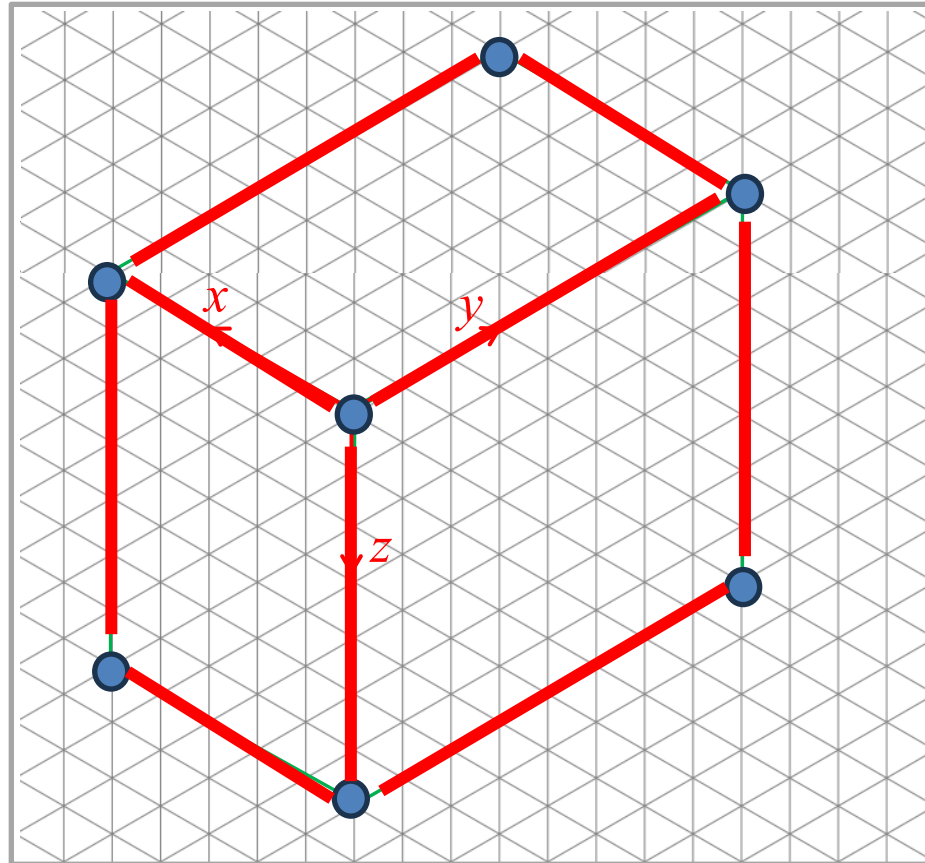


➤ Vistas ortogonais



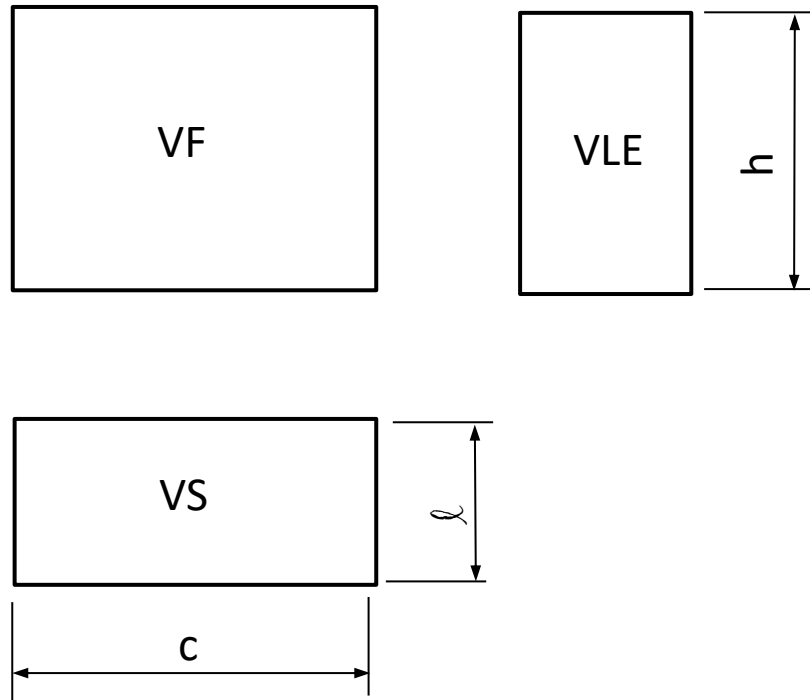
Papel com grade isométrica

As linhas são usadas para orientar o traçado da perspectiva isométrica

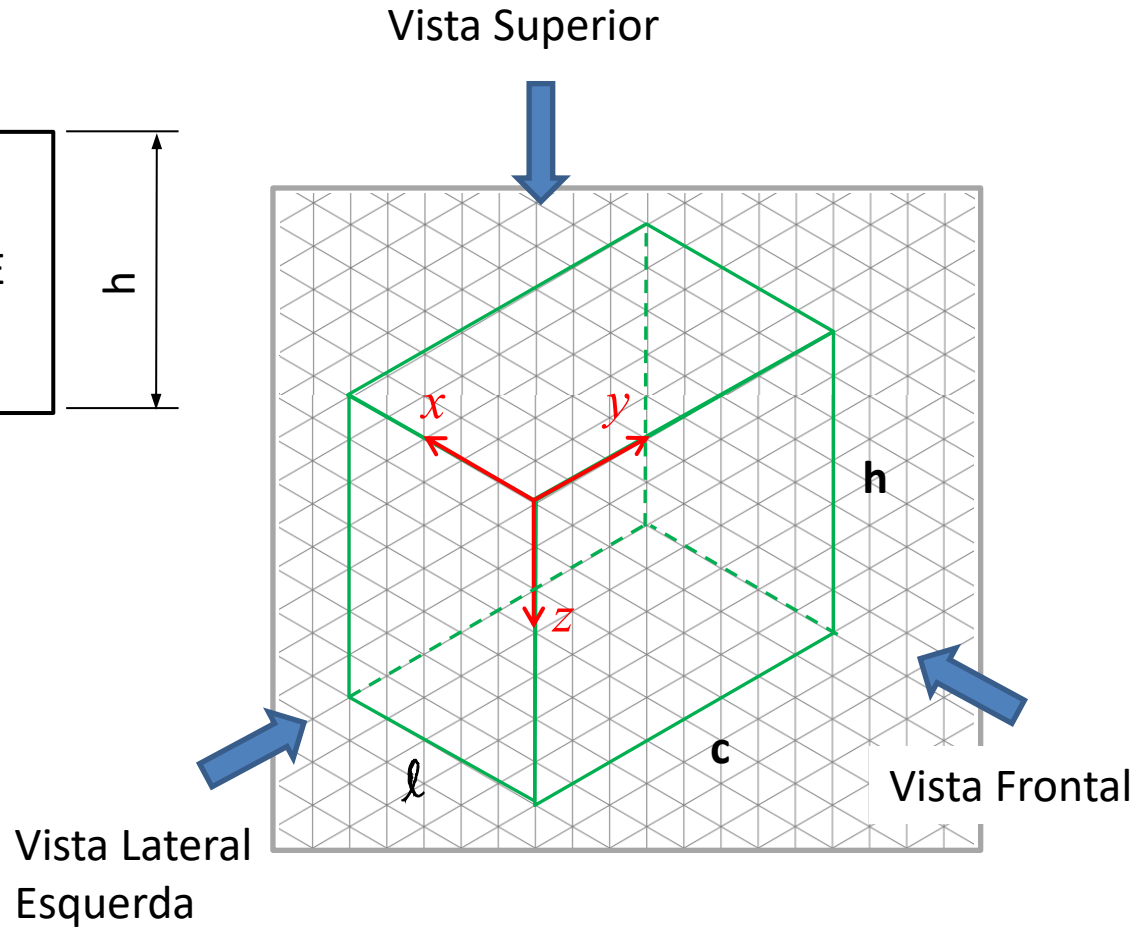


Eixos isométricos → formam entre si ângulos de 120°

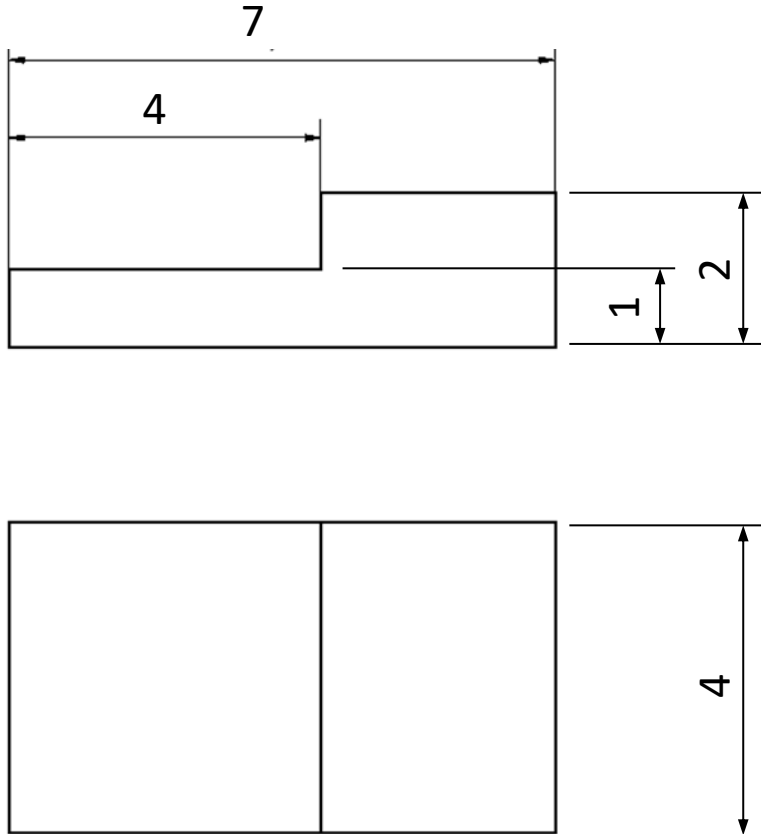
Vistas ortográficas



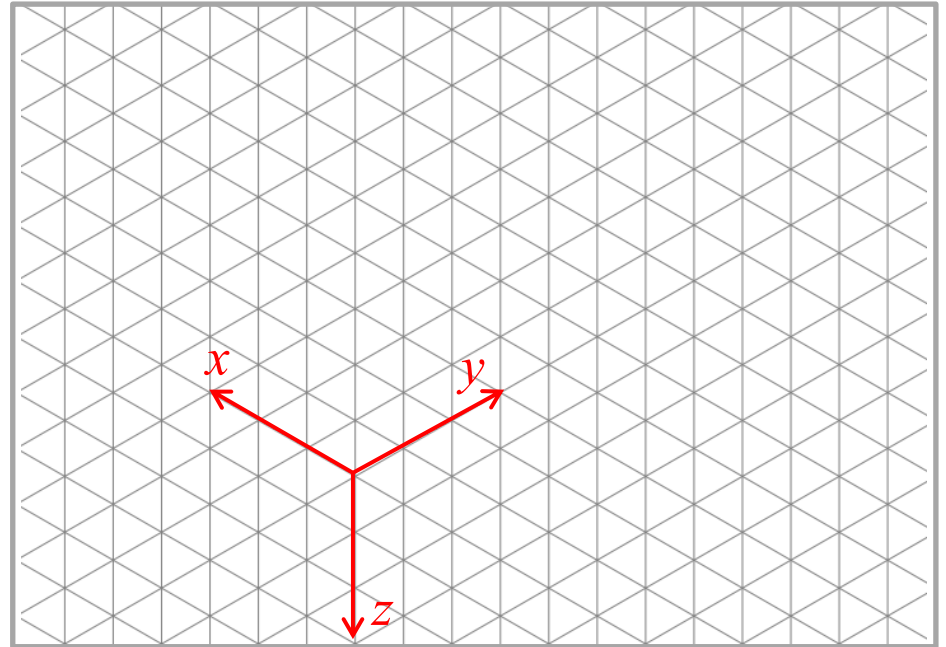
Perspectiva Isométrica



Vistas ortográficas

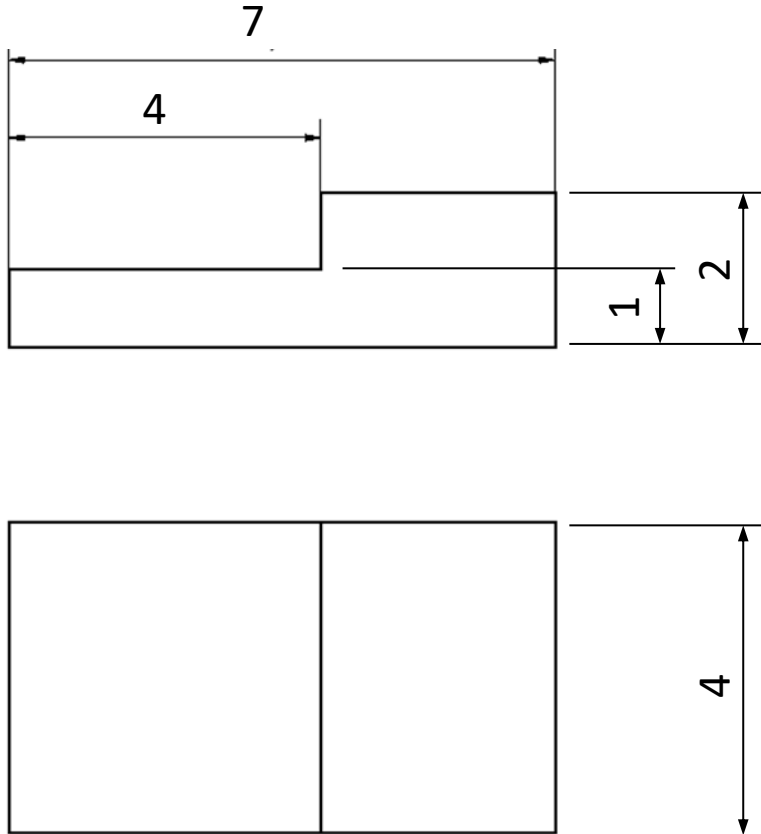


Perspectiva Isométrica - 1

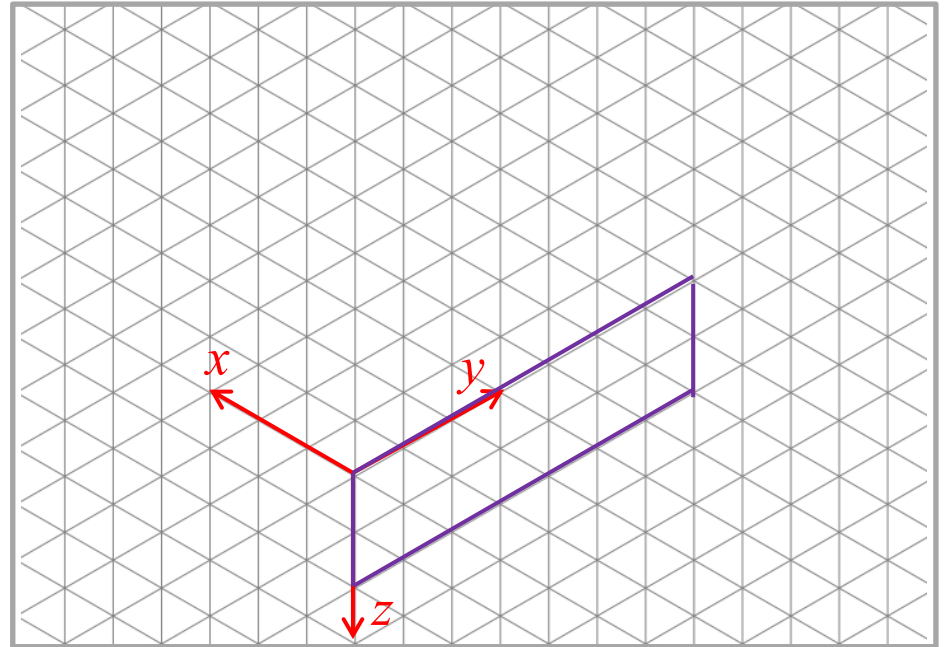


Eixos isométricos

Vistas ortográficas

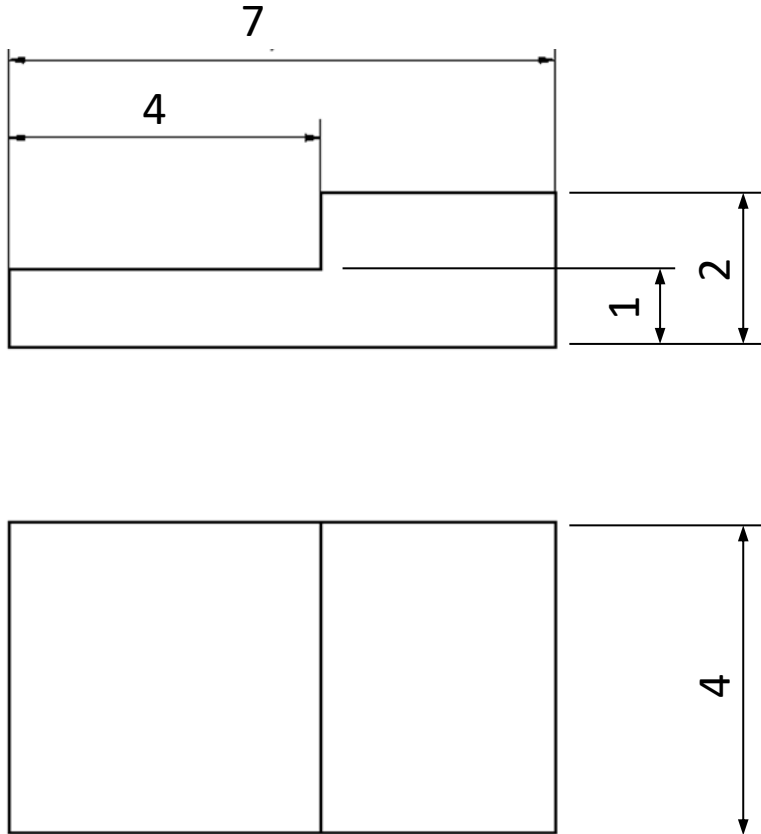


Perspectiva Isométrica - 1

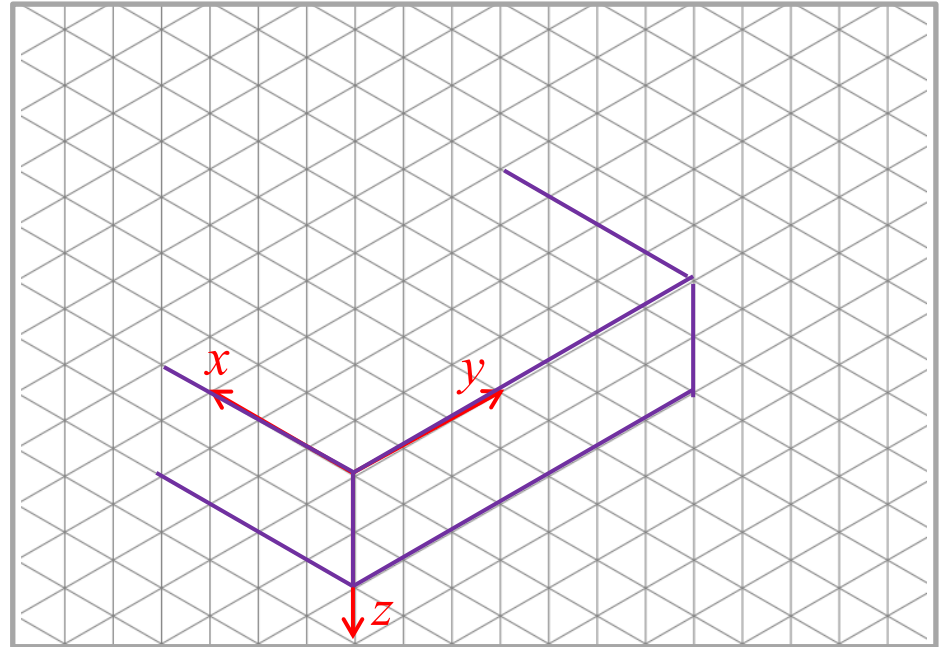


Desenhando o prisma auxiliar

Vistas ortográficas

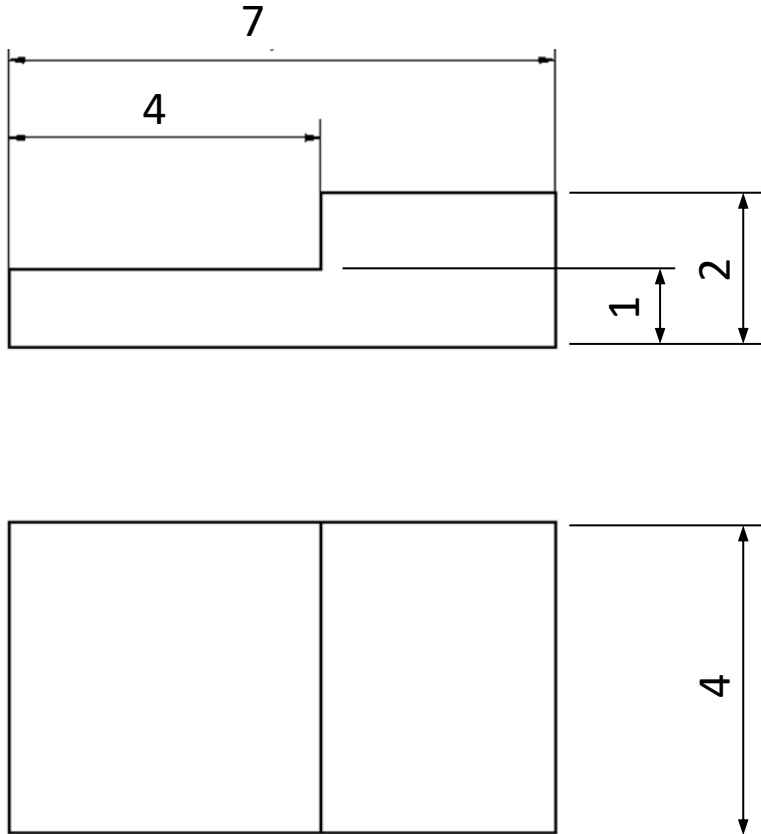


Perspectiva Isométrica - 1

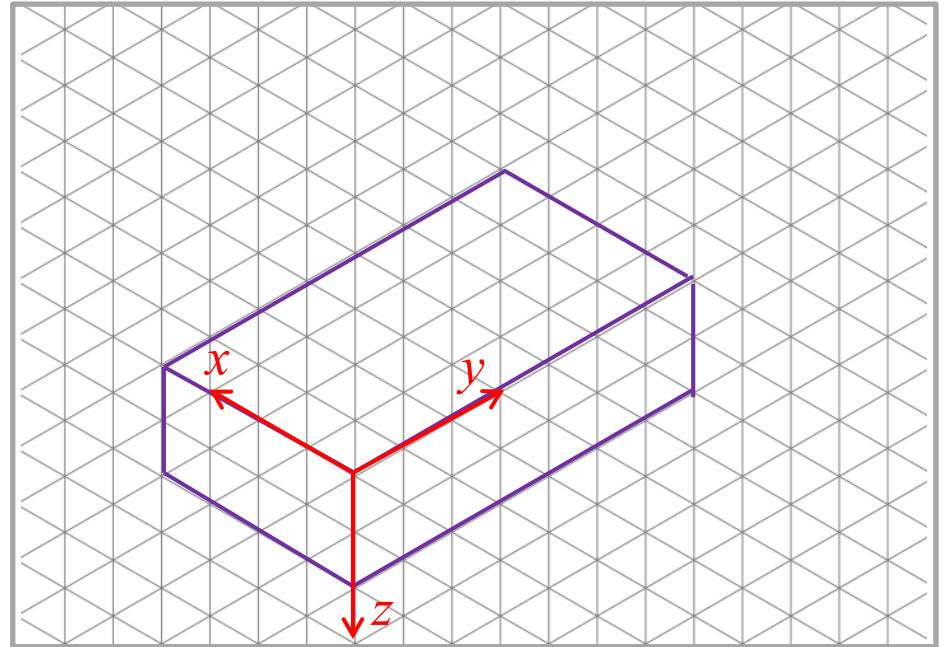


Desenhando o prisma auxiliar

Vistas ortográficas

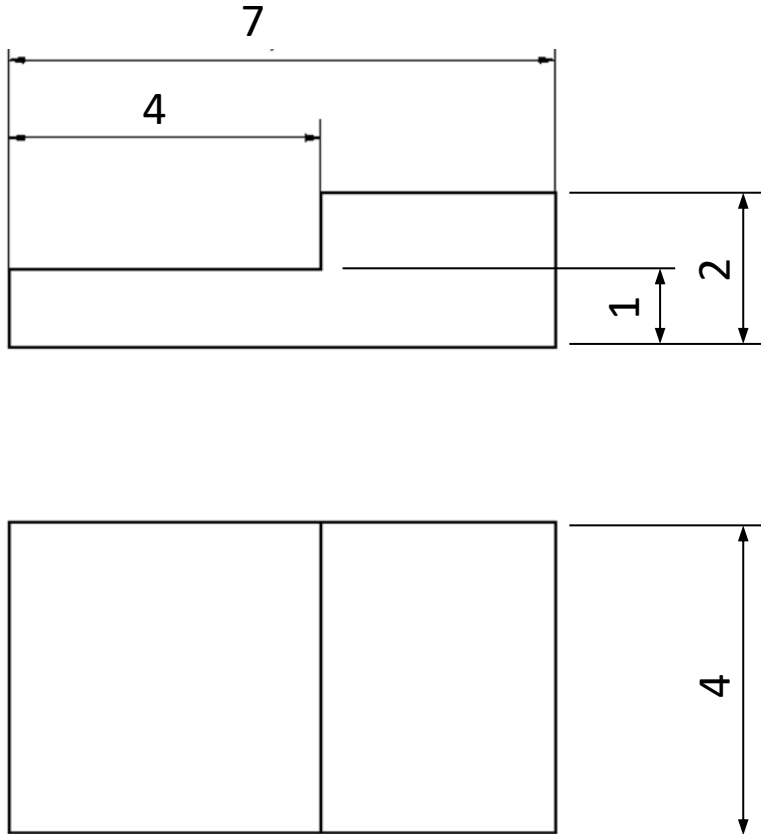


Perspectiva Isométrica - 1

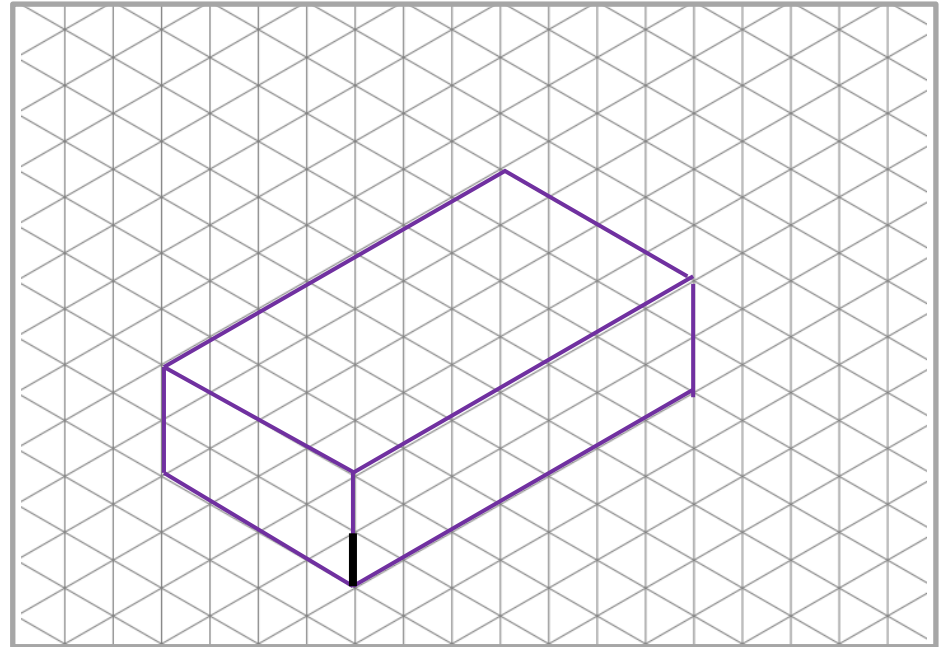


Desenhando o prisma auxiliar

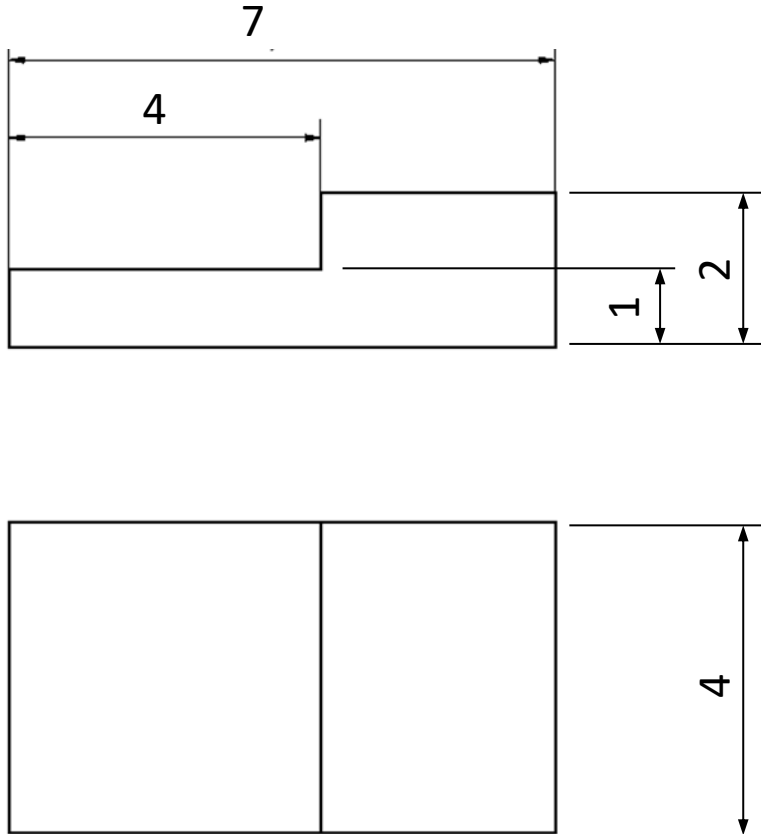
Vistas ortográficas



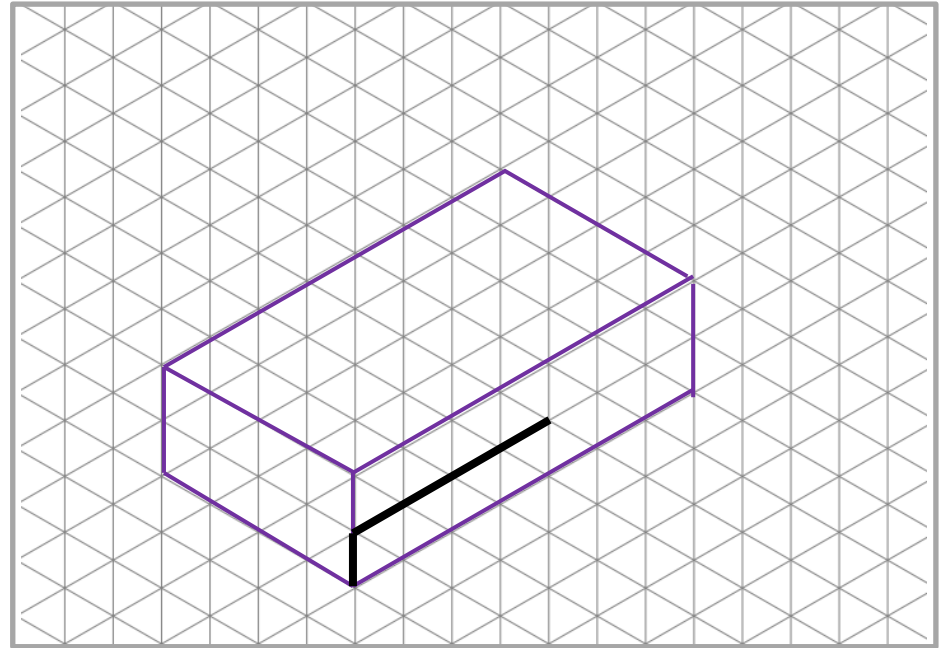
Perspectiva Isométrica - 1



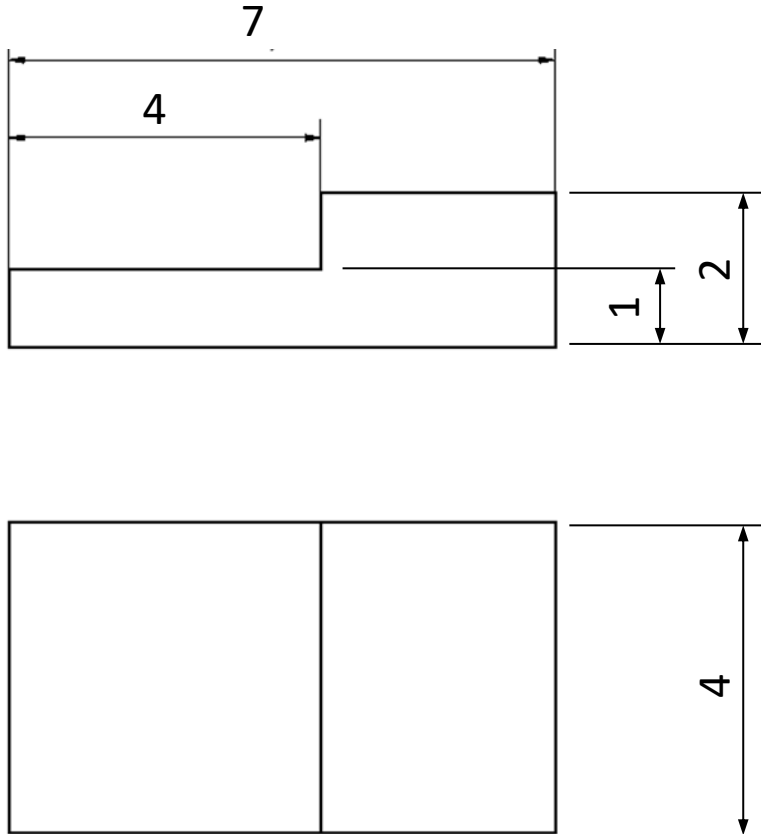
Vistas ortográficas



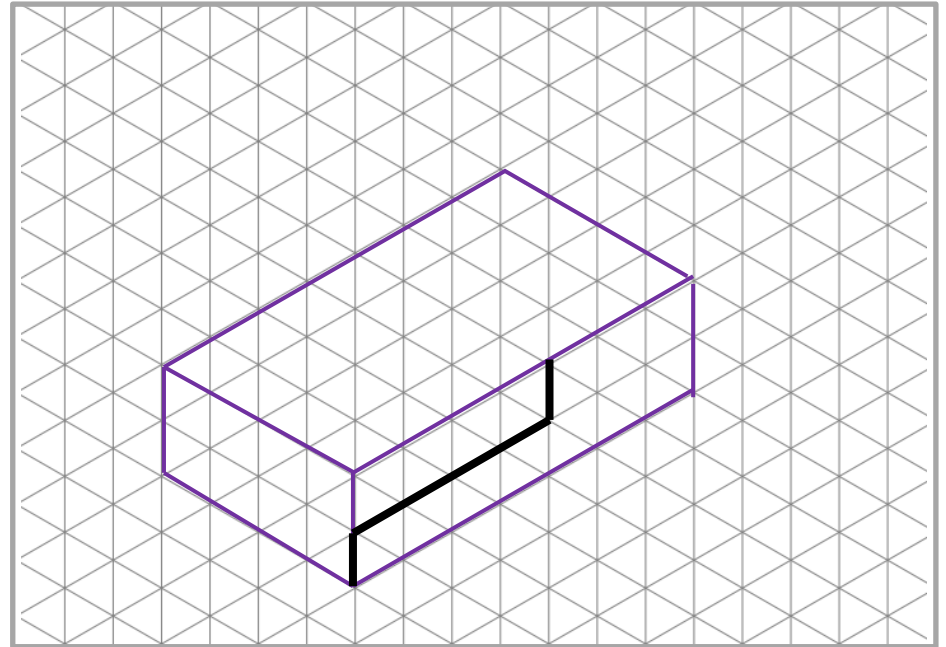
Perspectiva Isométrica - 1



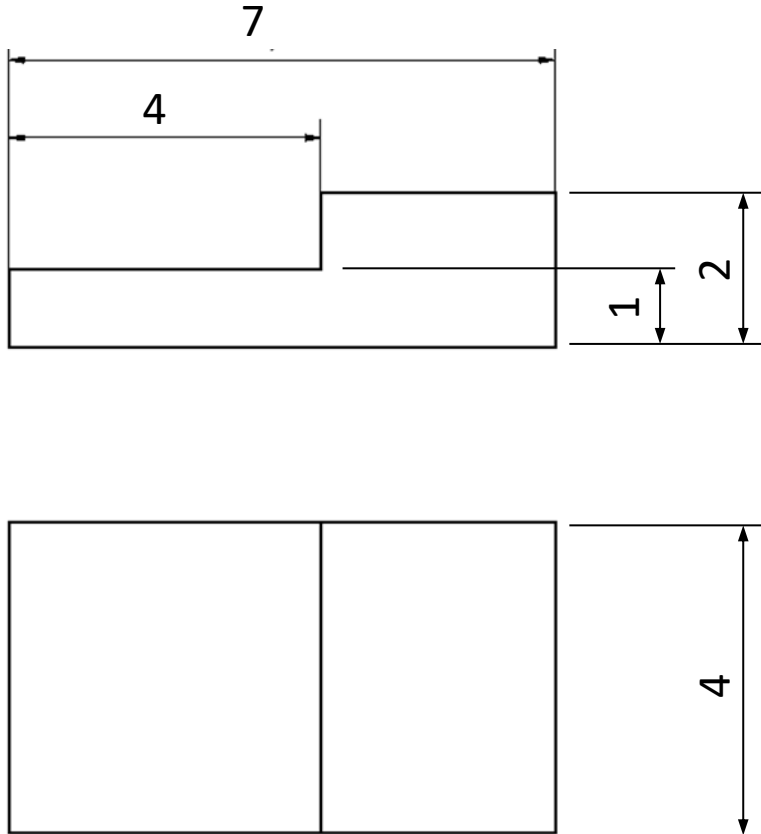
Vistas ortográficas



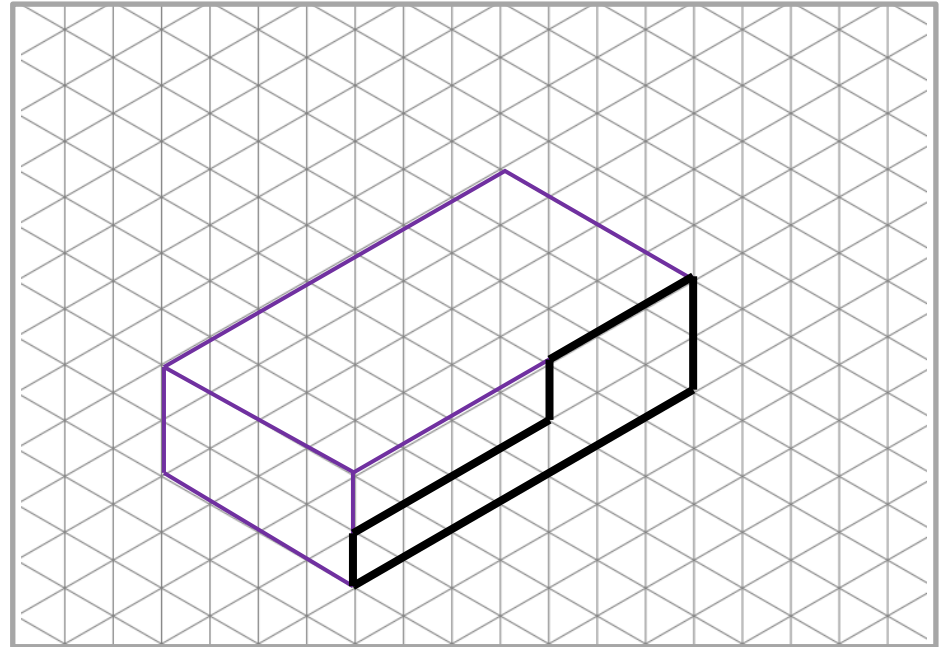
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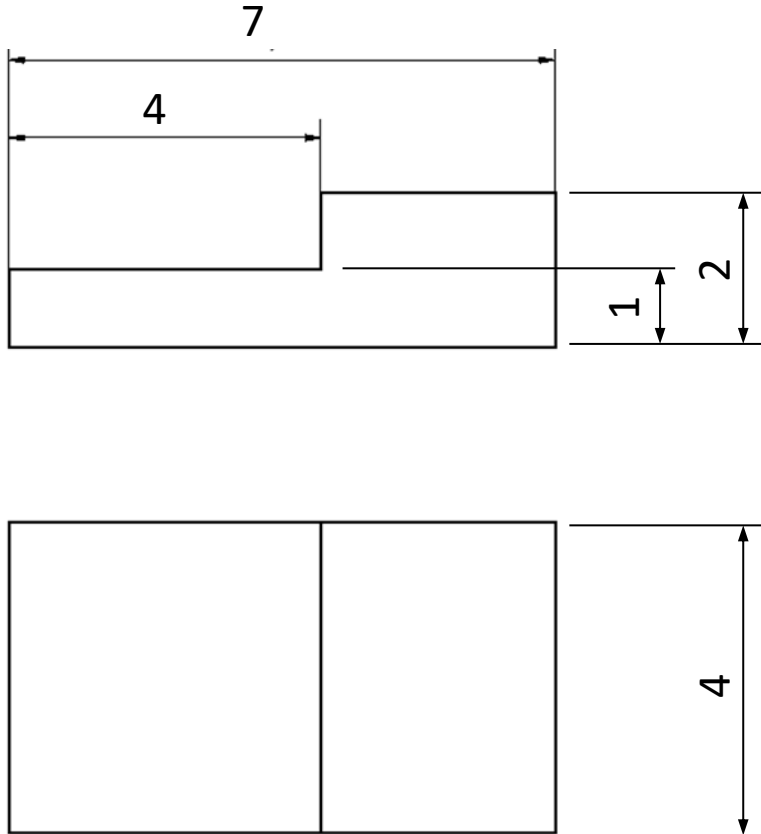
Vistas ortográficas



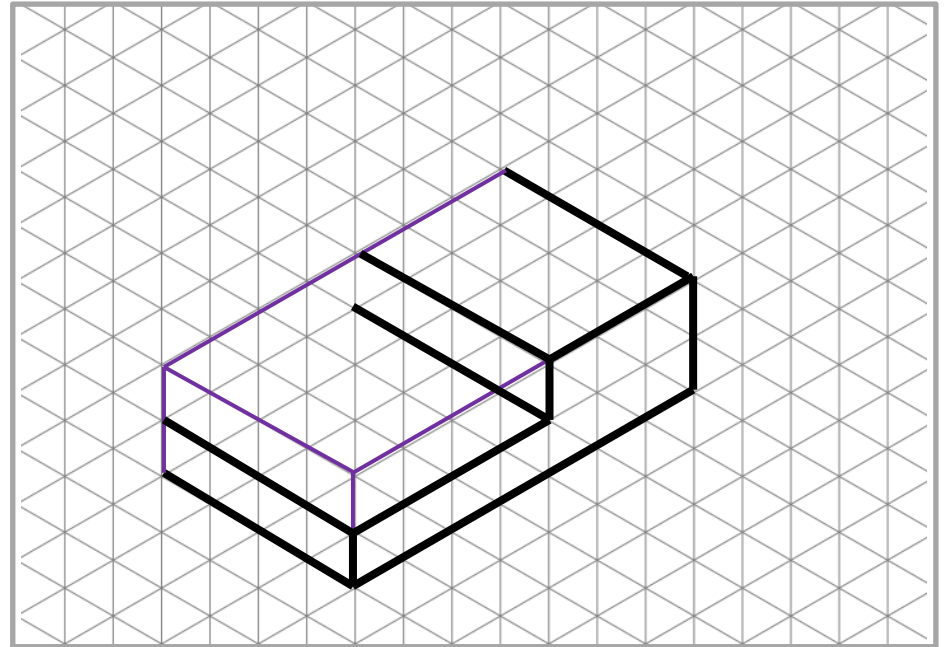
Perspectiva Isométrica - 1



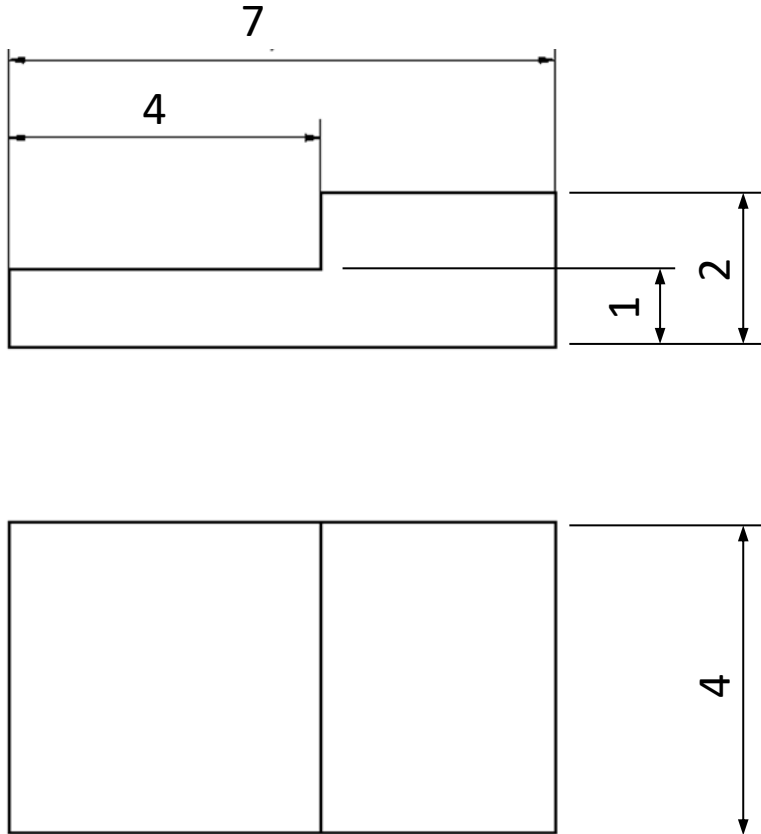
Vistas ortográficas



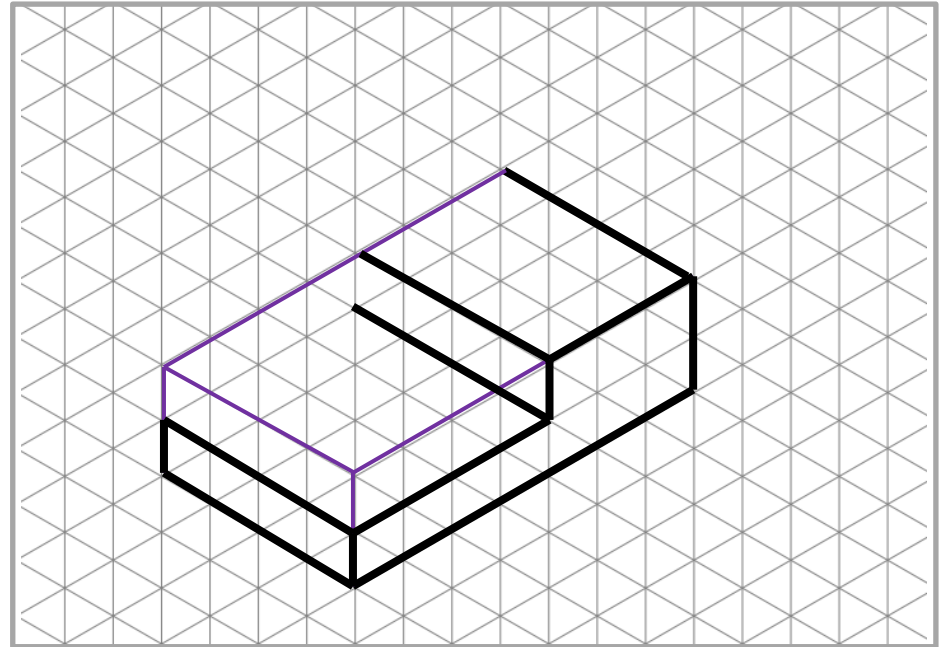
Perspectiva Isométrica - 1



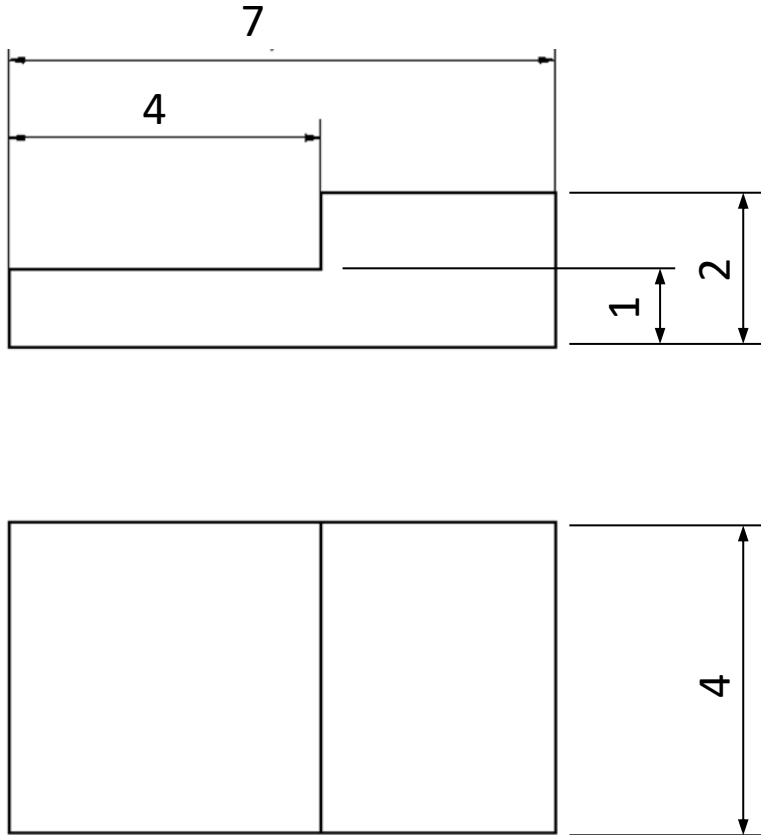
Vistas ortográficas



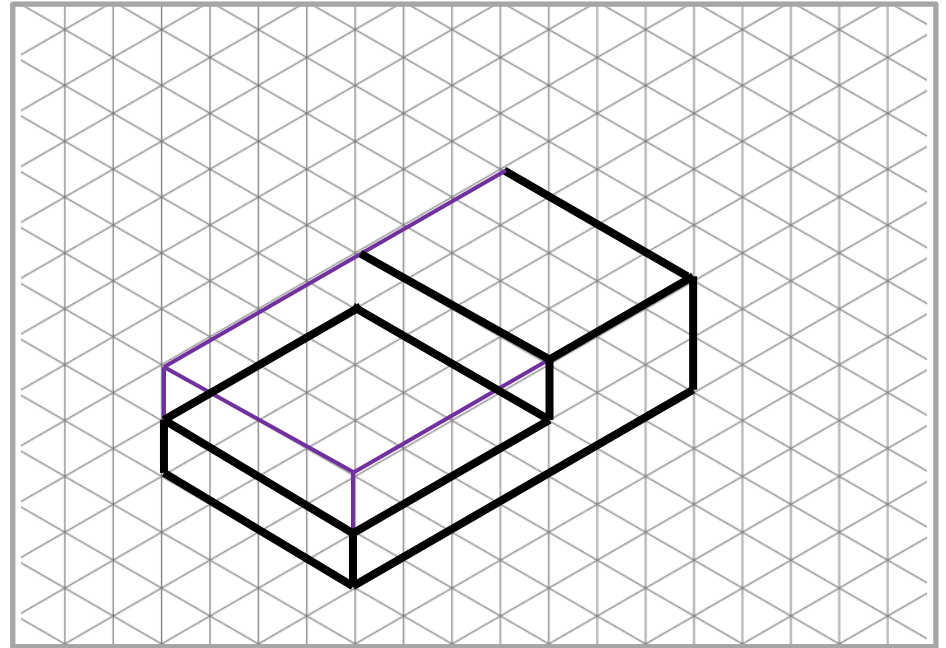
Perspectiva Isométrica - 1



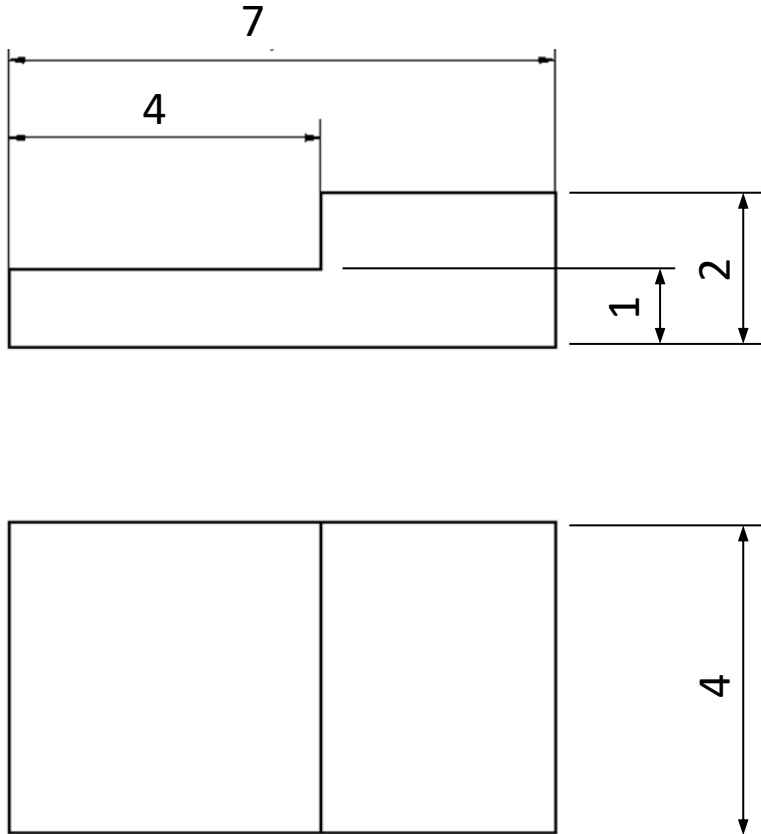
Vistas ortográficas



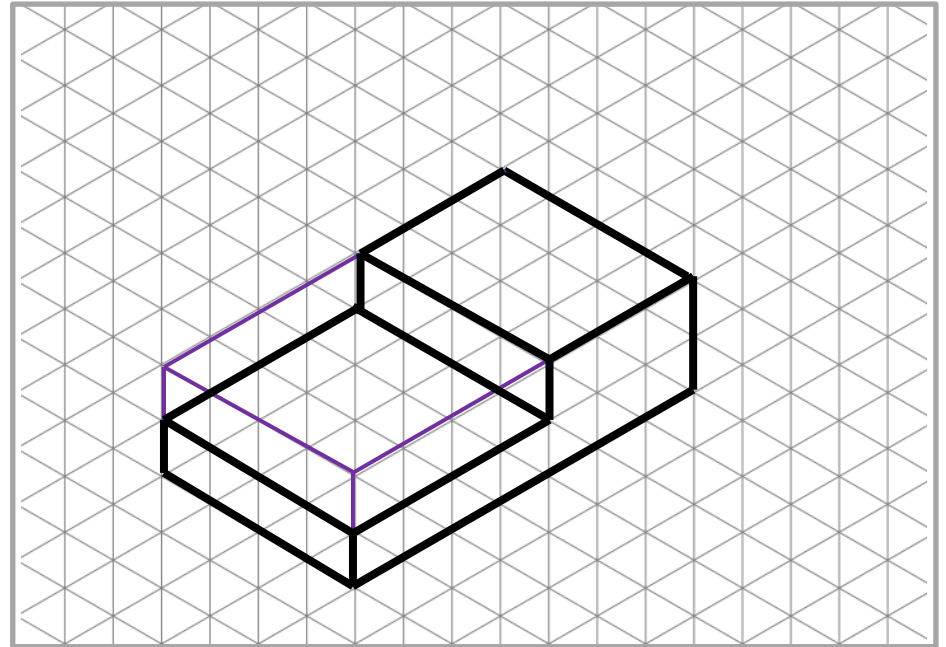
Perspectiva Isométrica - 1



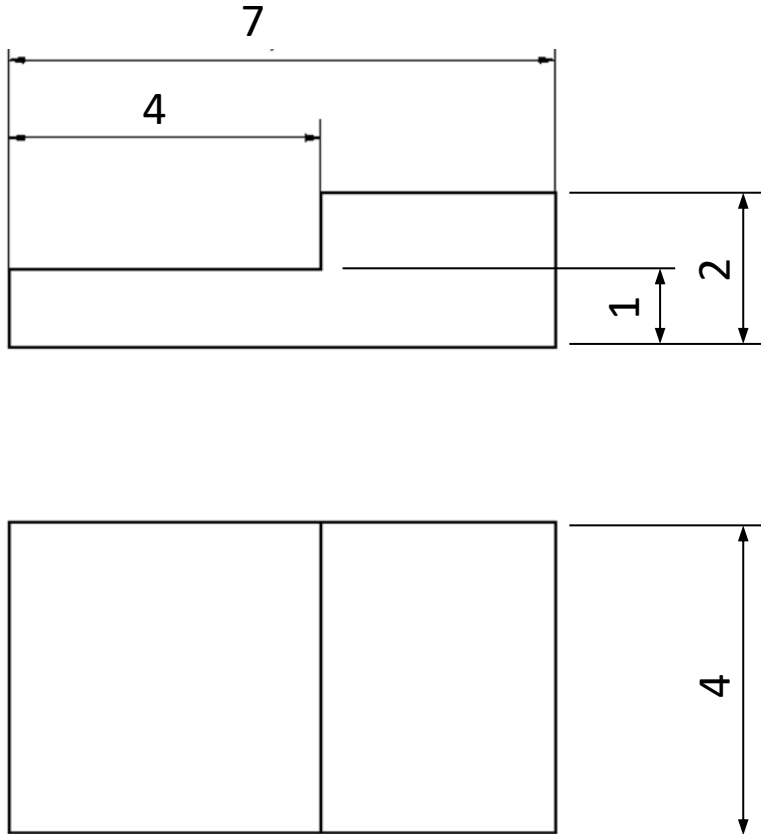
Vistas ortográficas



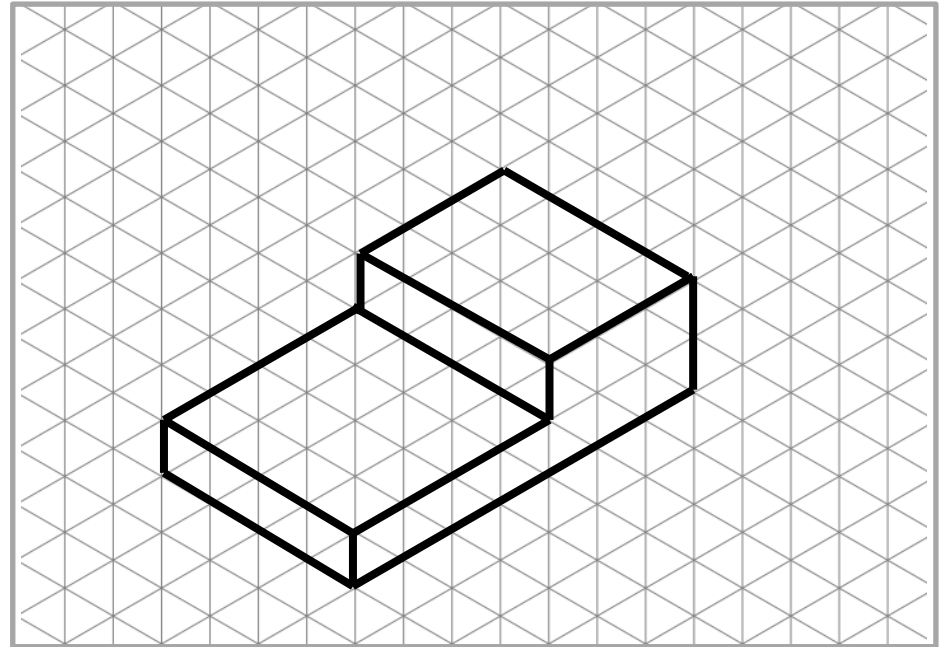
Perspectiva Isométrica - 1



Vistas ortográficas

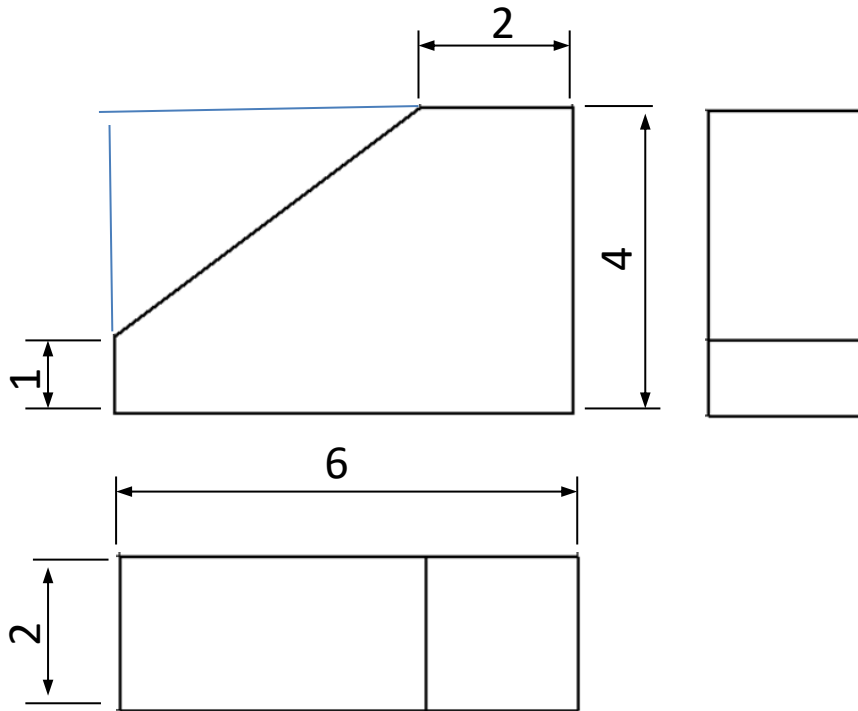


Perspectiva Isométrica - 1

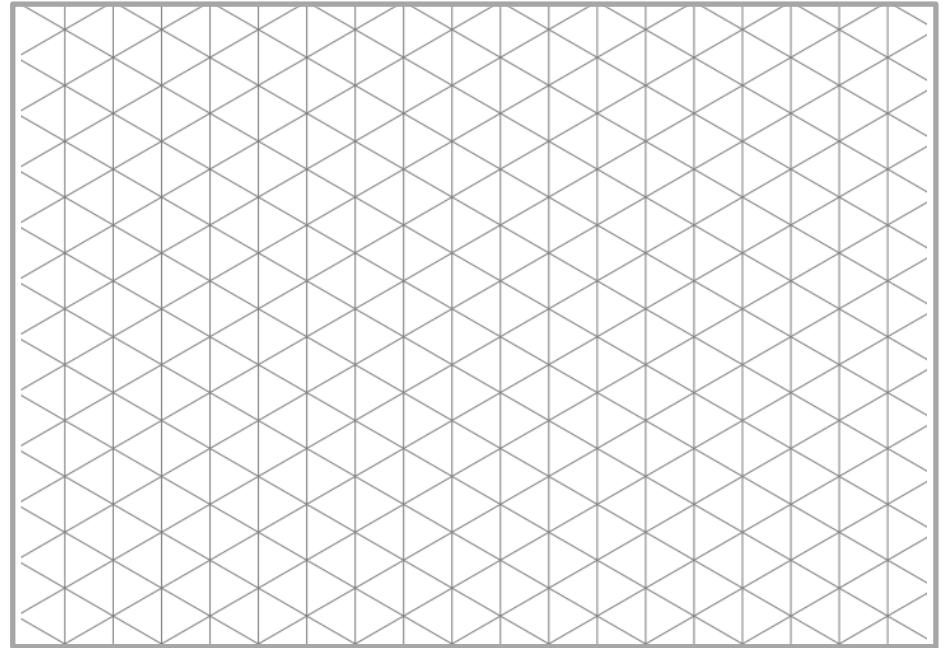


Apagando o prisma auxiliar

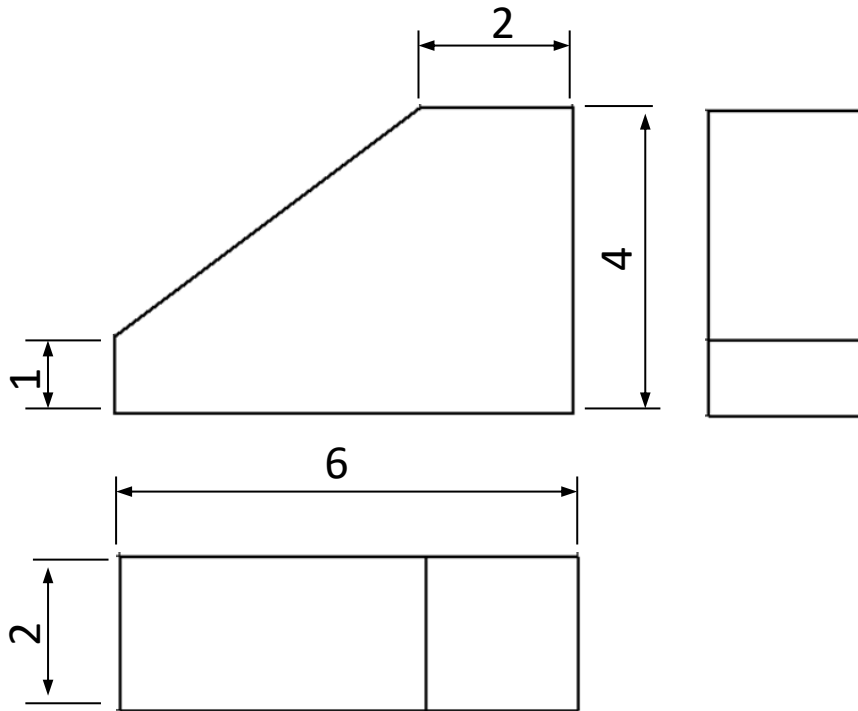
Vistas ortográficas



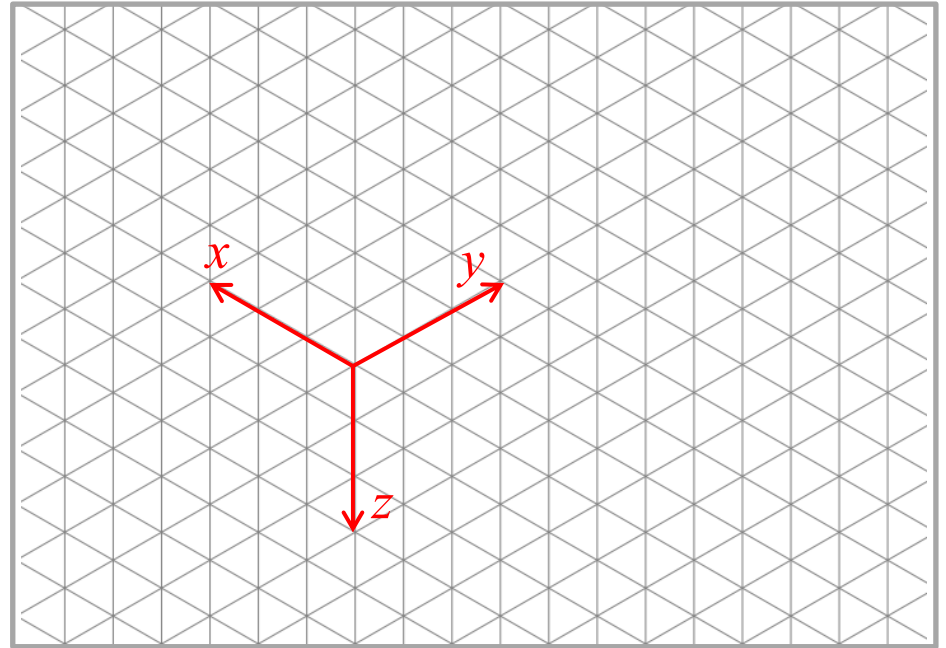
Perspectiva Isométrica - 2



Vistas ortográficas

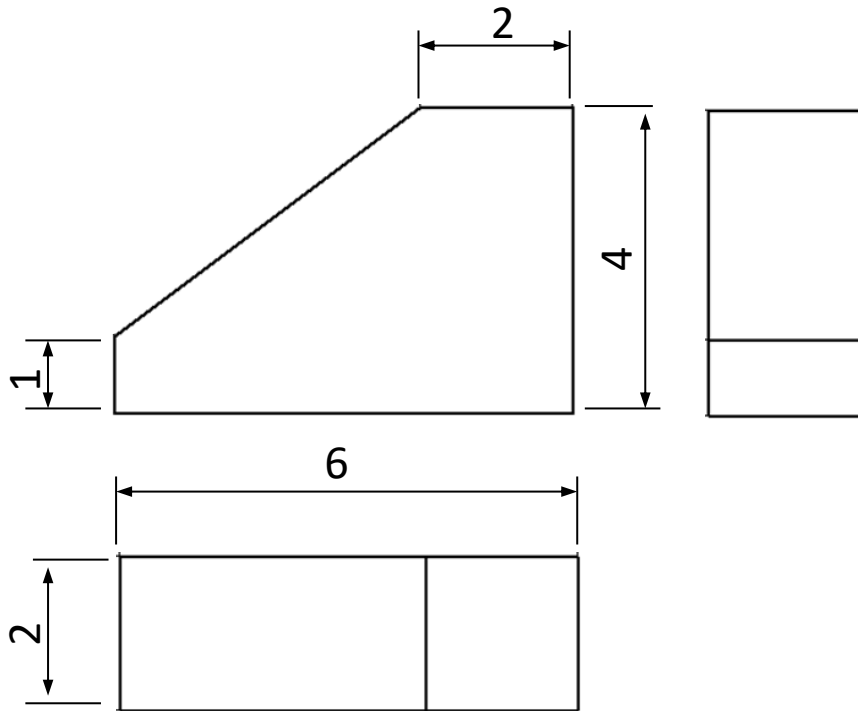


Perspectiva Isométrica - 2

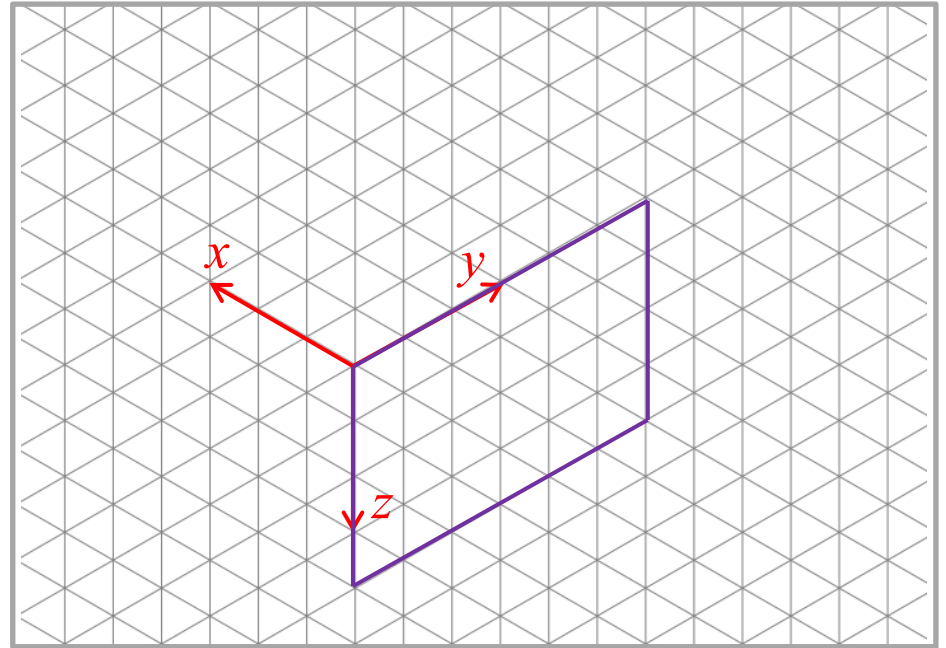


Eixos isométricos

Vistas ortográficas

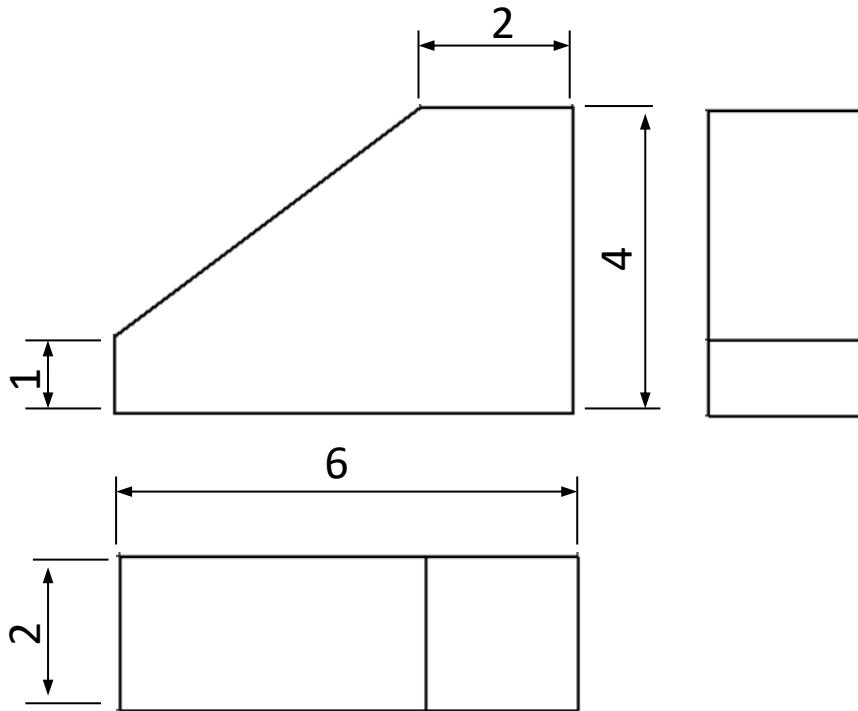


Perspectiva Isométrica - 2

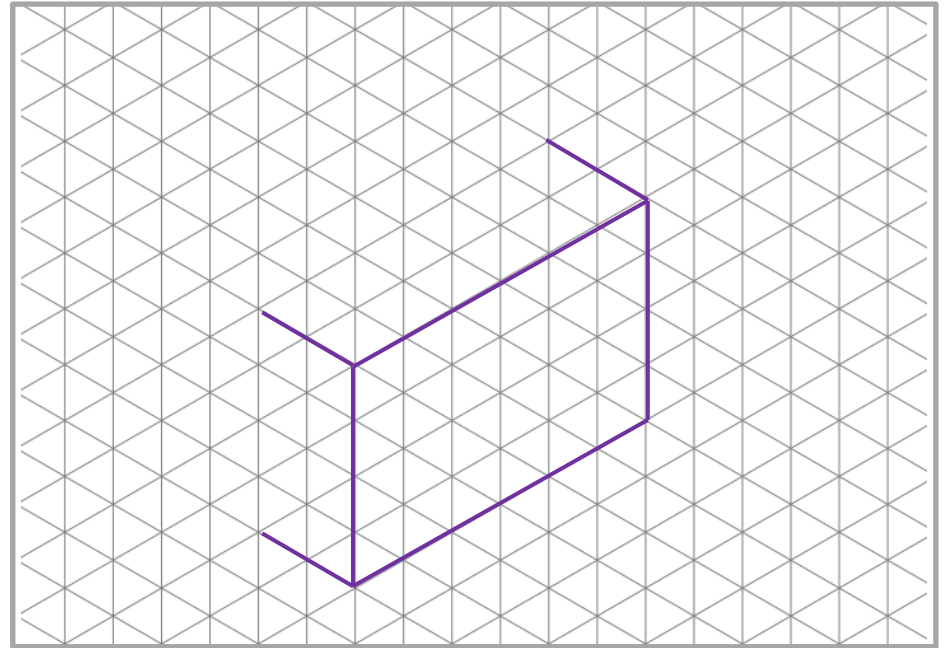


Desenhando o prisma auxiliar

Vistas ortográficas

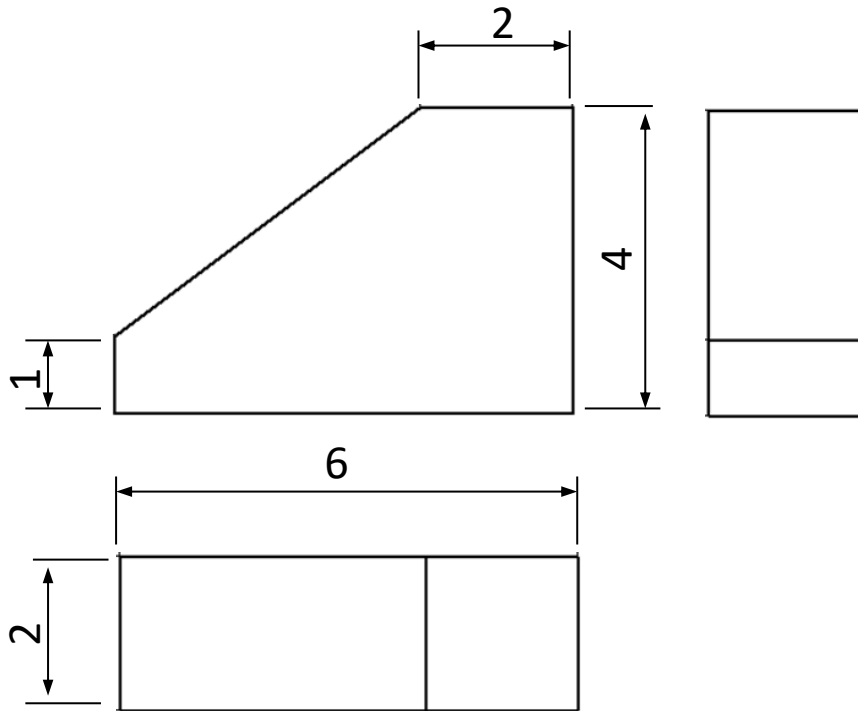


Perspectiva Isométrica - 2

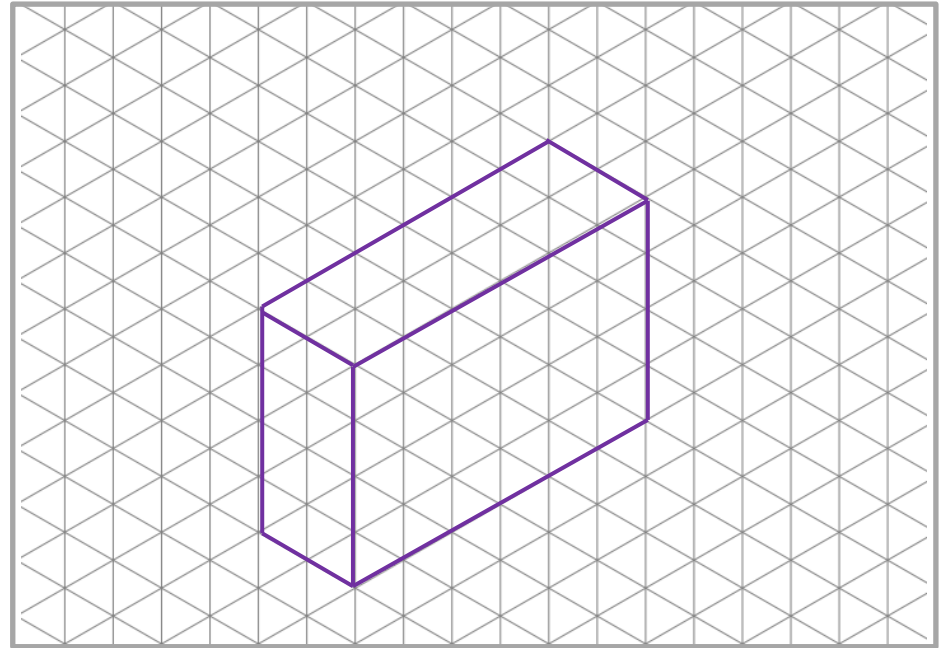


Desenhando o prisma auxiliar

Vistas ortográficas

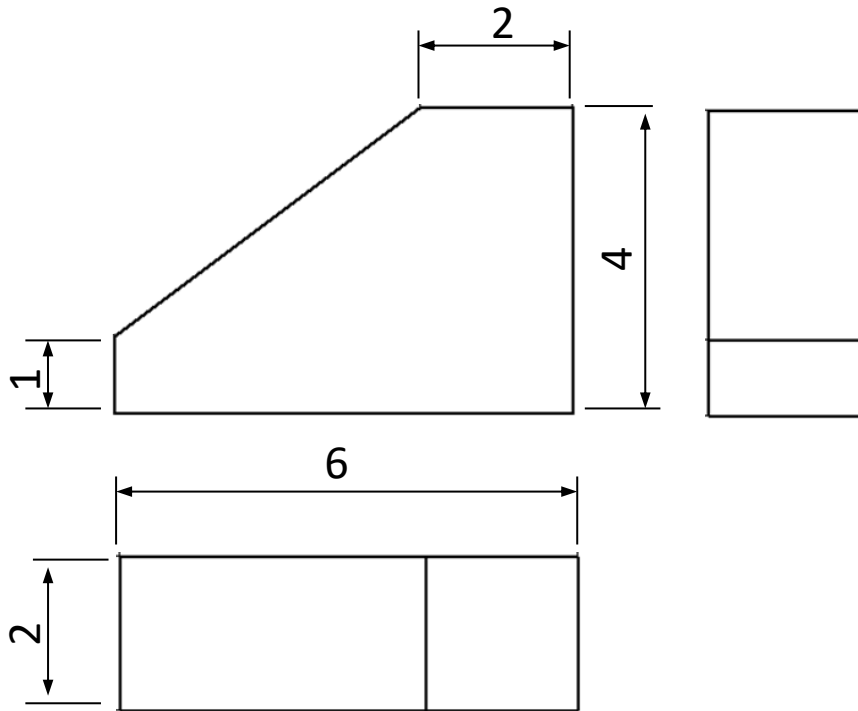


Perspectiva Isométrica - 2

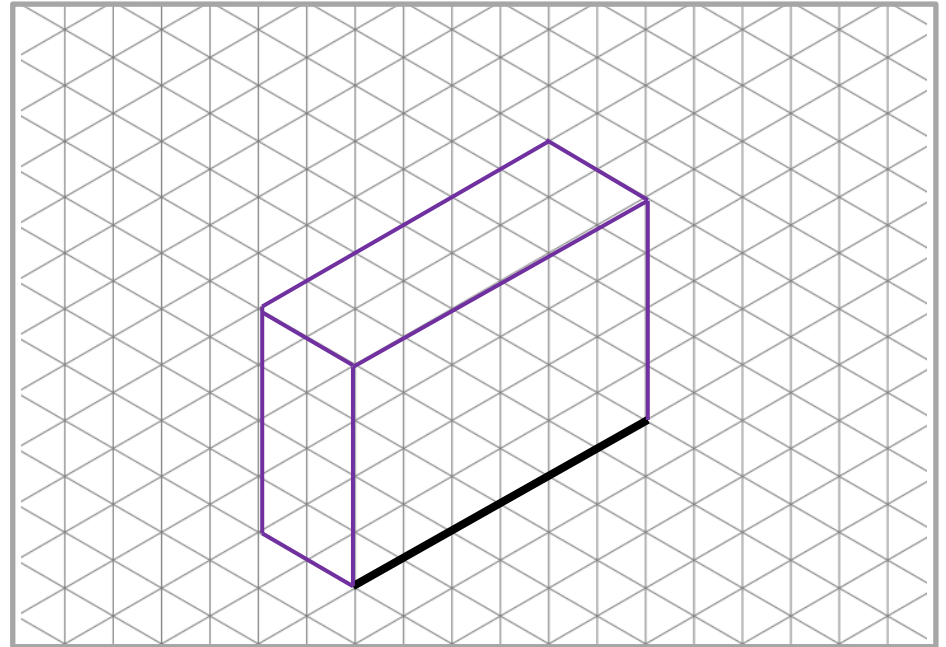


Desenhando o prisma auxiliar

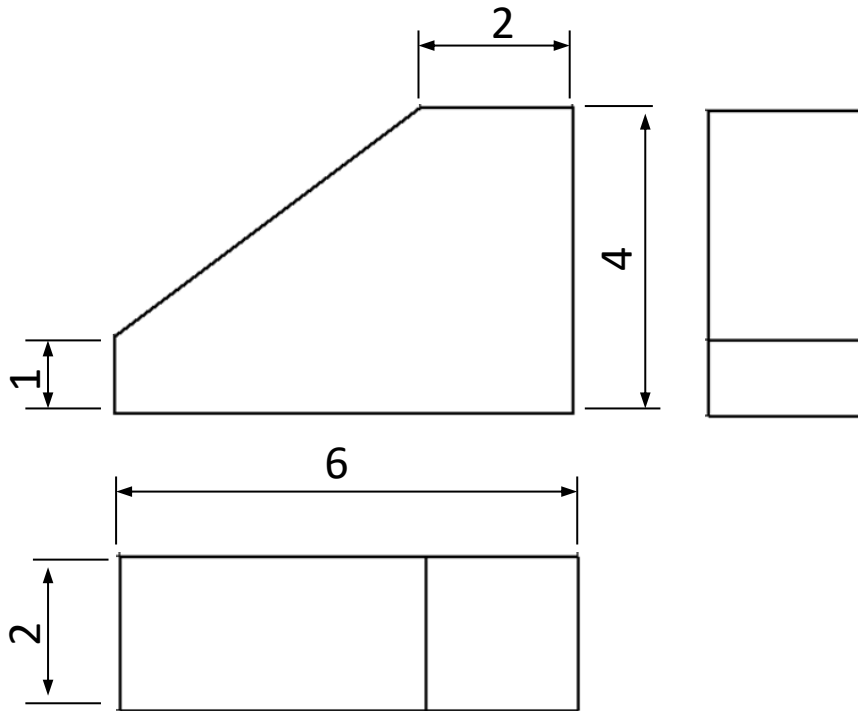
Vistas ortográficas



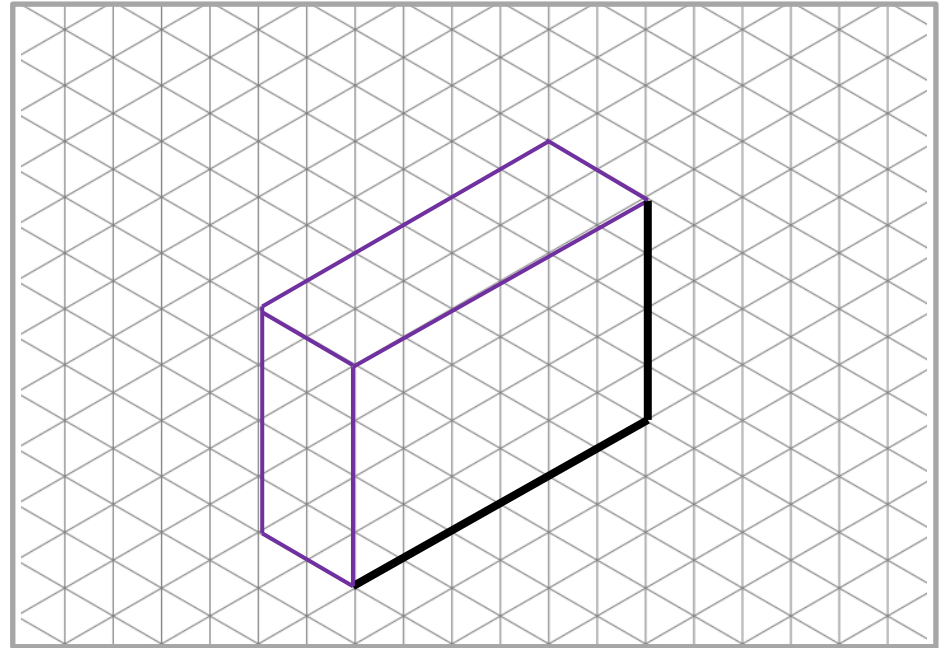
Perspectiva Isométrica - 2



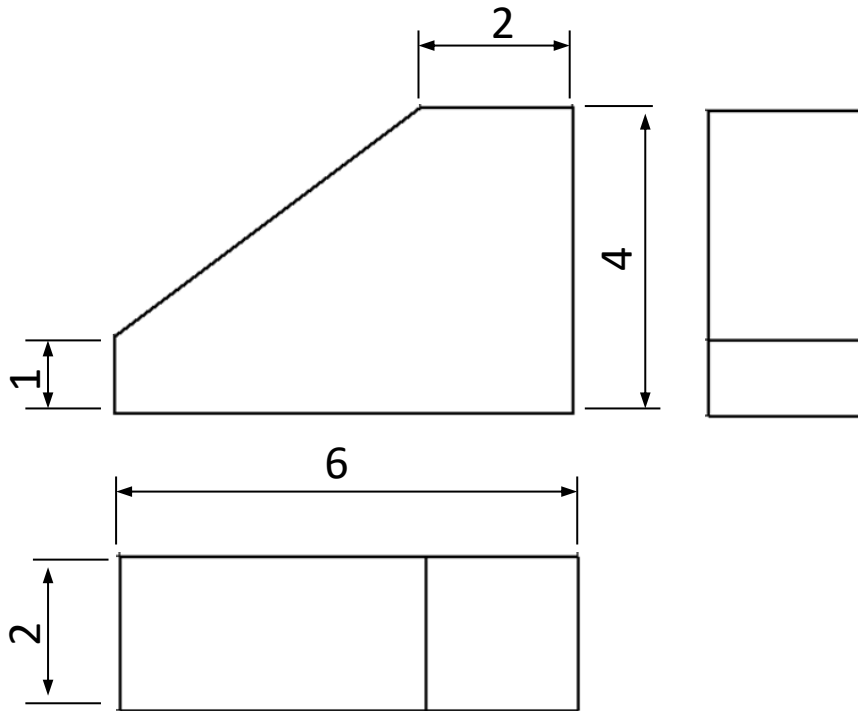
Vistas ortográficas



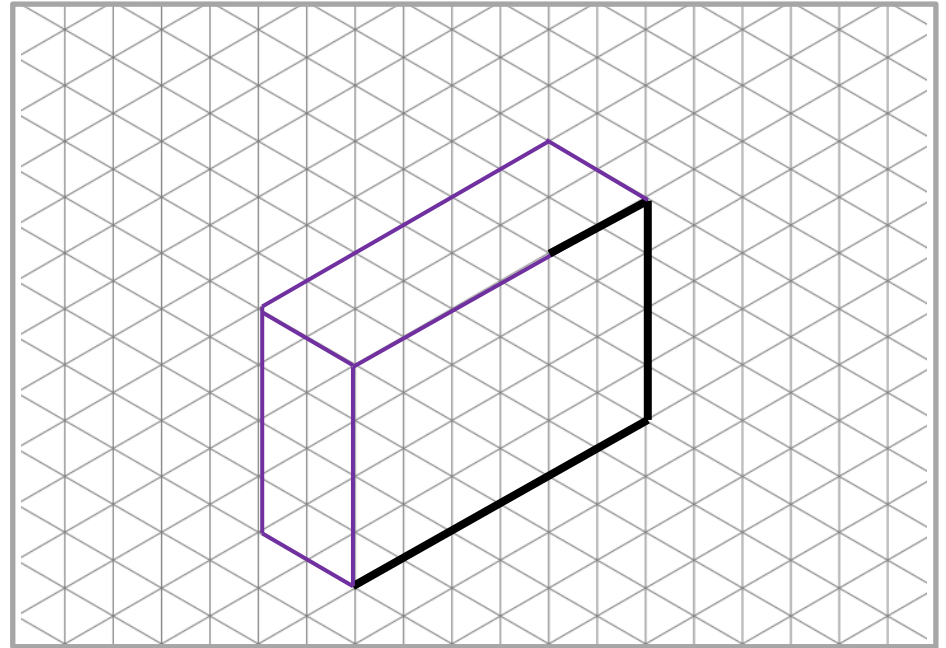
Perspectiva Isométrica - 2



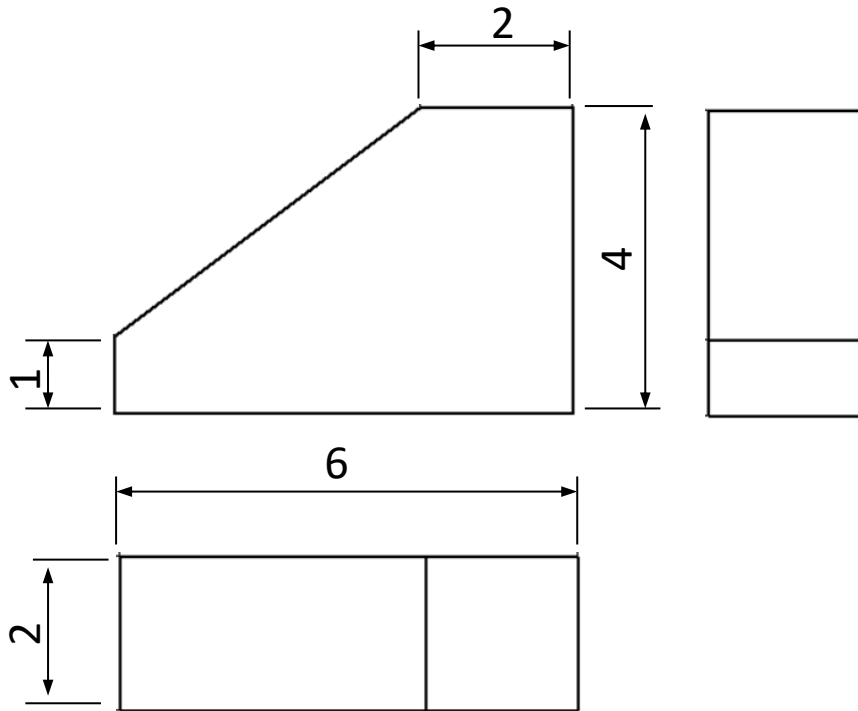
Vistas ortográficas



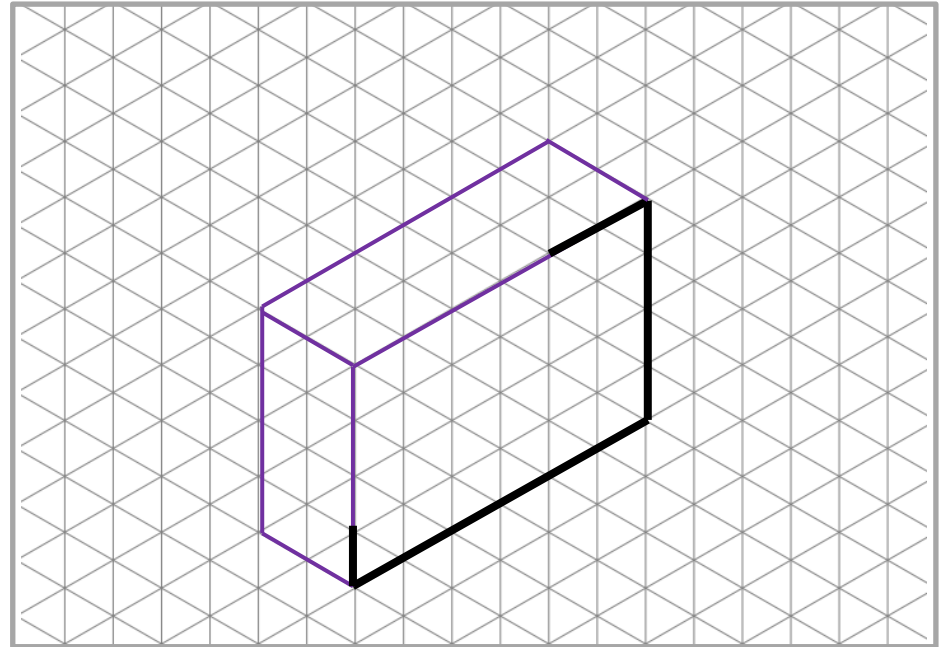
Perspectiva Isométrica - 2



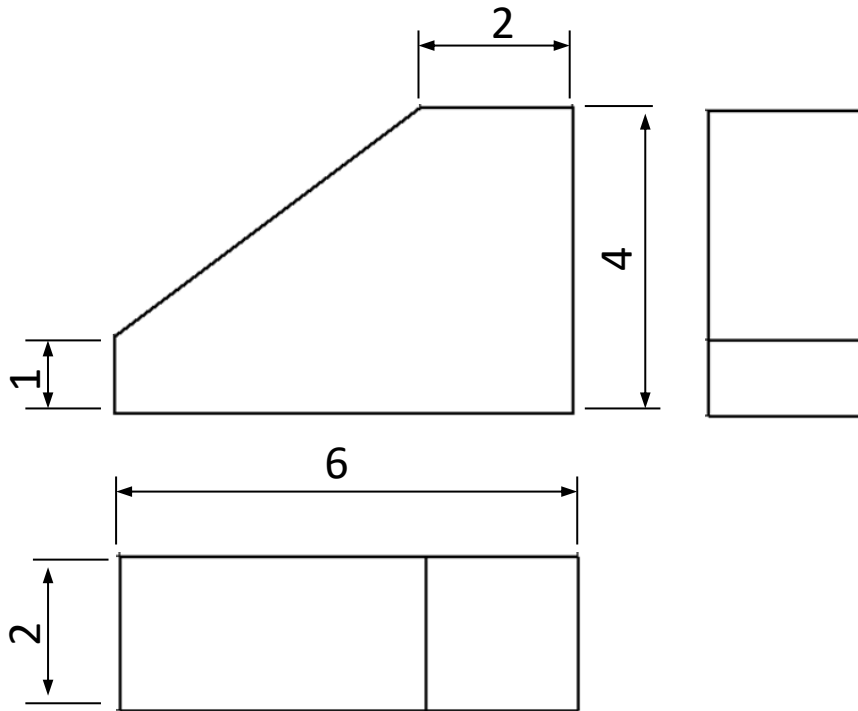
Vistas ortográficas



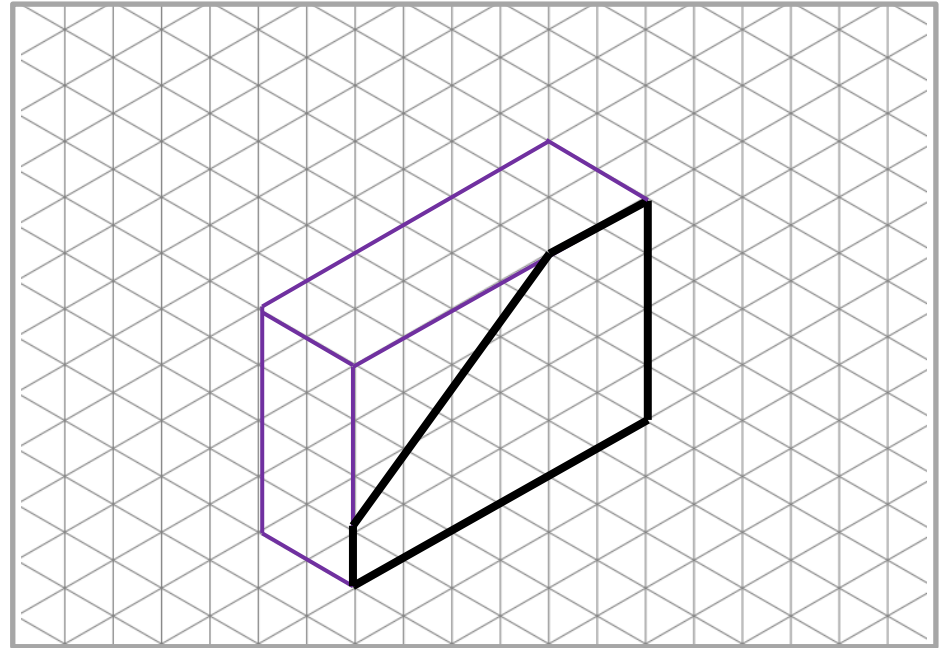
Perspectiva Isométrica - 2



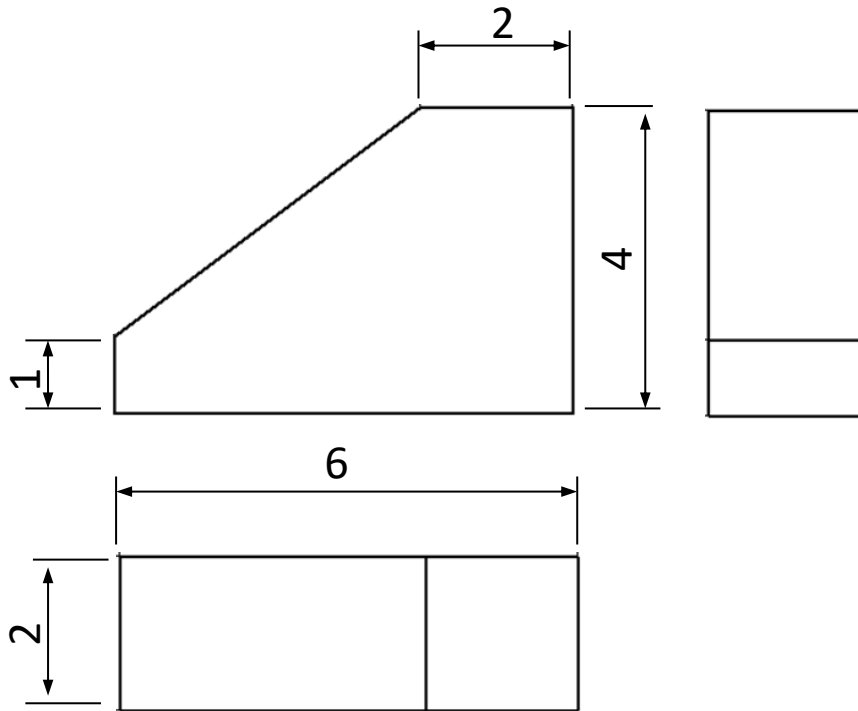
Vistas ortográficas



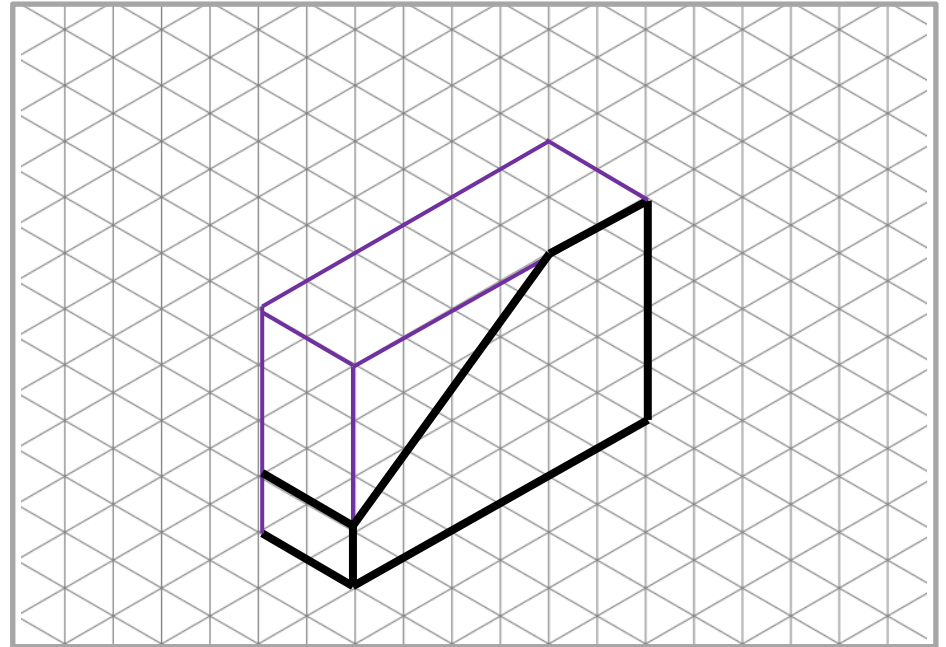
Perspectiva Isométrica - 2



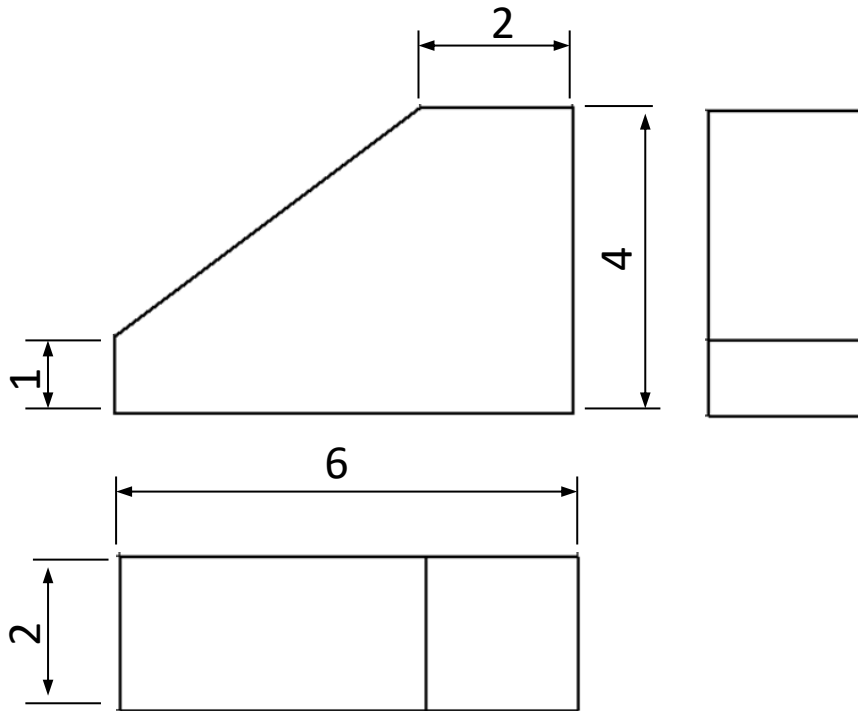
Vistas ortográficas



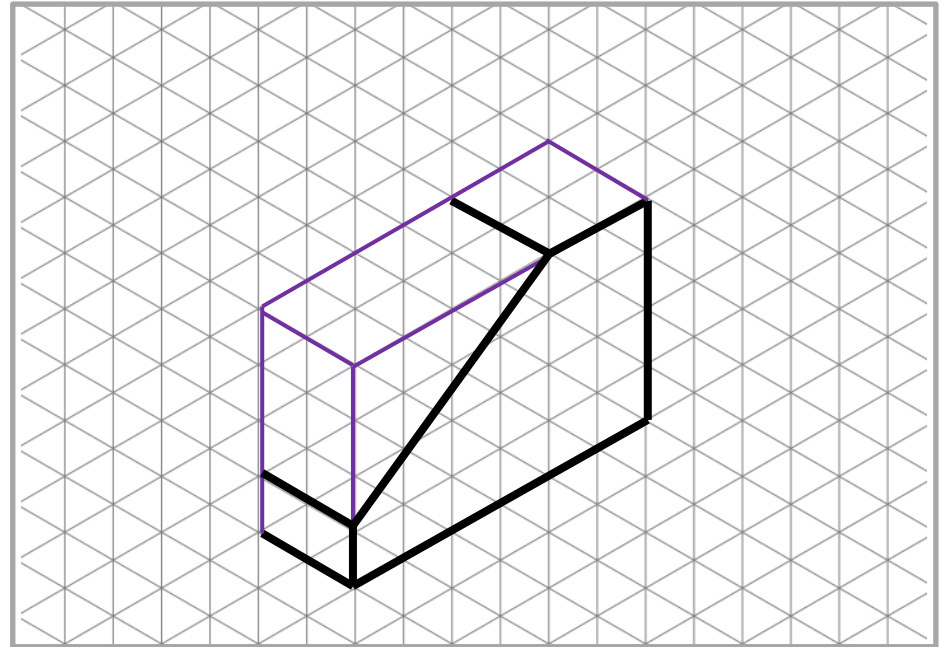
Perspectiva Isométrica - 2



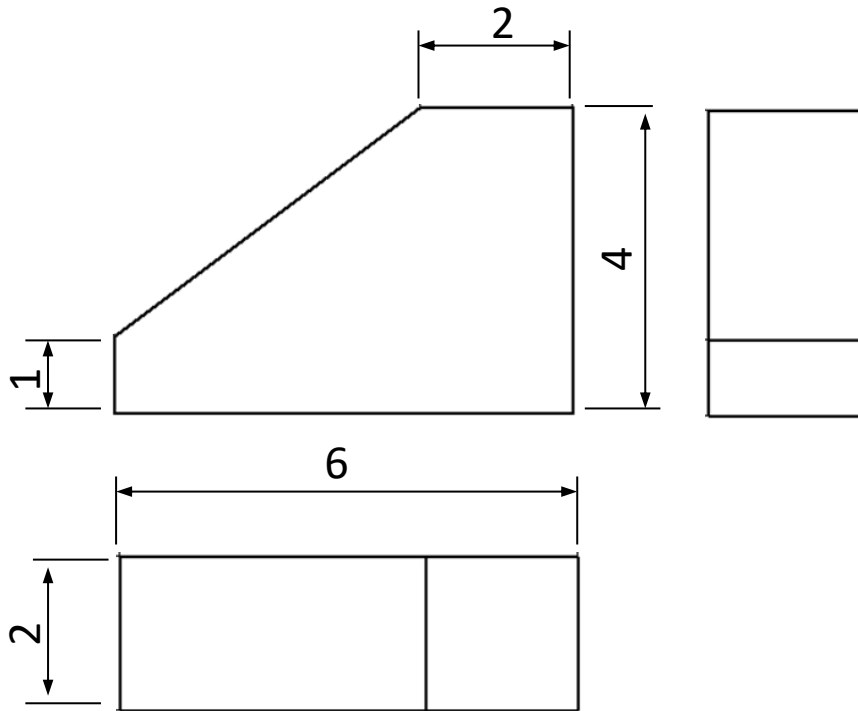
Vistas ortográficas



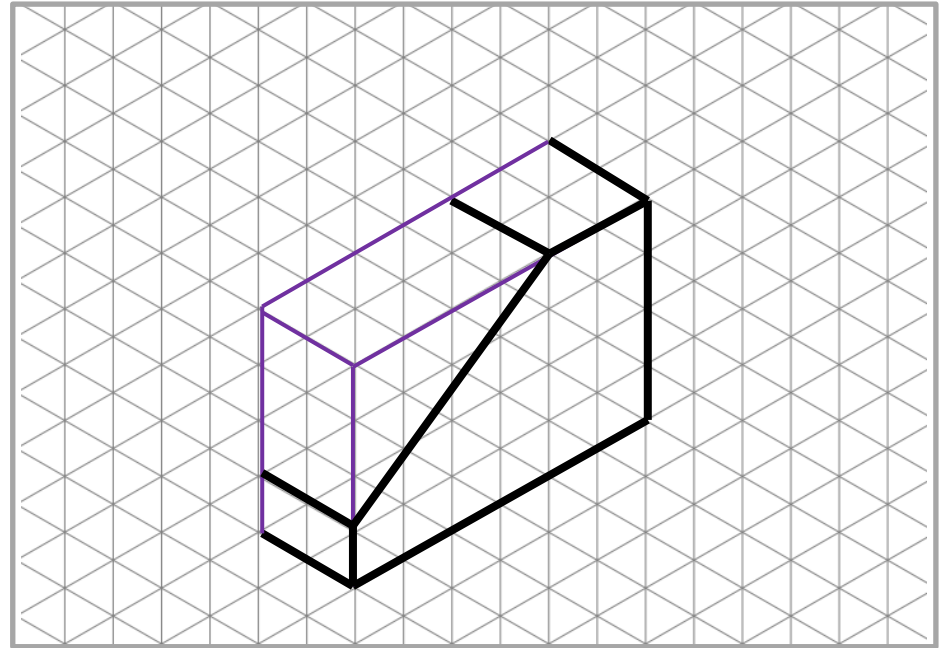
Perspectiva Isométrica - 2



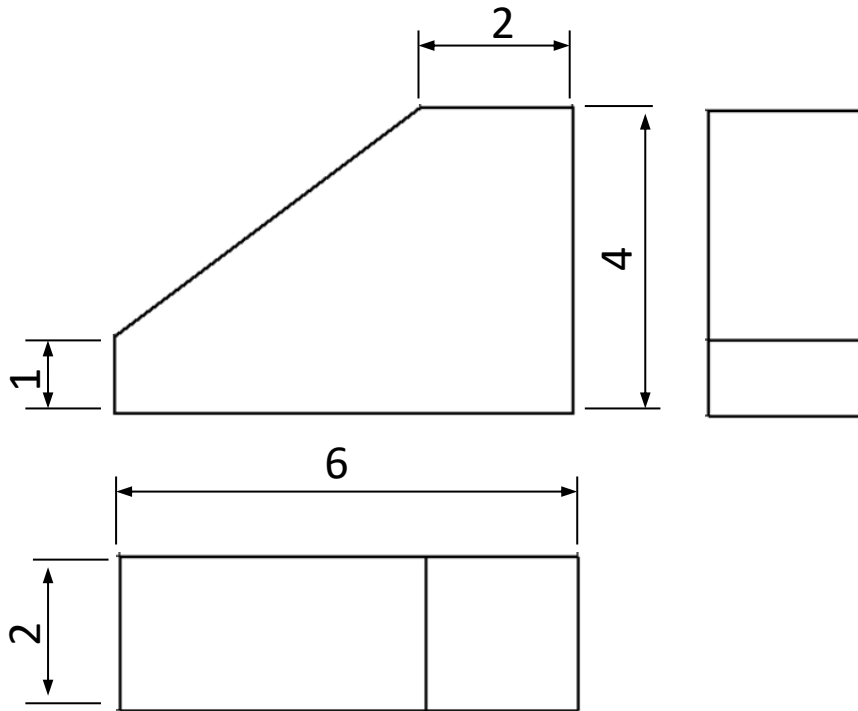
Vistas ortográficas



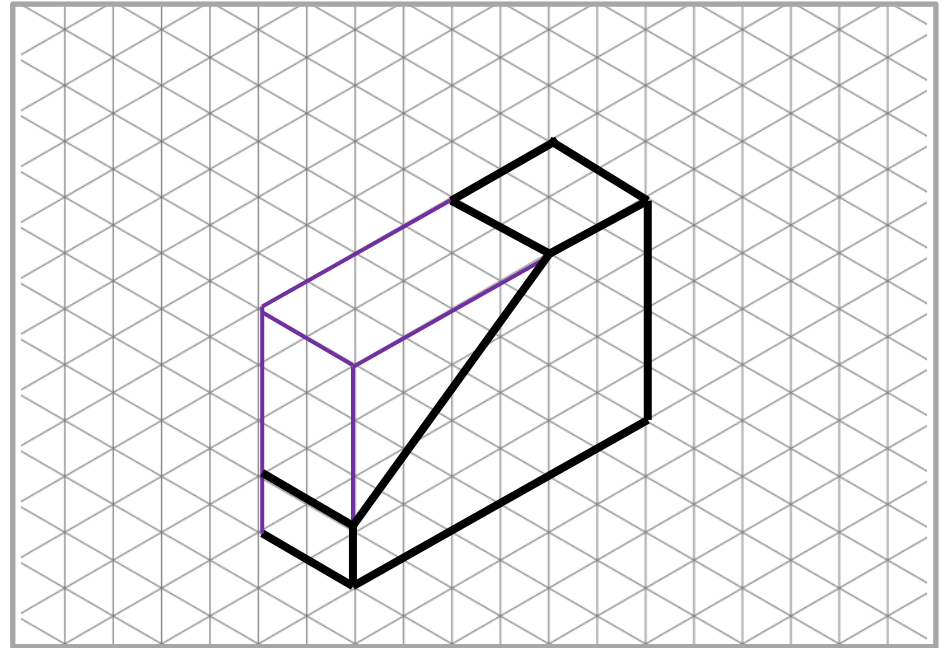
Perspectiva Isométrica - 2



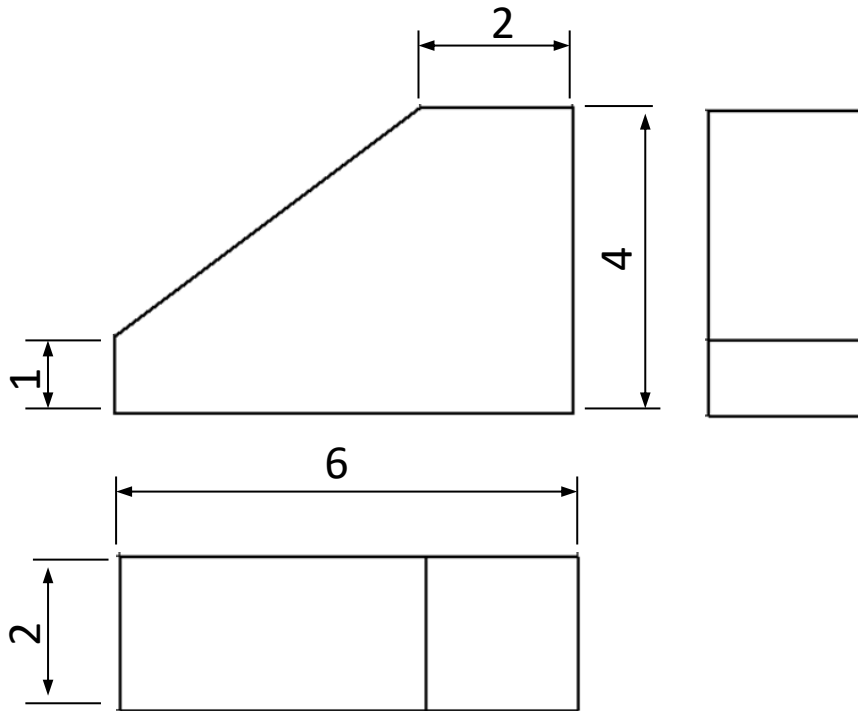
Vistas ortográficas



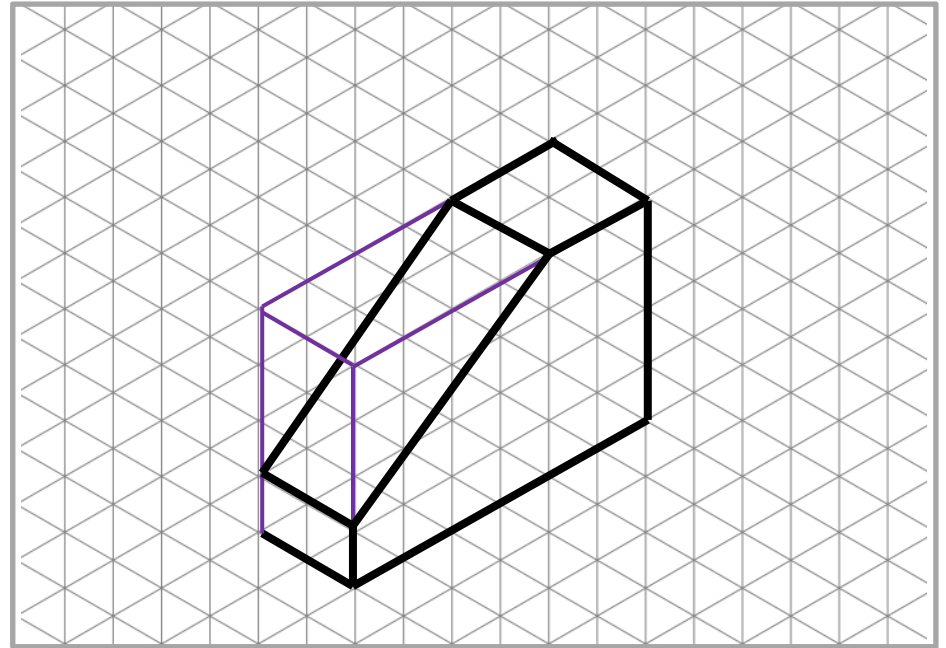
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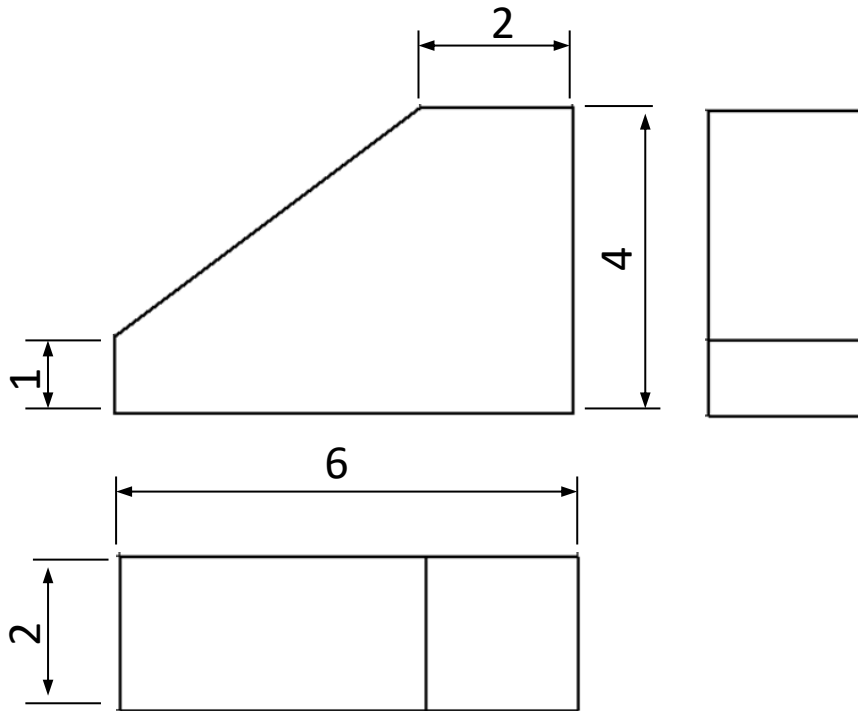
Vistas ortográficas



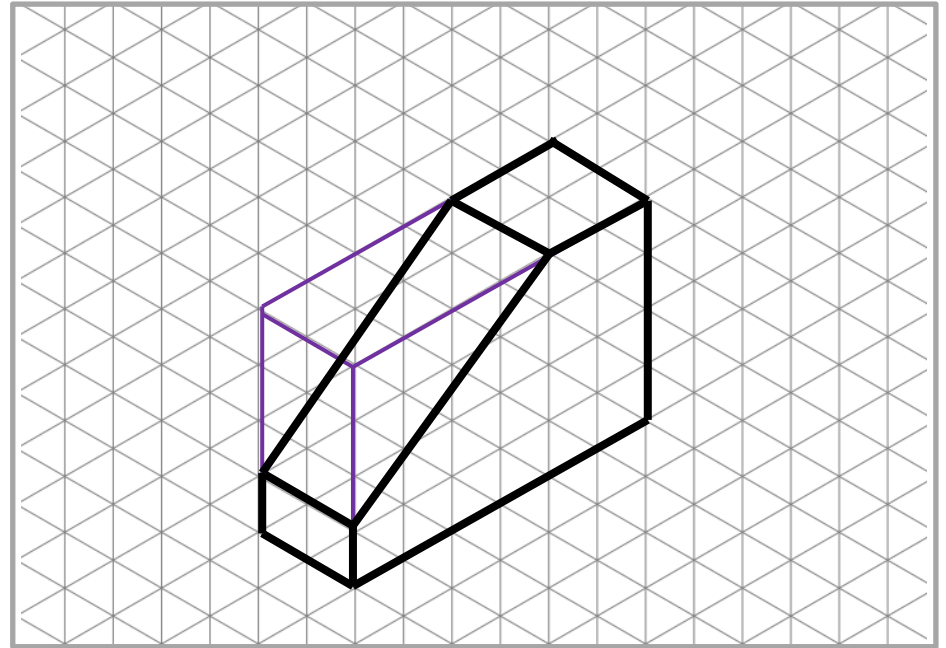
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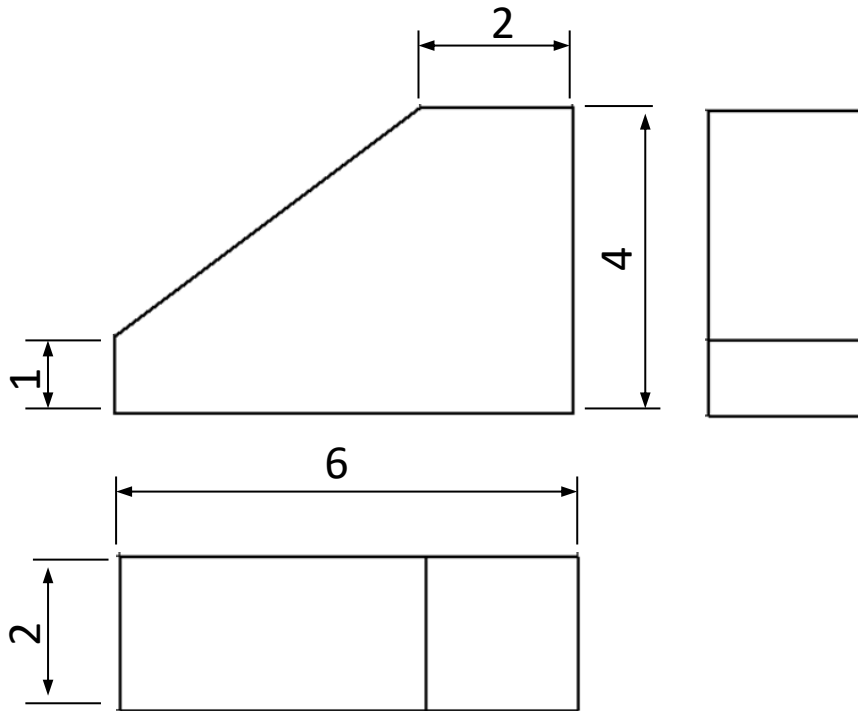
Vistas ortográficas



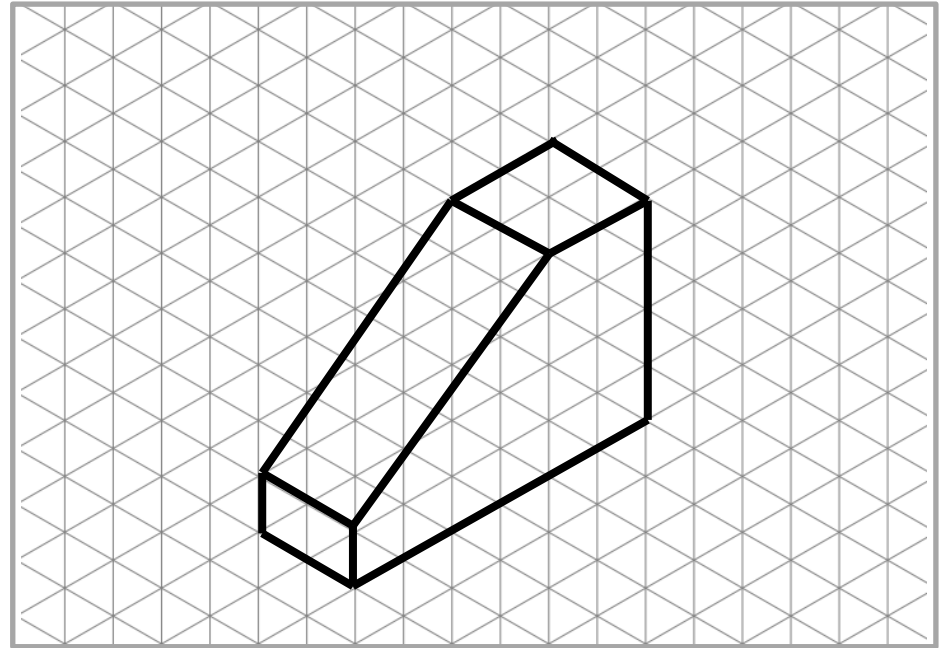
Perspectiva Isométrica - 2



Vistas ortográficas

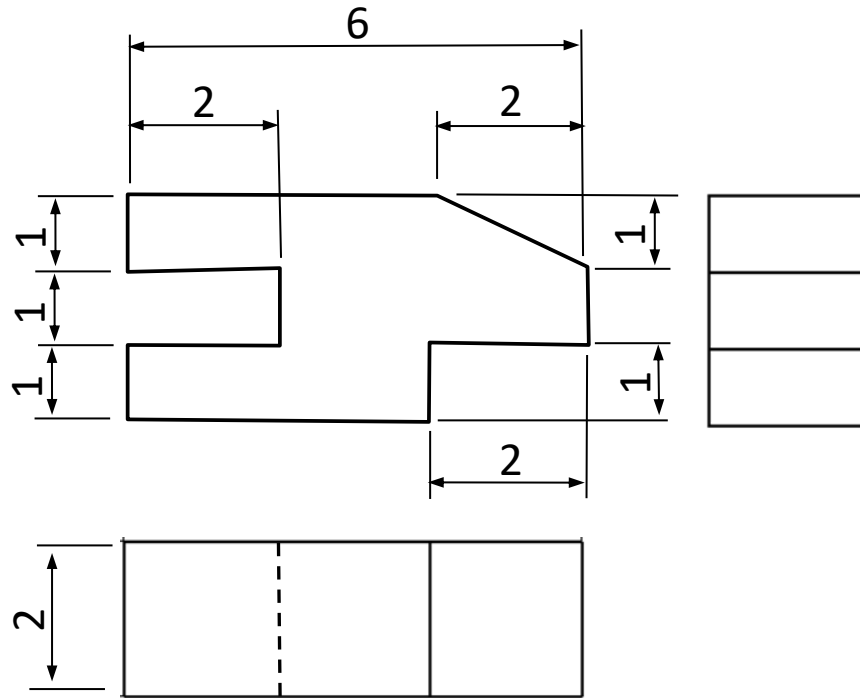


Perspectiva Isométrica - 2

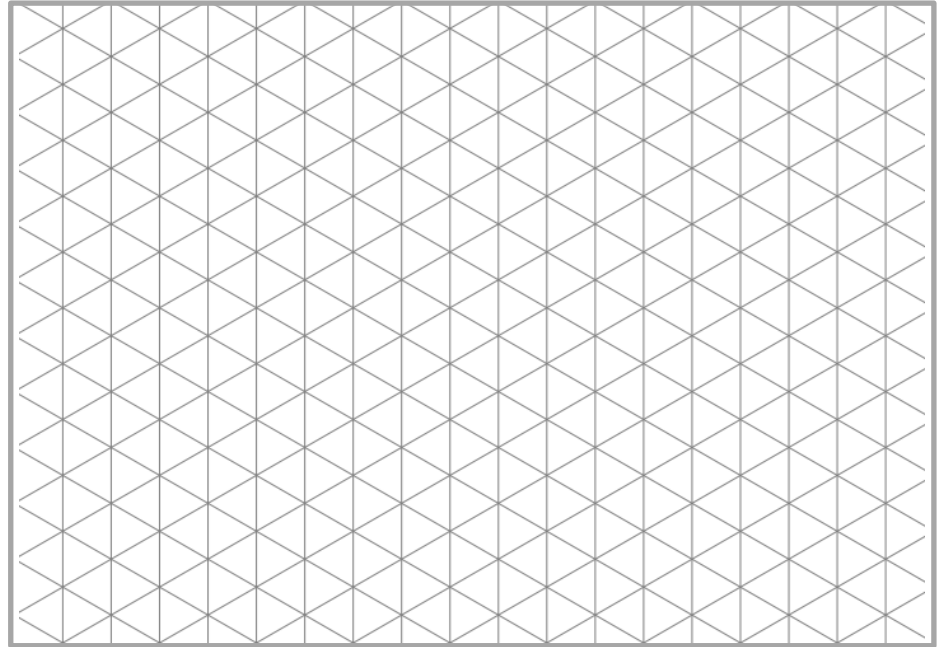


Apagando o prisma auxiliar

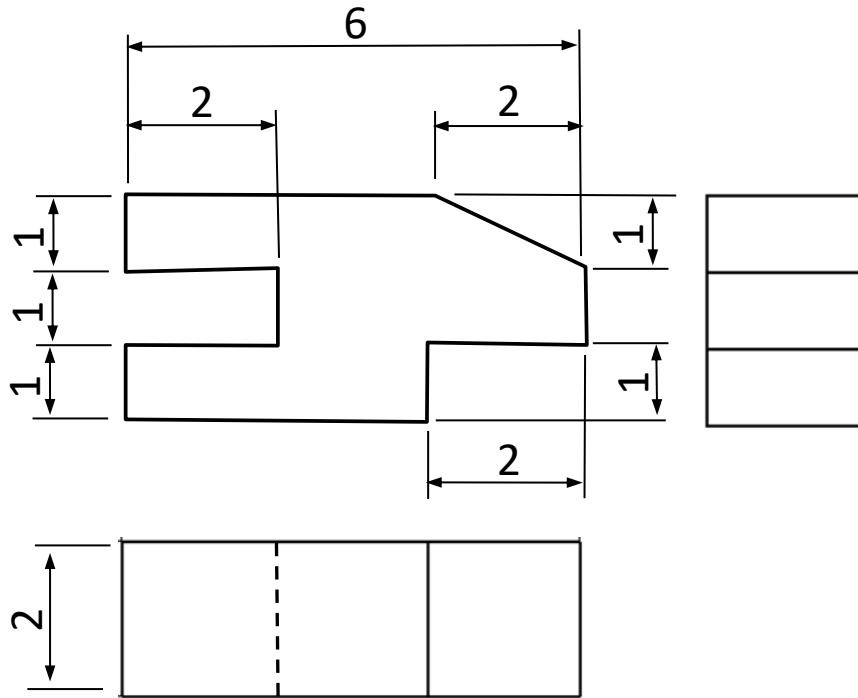
Vistas ortográficas



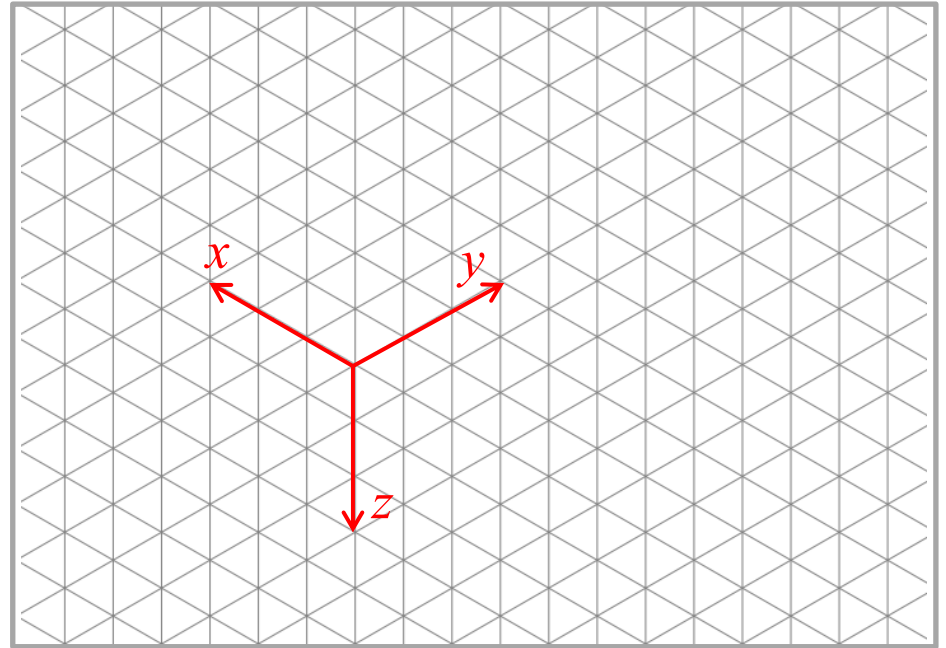
Perspectiva Isométrica - 3



Vistas ortográficas

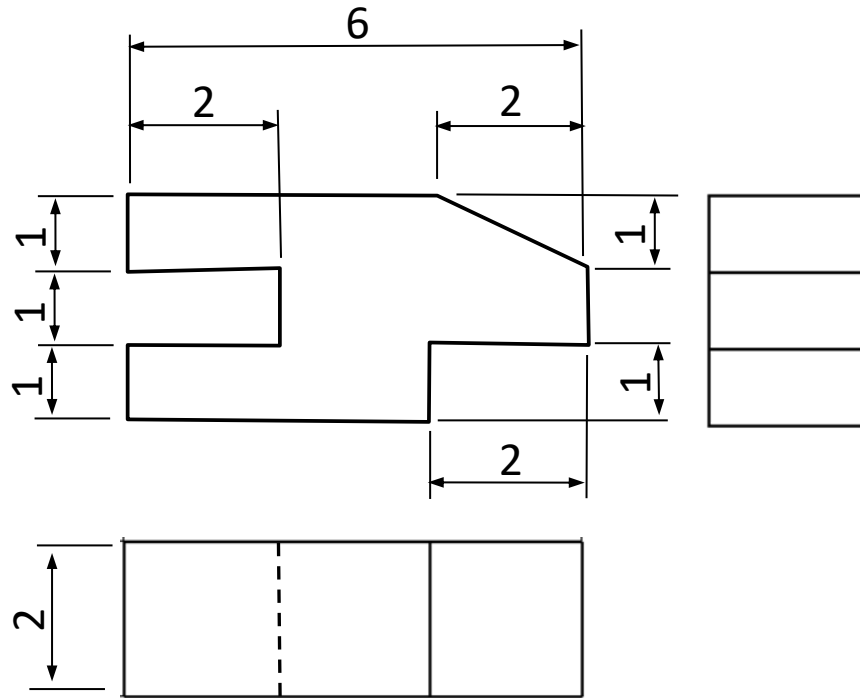


Perspectiva Isométrica - 3

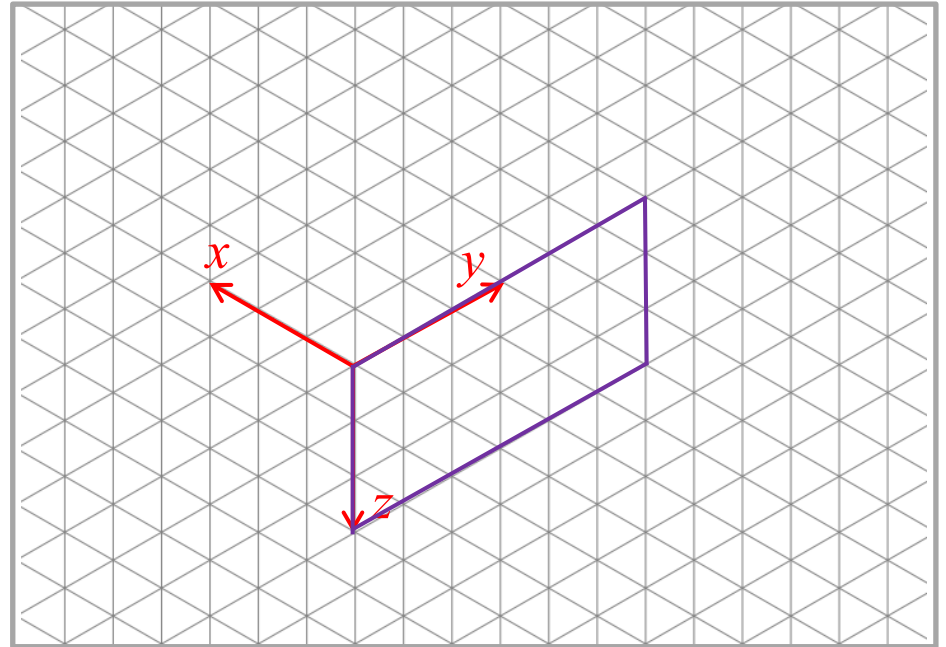


Eixos isométricos

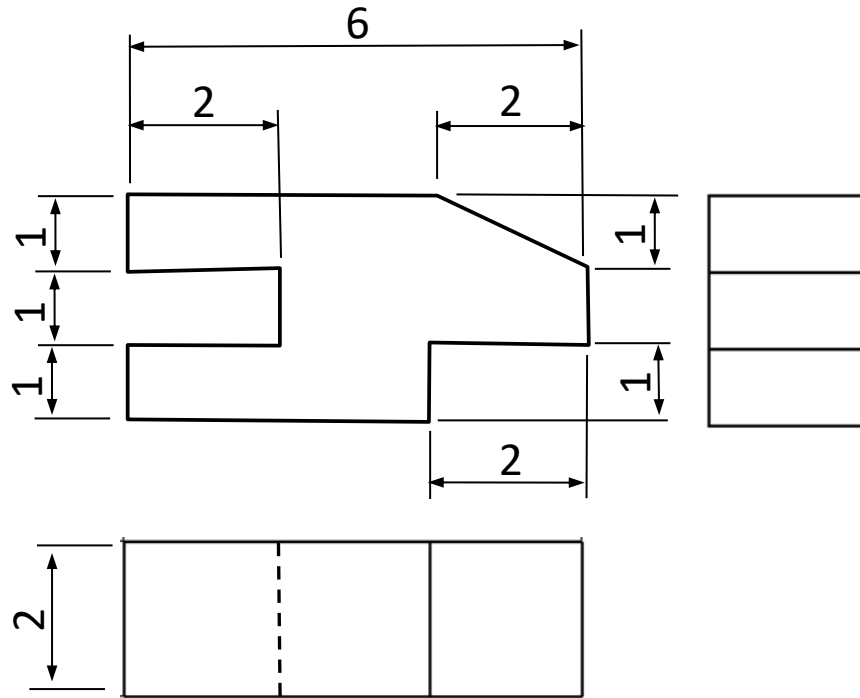
Vistas ortográficas



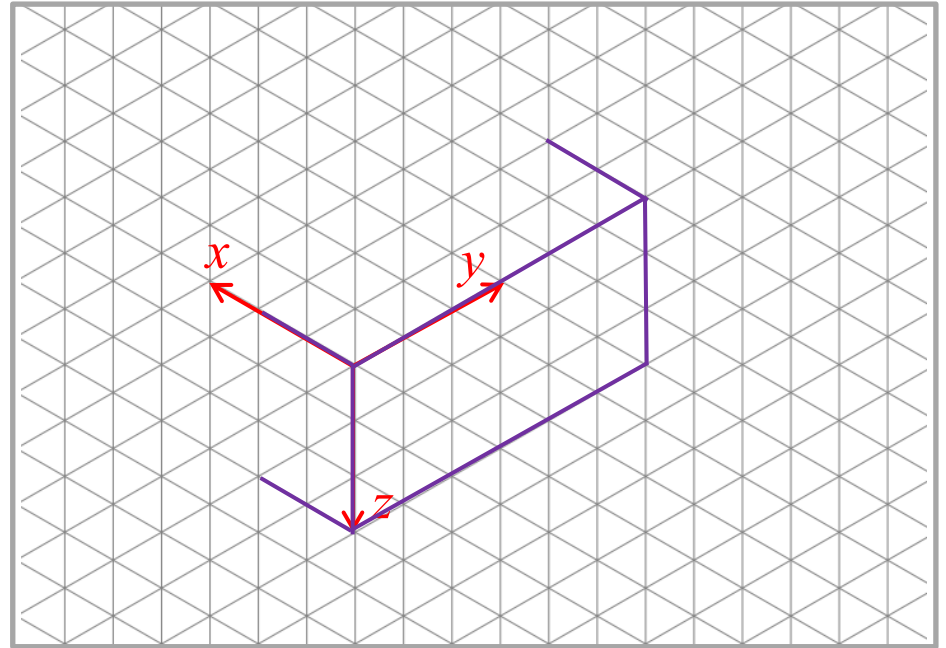
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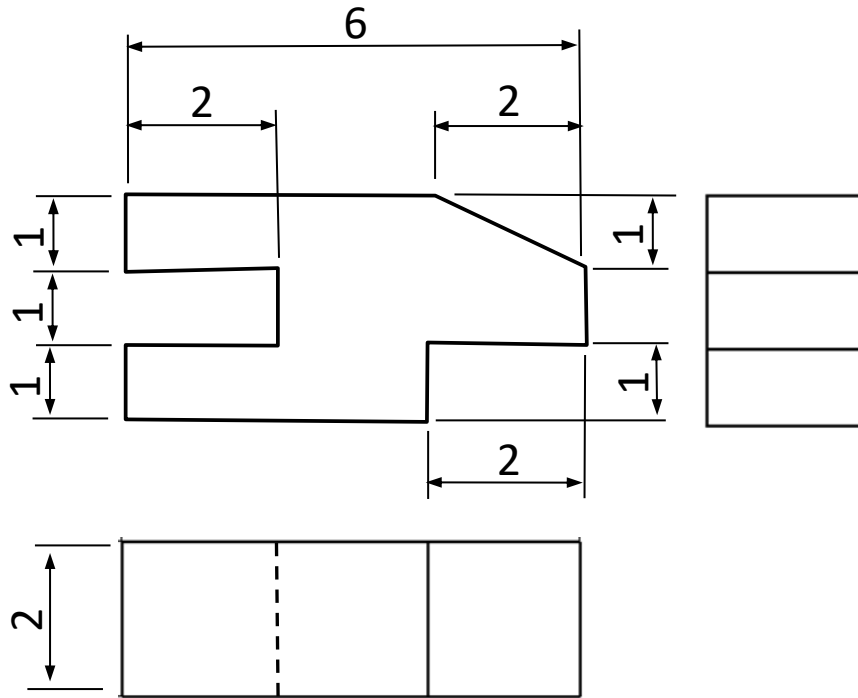
Vistas ortográficas



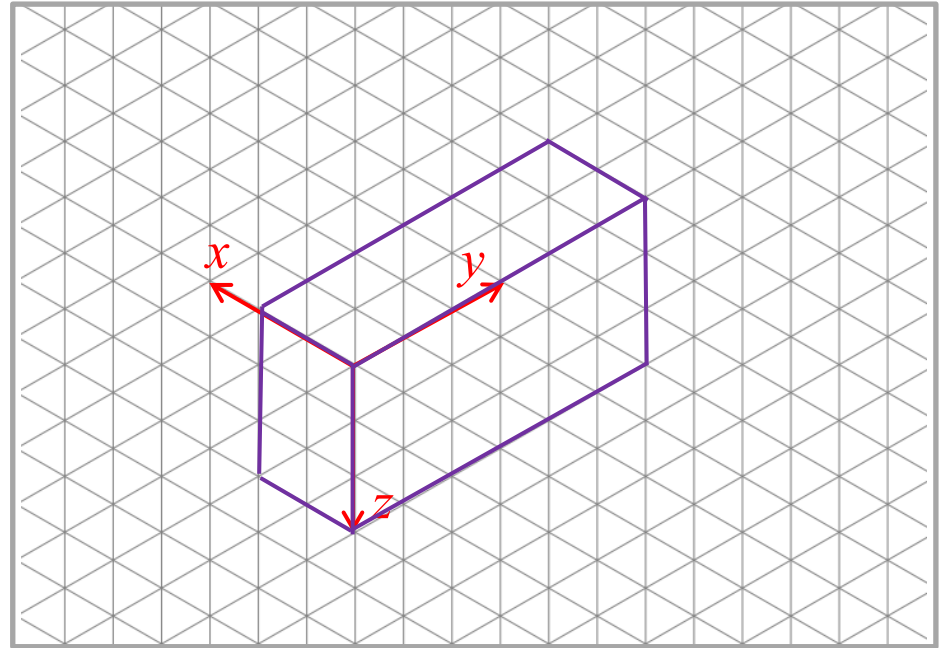
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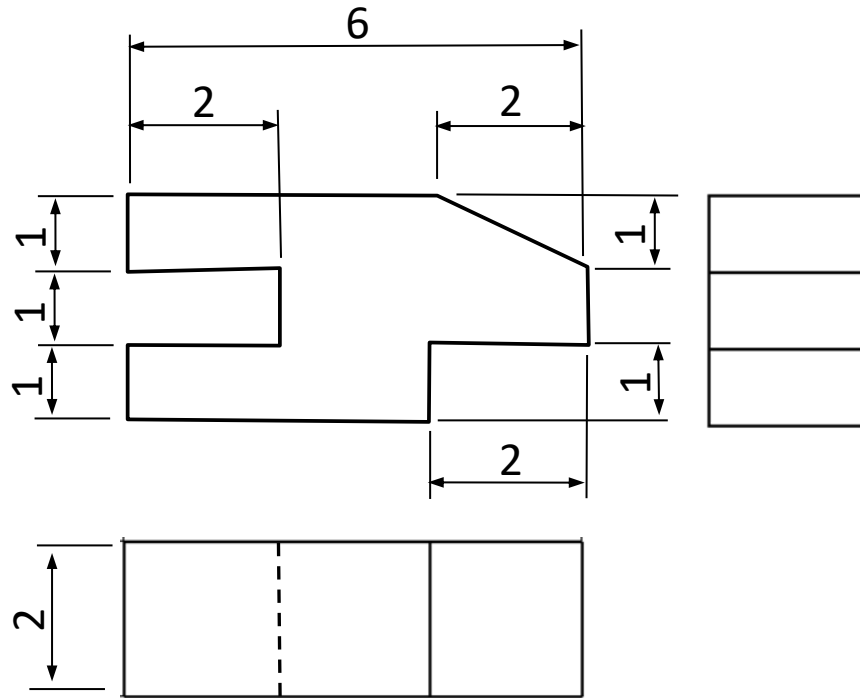
Vistas ortográficas



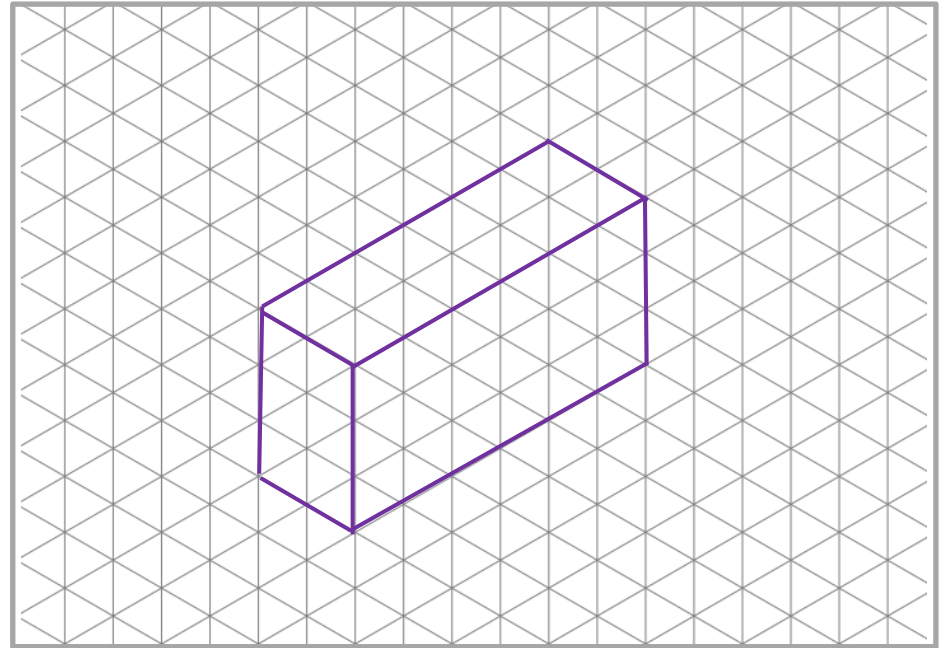
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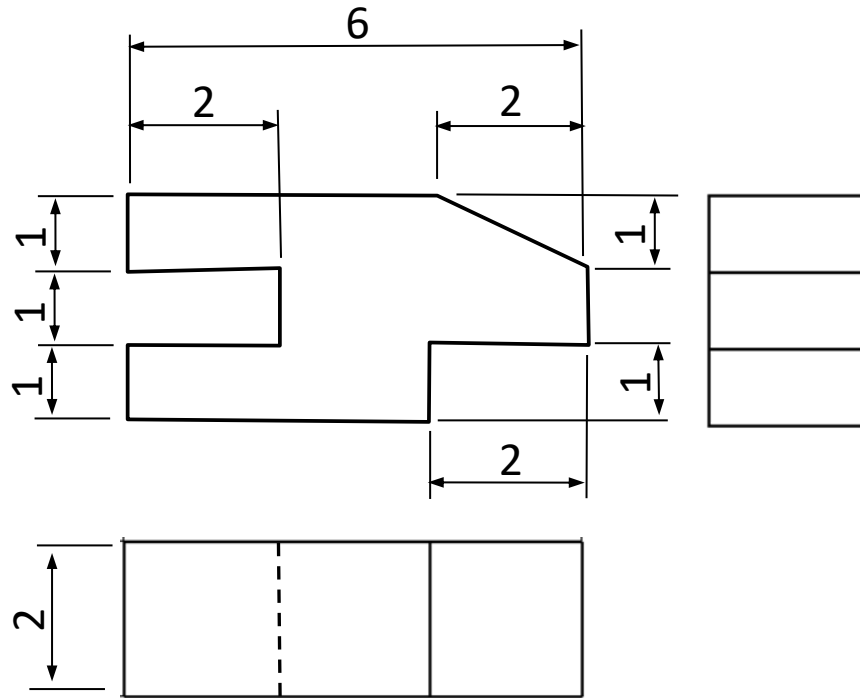
Vistas ortográficas



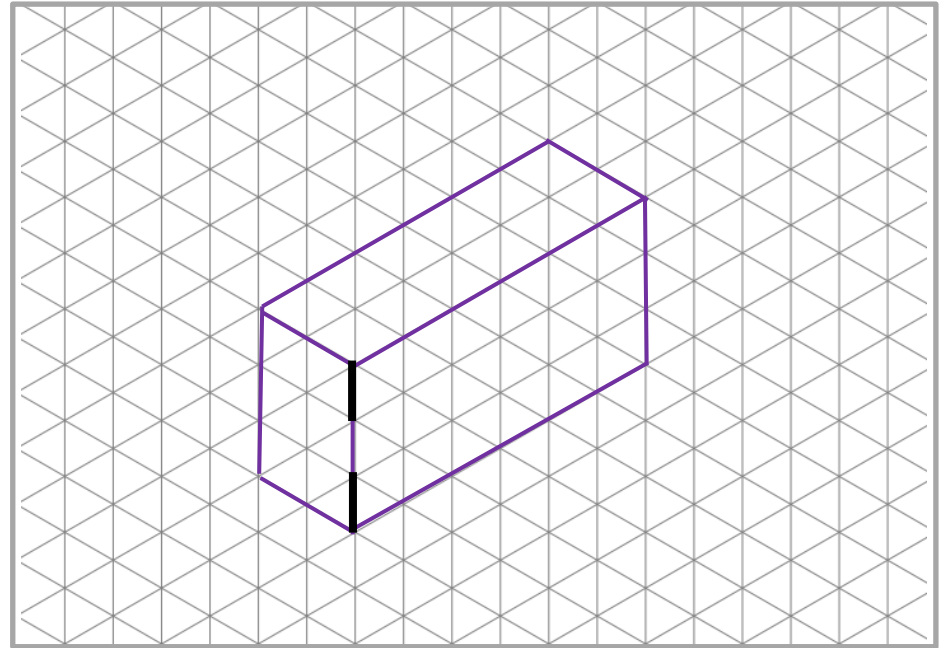
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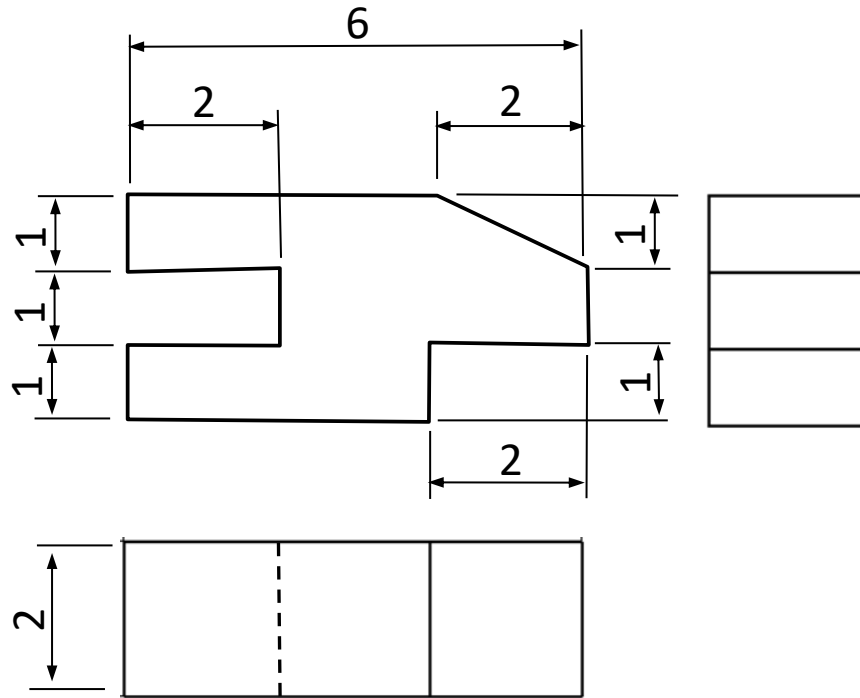
Vistas ortográficas



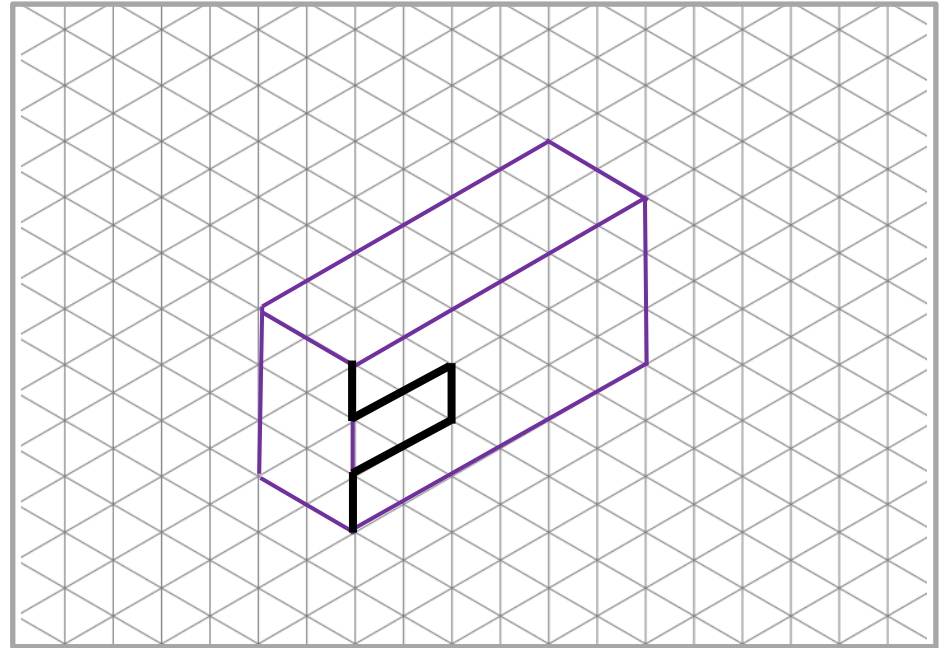
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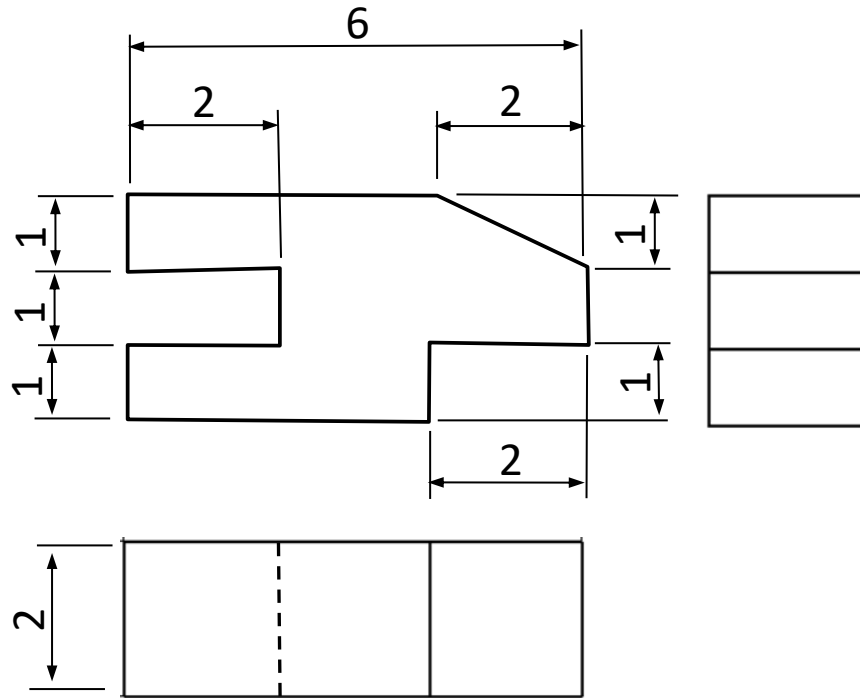
Vistas ortográficas



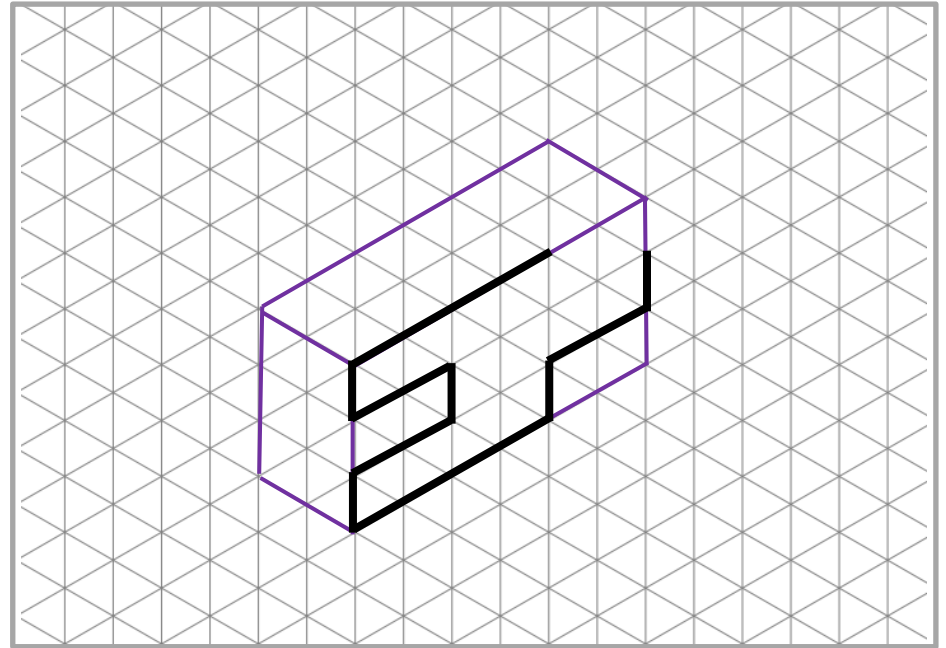
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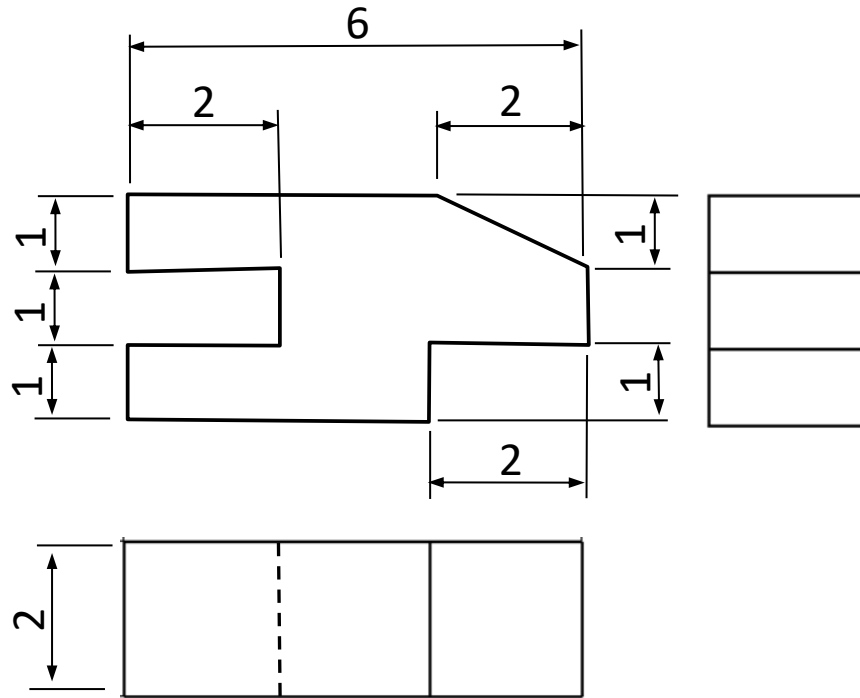
Vistas ortográficas



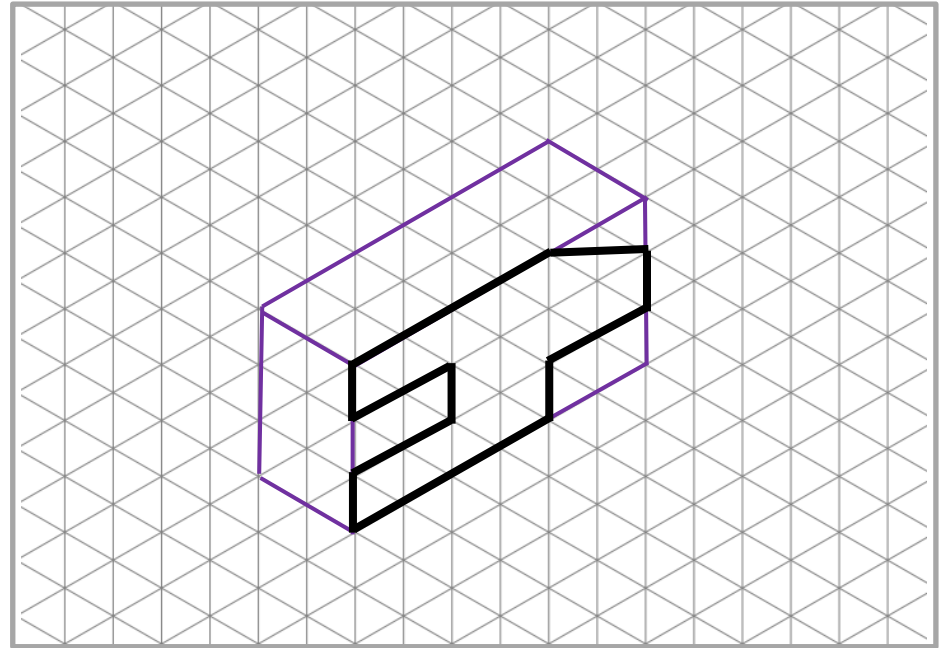
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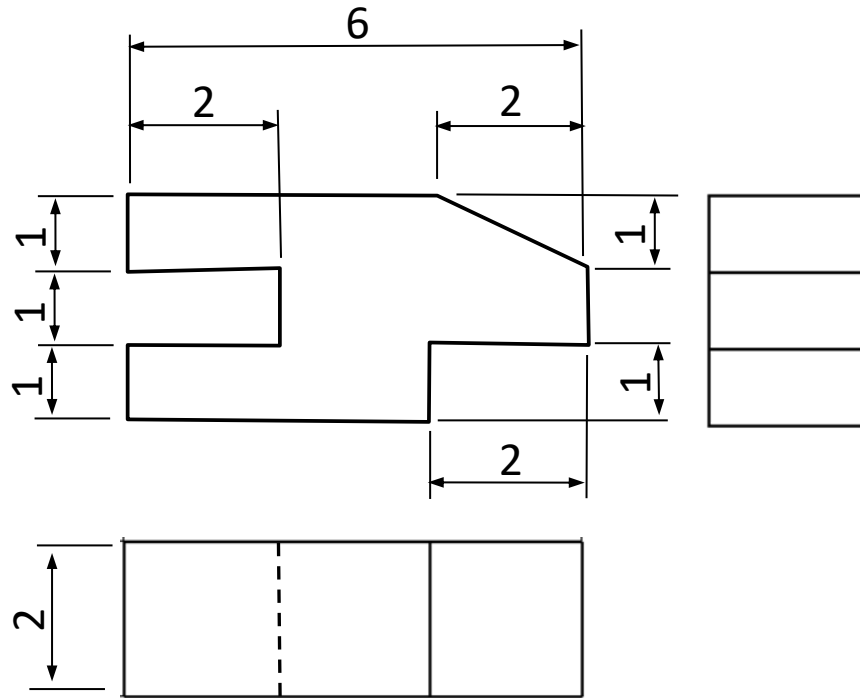
Vistas ortográficas



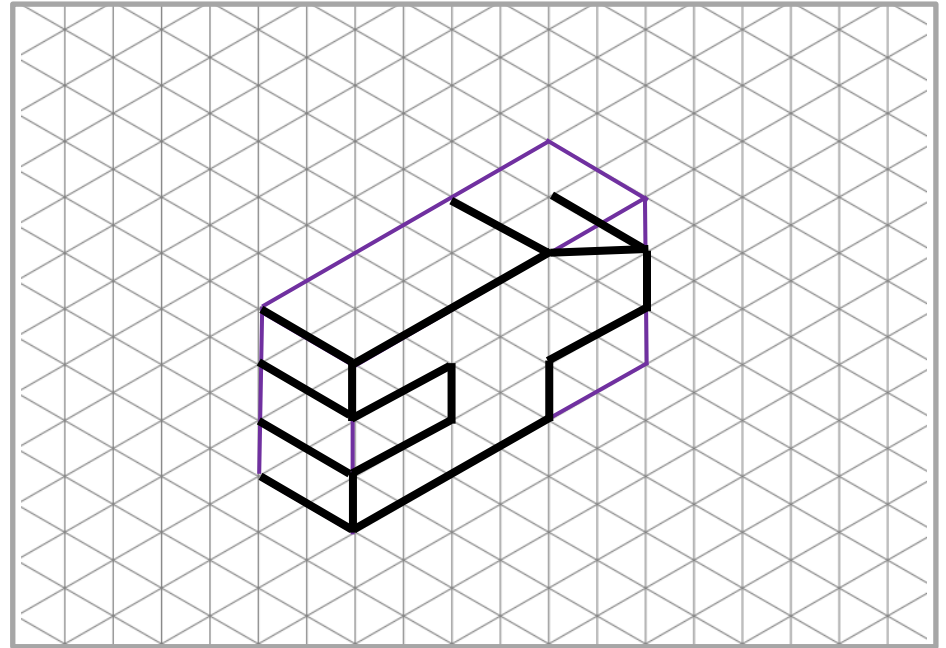
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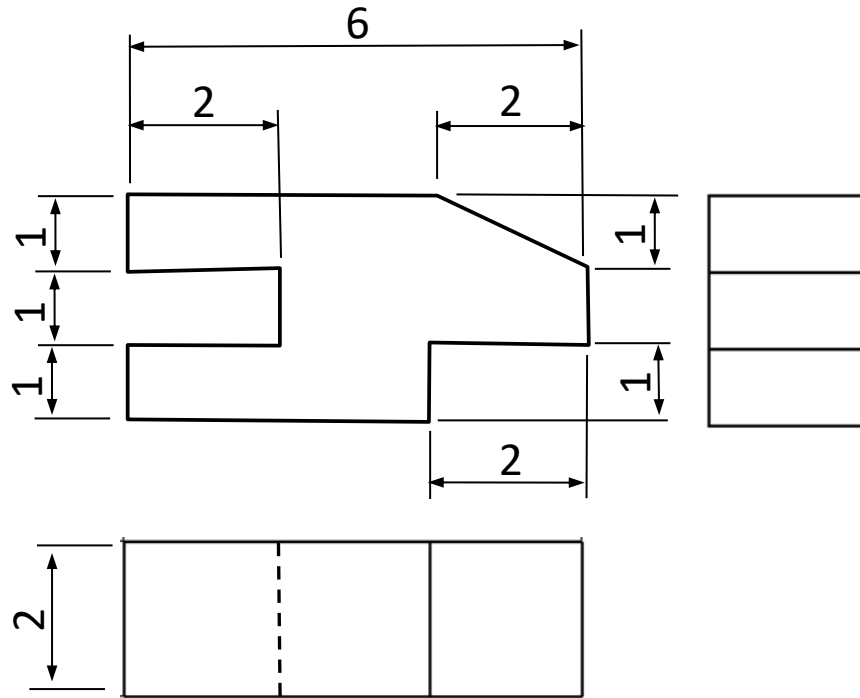
Vistas ortográficas



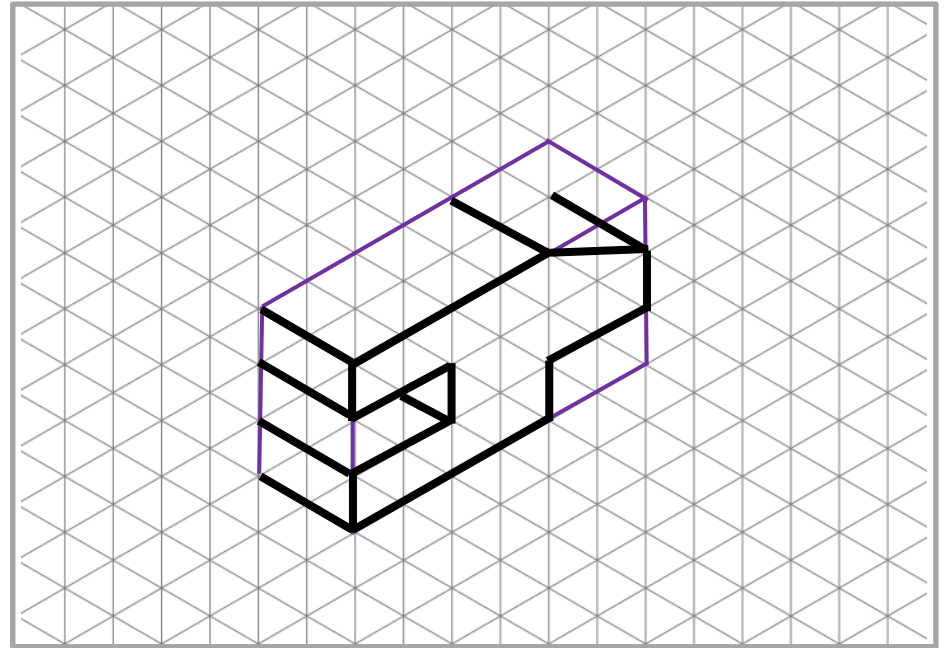
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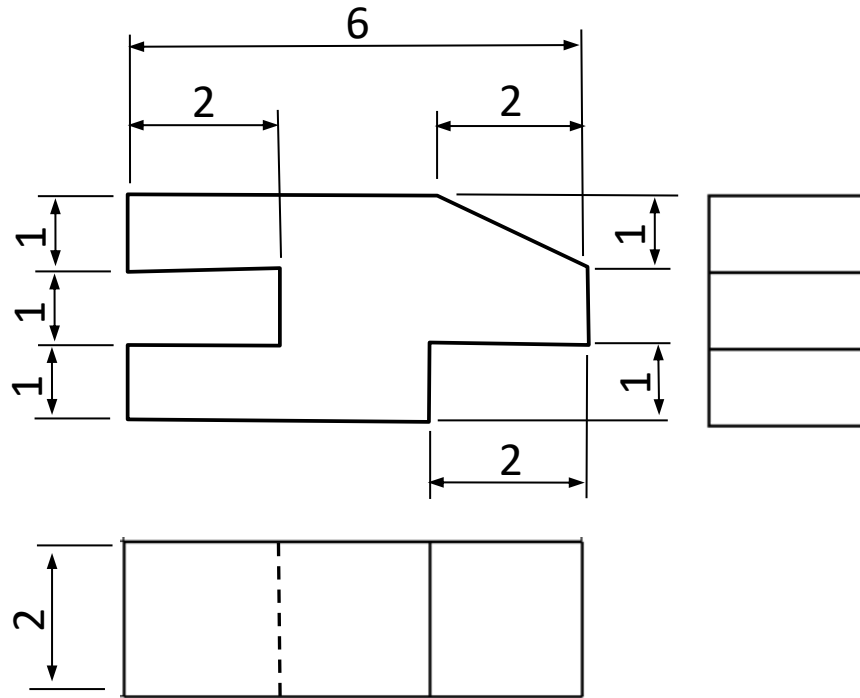
Vistas ortográficas



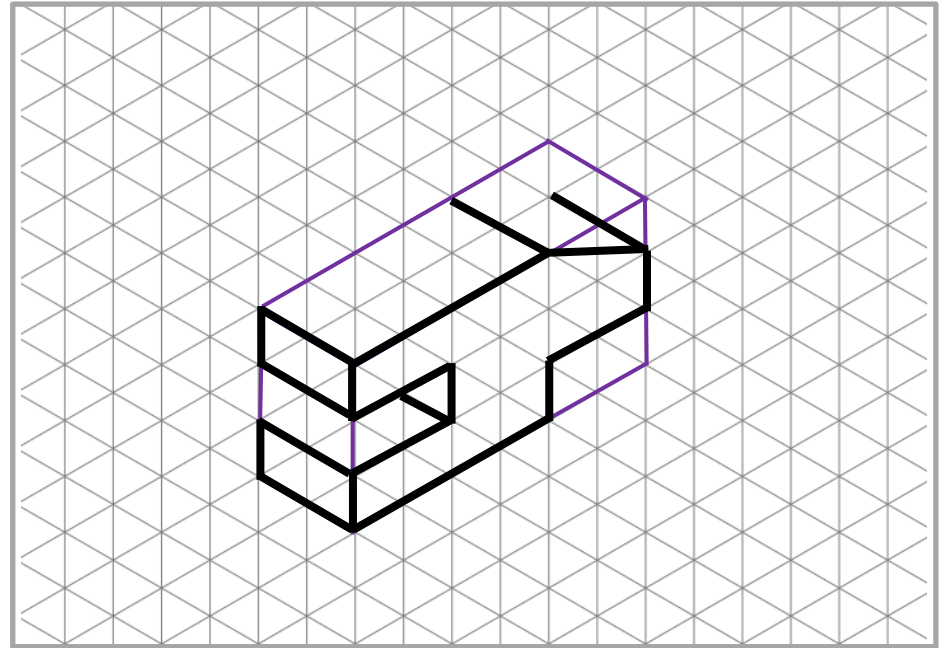
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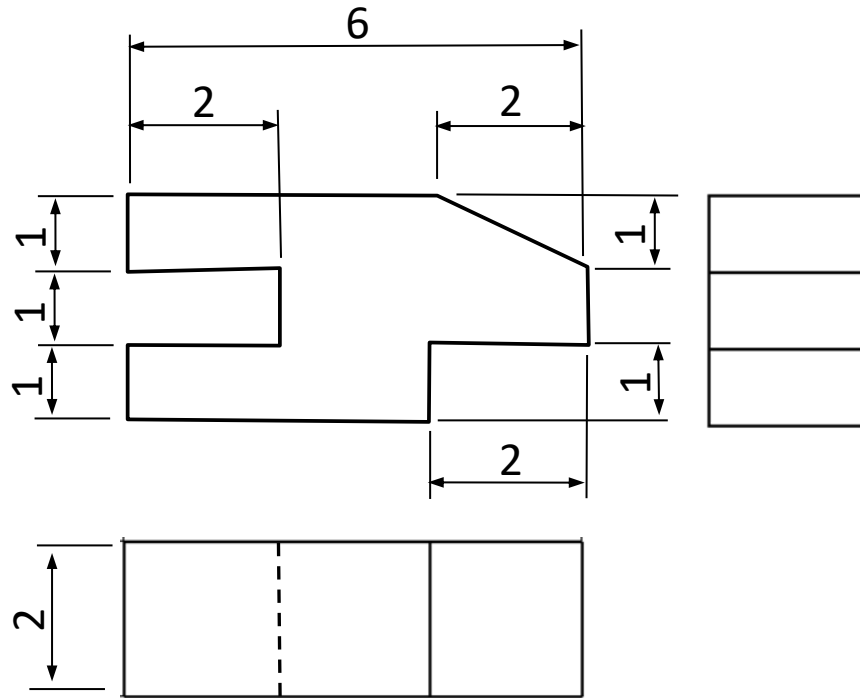
Vistas ortográficas



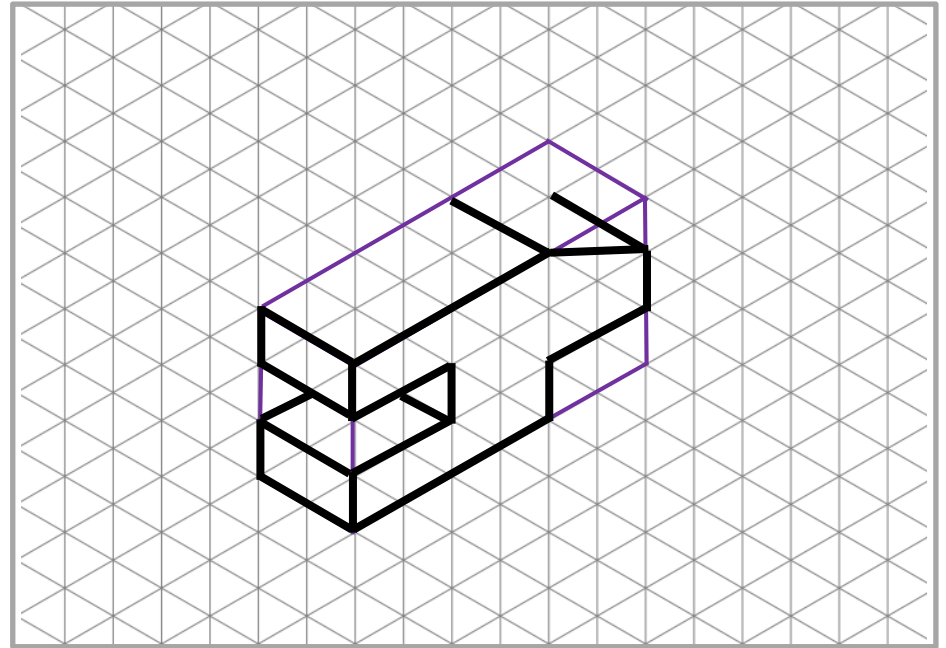
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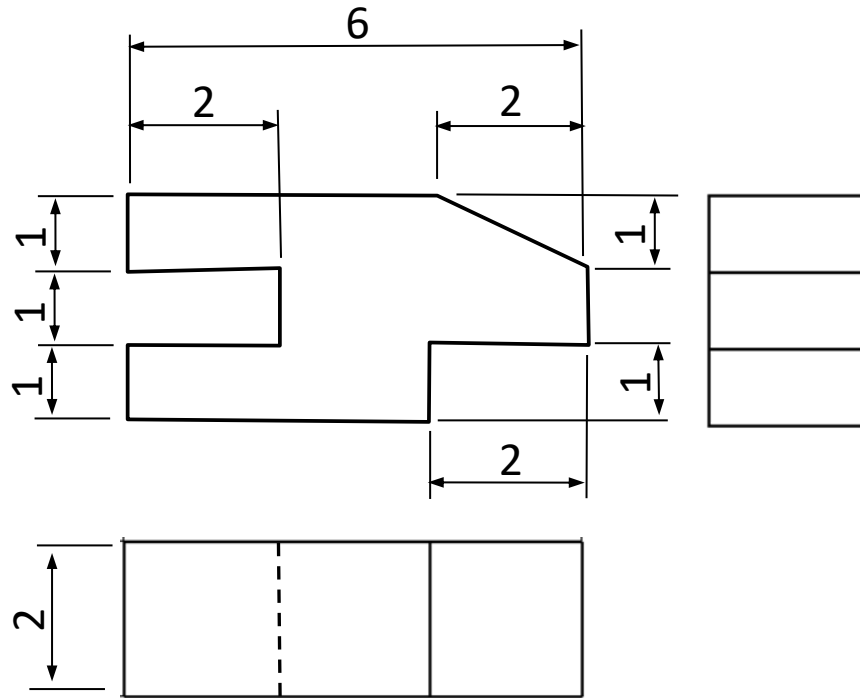
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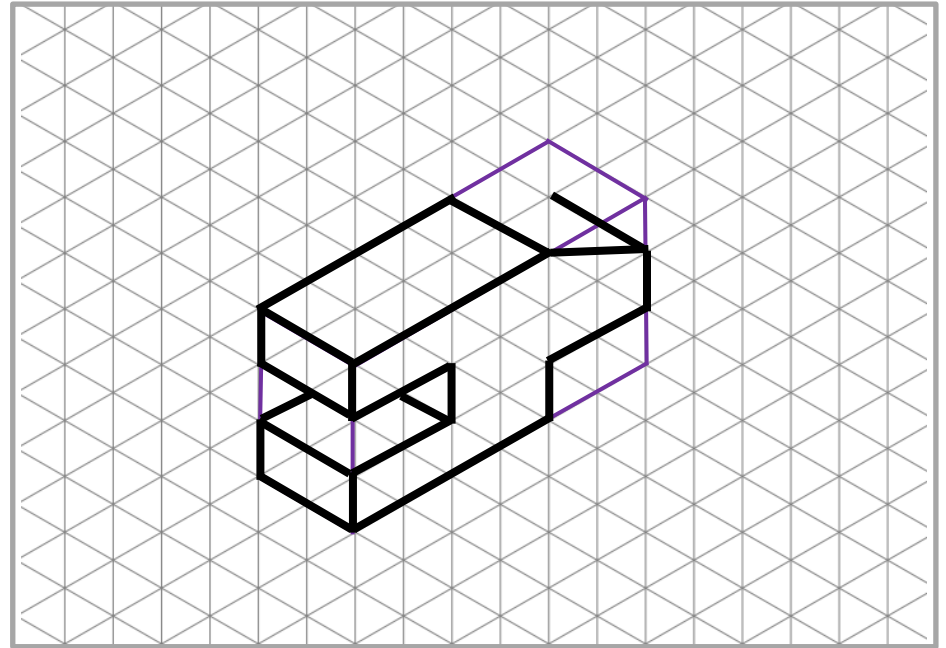
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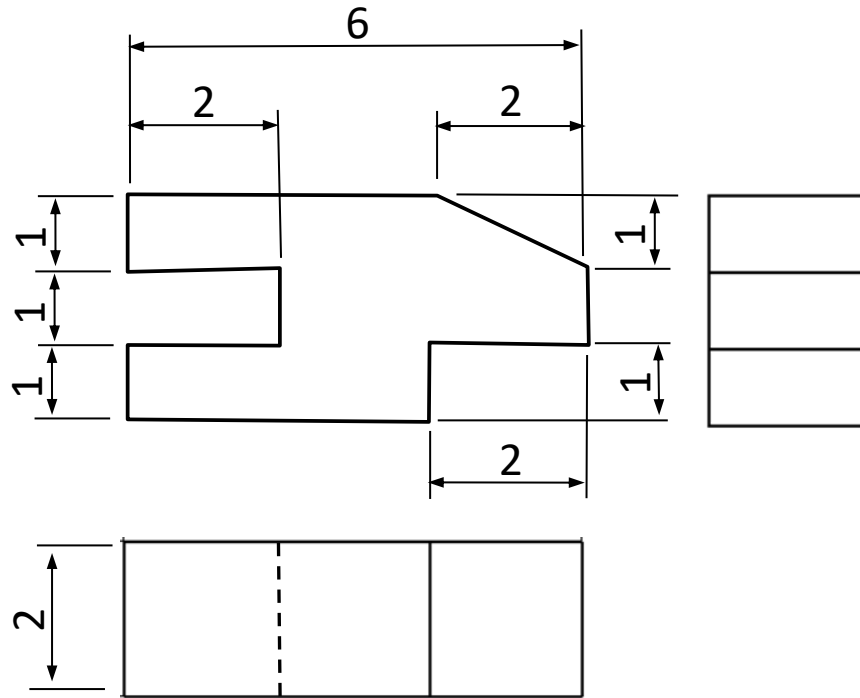
Vistas ortográficas



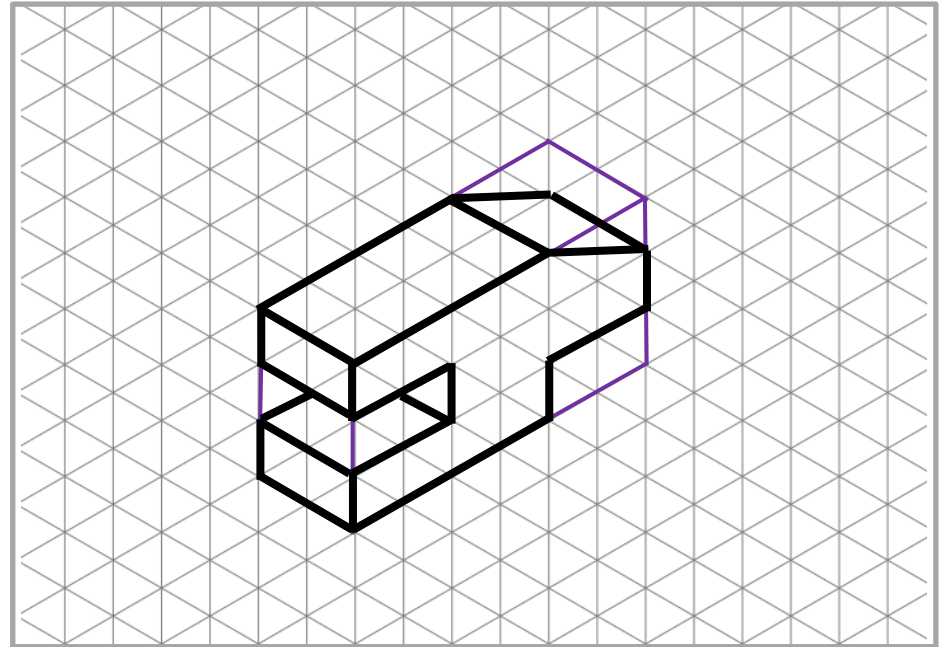
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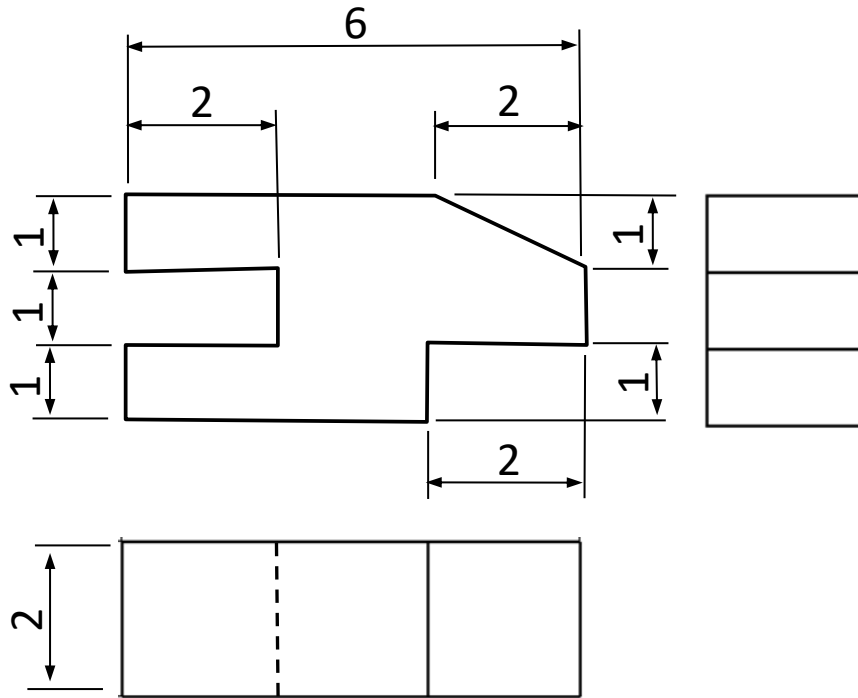
Vistas ortográficas



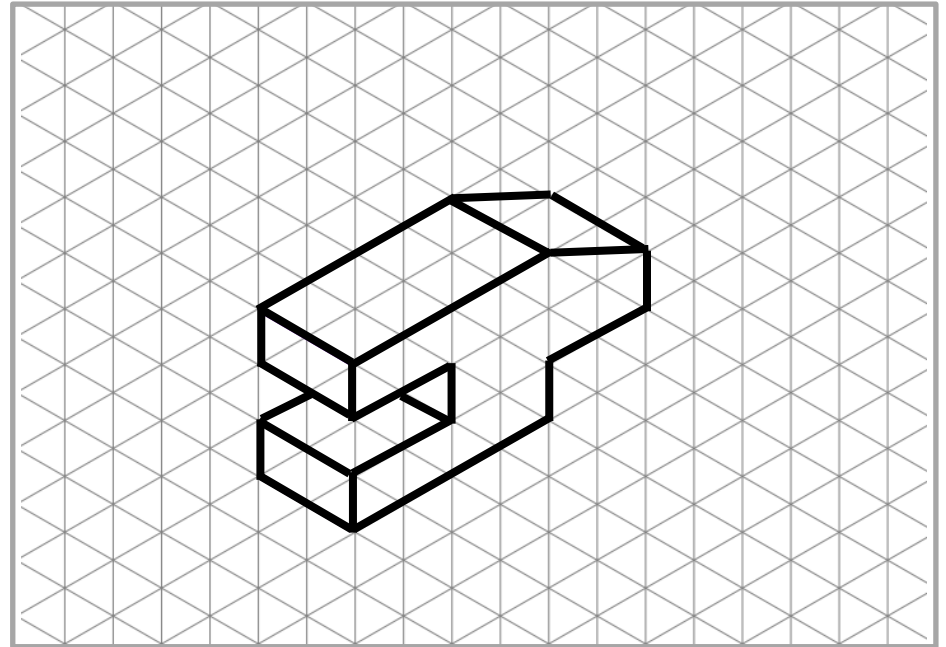
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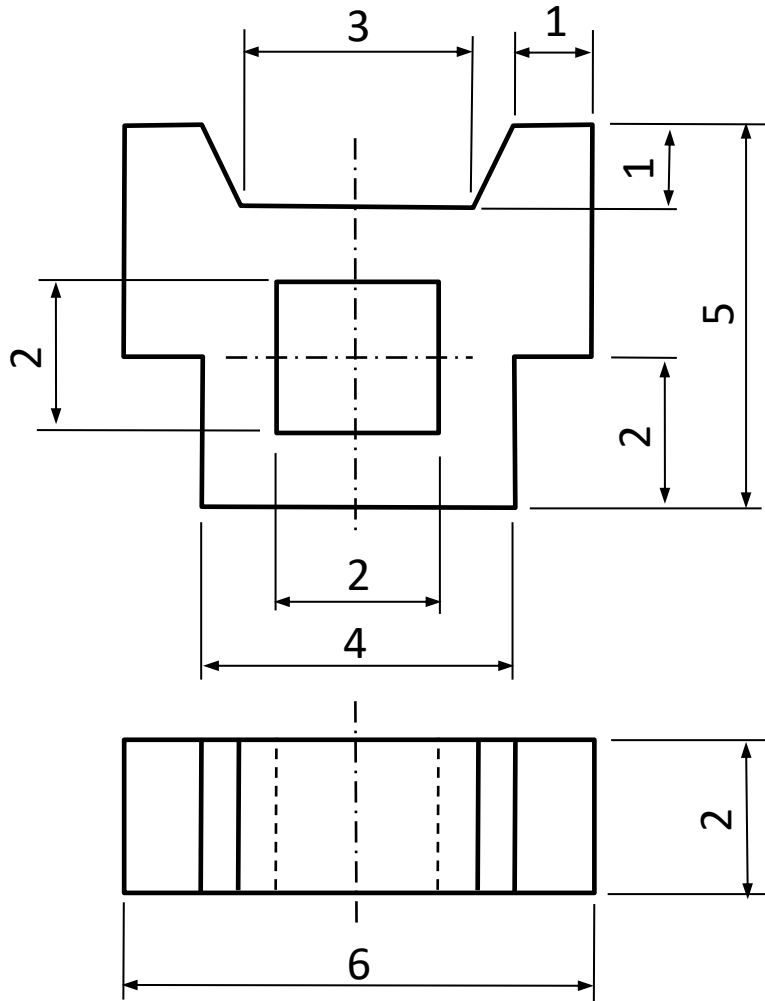
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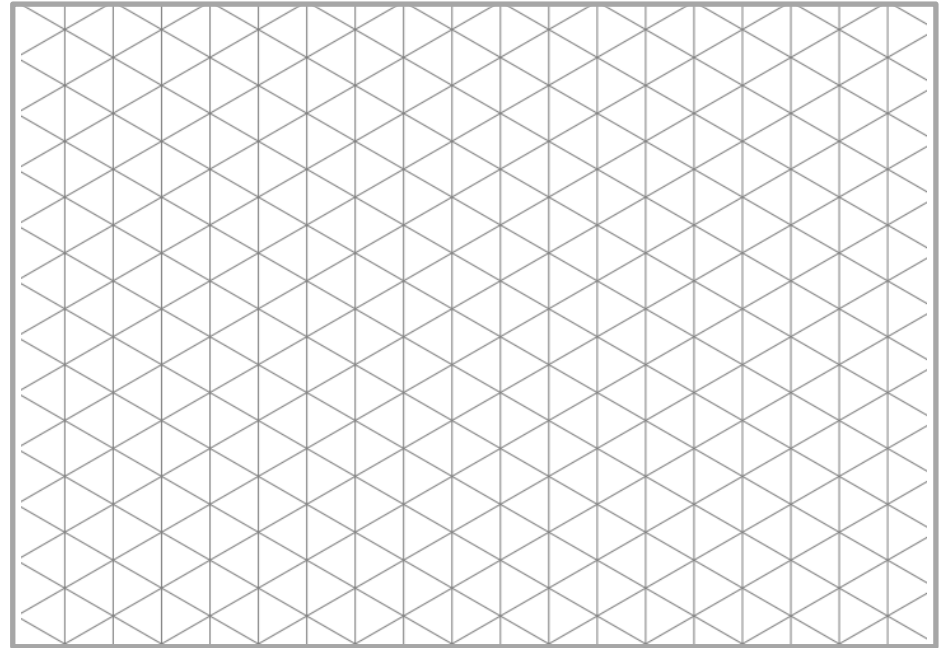
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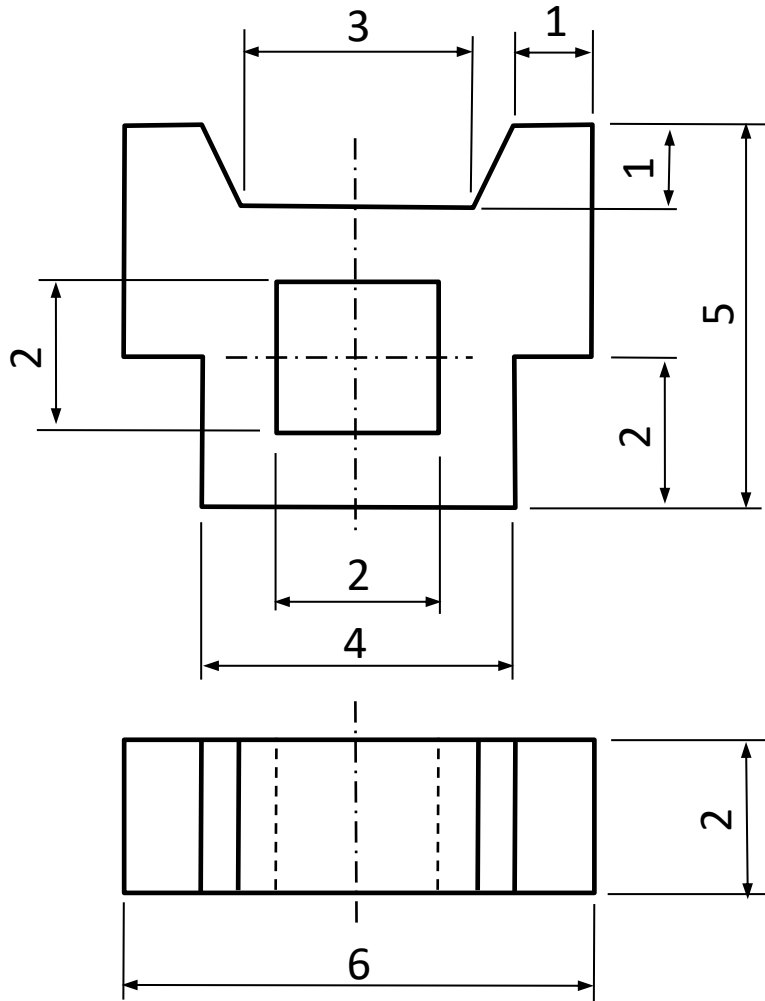
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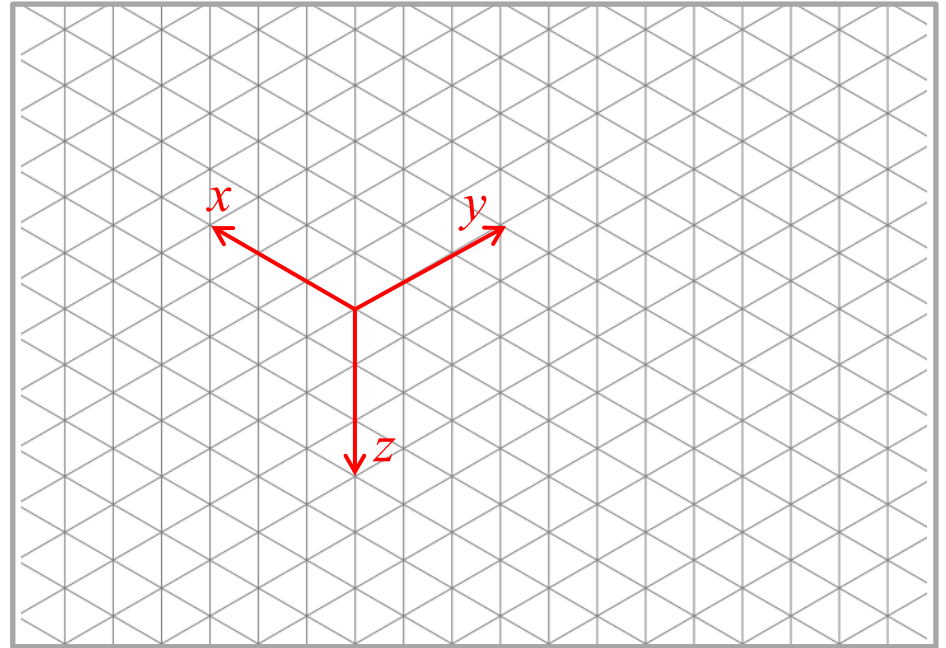
Perspectiva Isométrica - 4



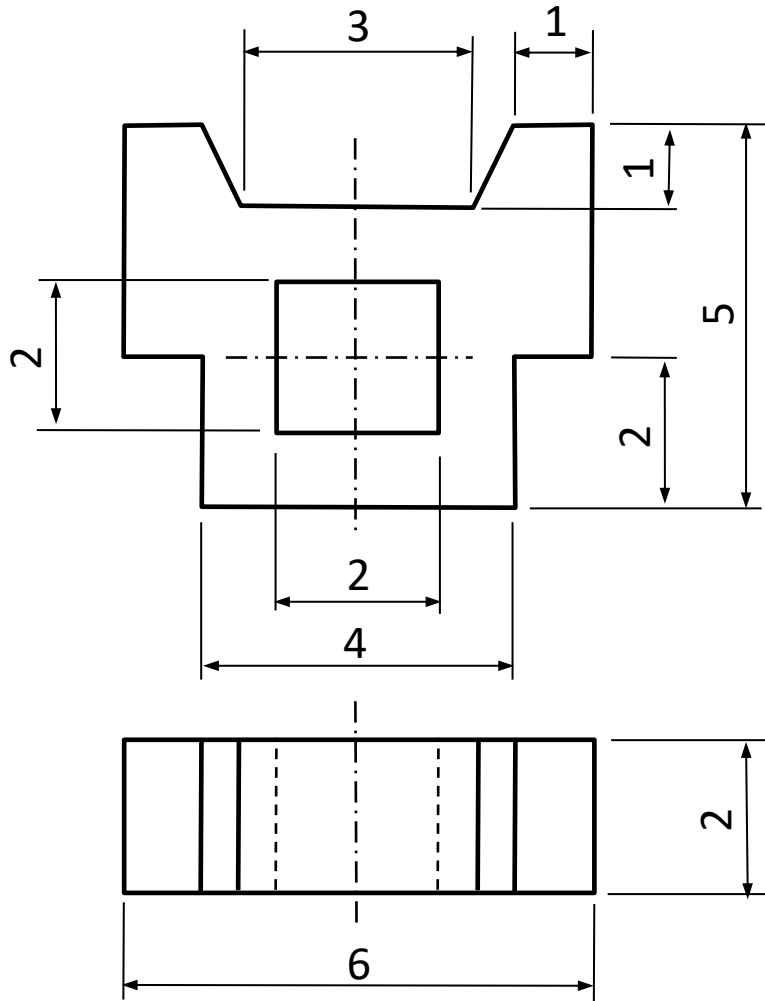
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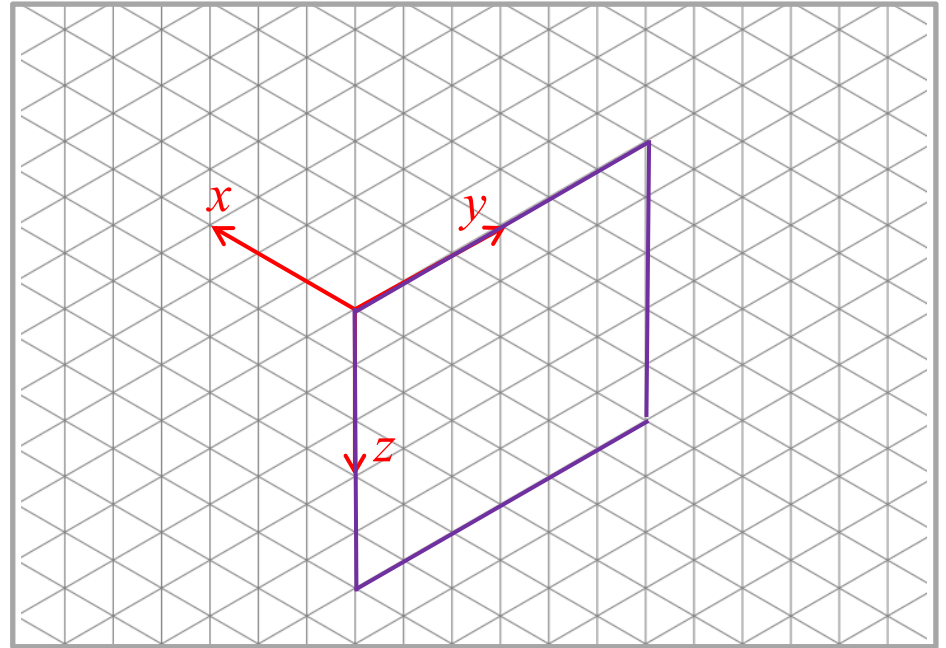
Perspectiva Isométrica - 4



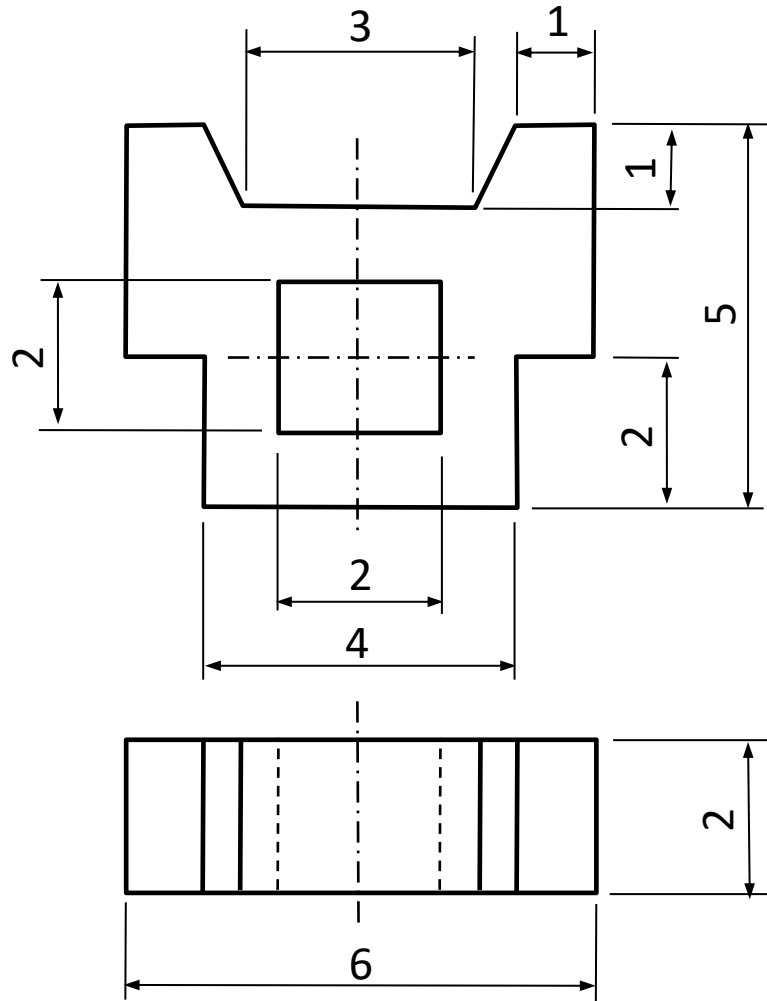
Vistas ortográficas



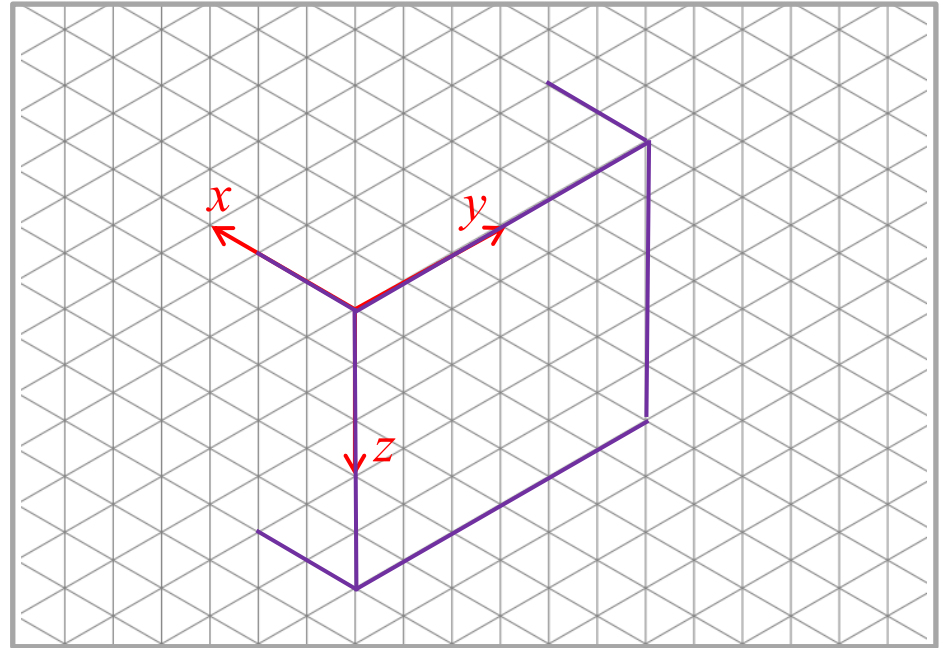
Perspectiva Isométrica - 4



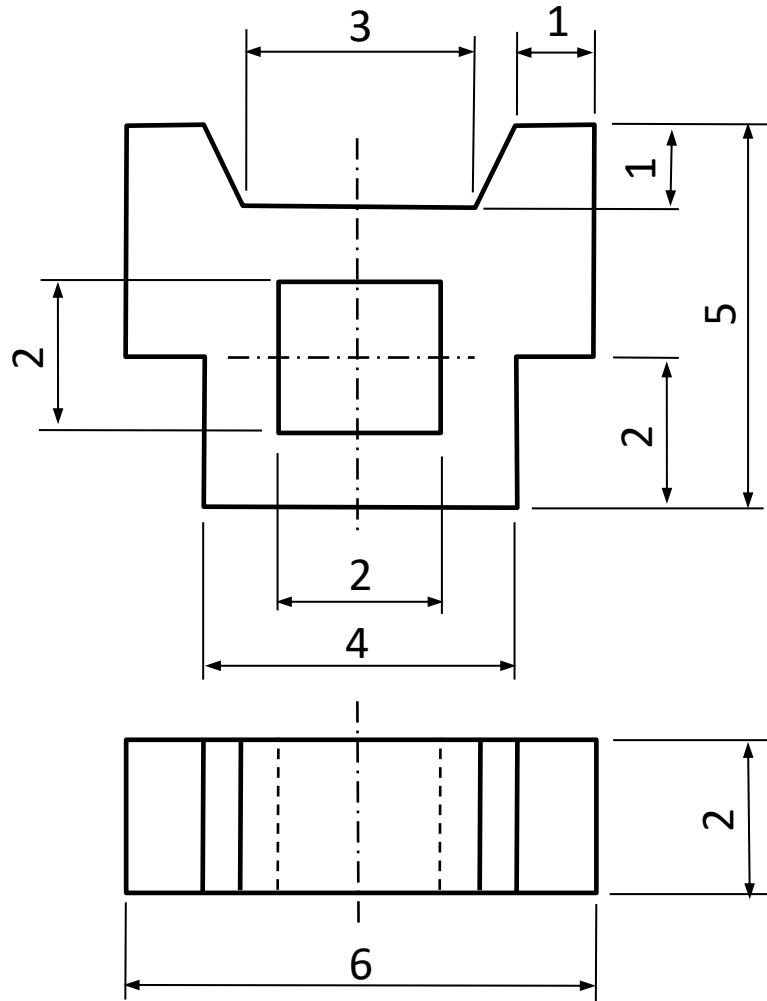
Vistas ortográficas



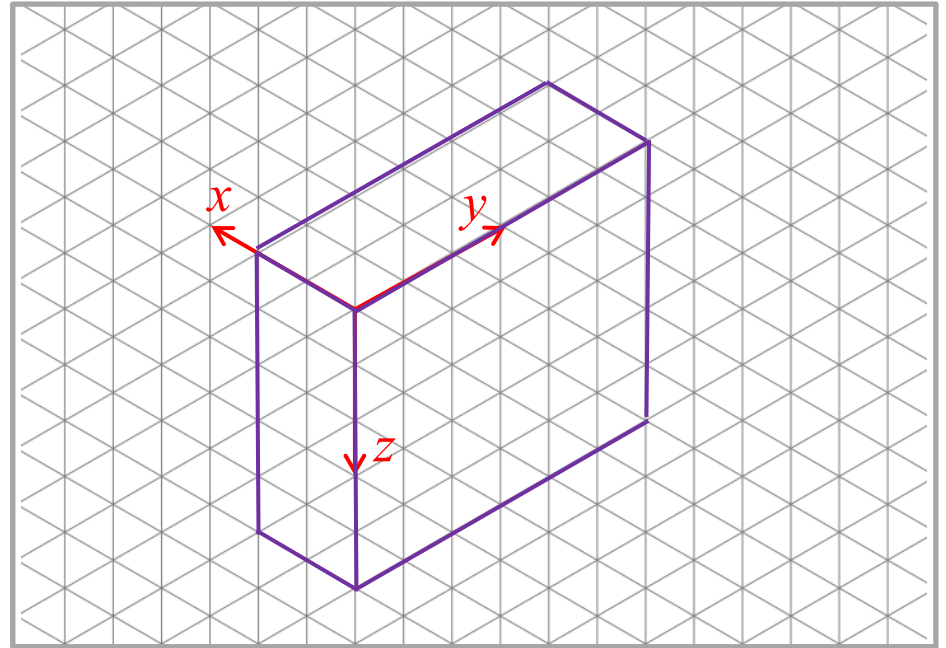
Perspectiva Isométrica - 4



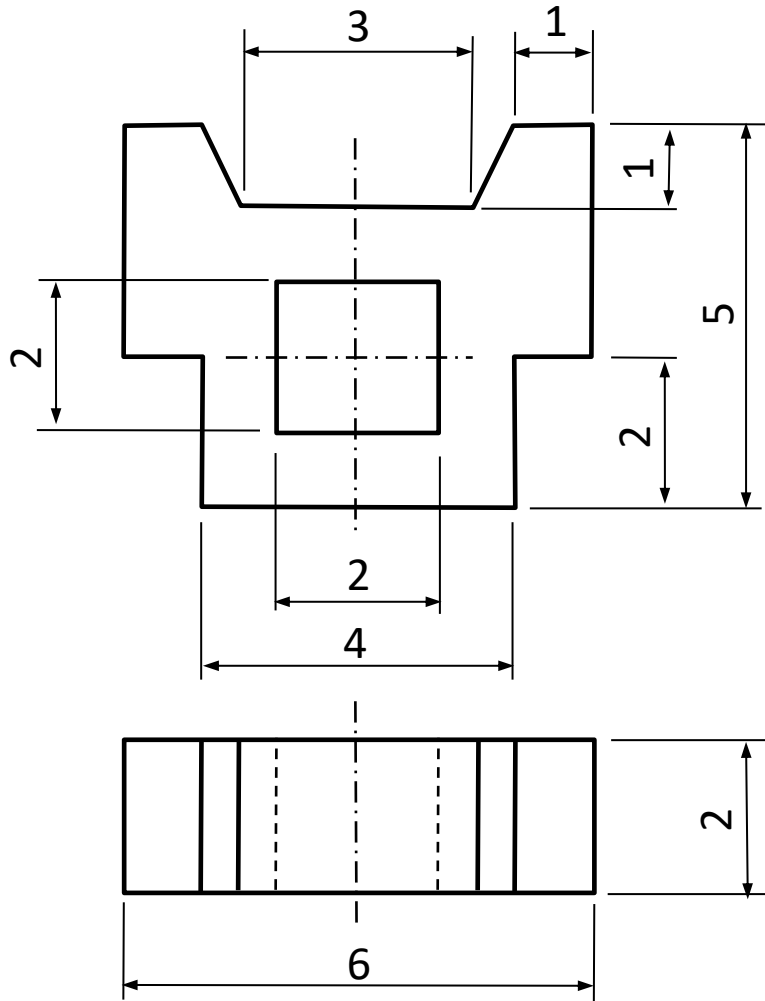
Vistas ortográficas



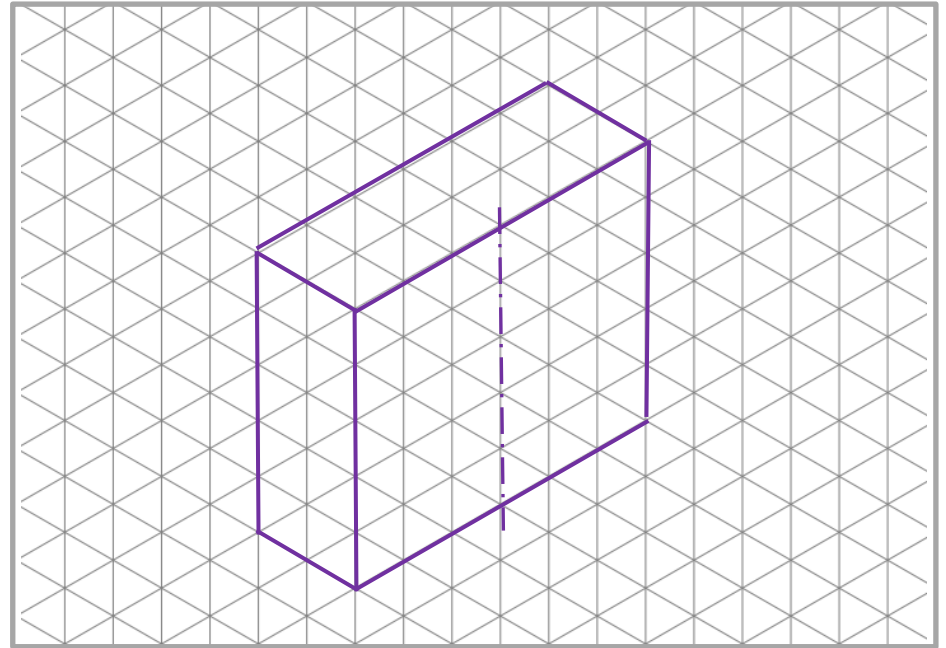
Perspectiva Isométrica - 4



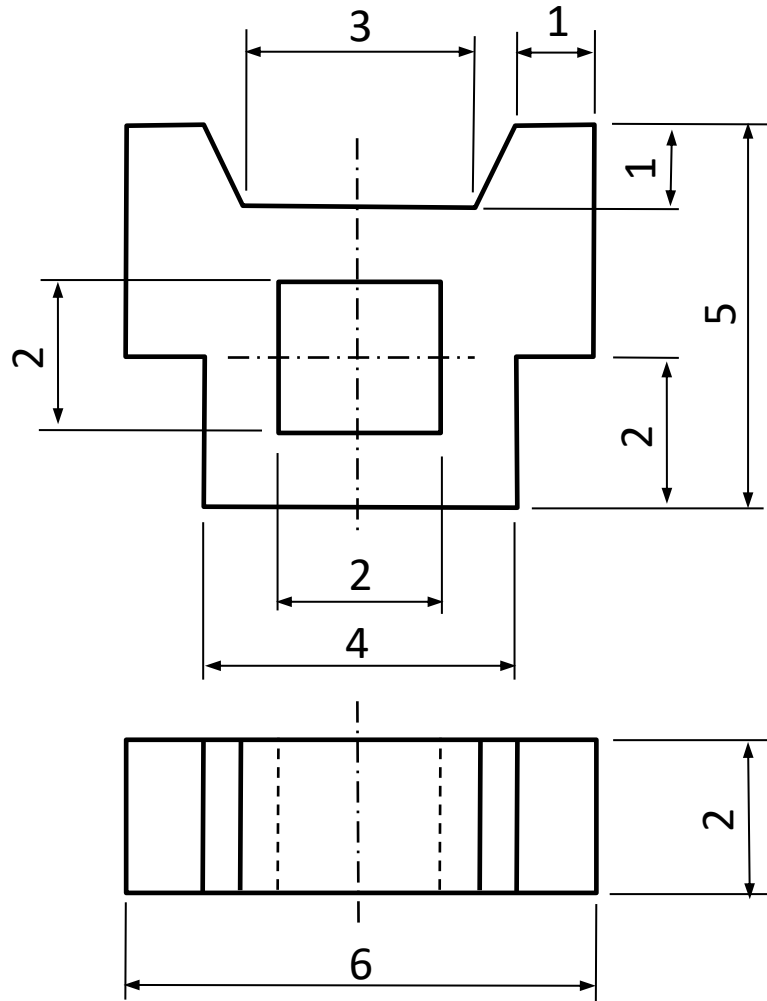
Vistas ortográficas



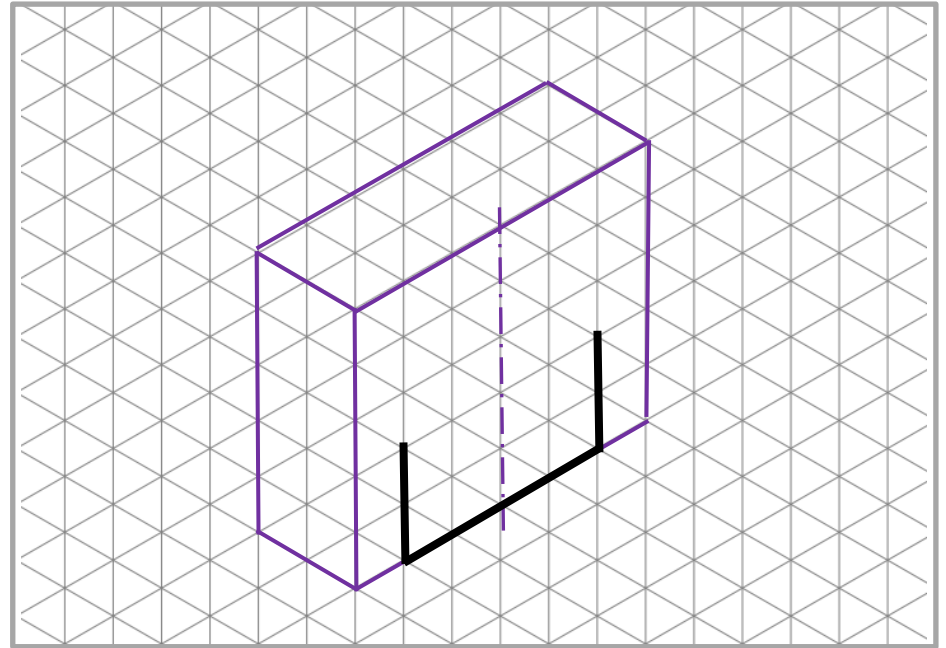
Perspectiva Isométrica - 4



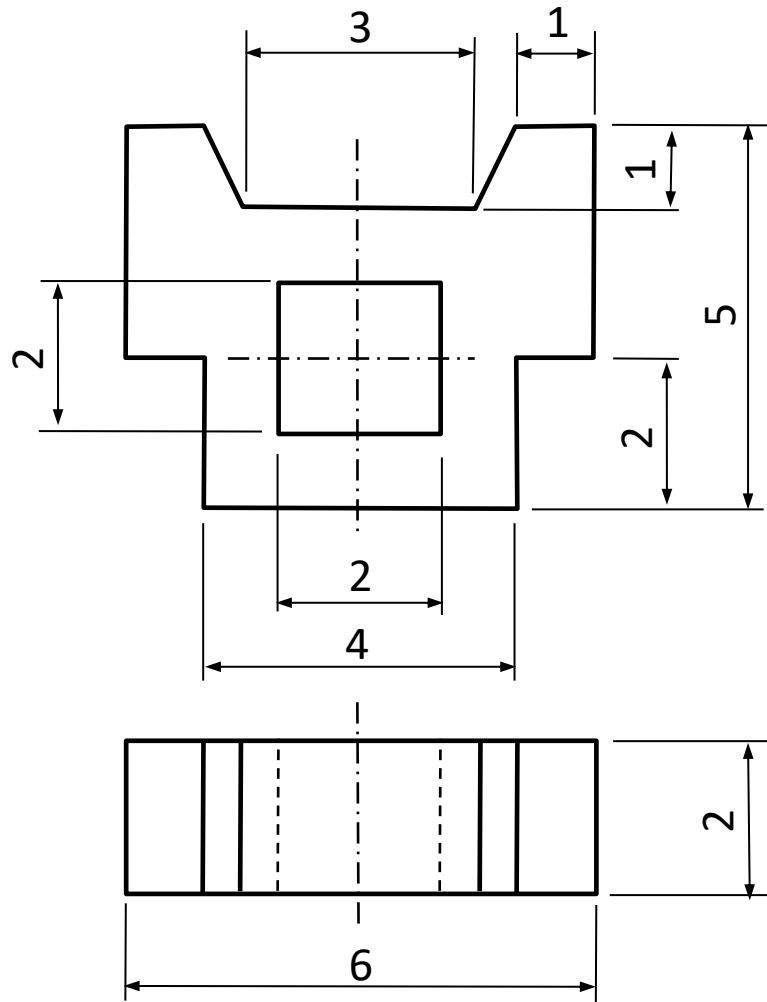
Vistas ortográficas



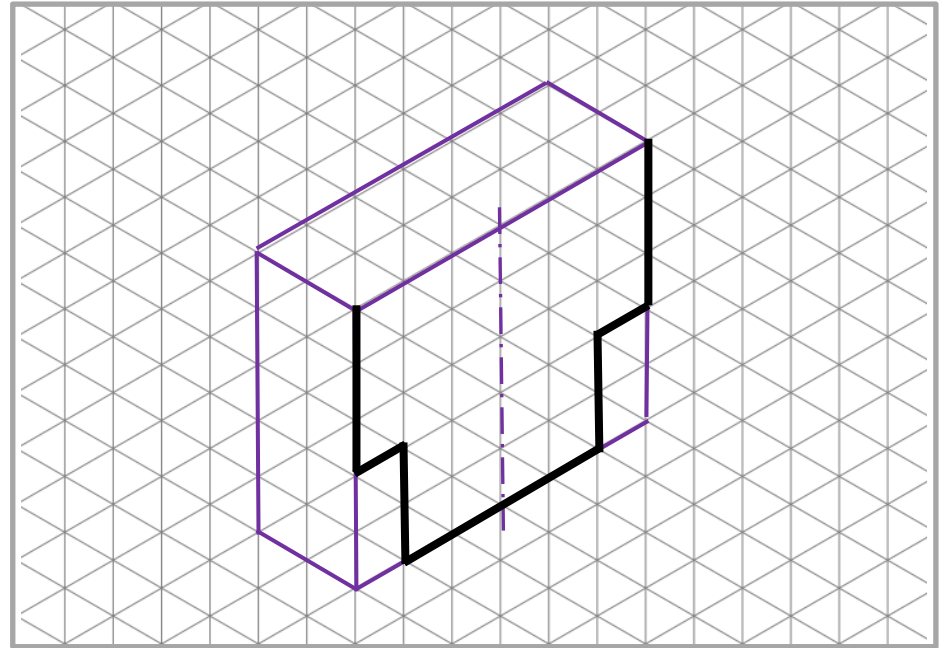
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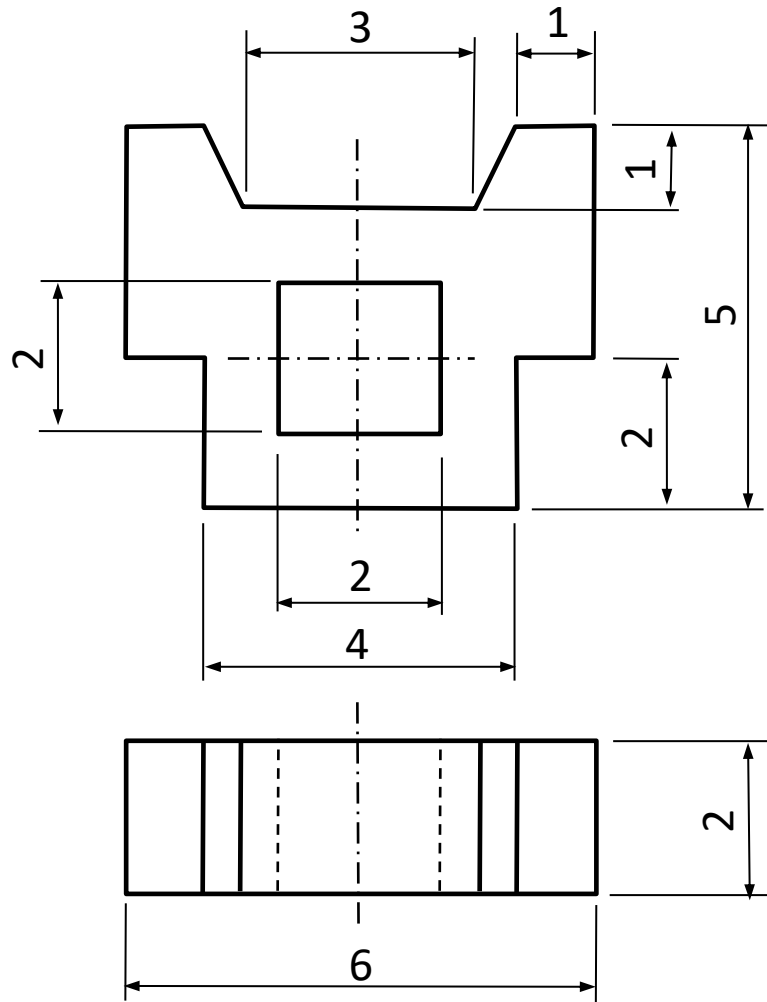
Vistas ortográficas



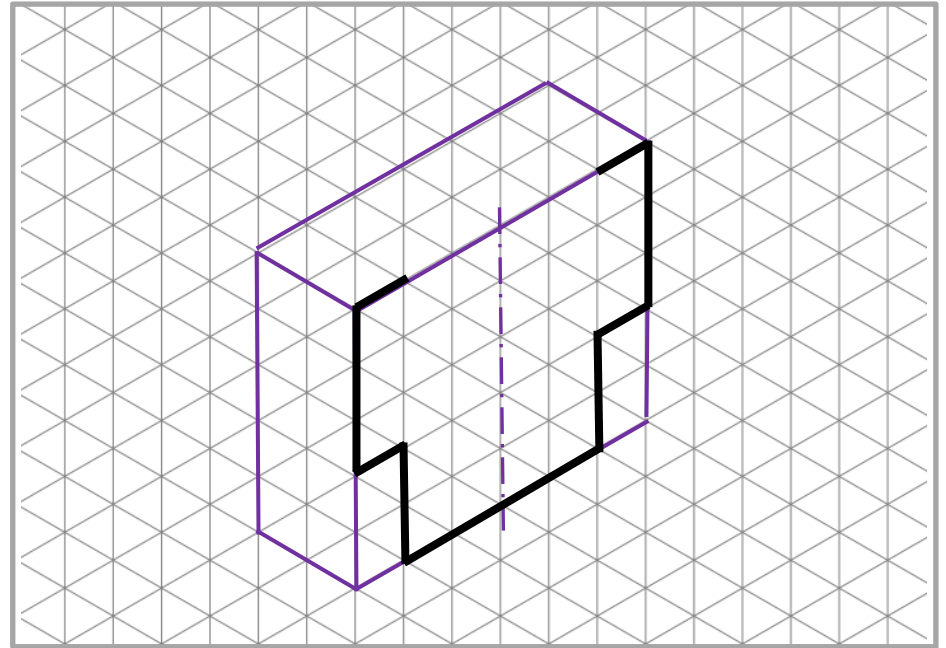
Perspectiva Isométrica - 4



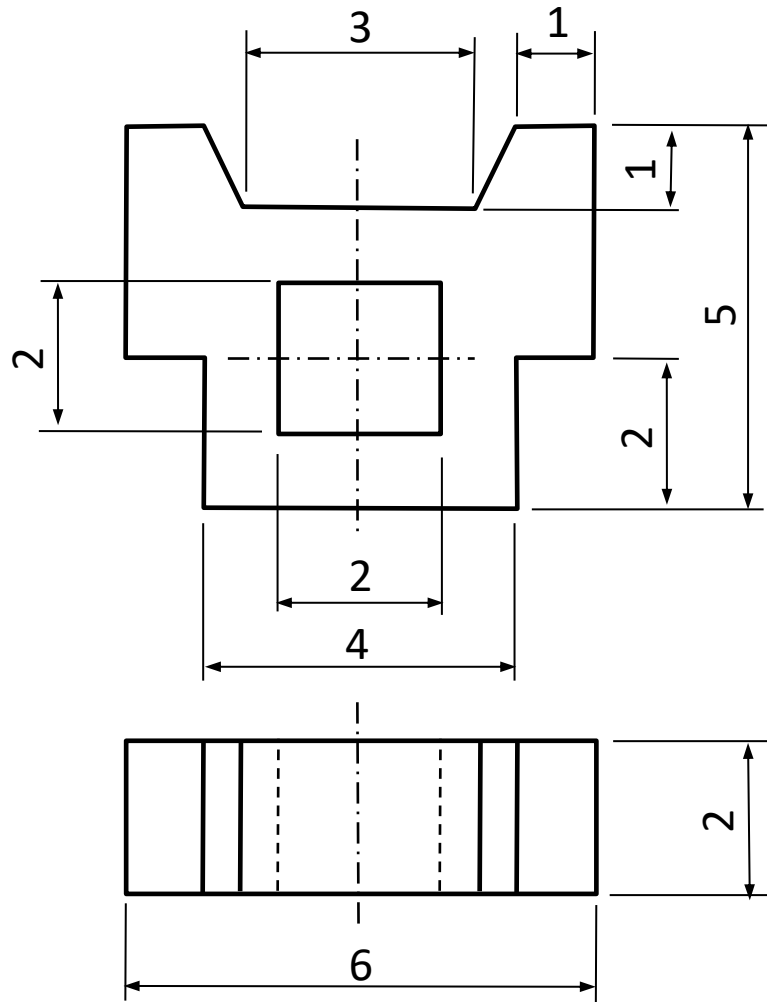
Vistas ortográficas



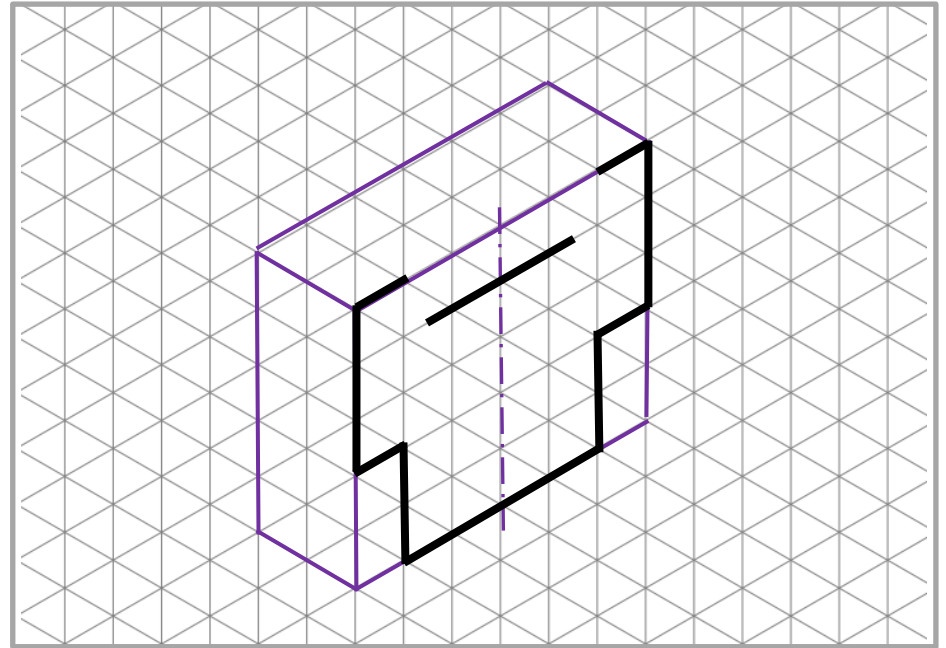
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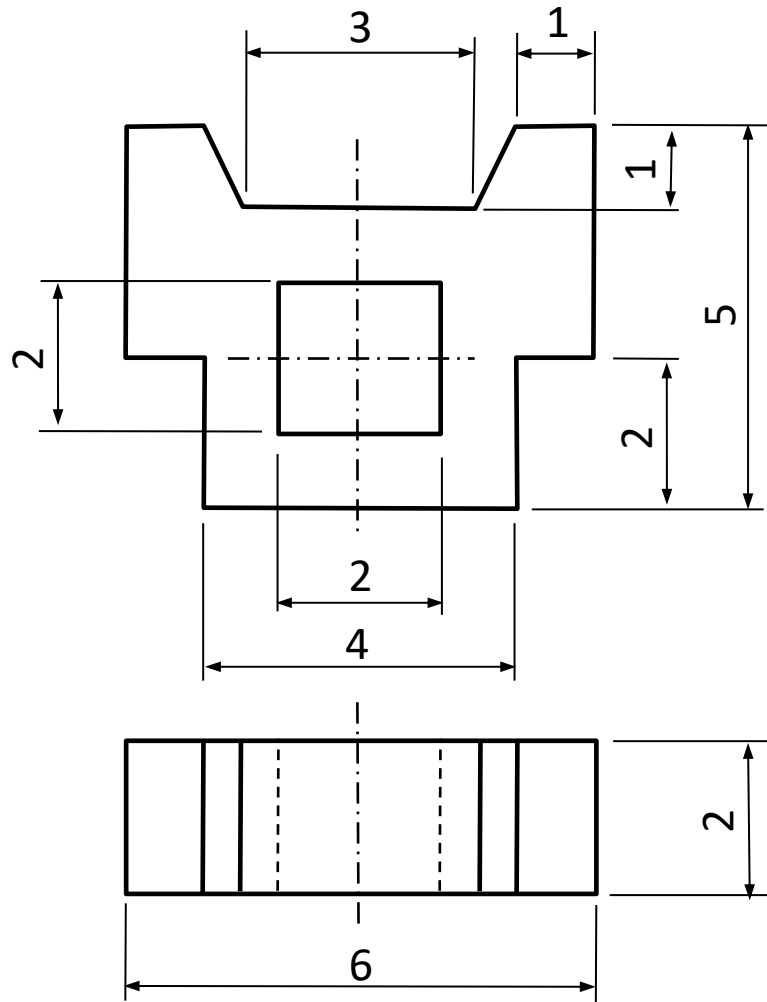
Vistas ortográficas



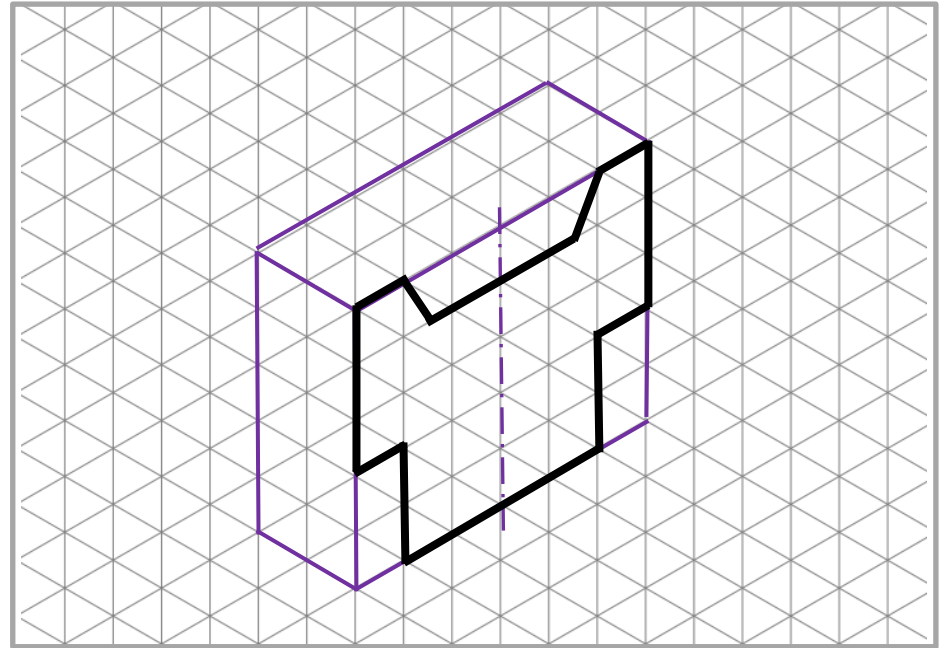
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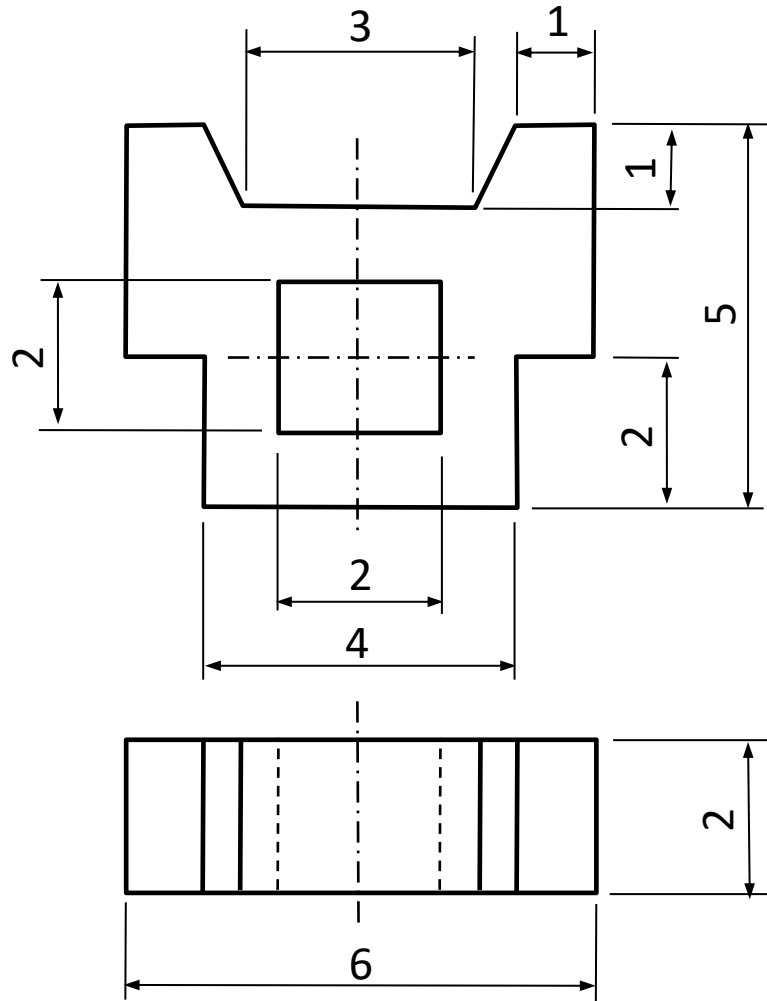
Vistas ortográficas



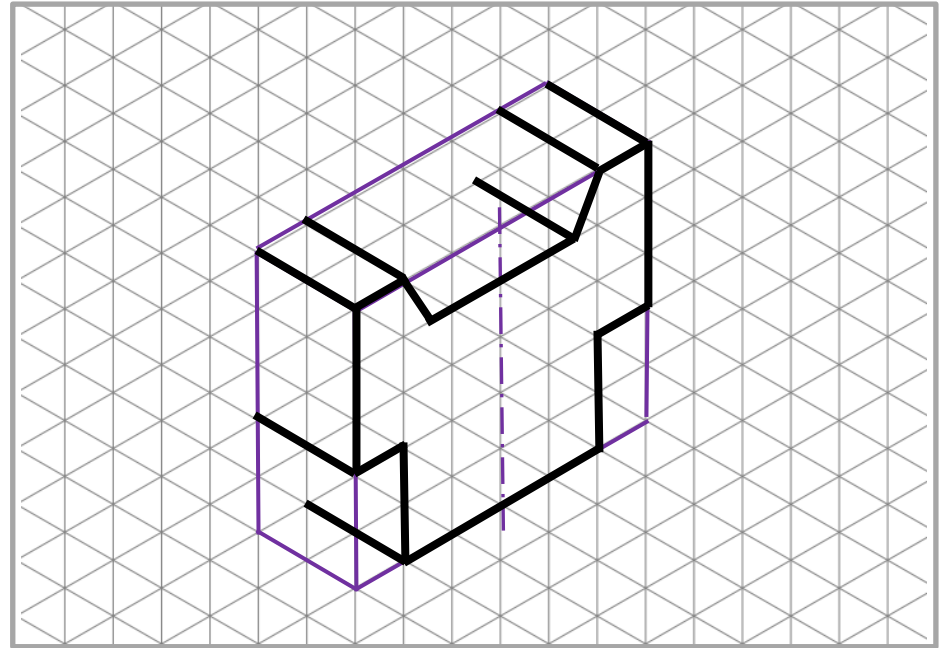
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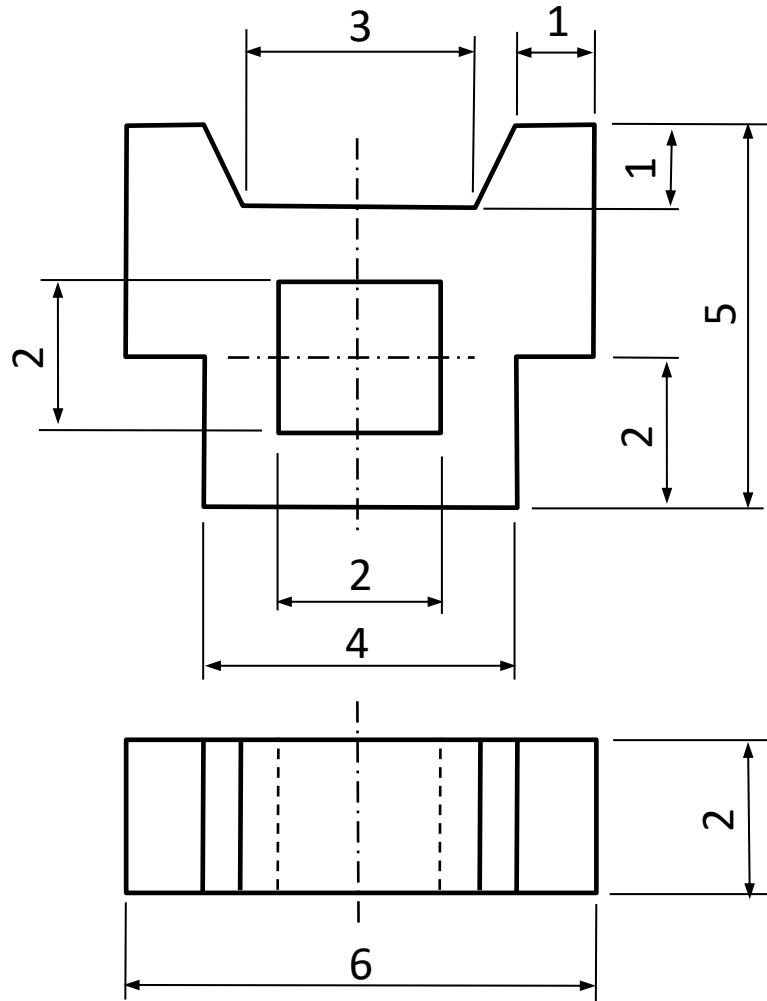
Vistas ortográficas



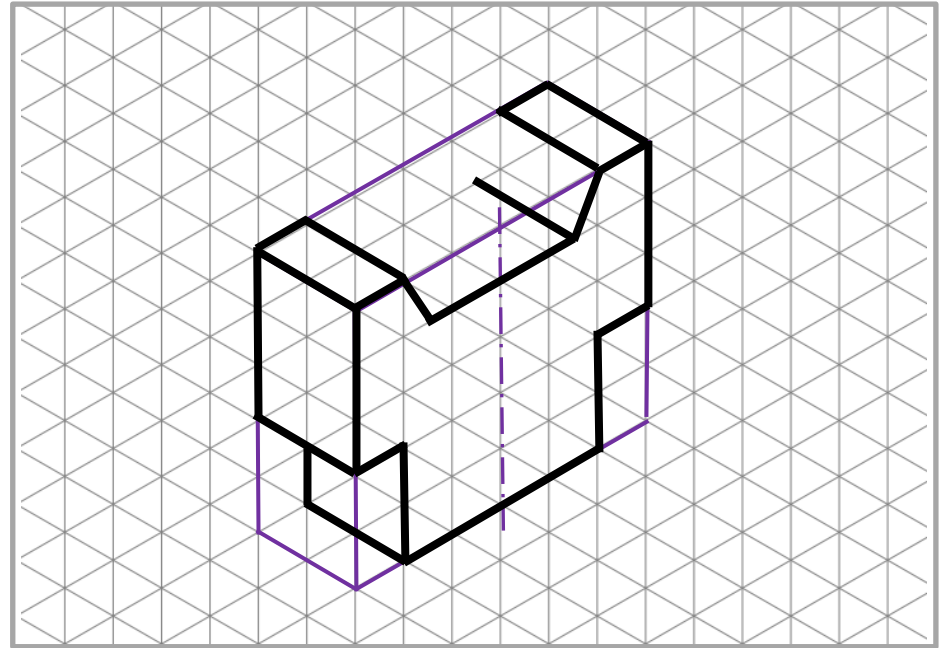
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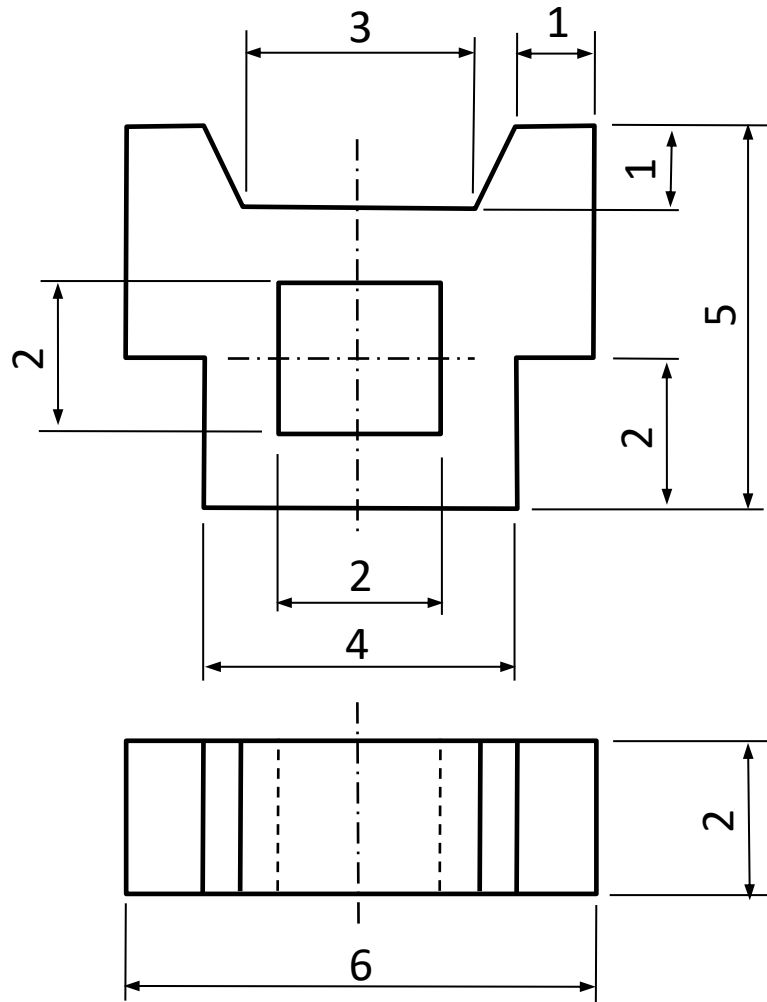
Vistas ortográficas



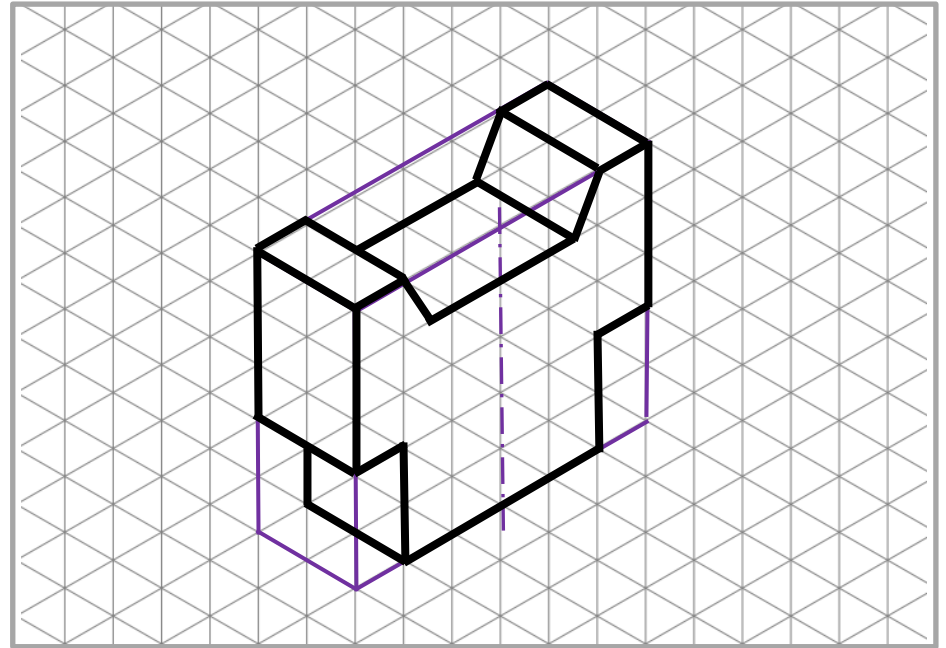
Perspectiva Isométrica - 4



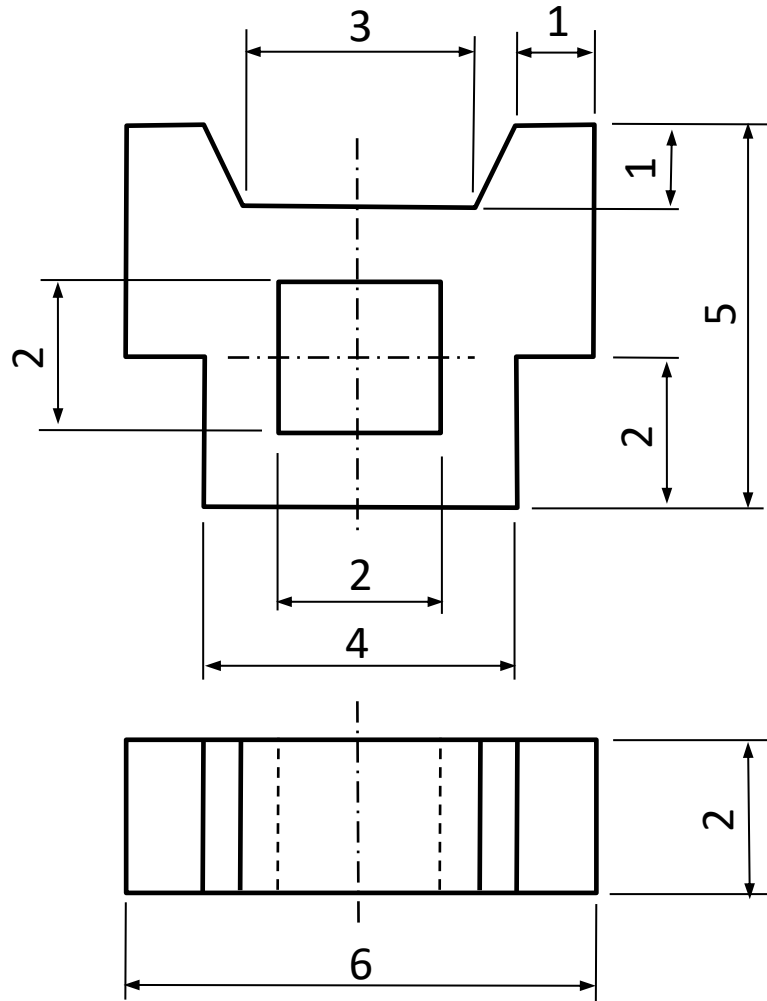
Vistas ortográficas



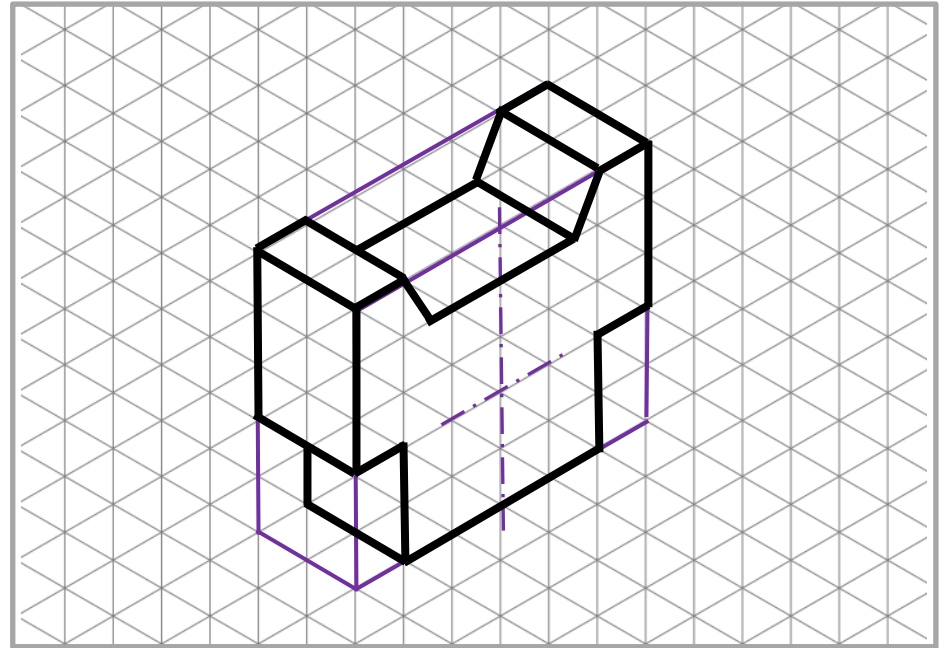
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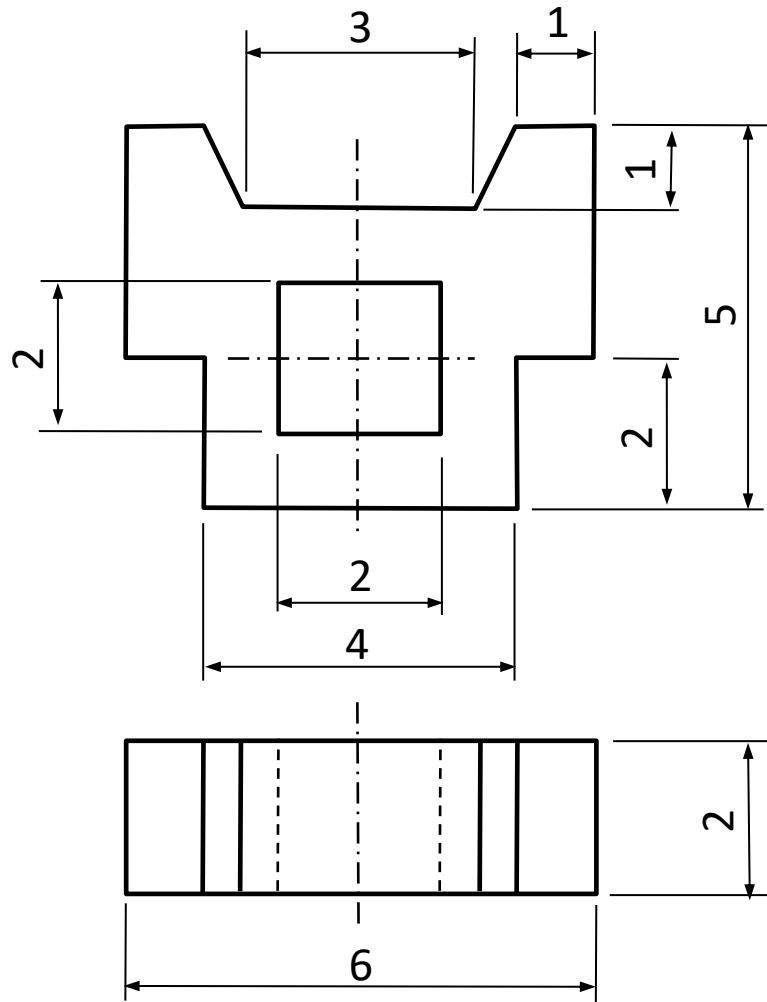
Vistas ortográficas



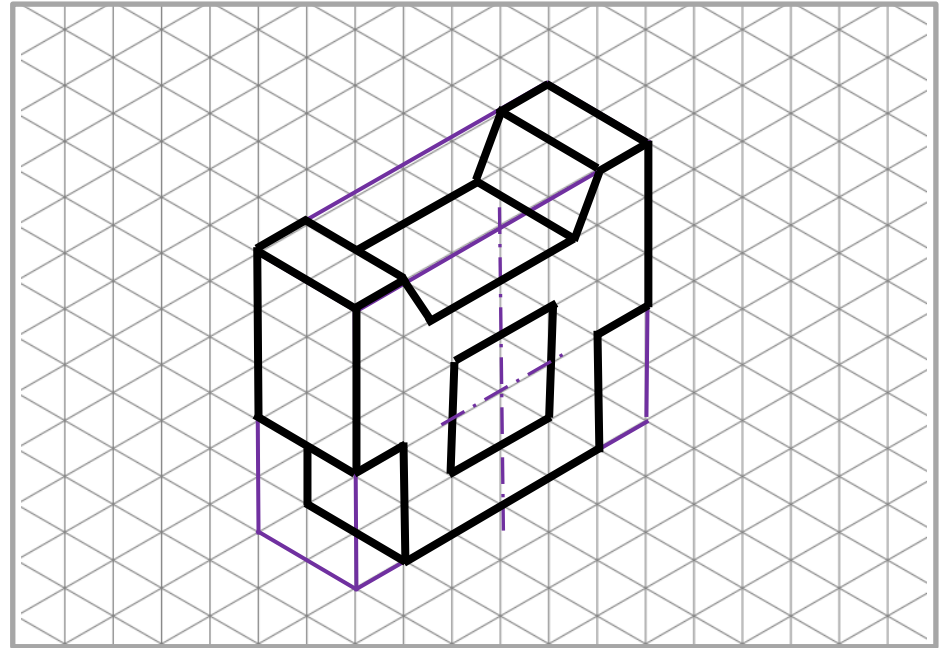
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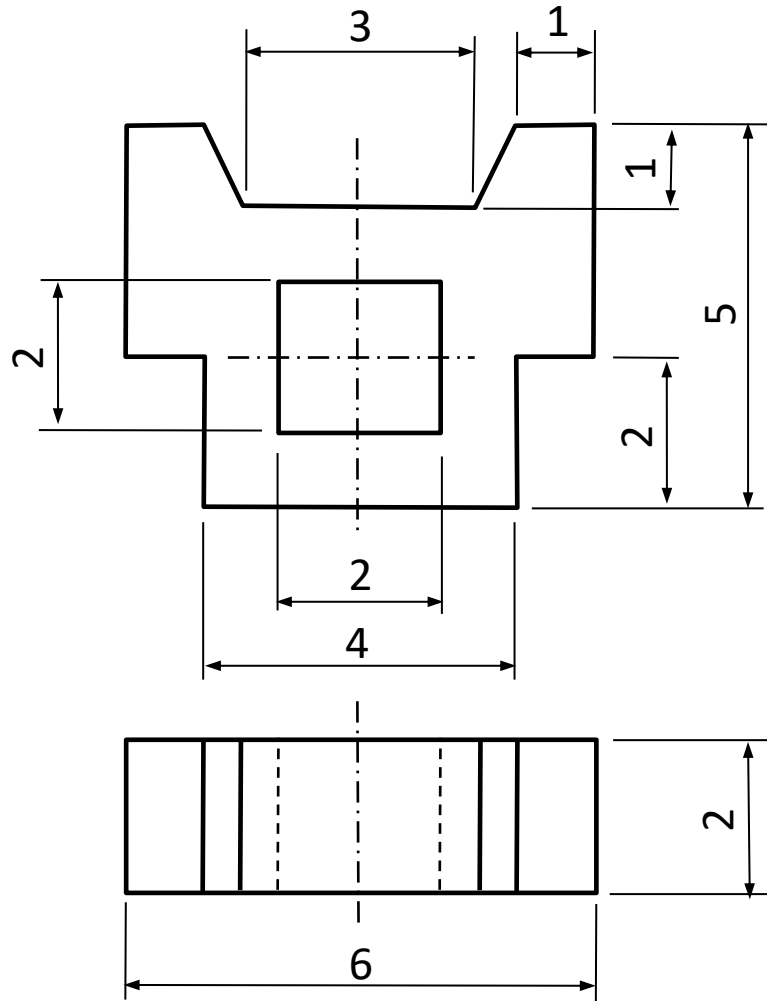
Vistas ortográficas



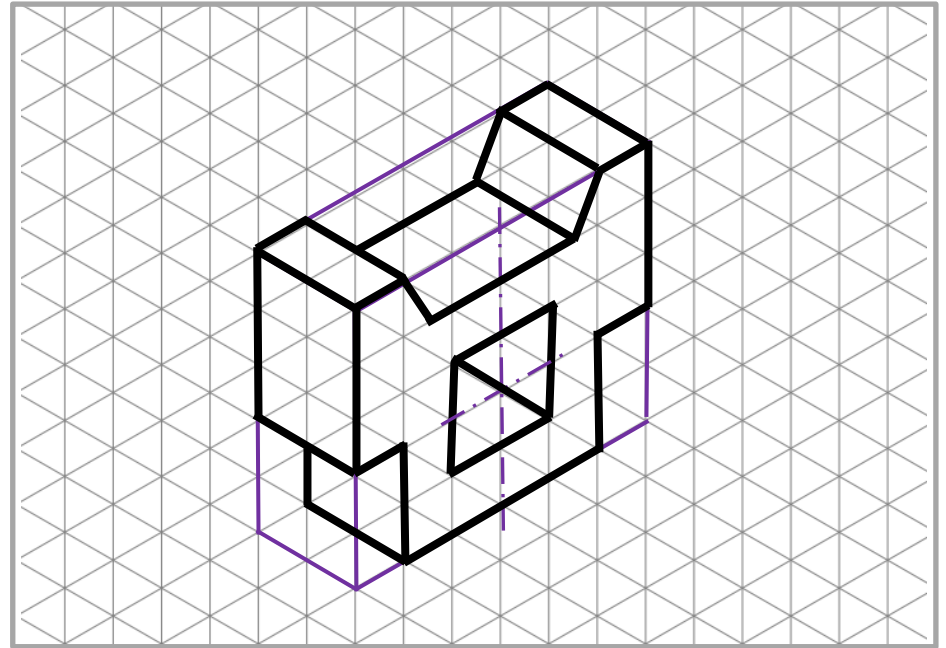
Perspectiva Isométrica - 4



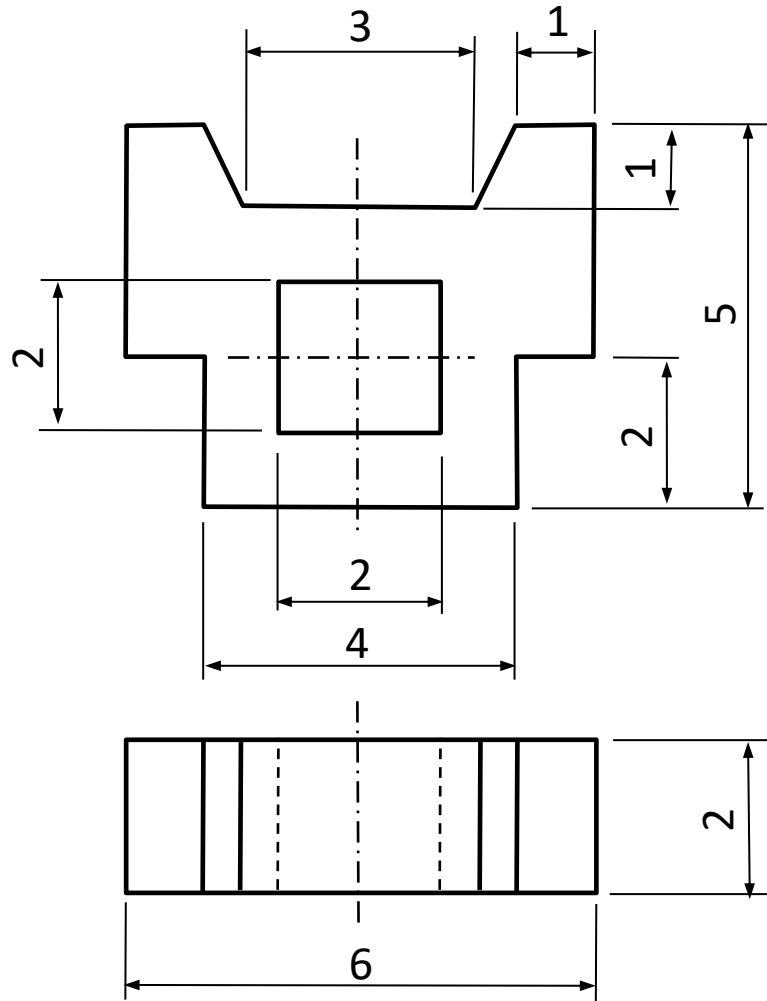
Vistas ortográficas



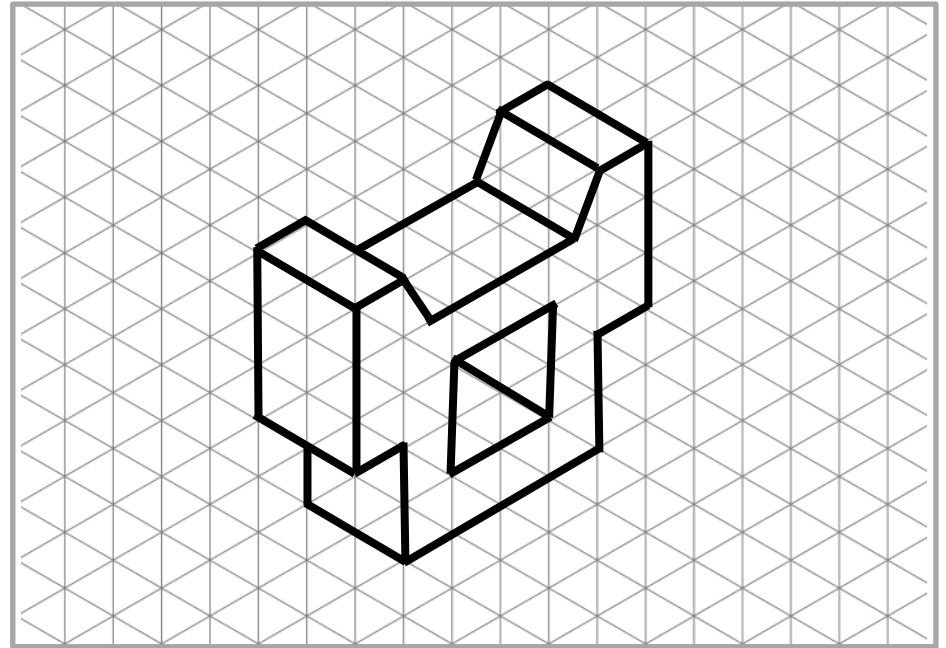
Perspectiva Isométrica - 4



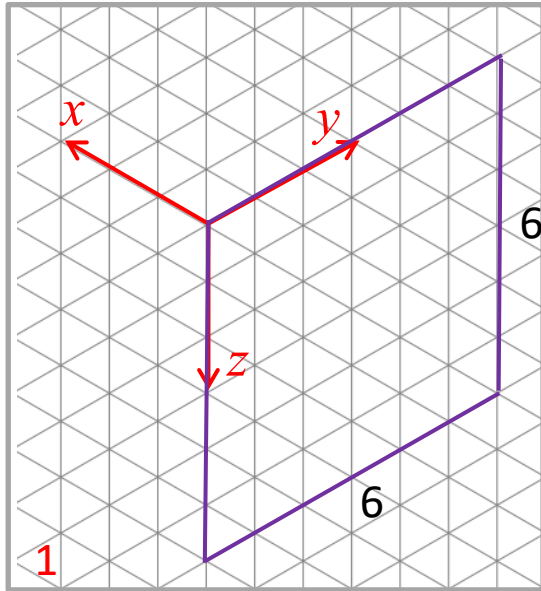
Vistas ortográficas



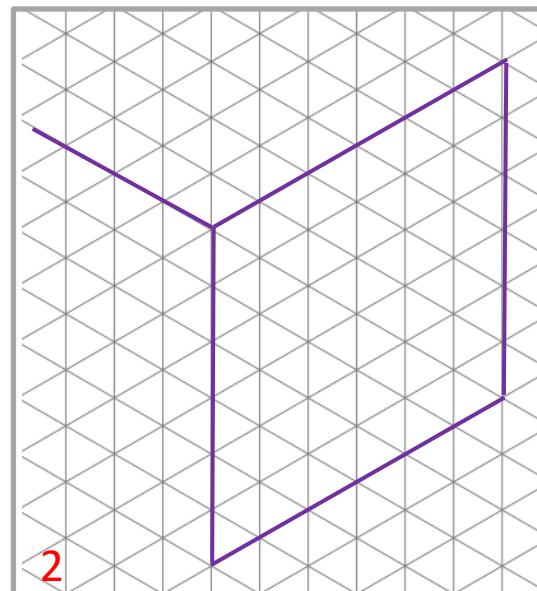
Perspectiva Isométrica - 4



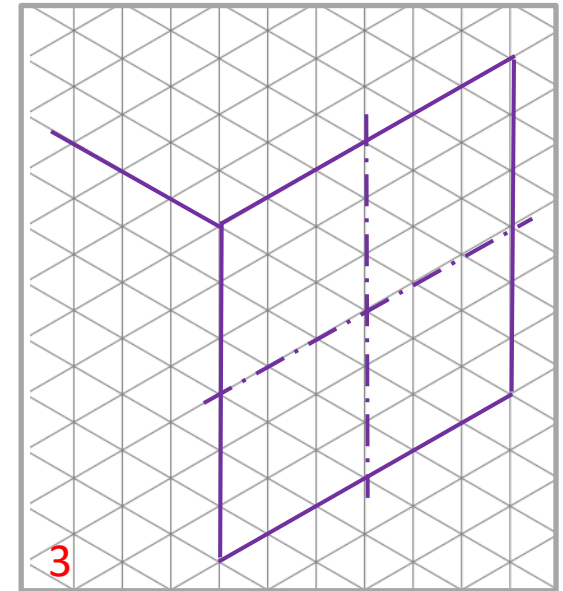
Perspectiva Isométrica de círculo (diâmetro 6 unidades)



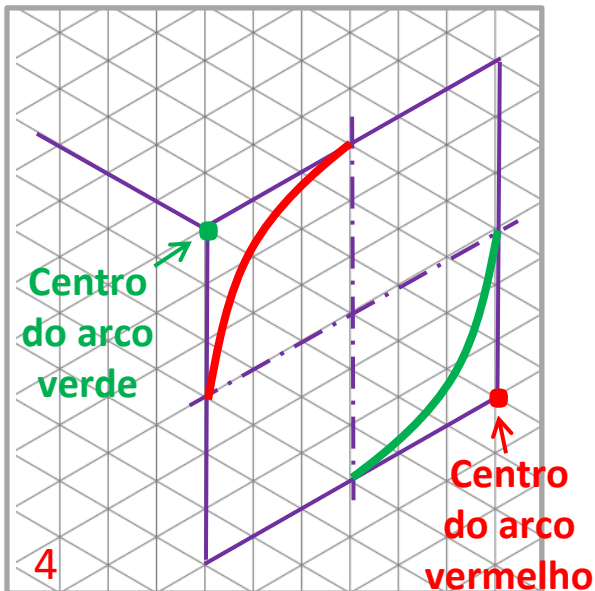
Desenhar entre os eixos y e z



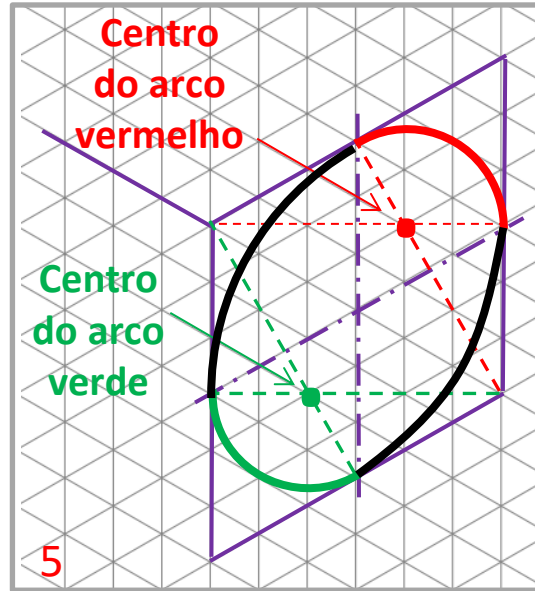
Desenhar quadrado 6 x 6



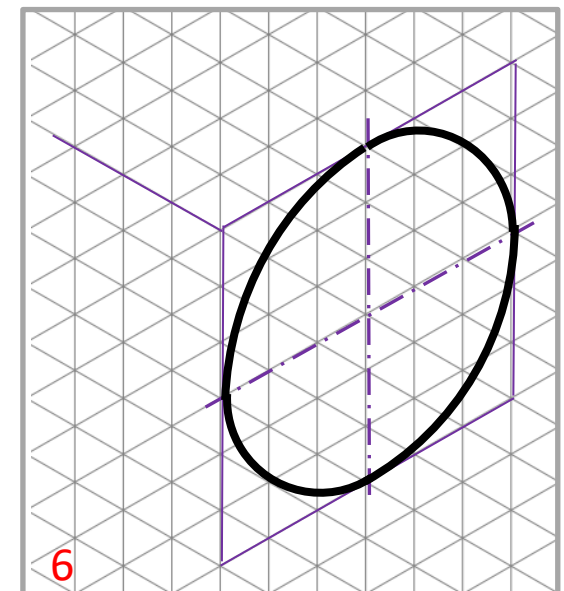
Desenhar linhas de centro



Desenhar arcos maiores



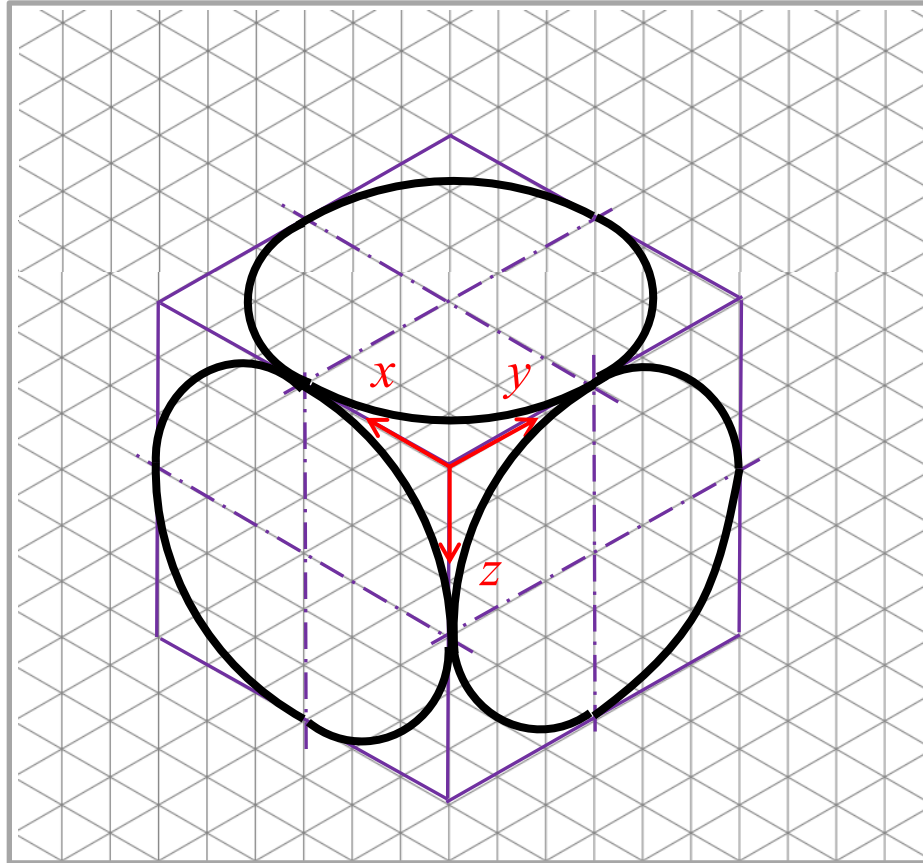
Desenhar arcos menores



Apagar linhas auxiliares

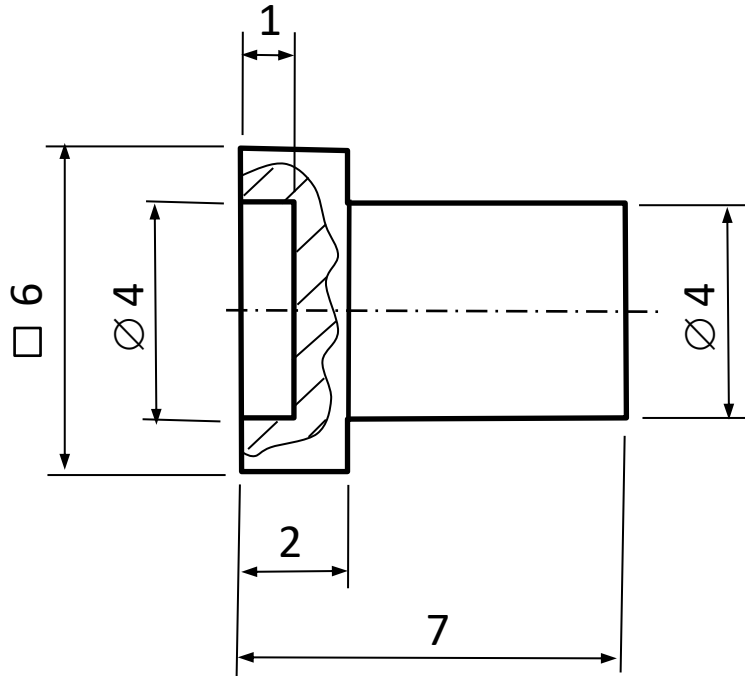
Perspectiva Isométrica

Círculos nos planos xy , xz e yz

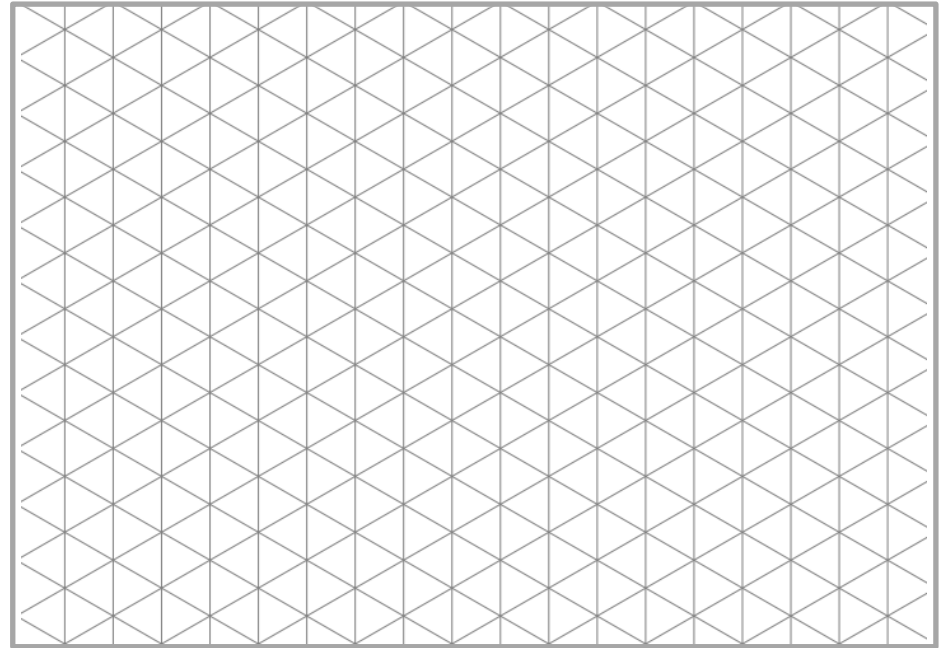


(diâmetro 6 unidades)

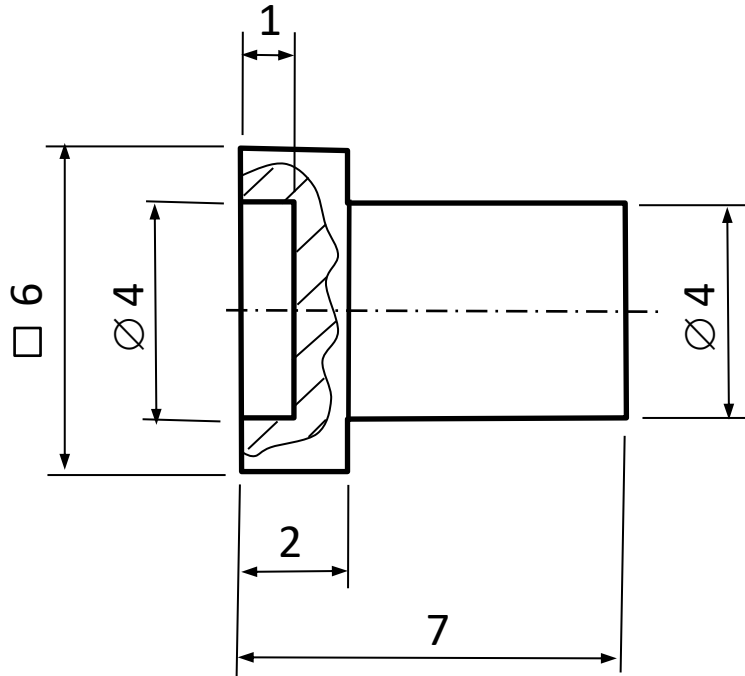
Vistas ortográficas



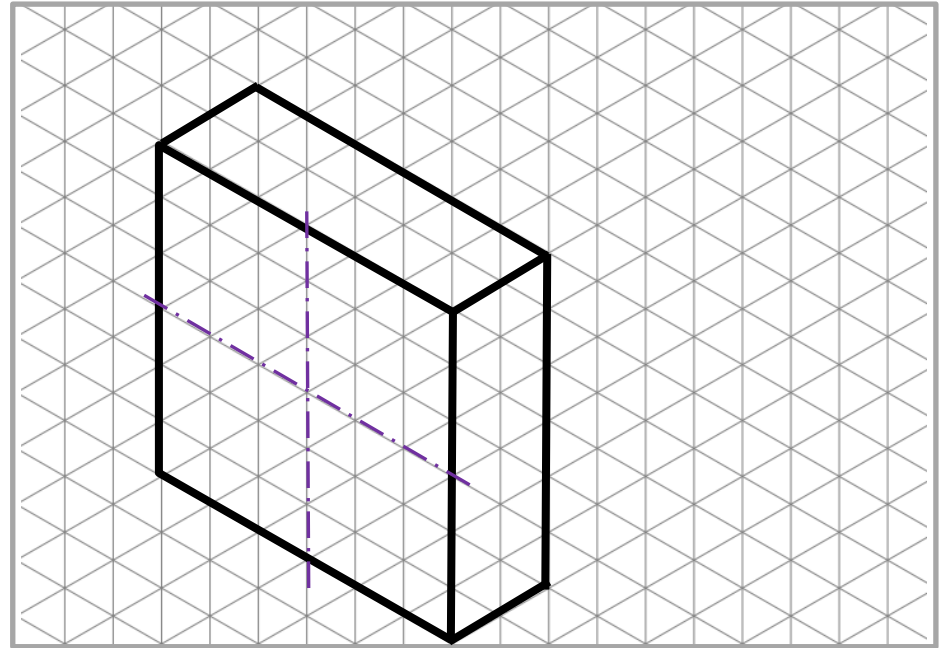
Perspectiva Isométrica - 5



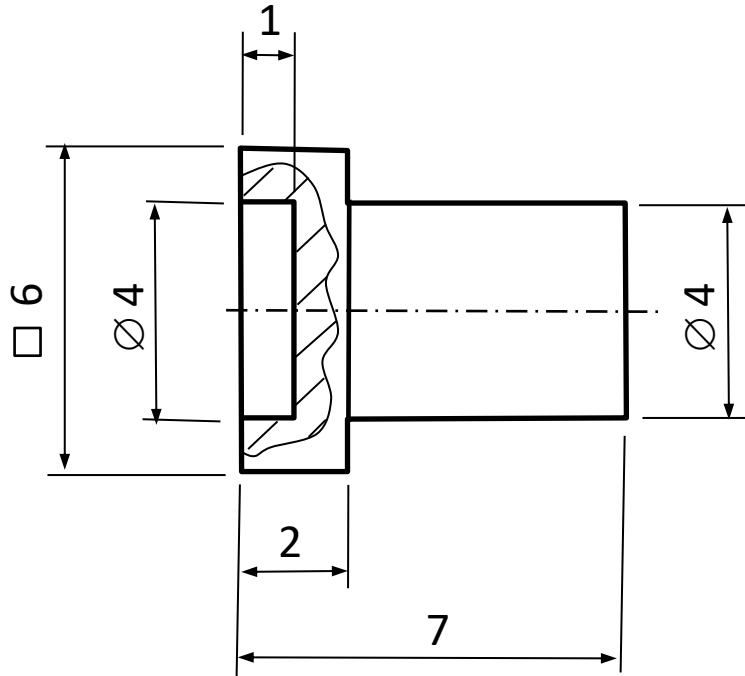
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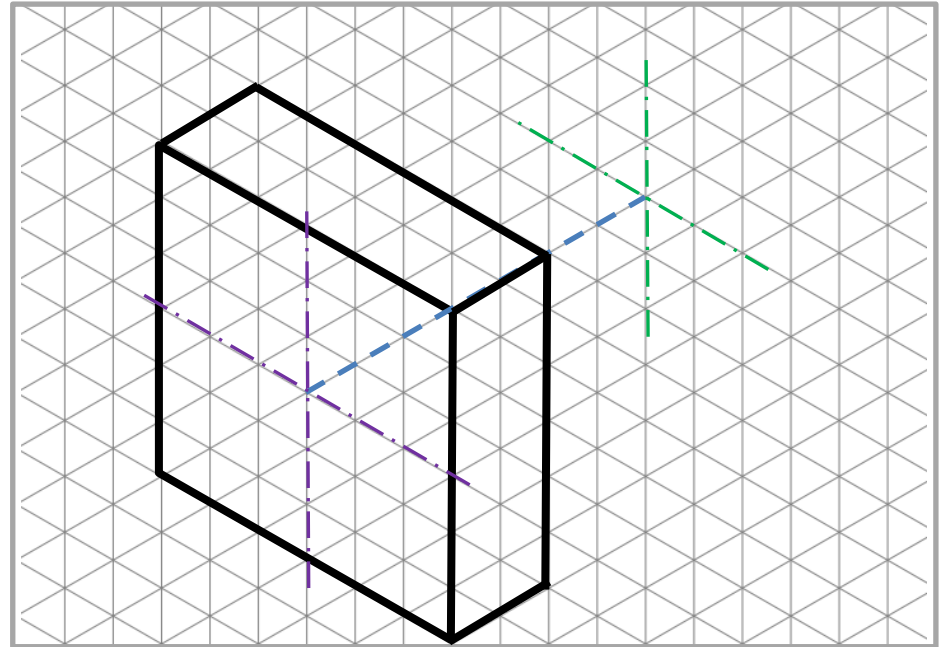
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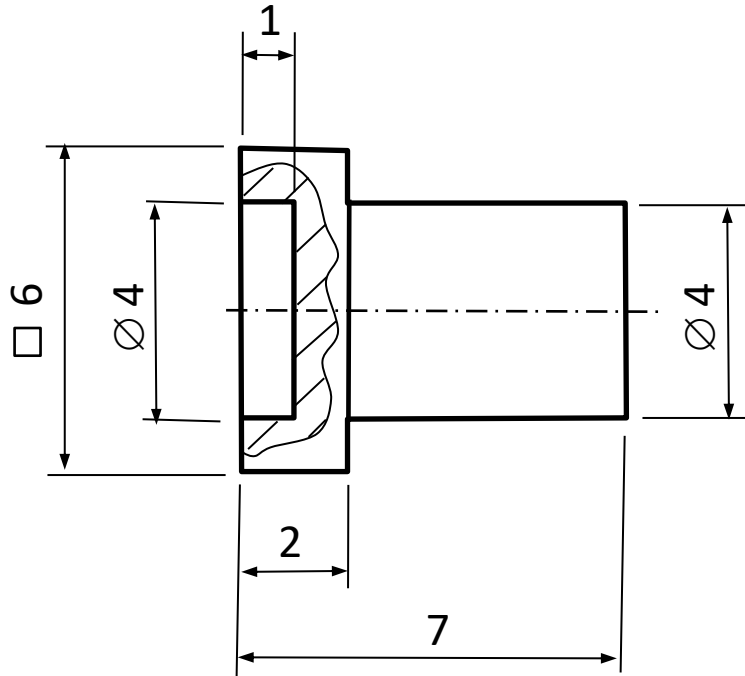
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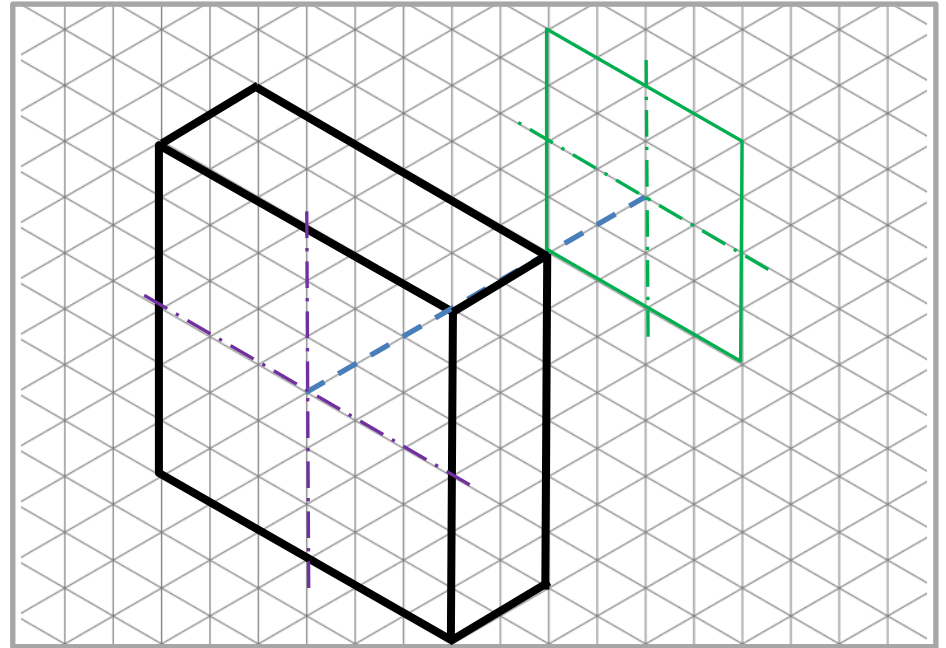
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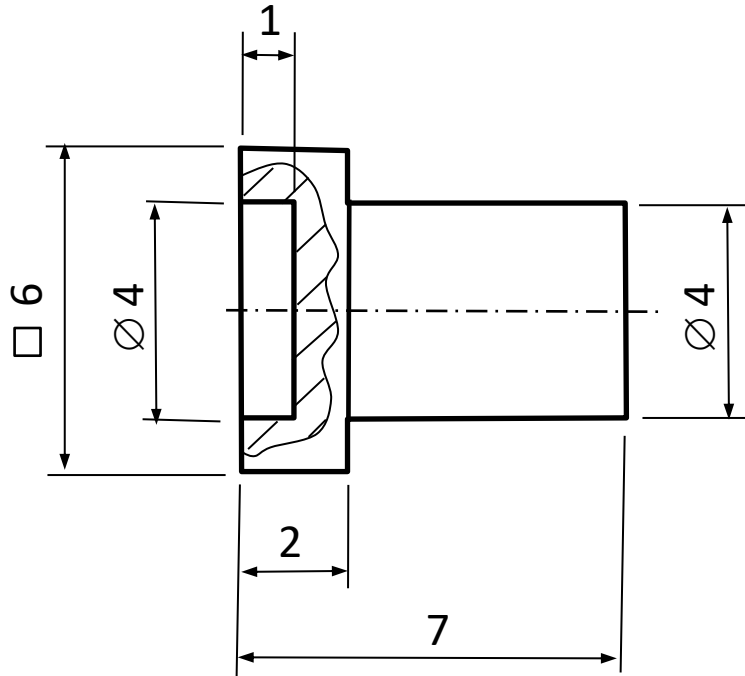
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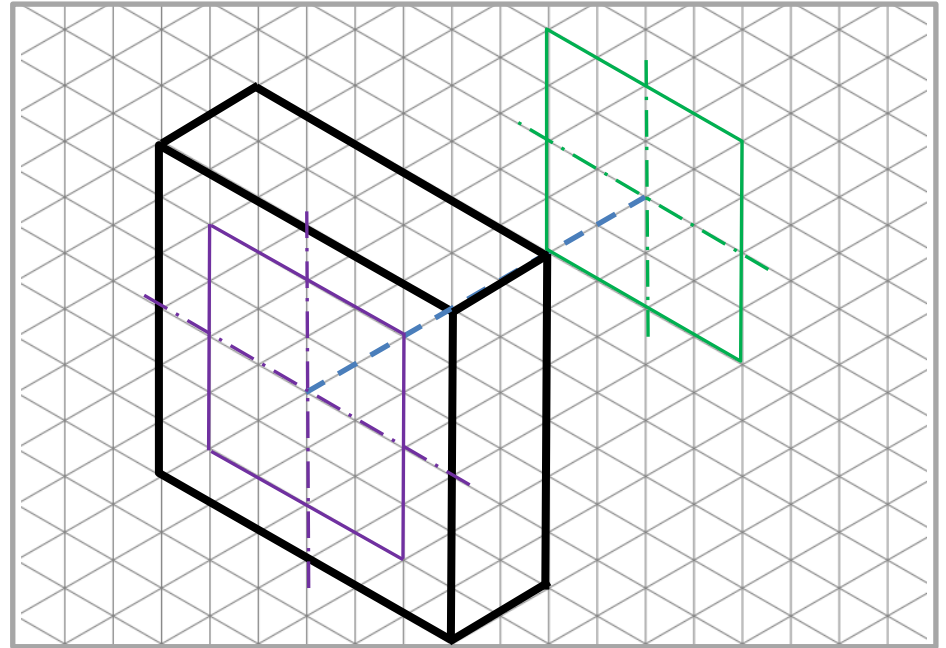
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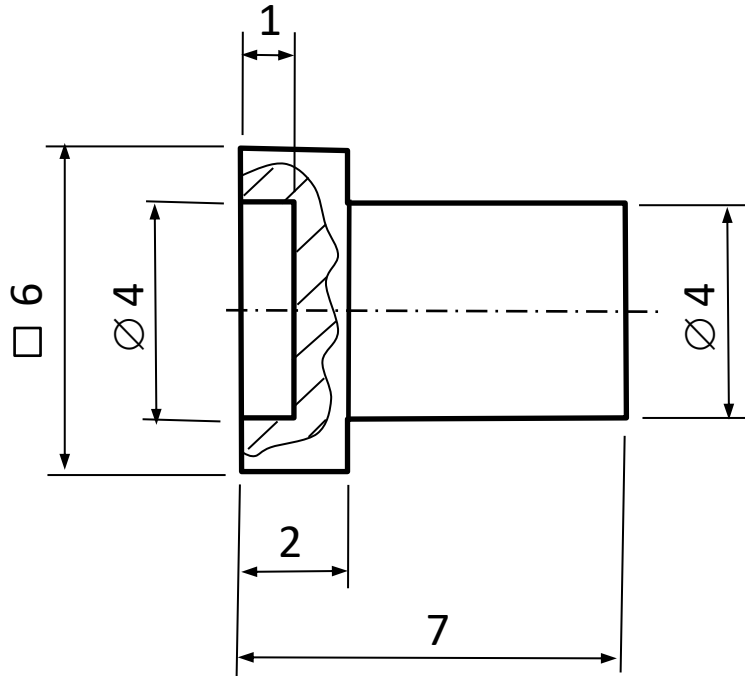
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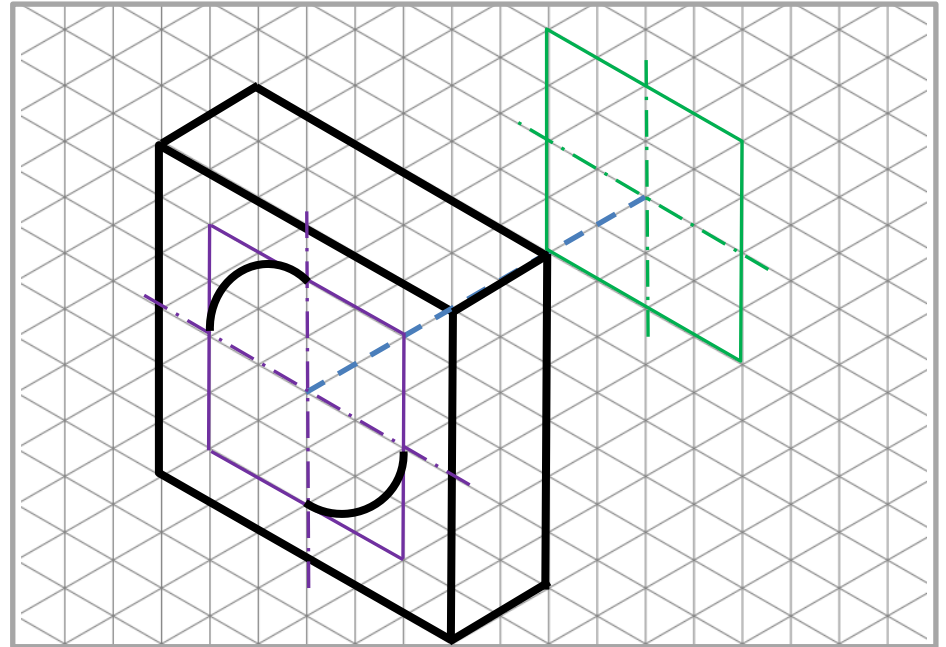
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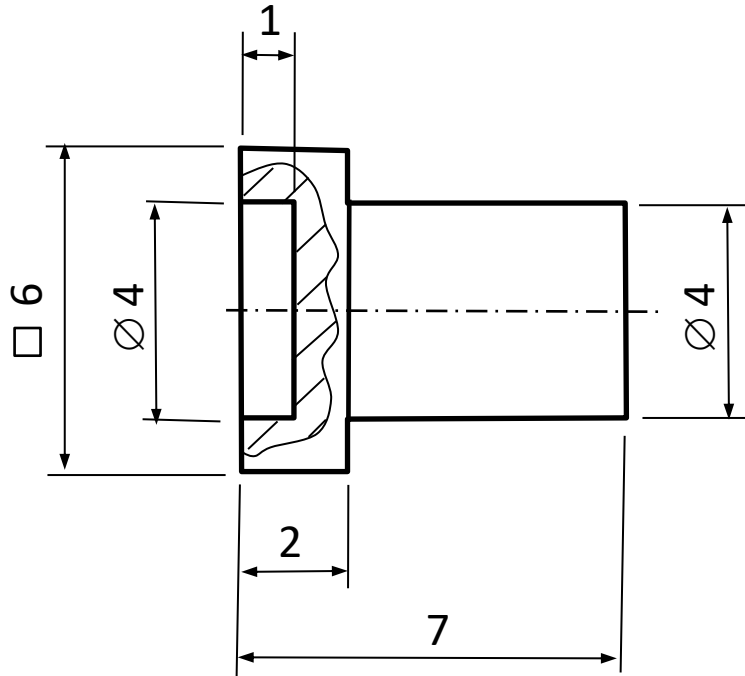
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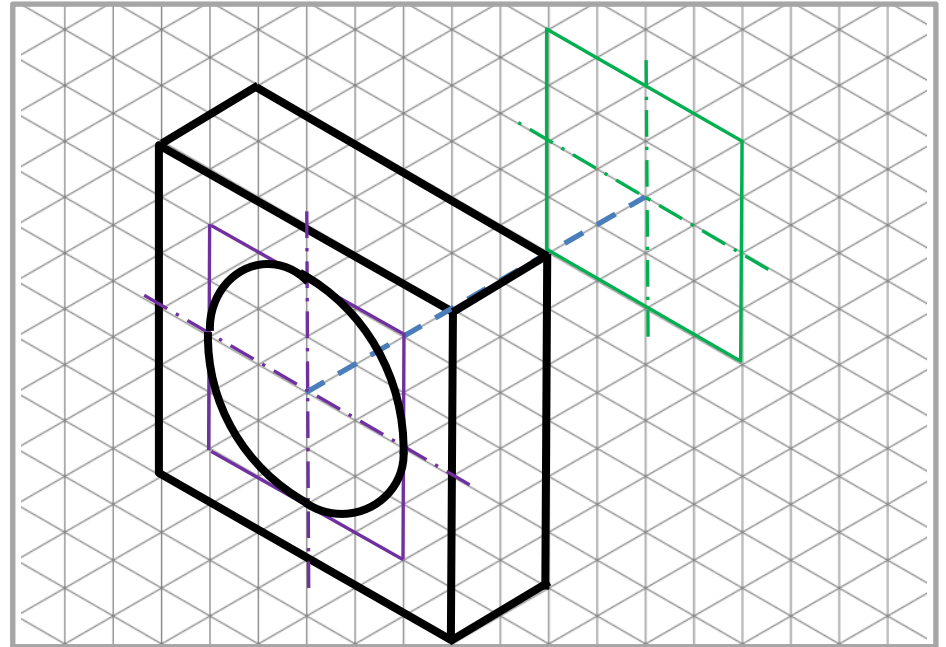
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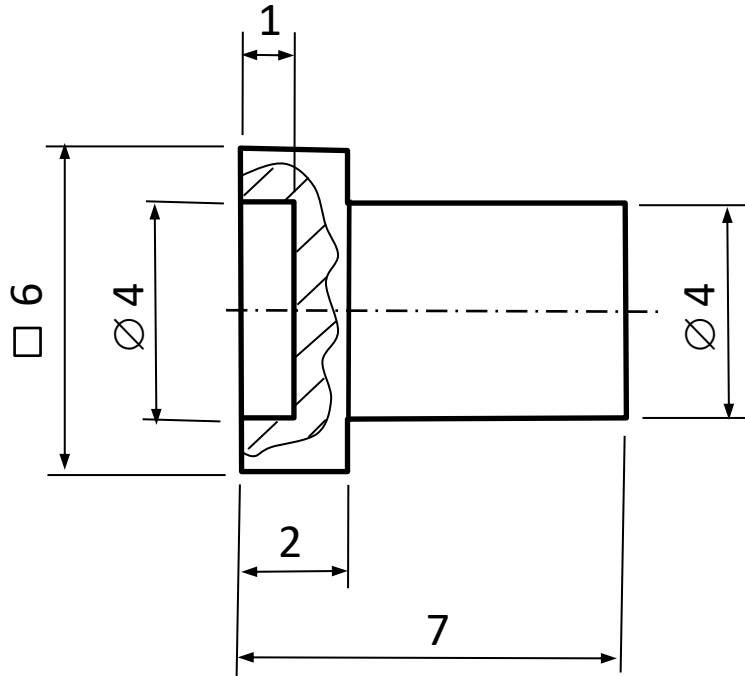
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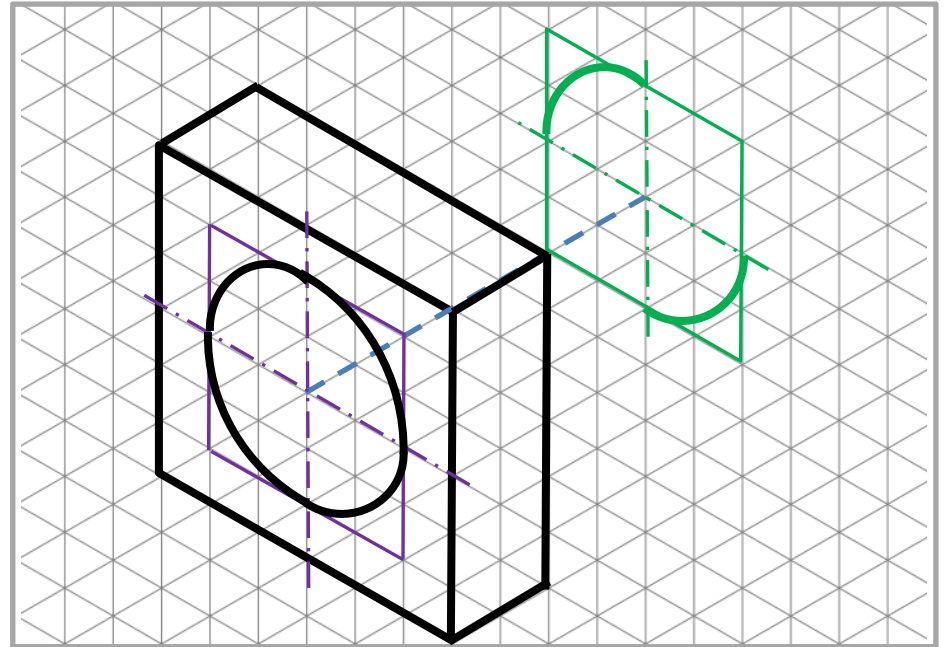
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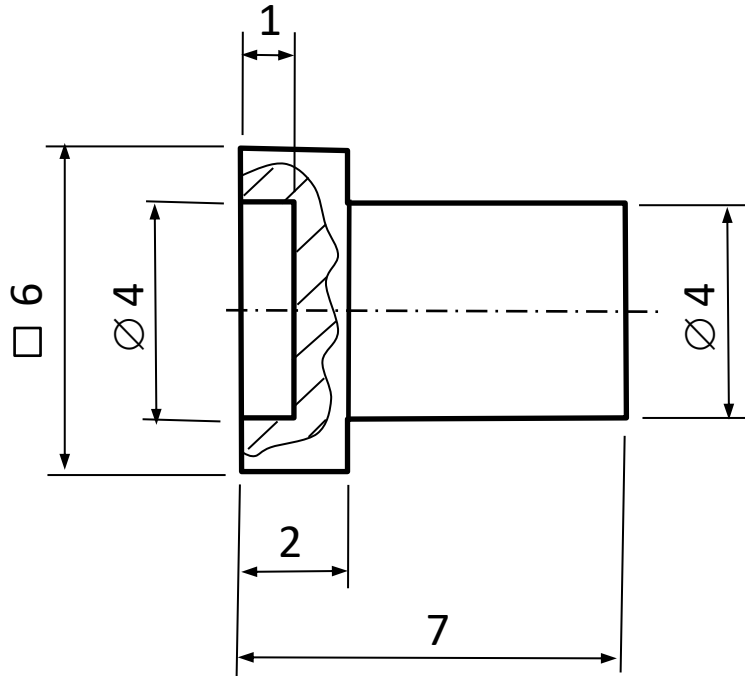
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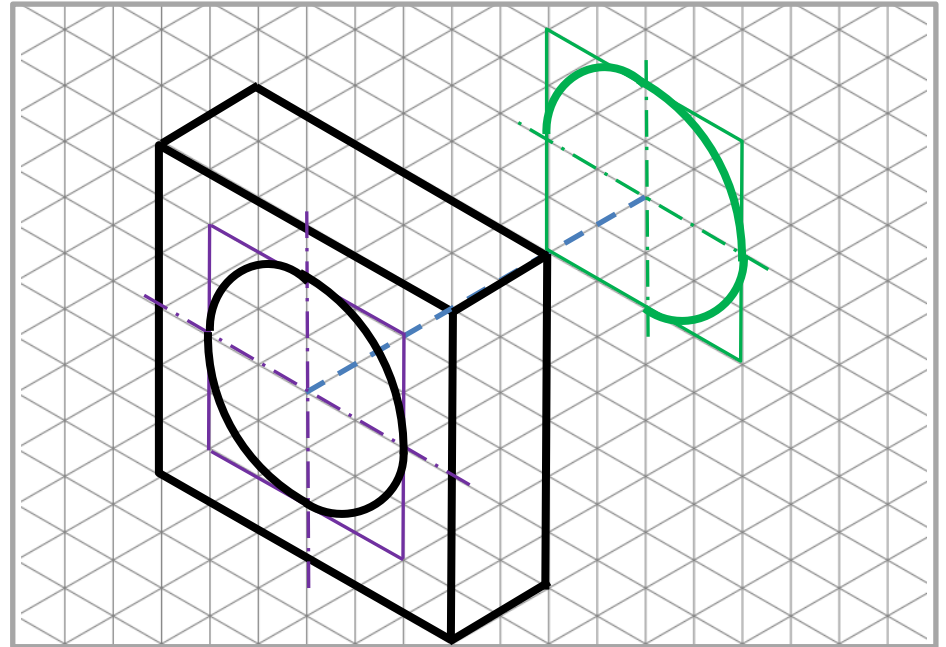
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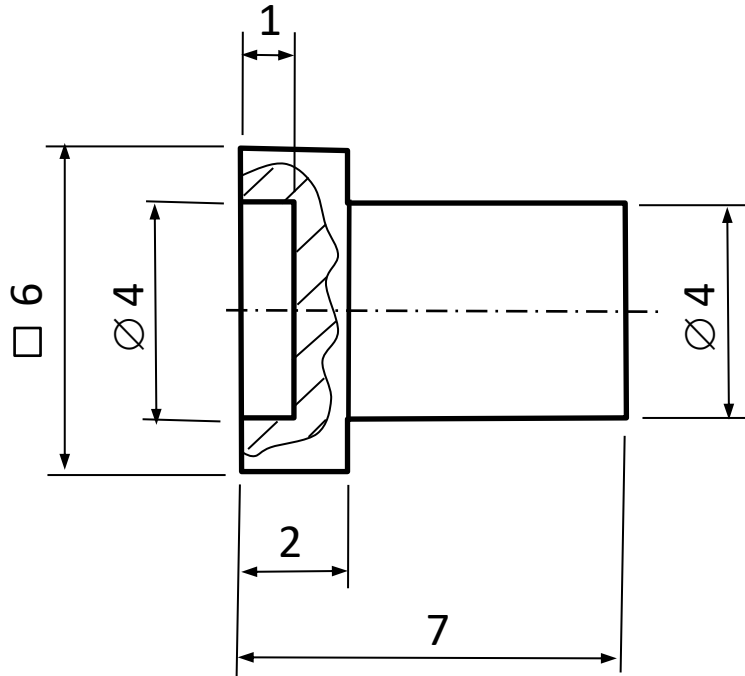
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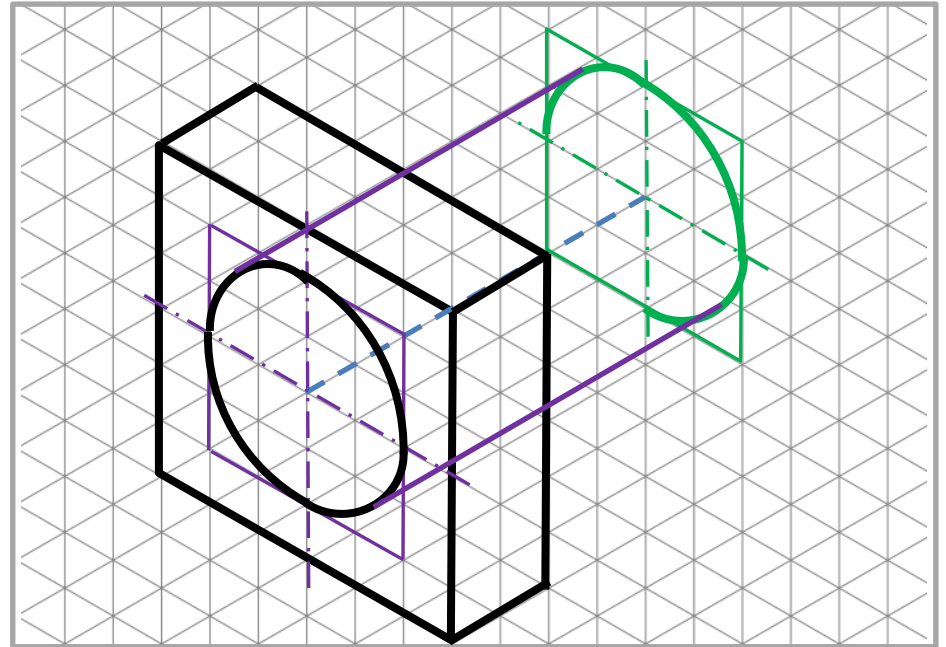
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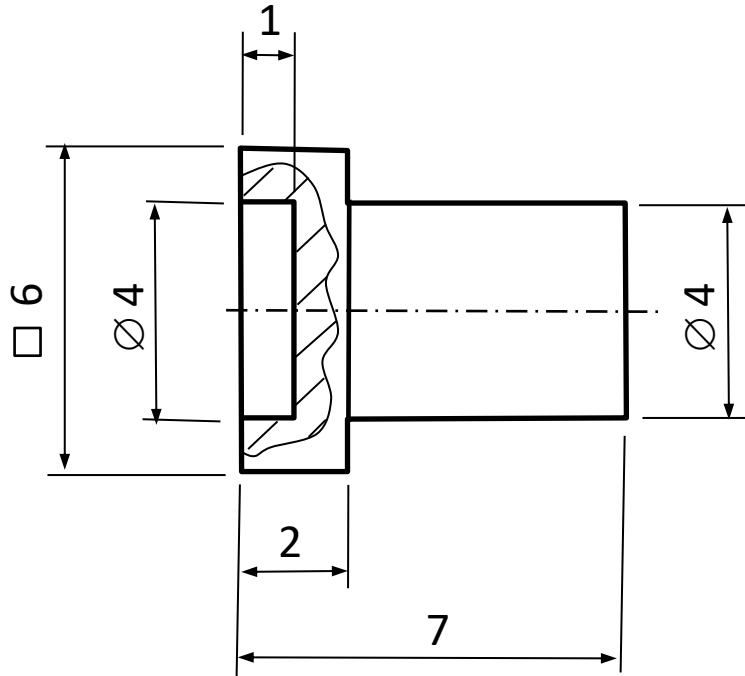
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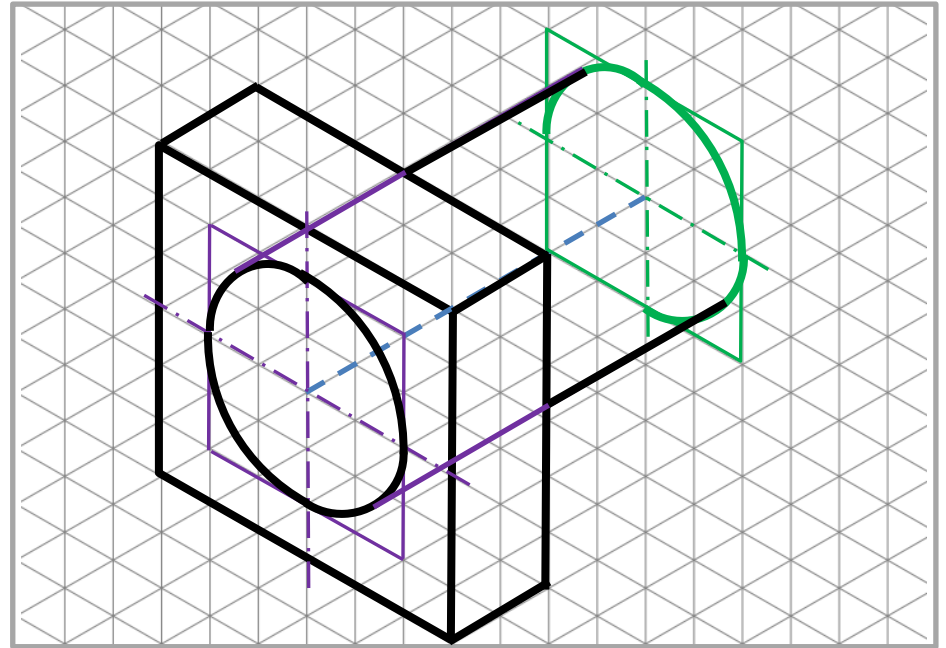
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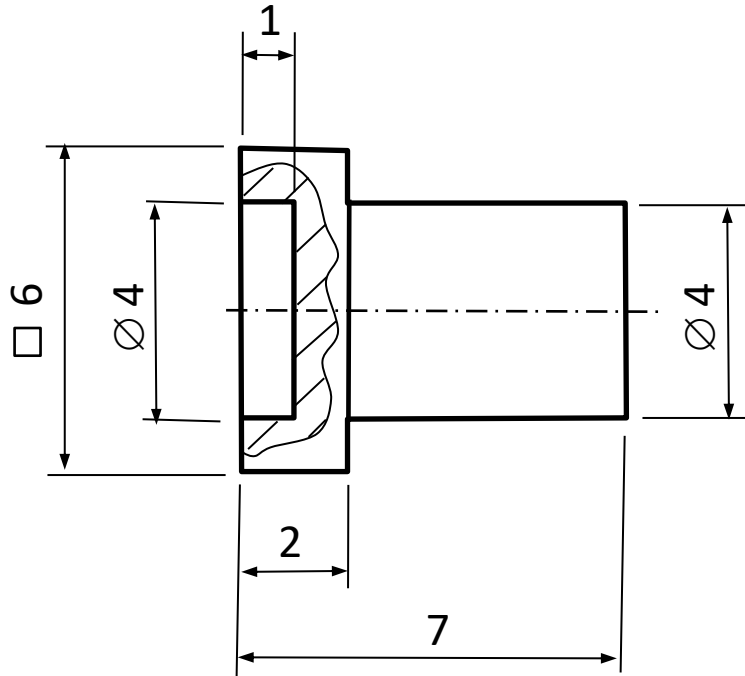
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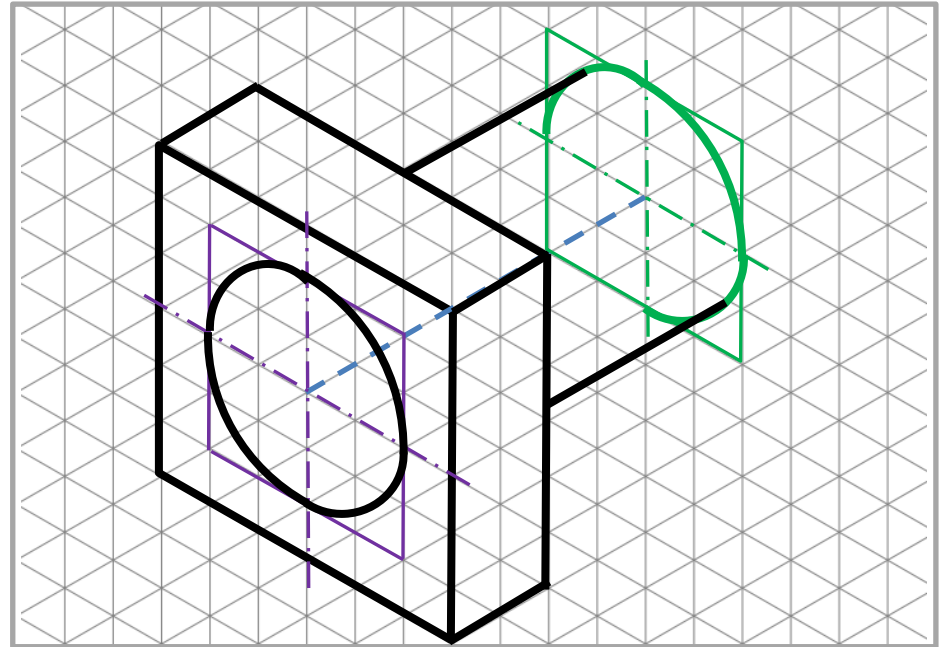
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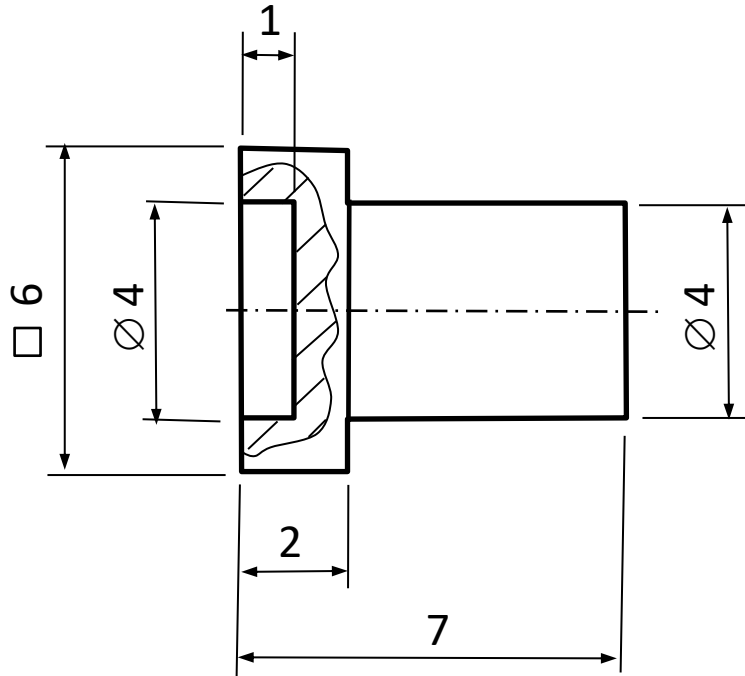
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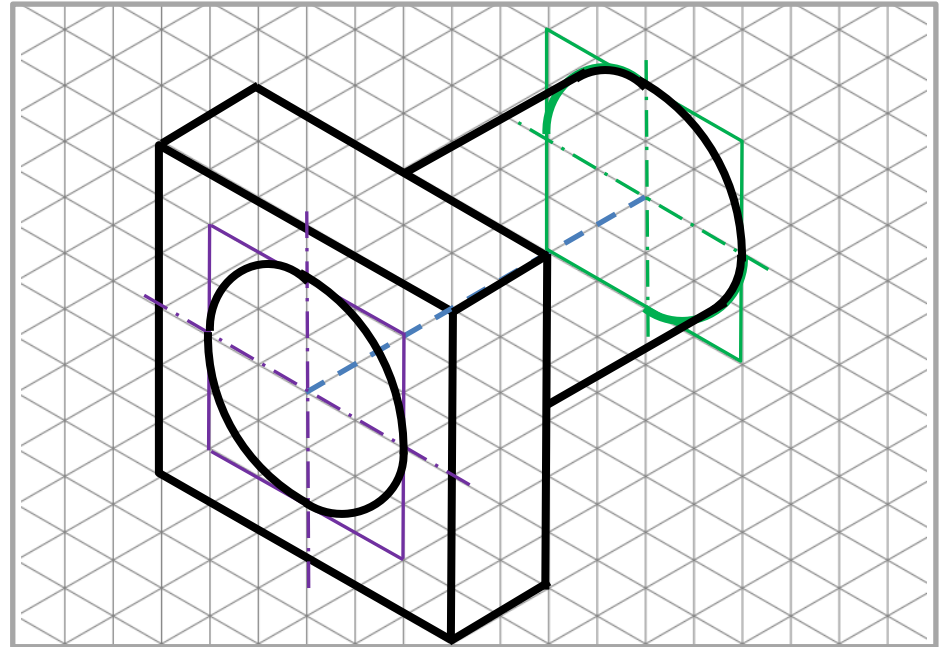
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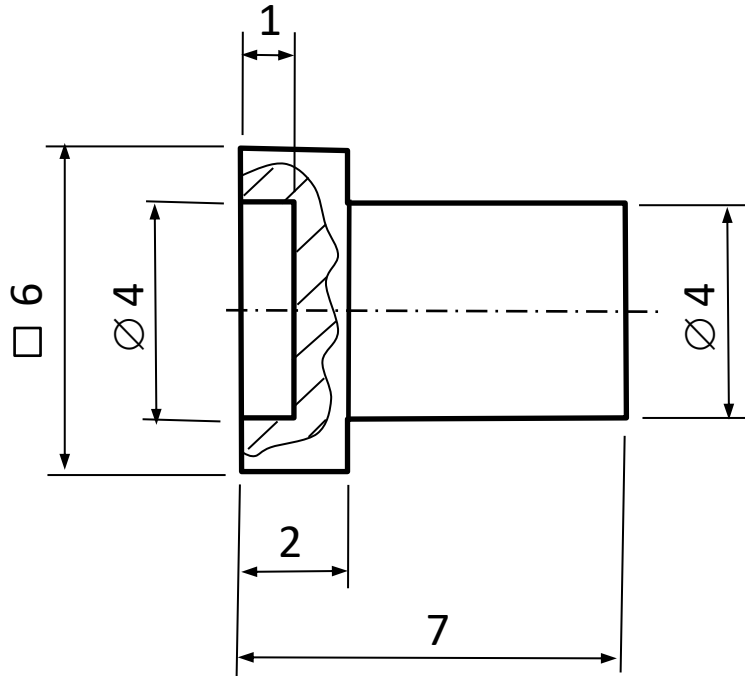
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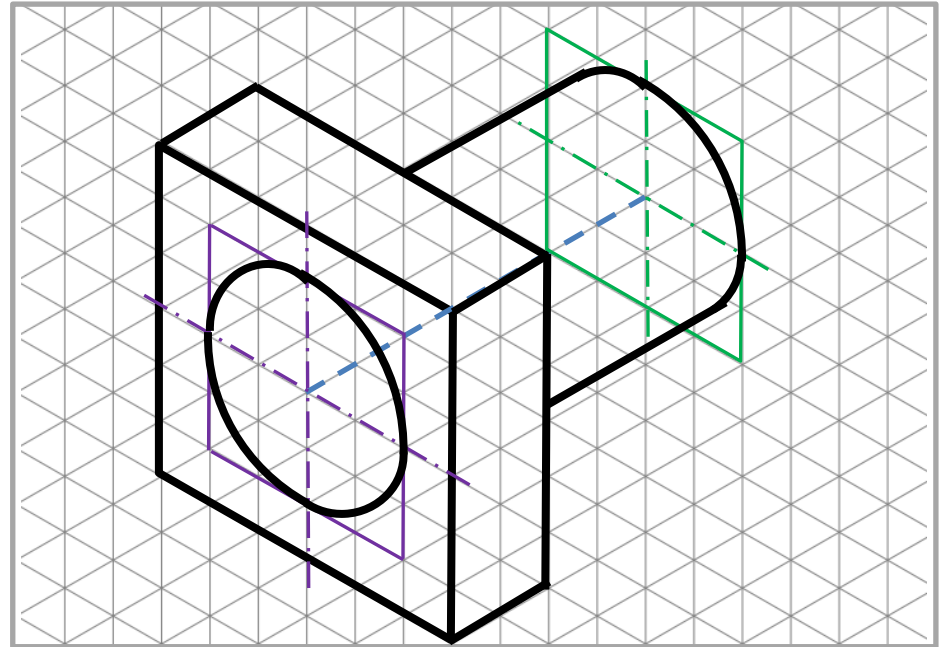
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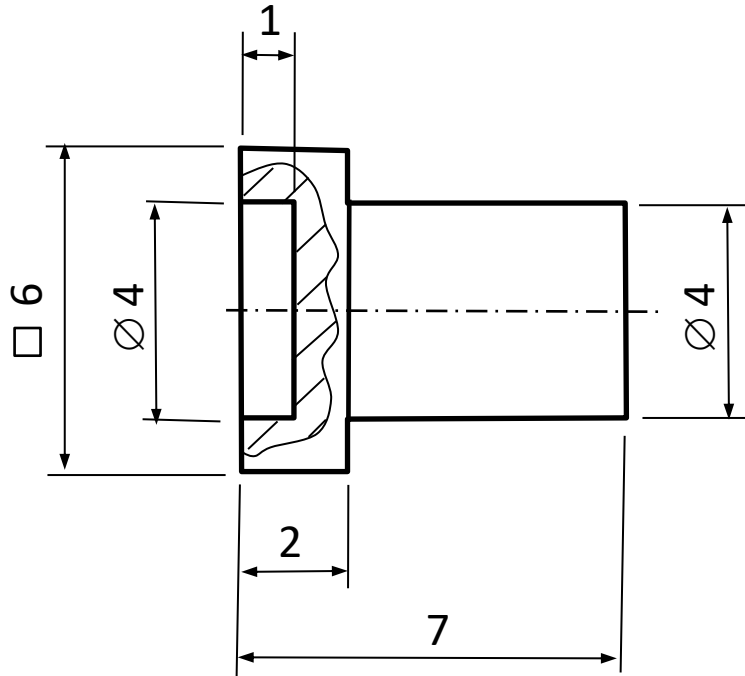
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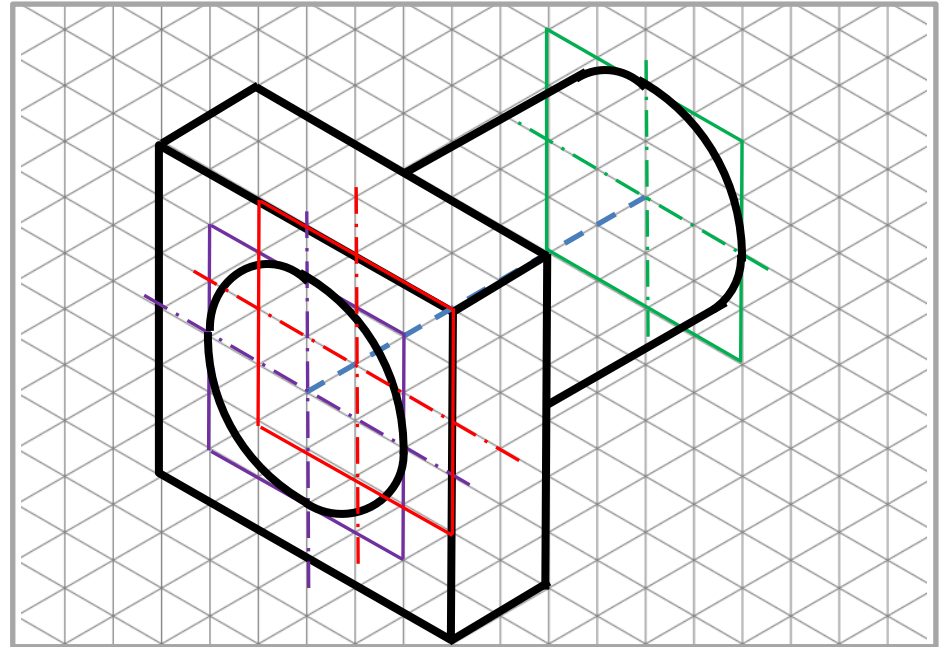
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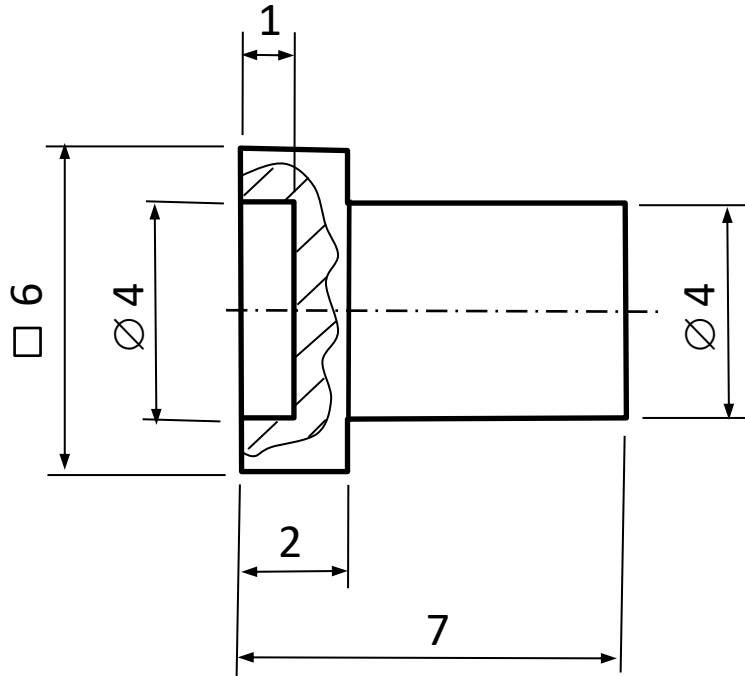
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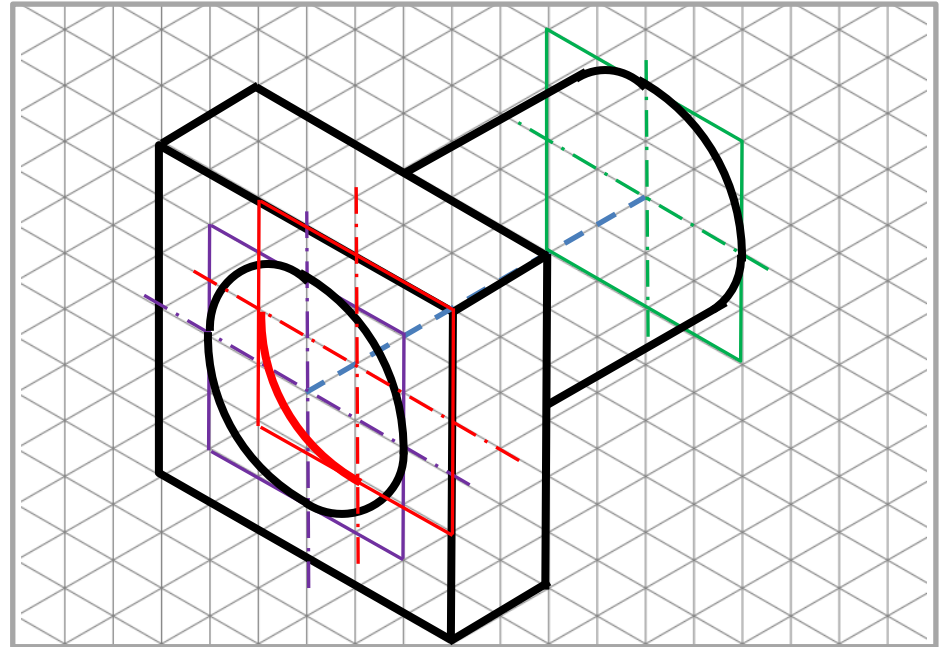
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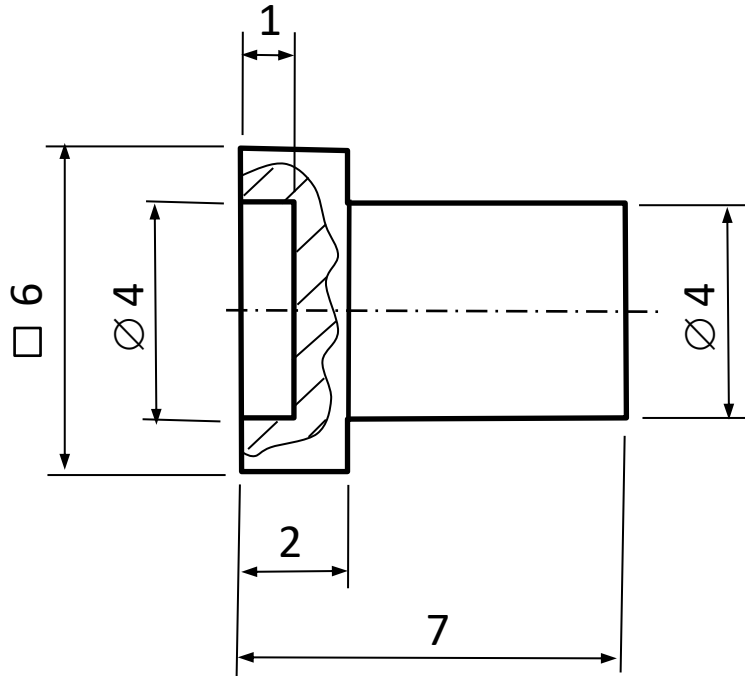
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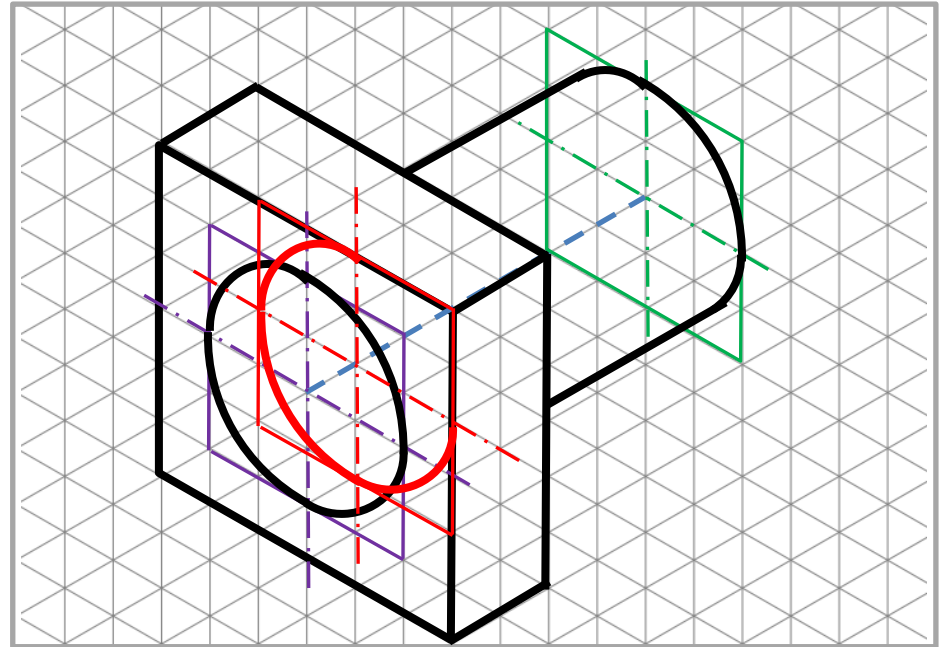
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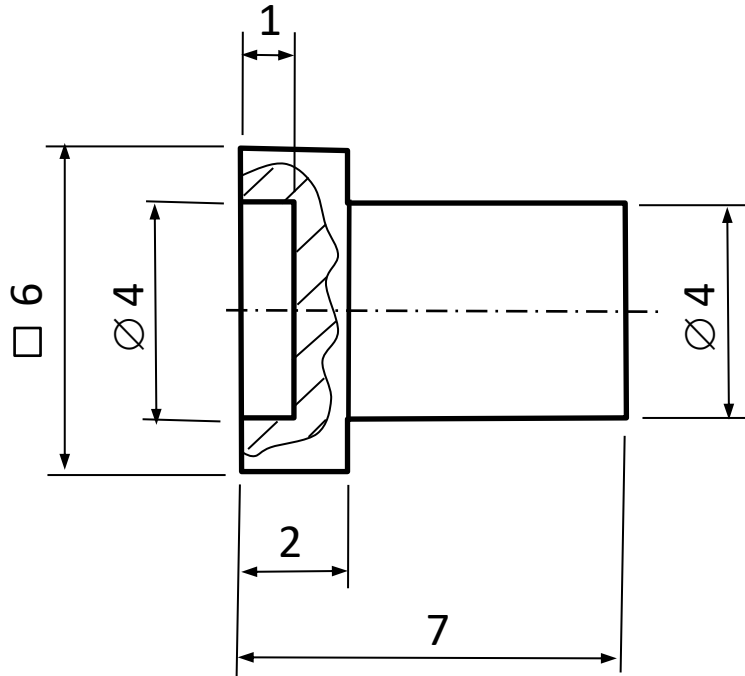
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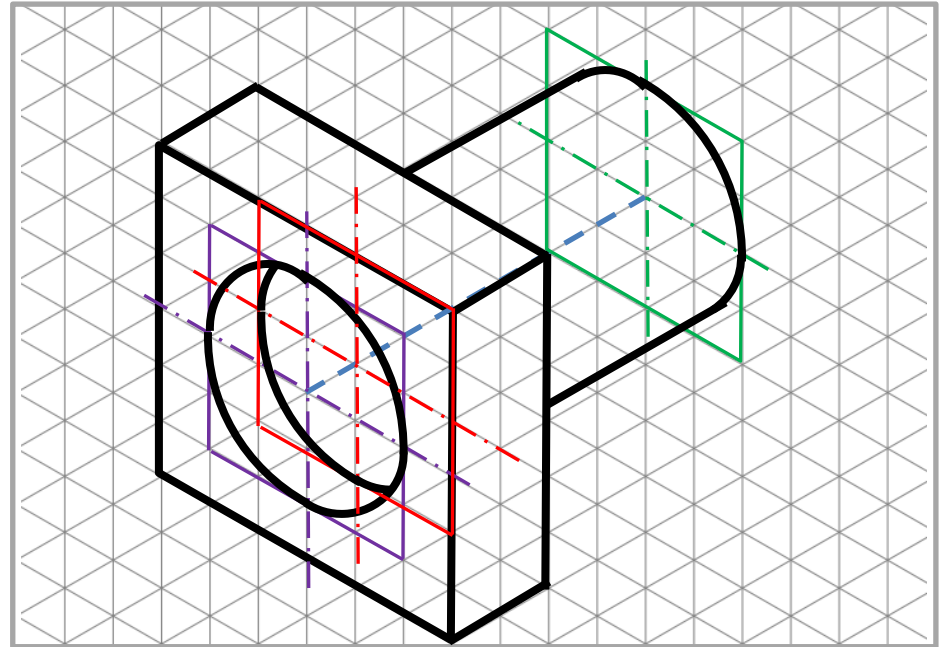
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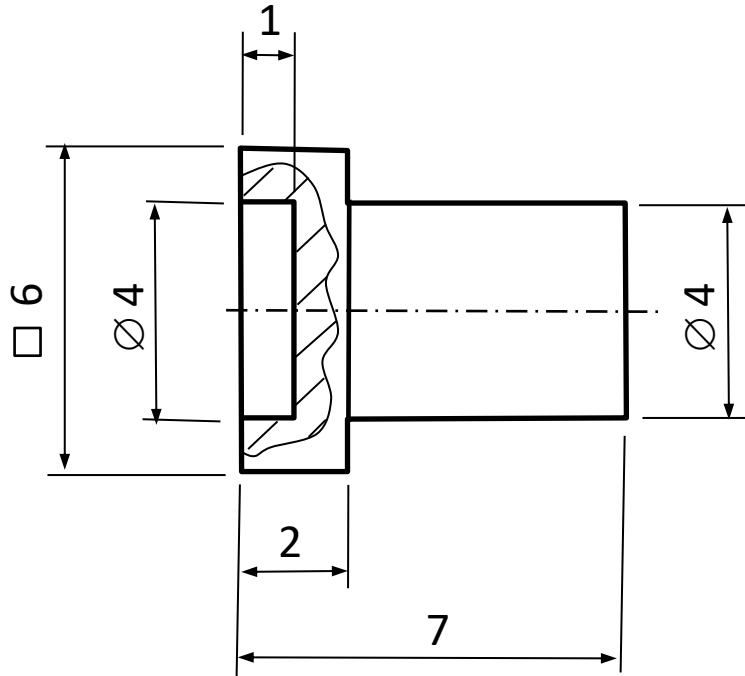
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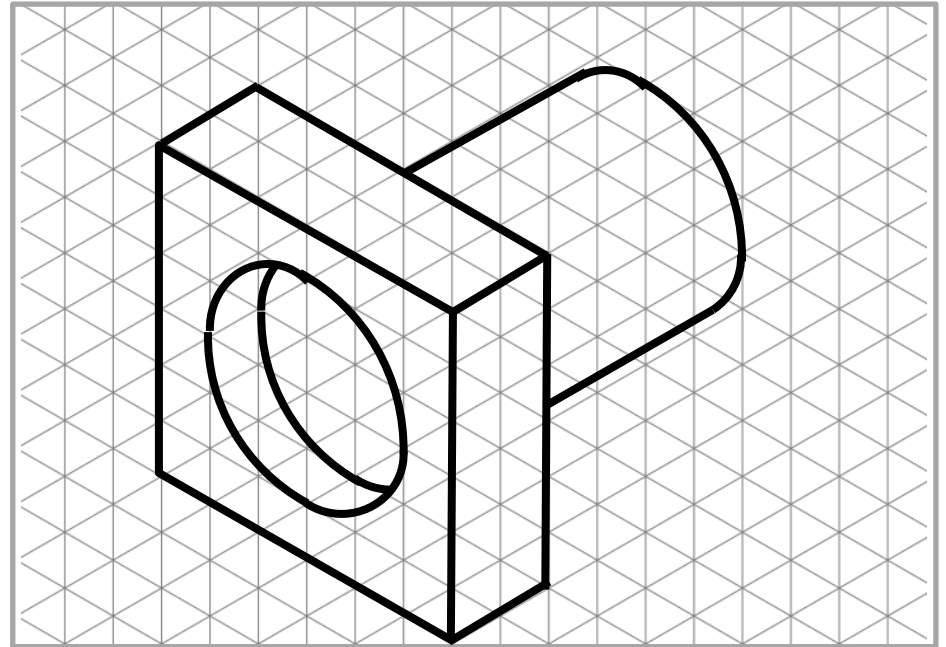
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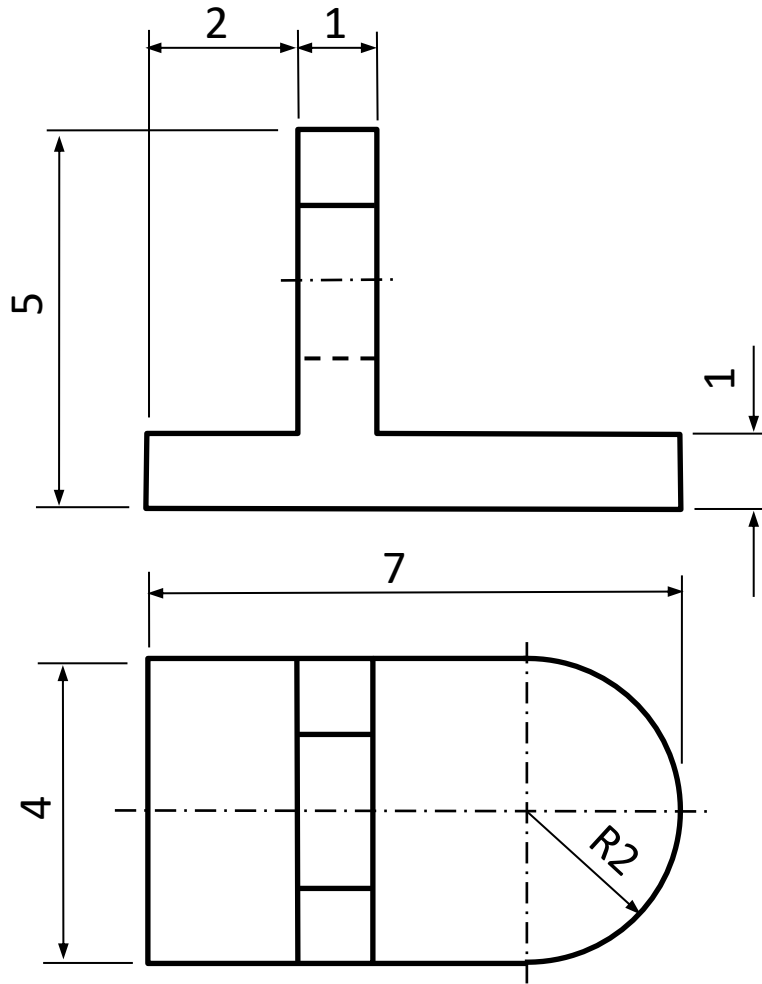
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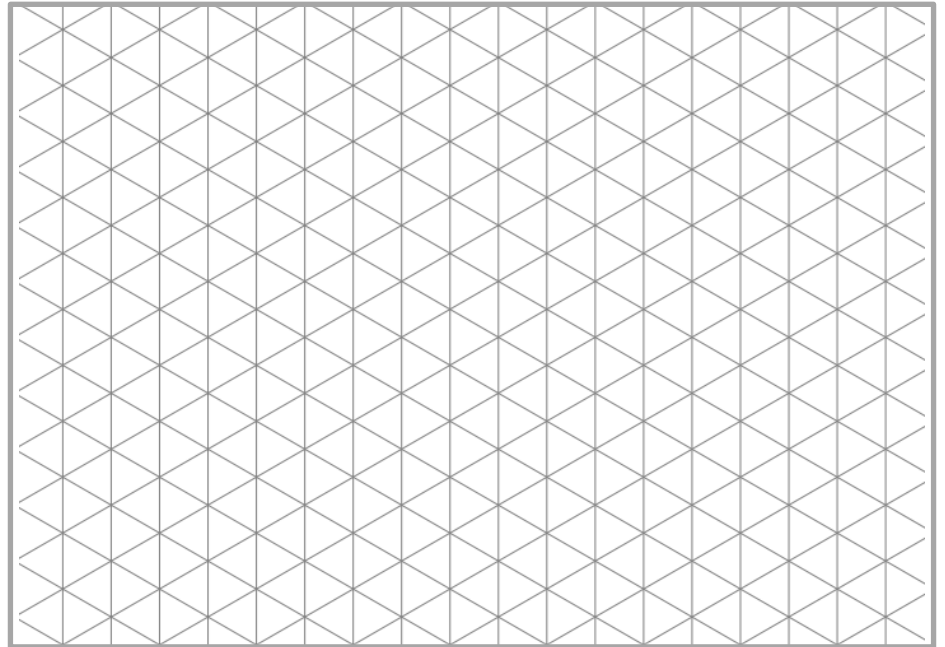
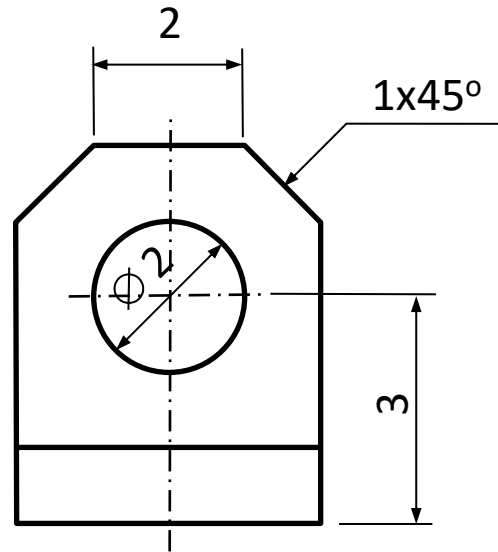
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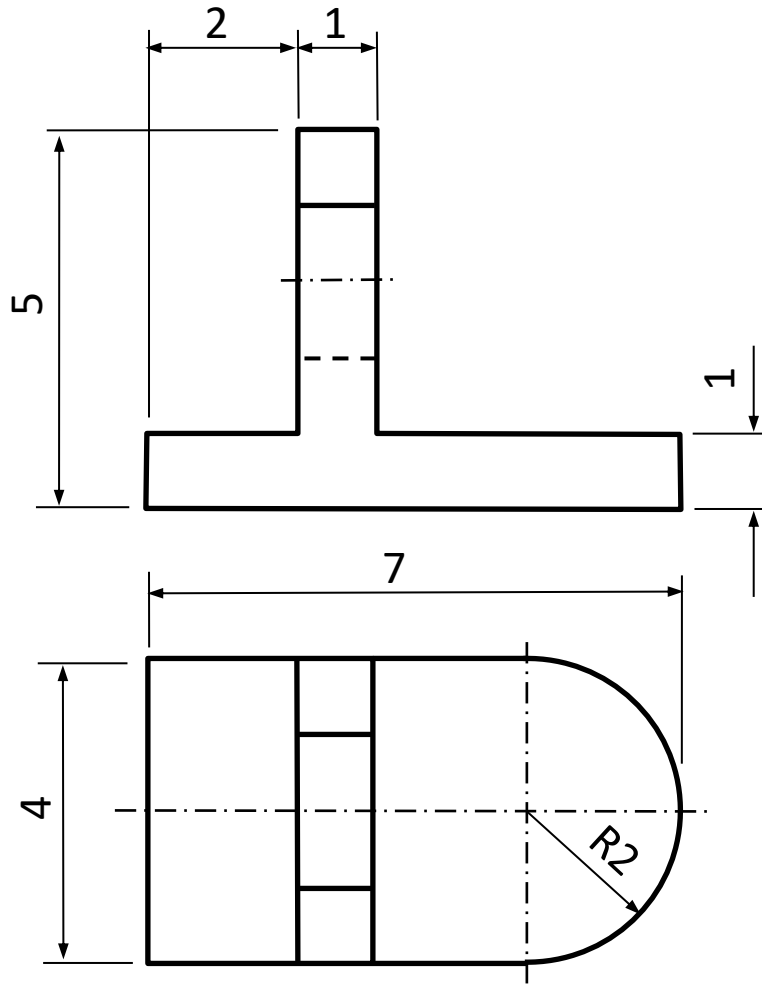
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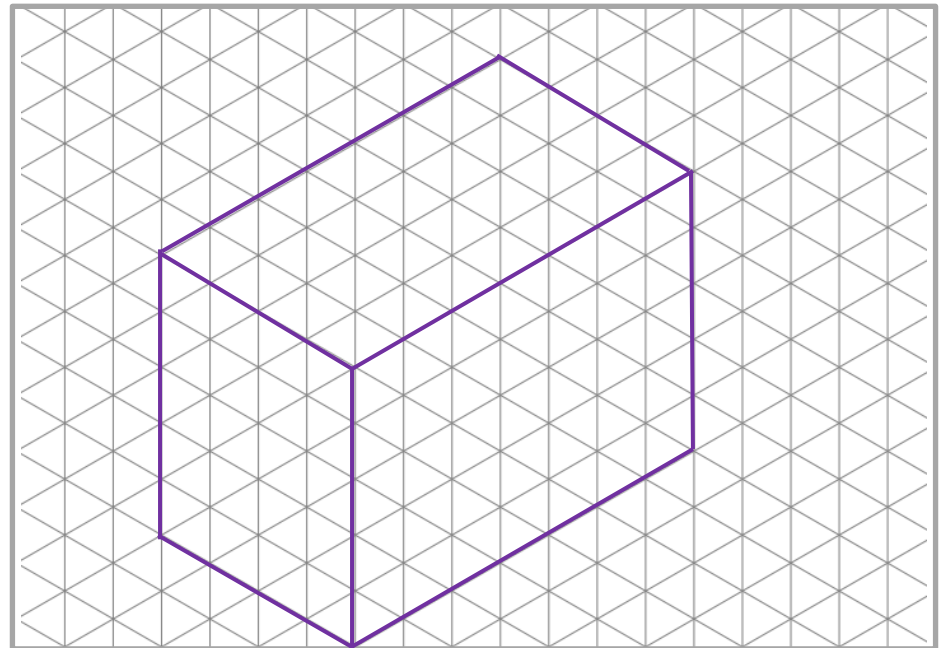
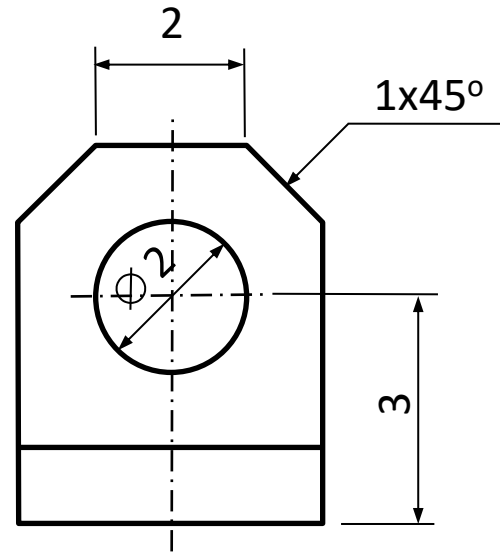
Perspectiva Isométrica - 6



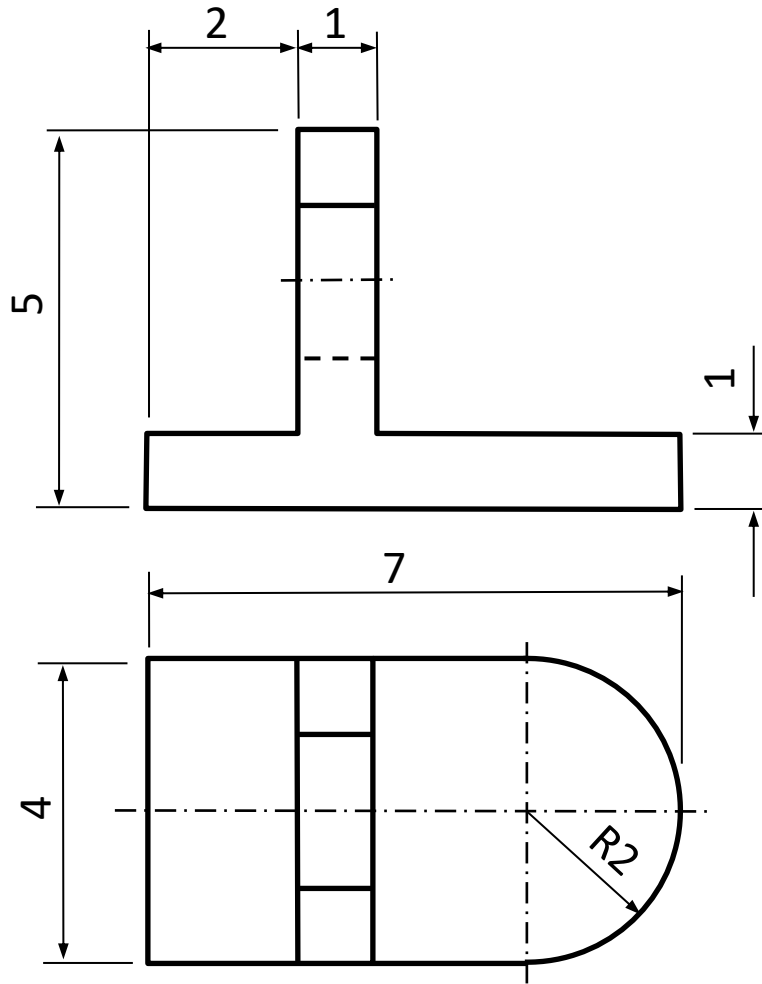
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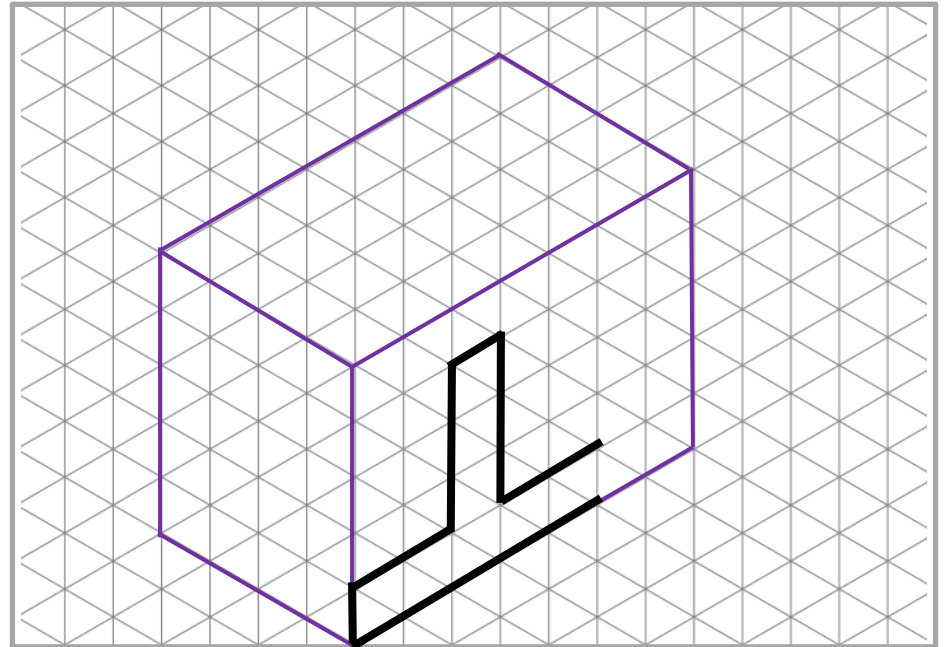
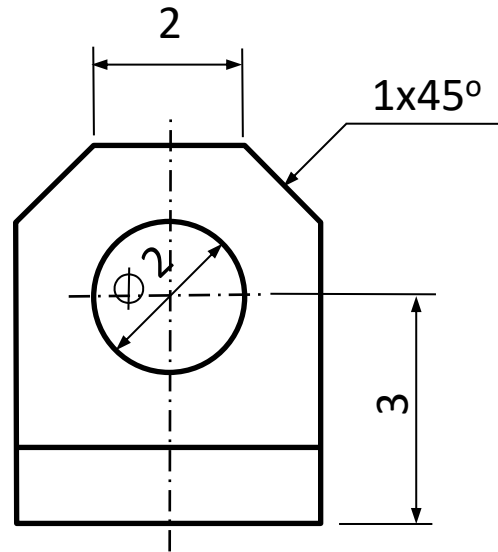
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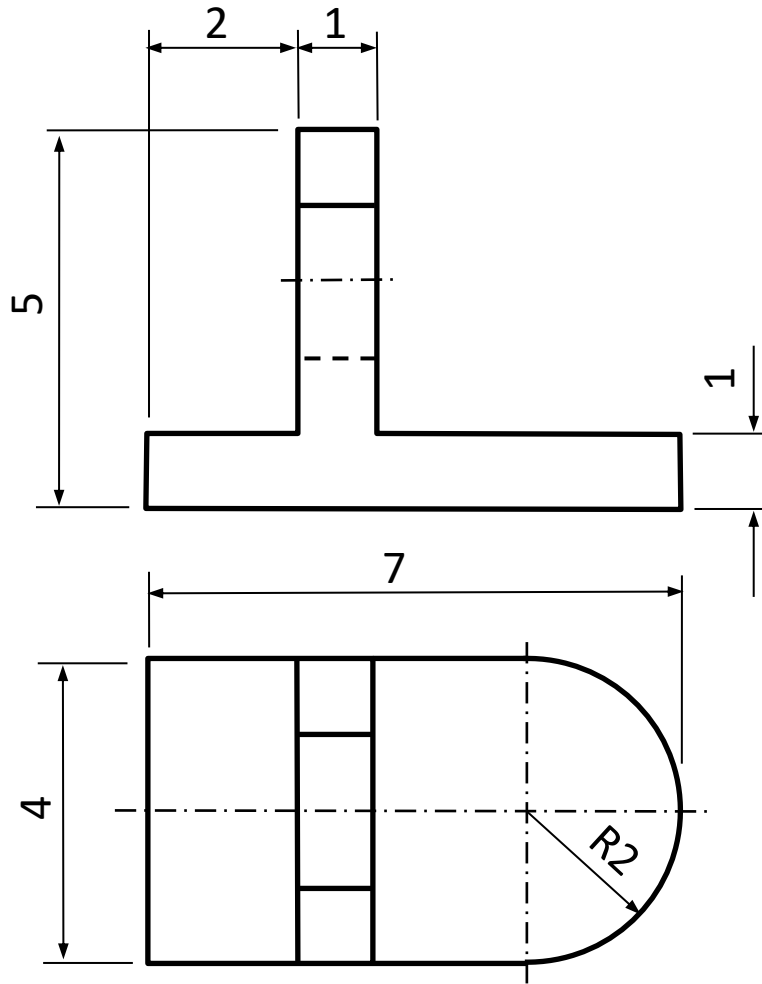
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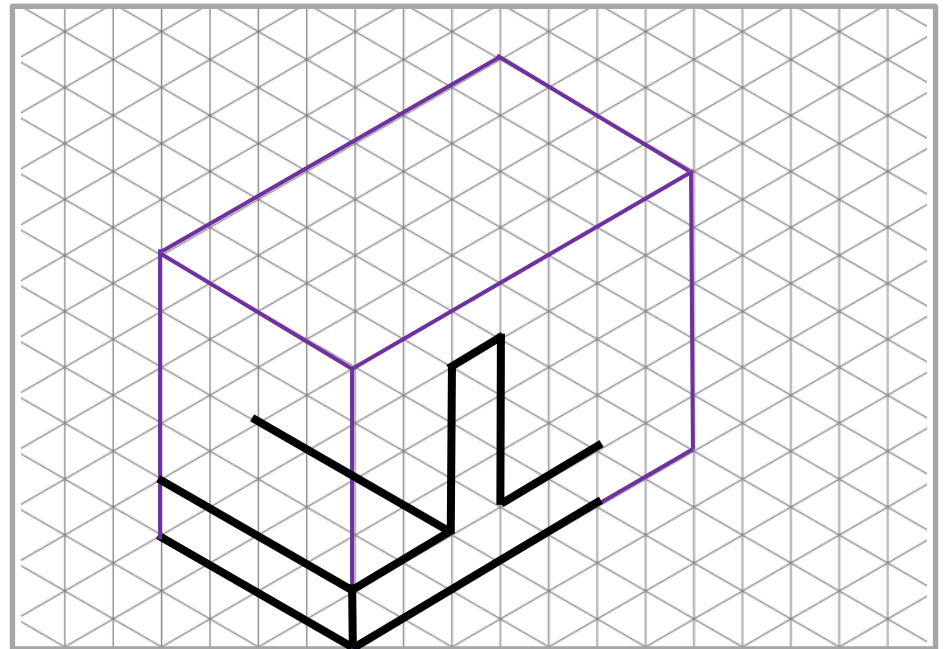
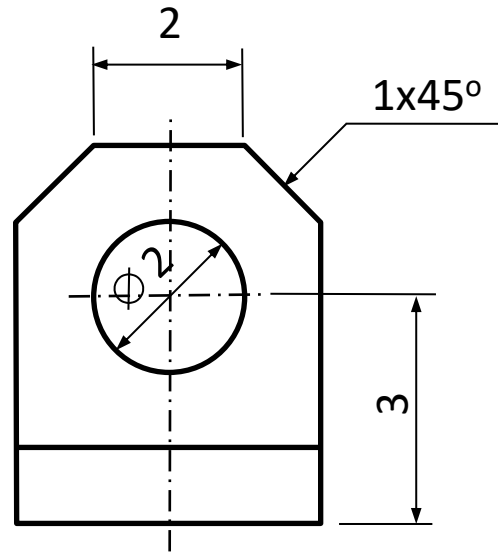
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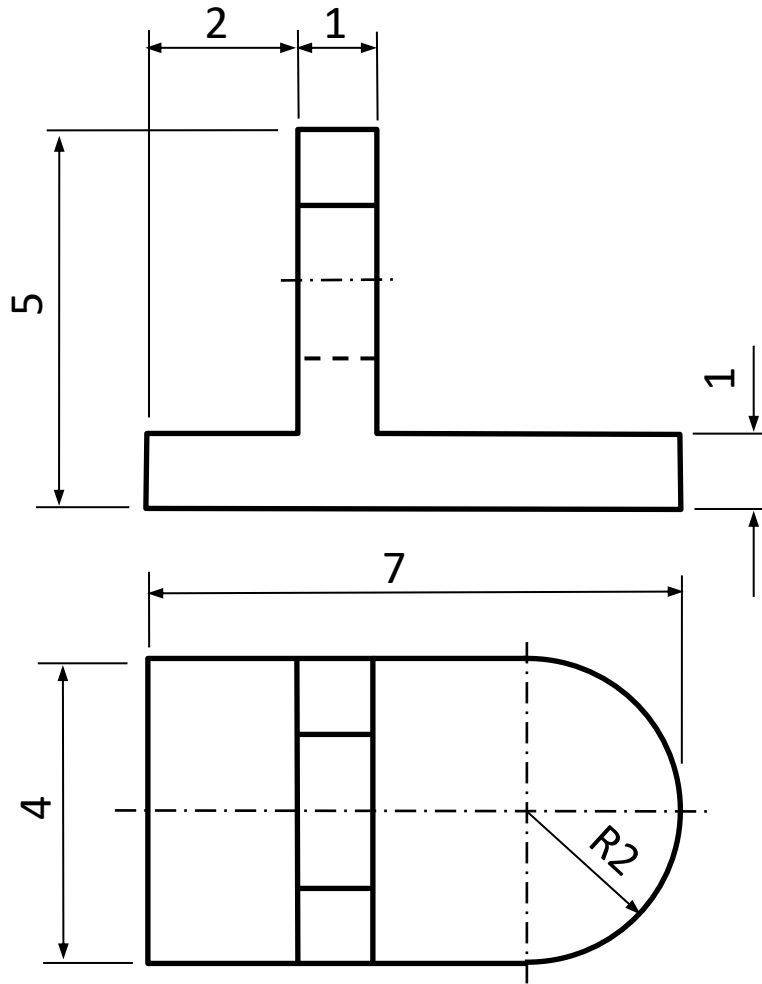
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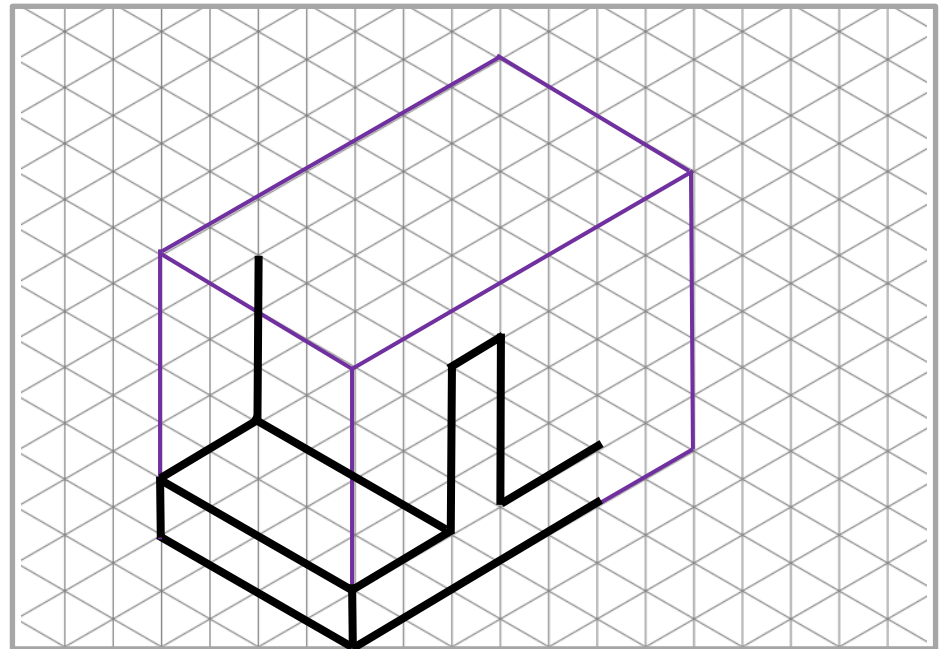
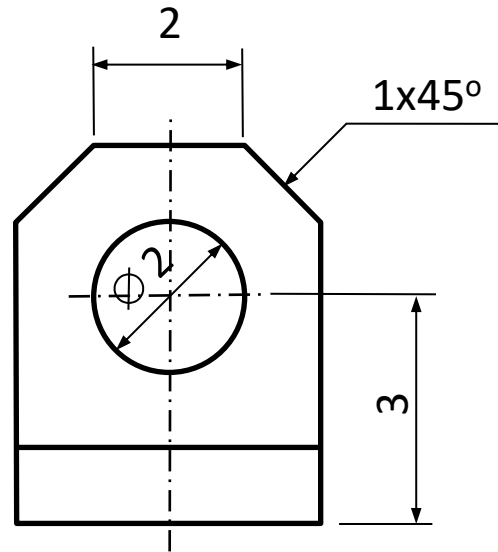
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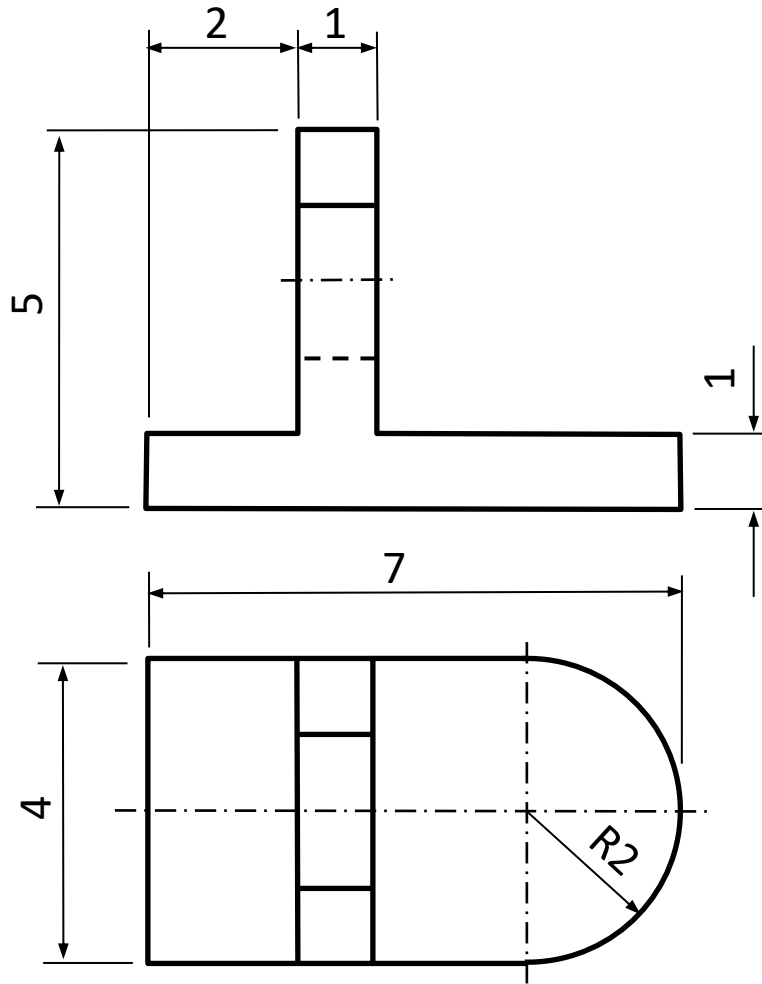
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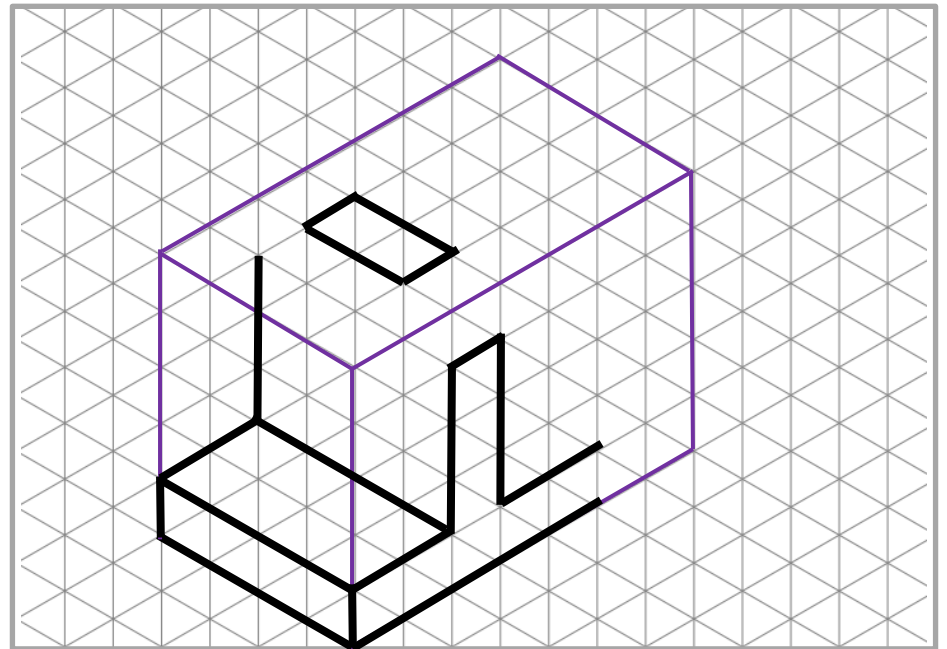
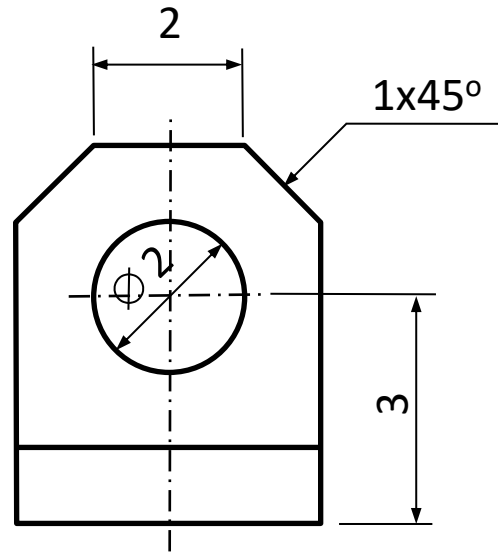
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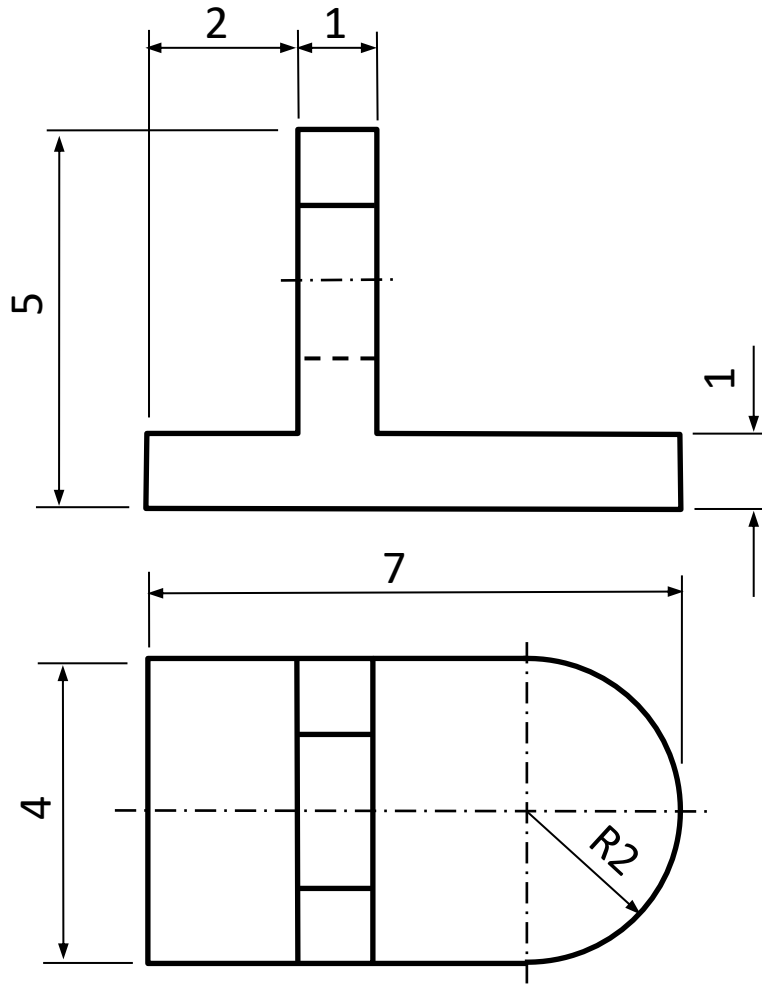
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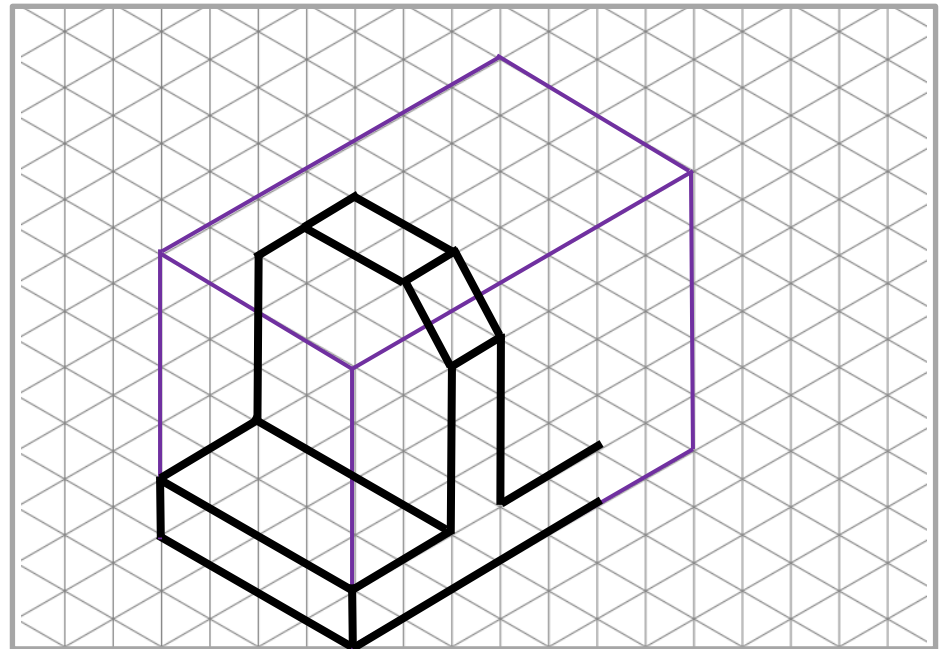
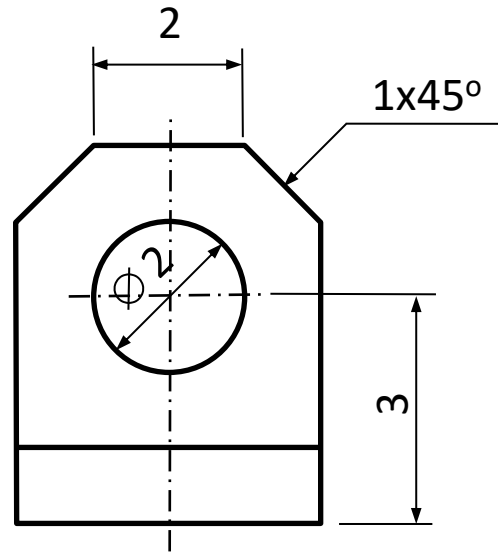
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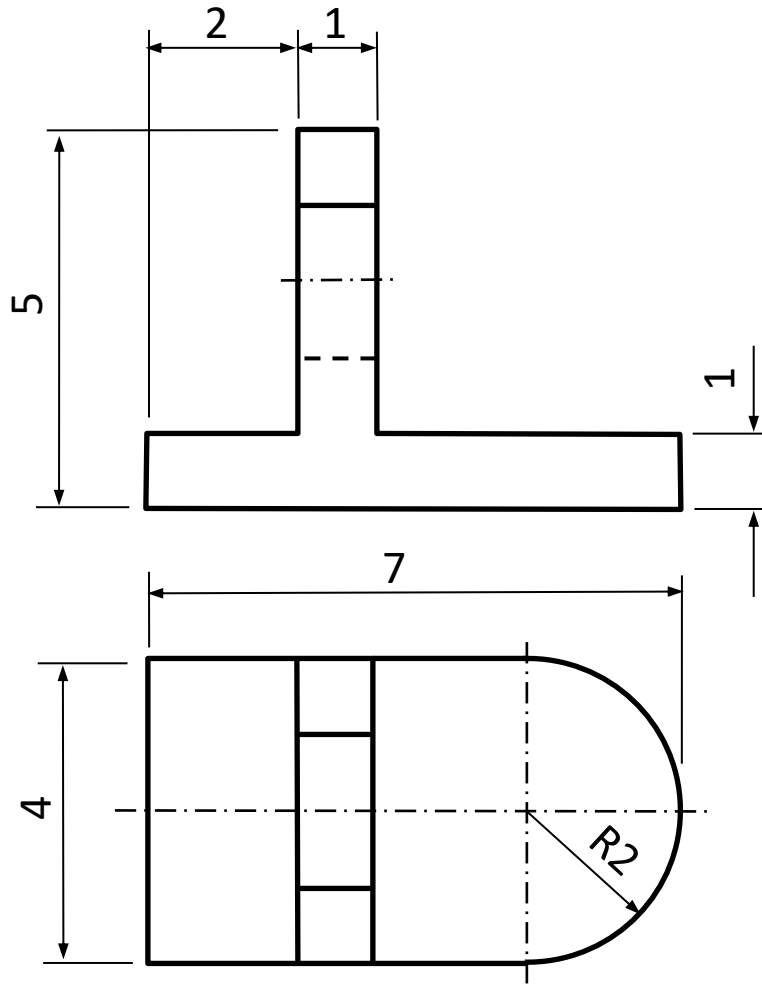
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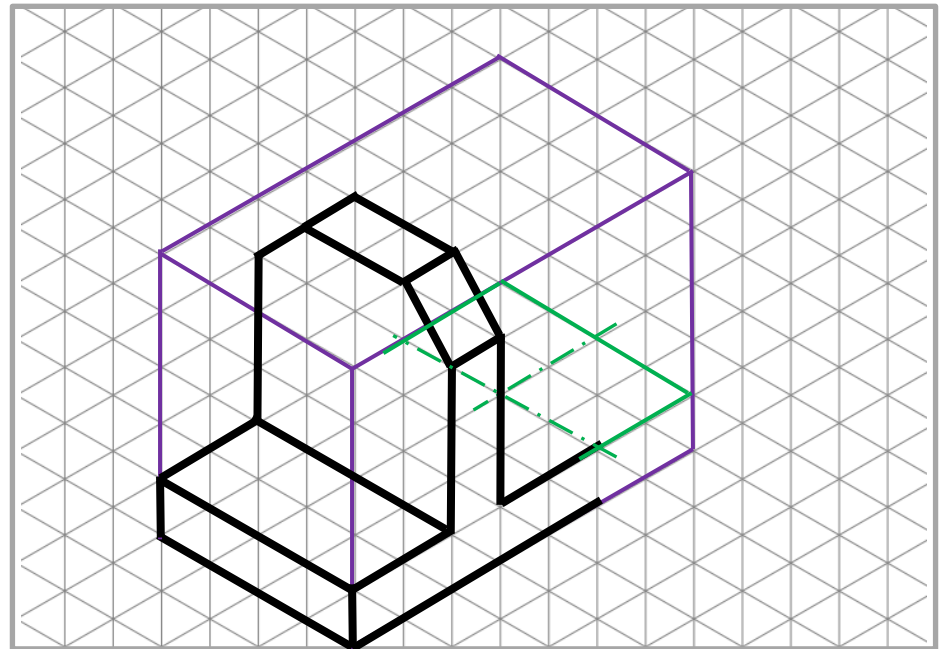
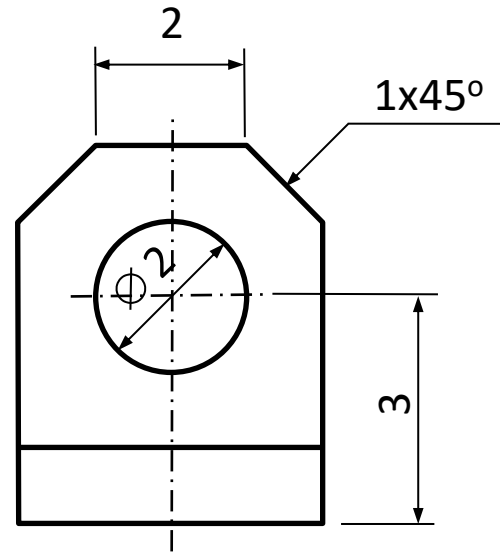
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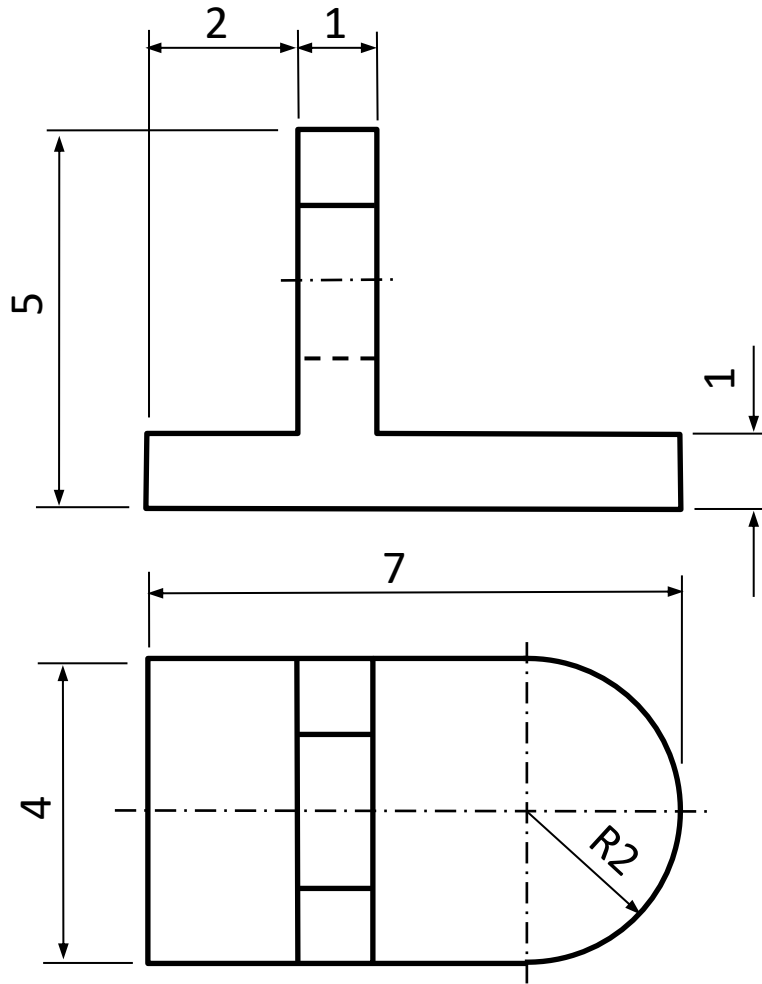
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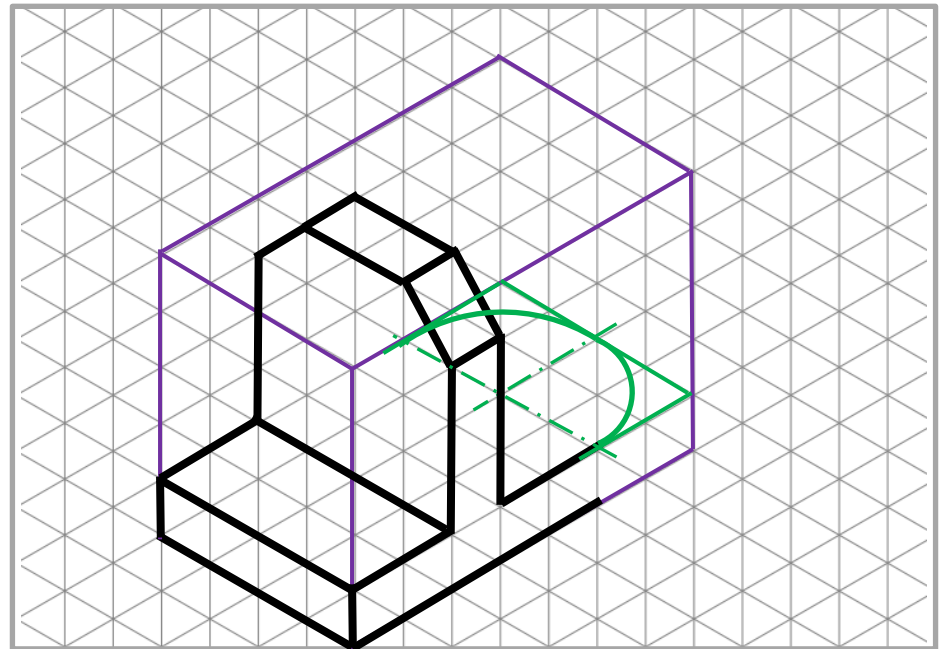
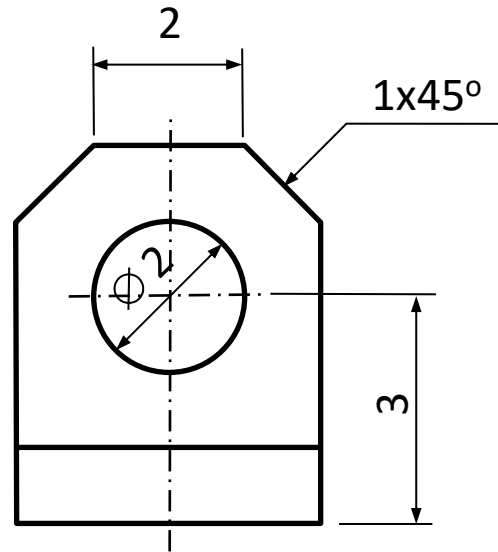
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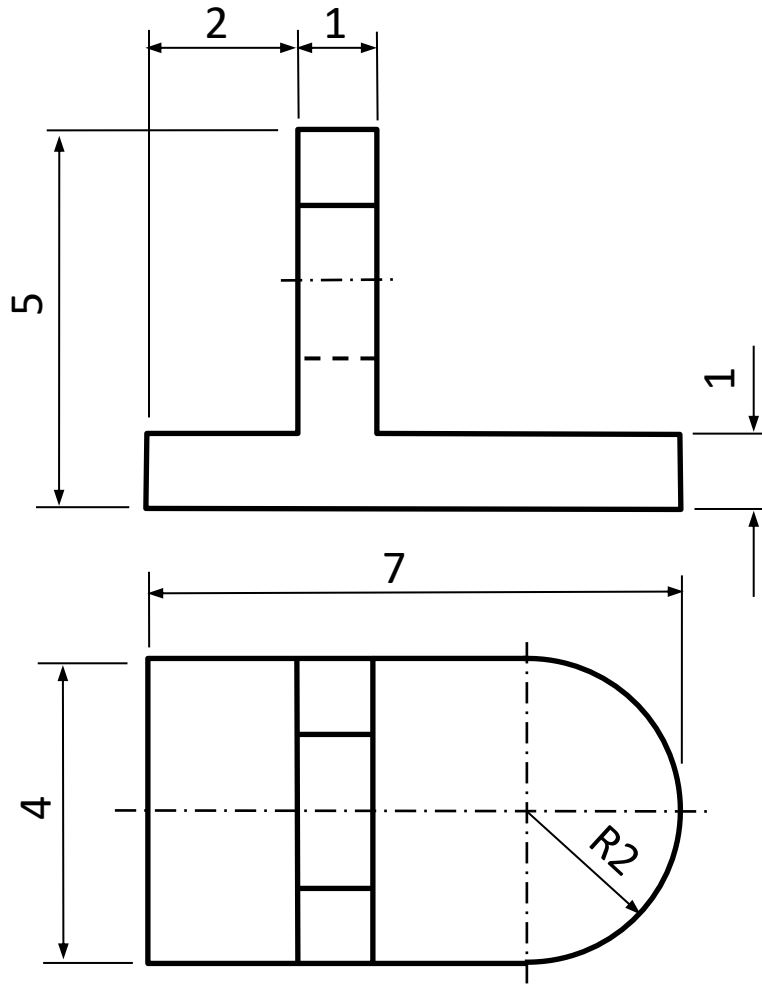
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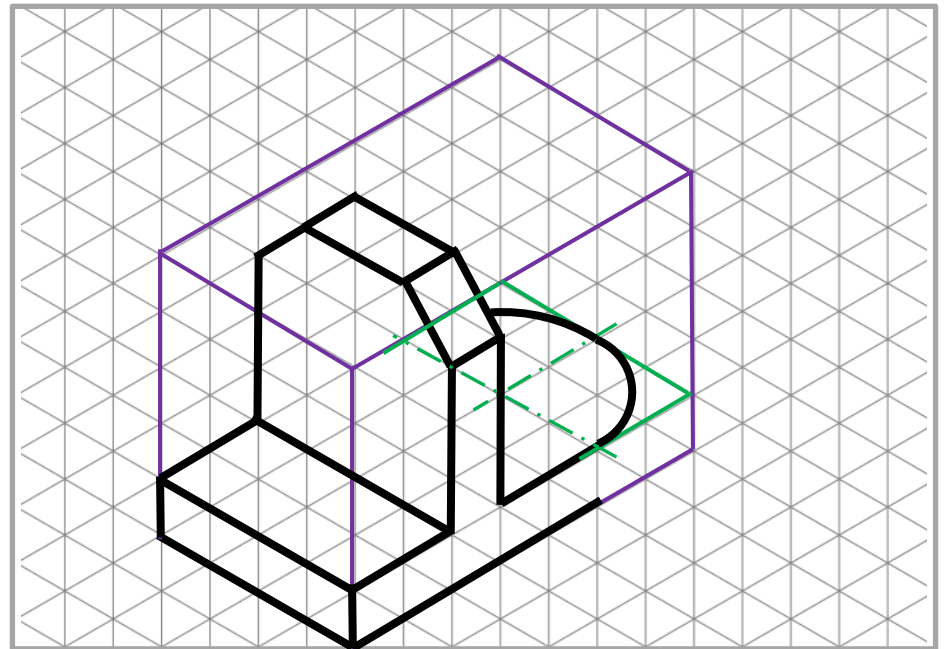
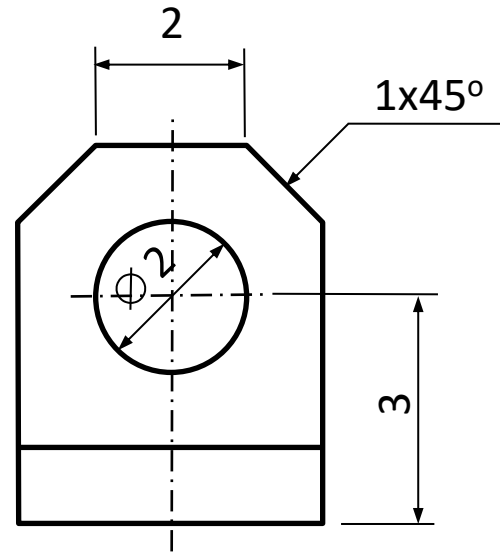
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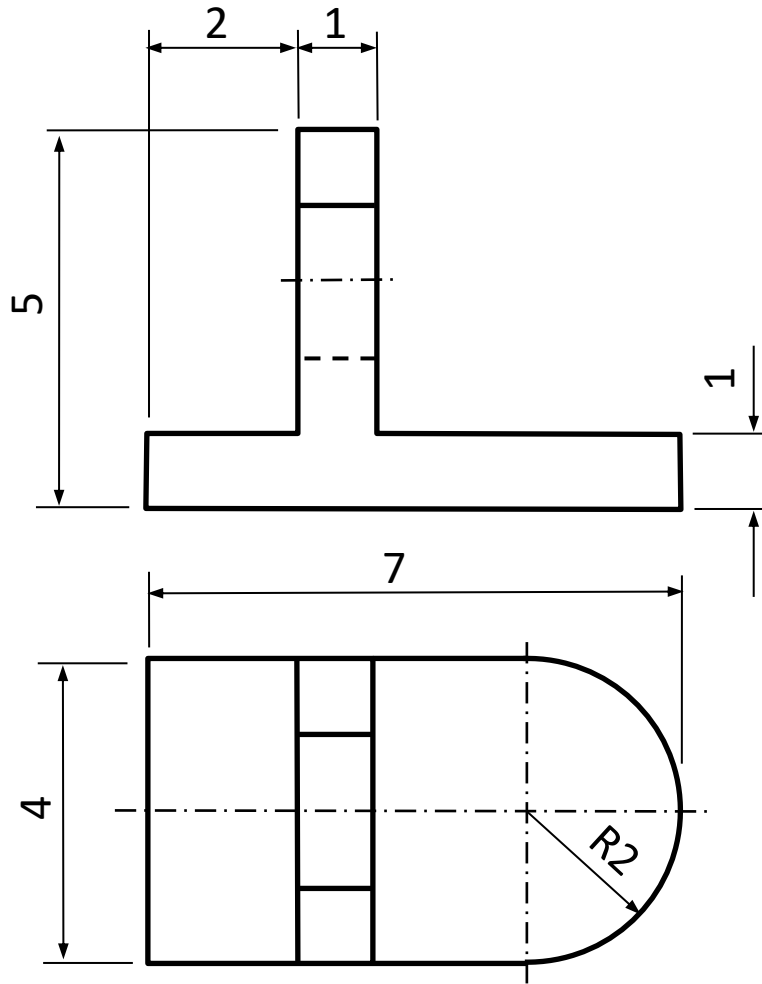
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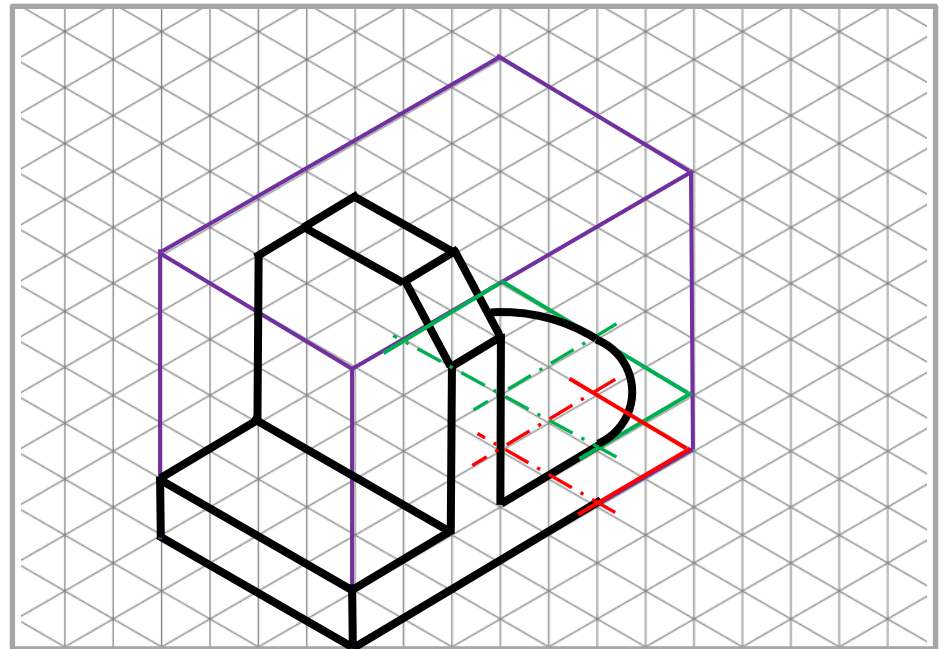
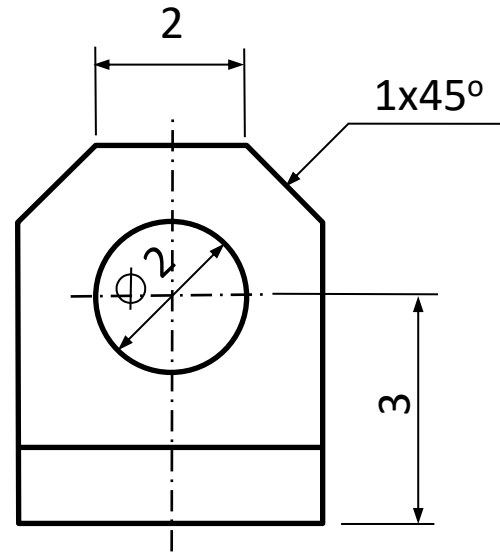
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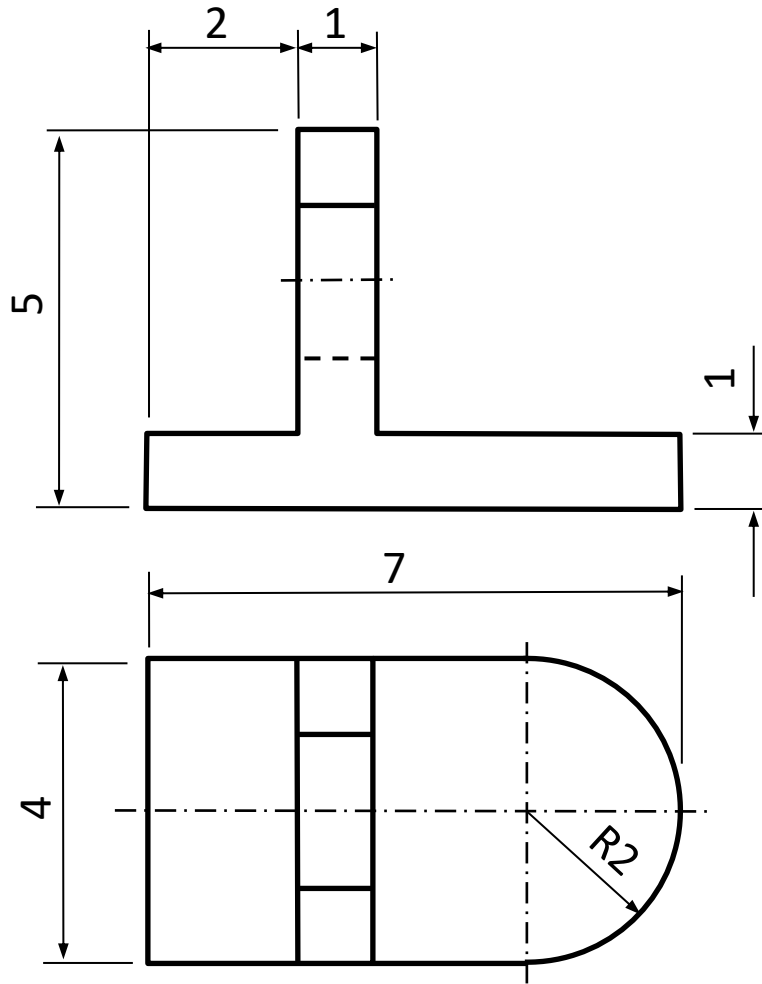
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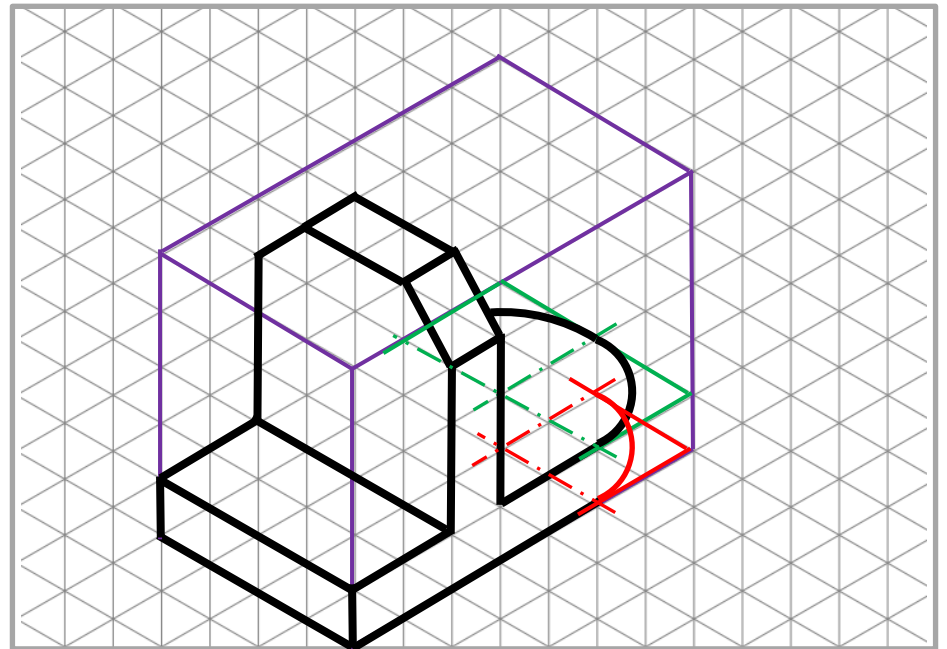
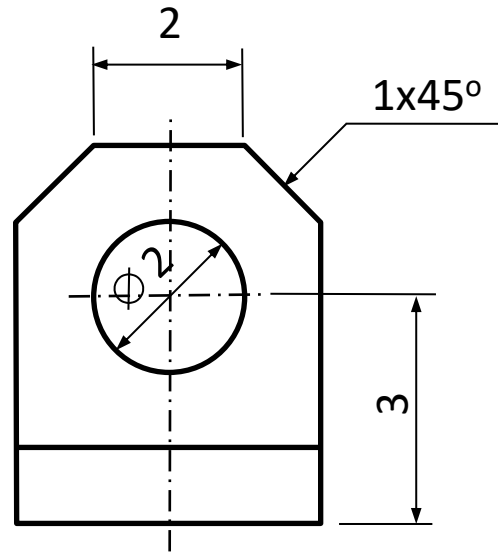
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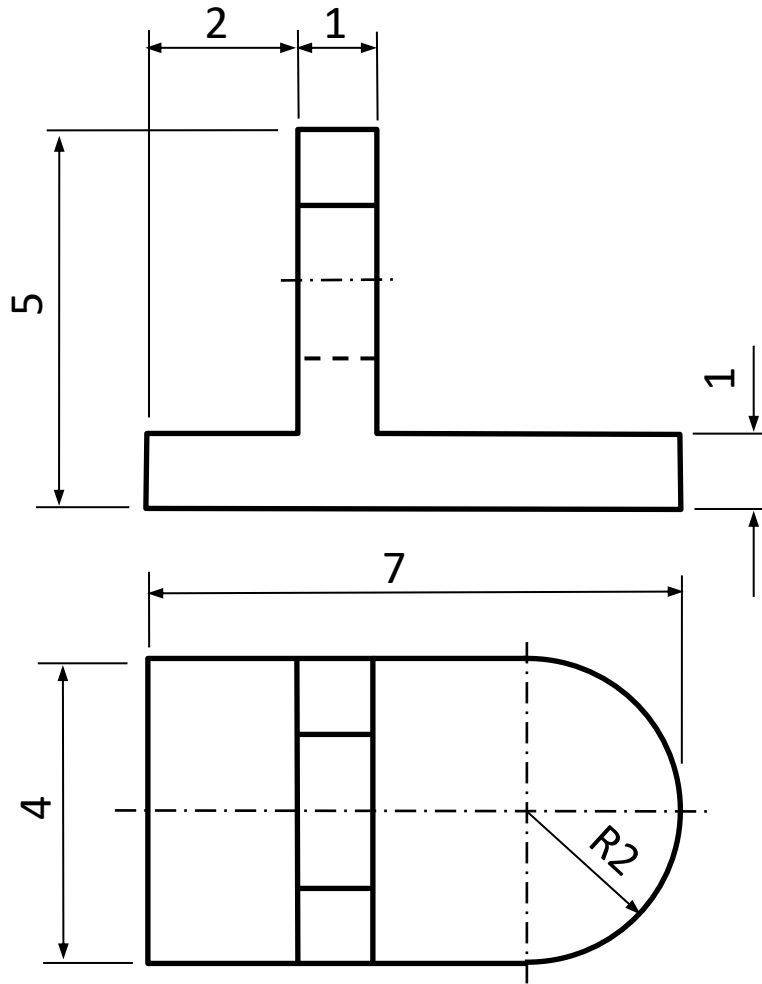
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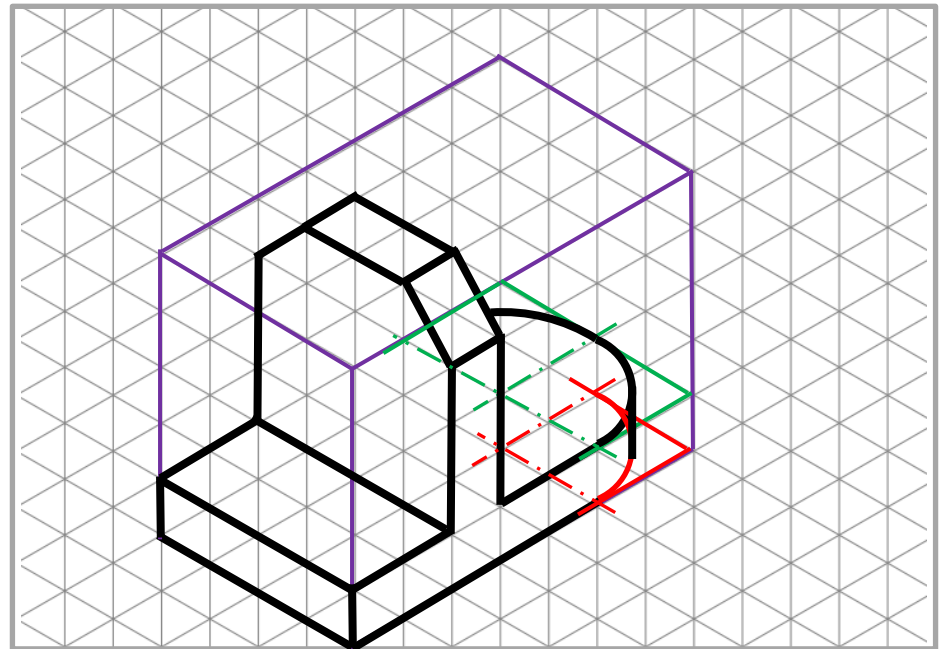
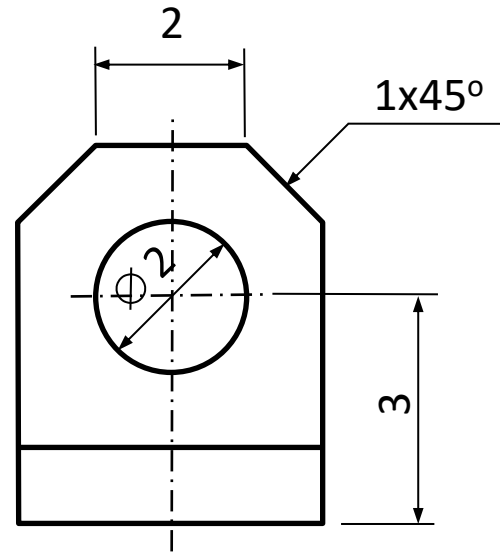
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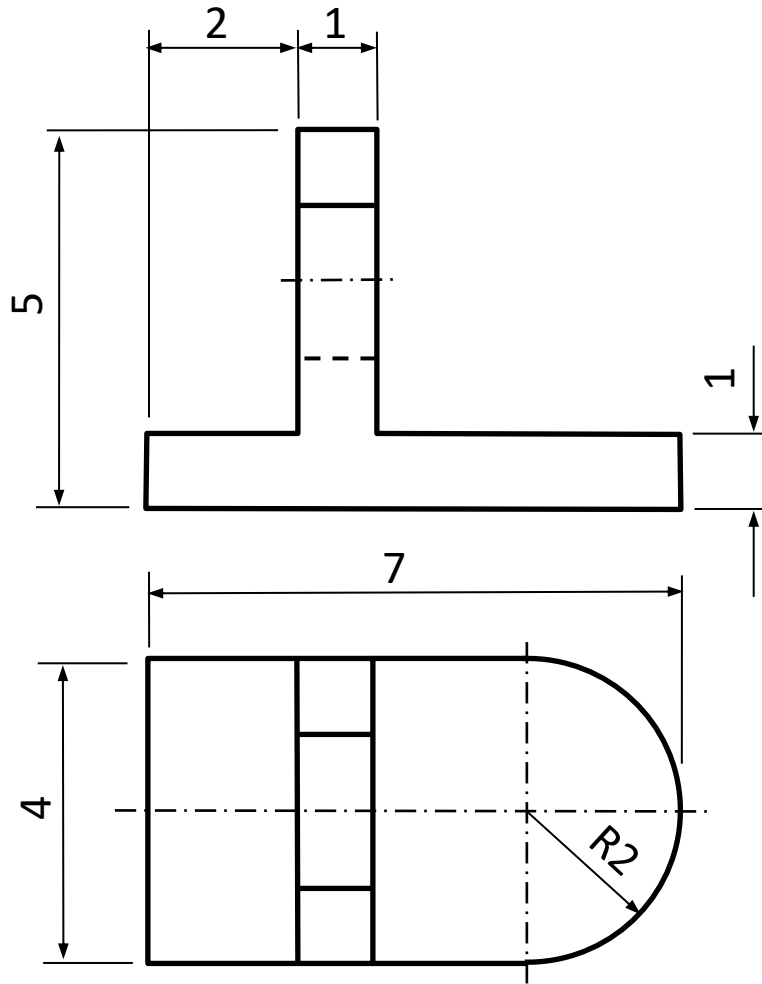
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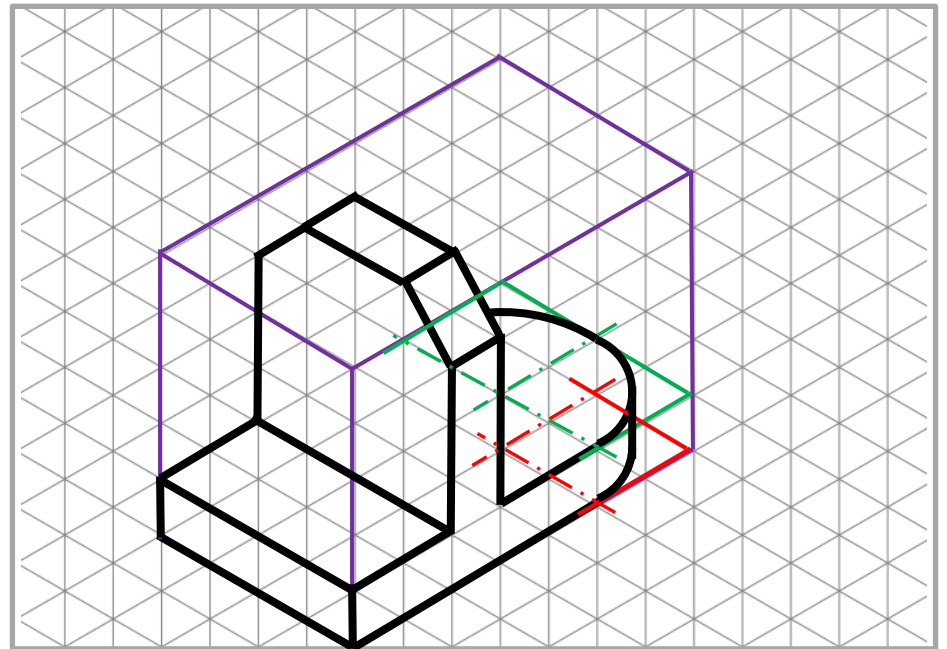
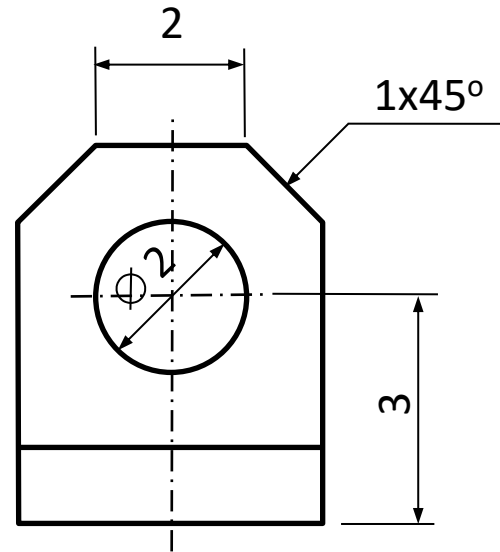
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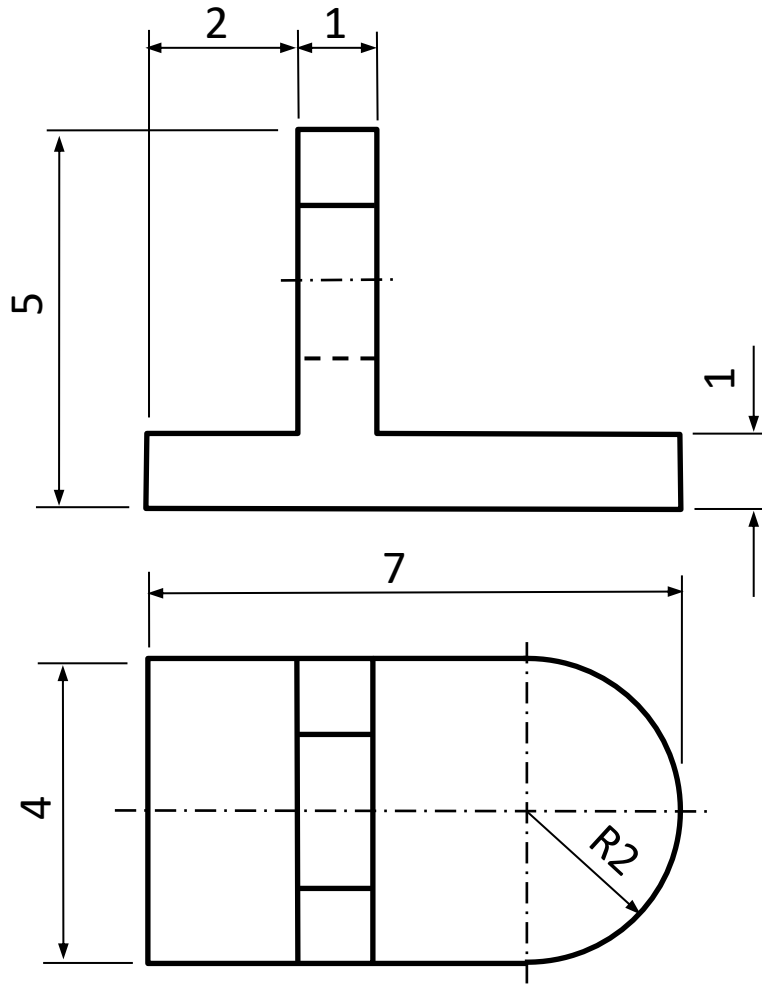
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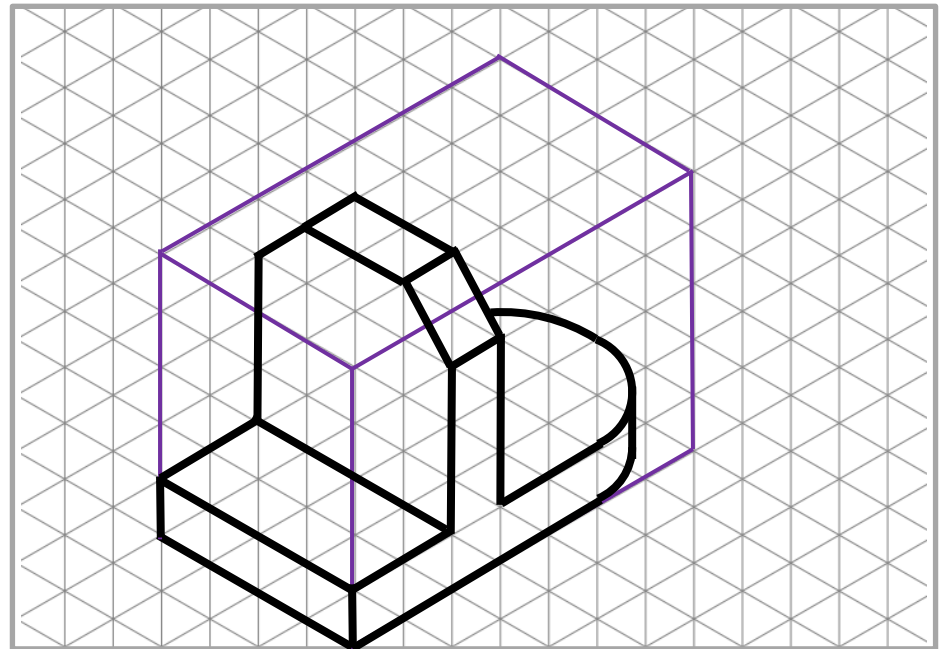
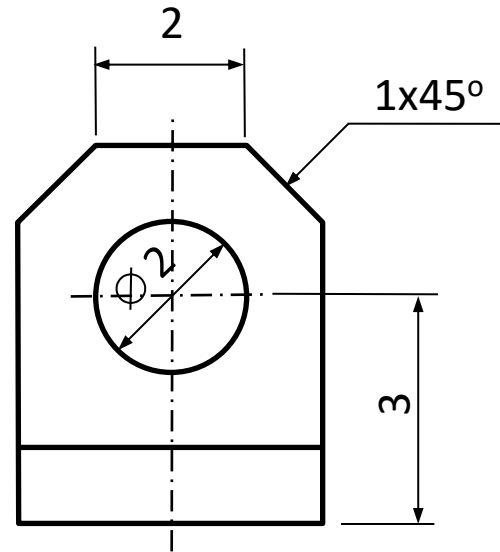
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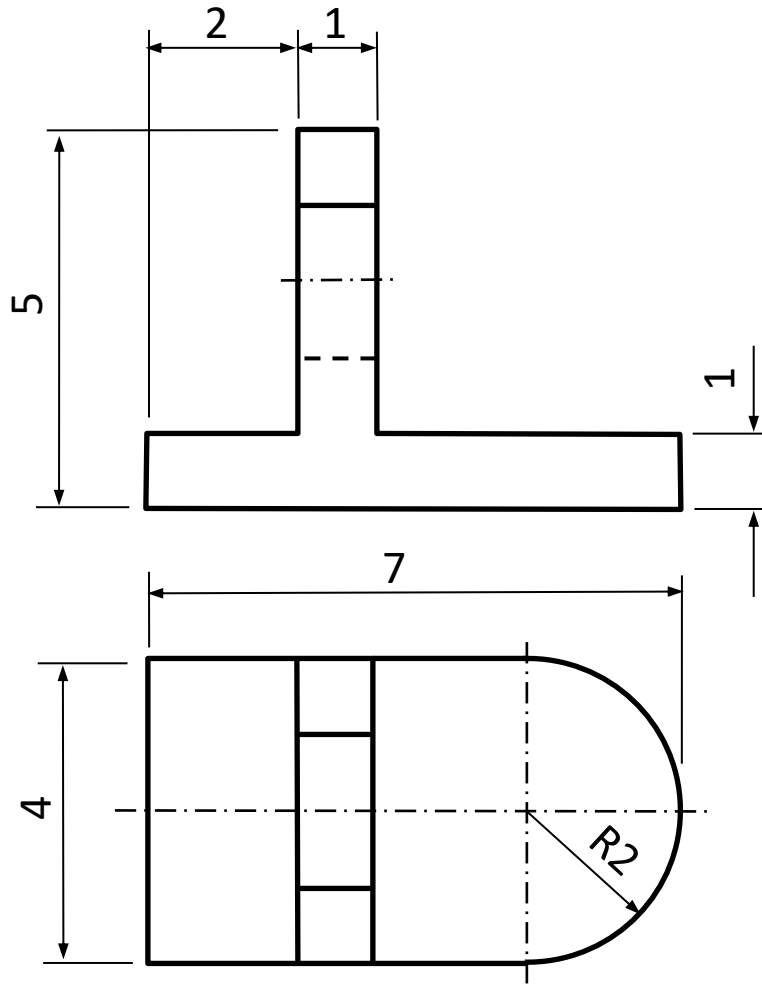
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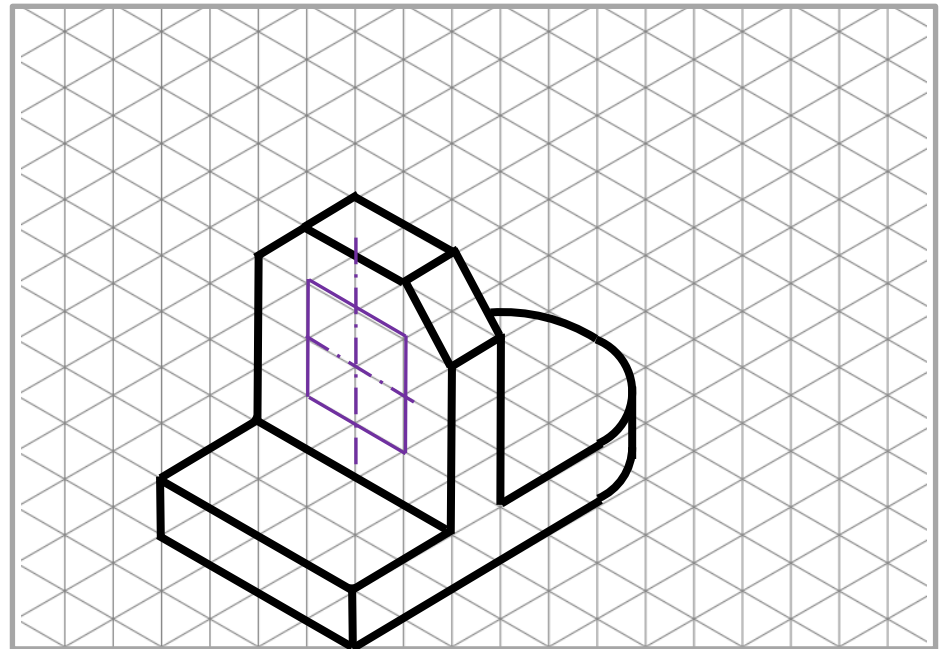
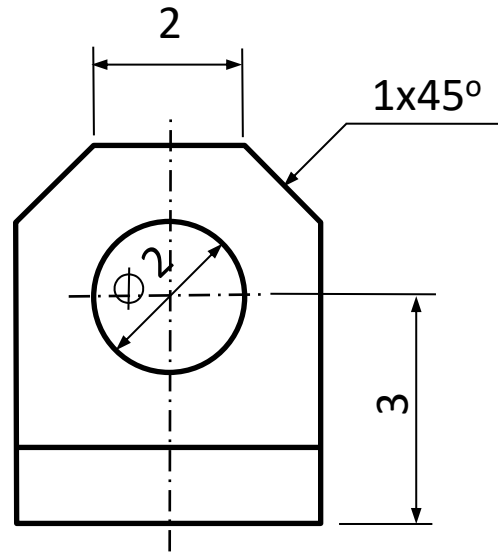
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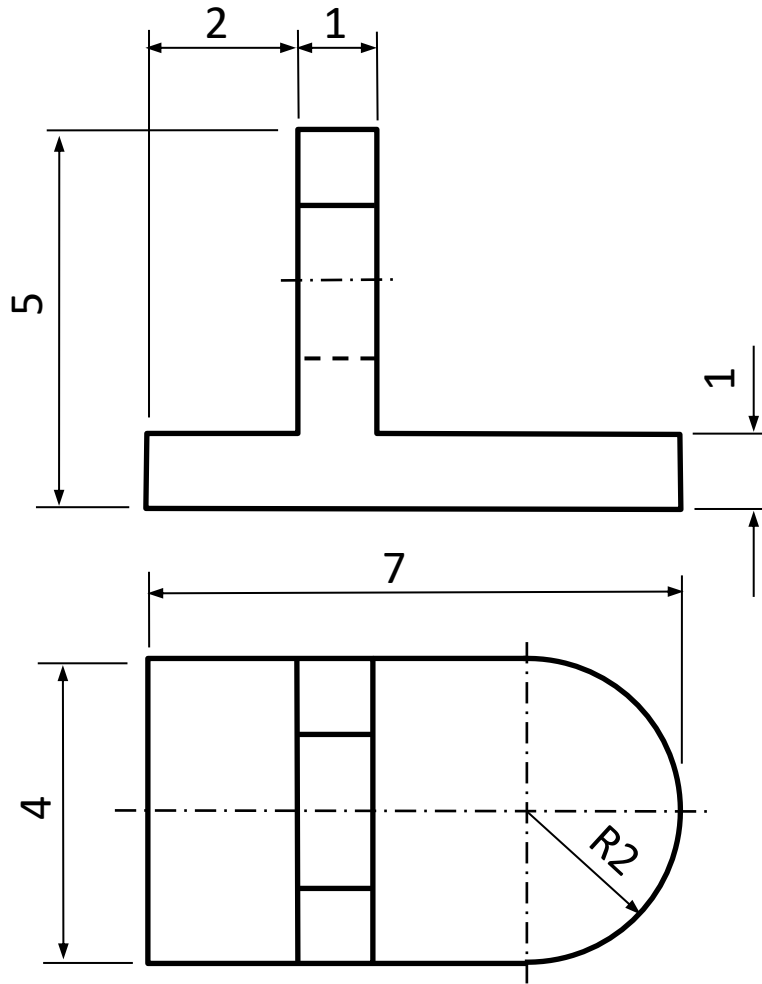
Vistas ortográficas



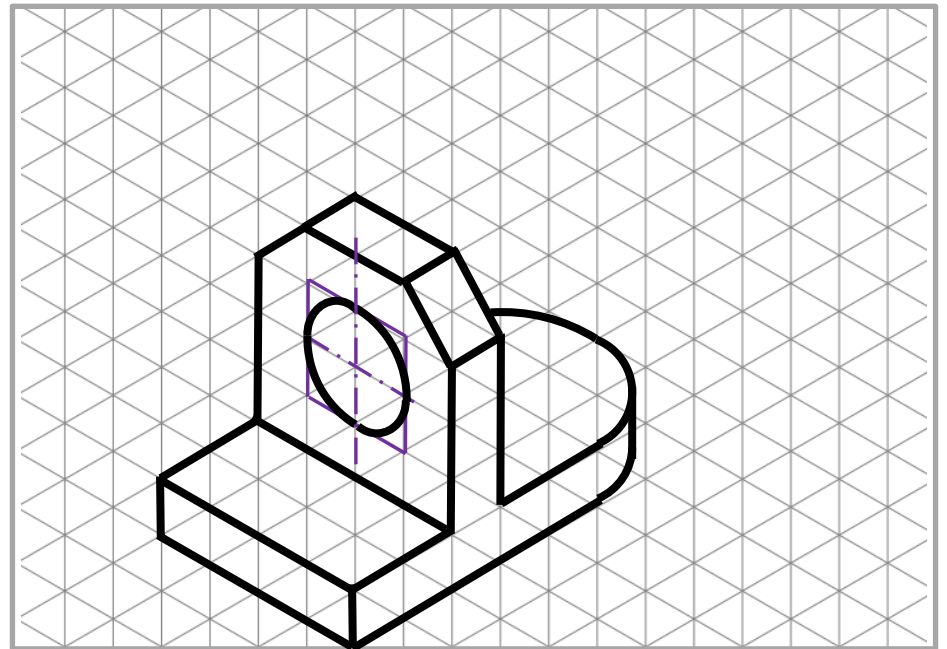
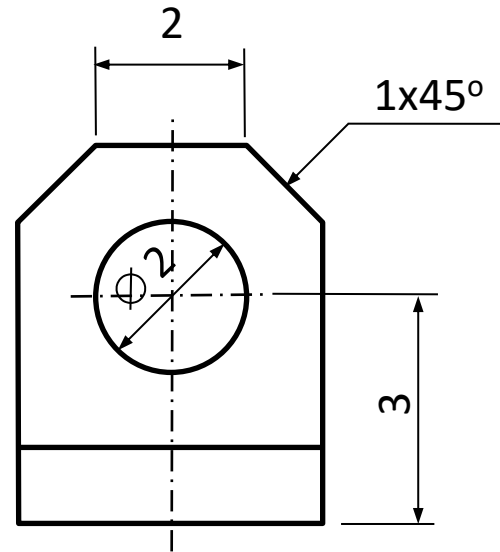
Perspectiva Isométrica - 6



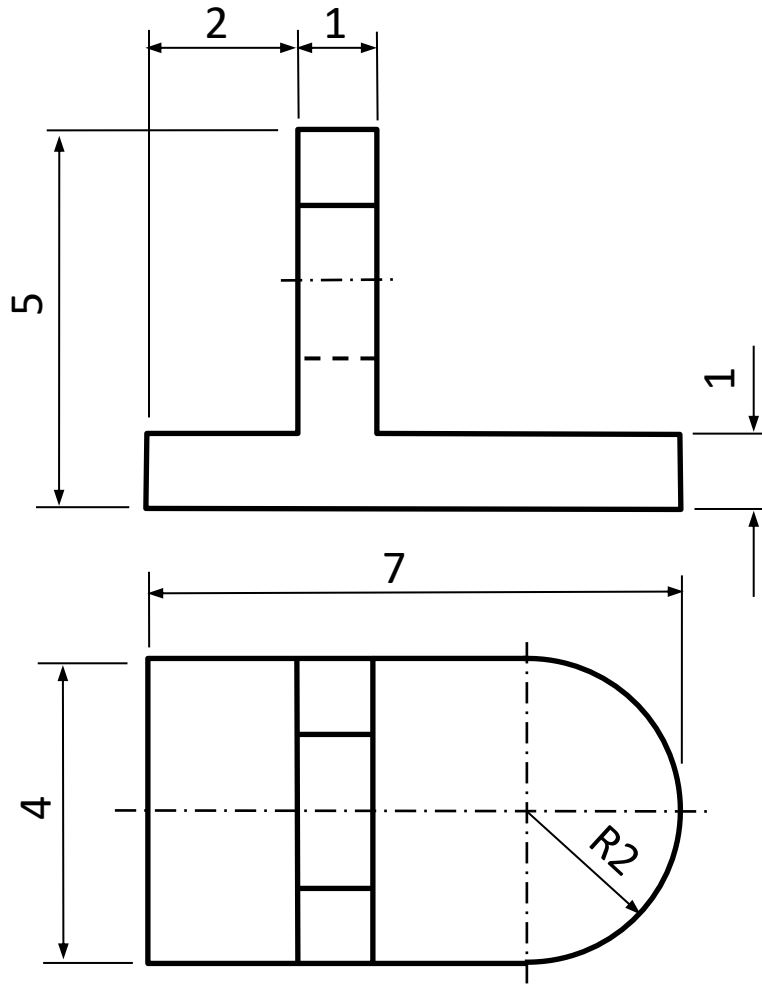
Vistas ortográficas



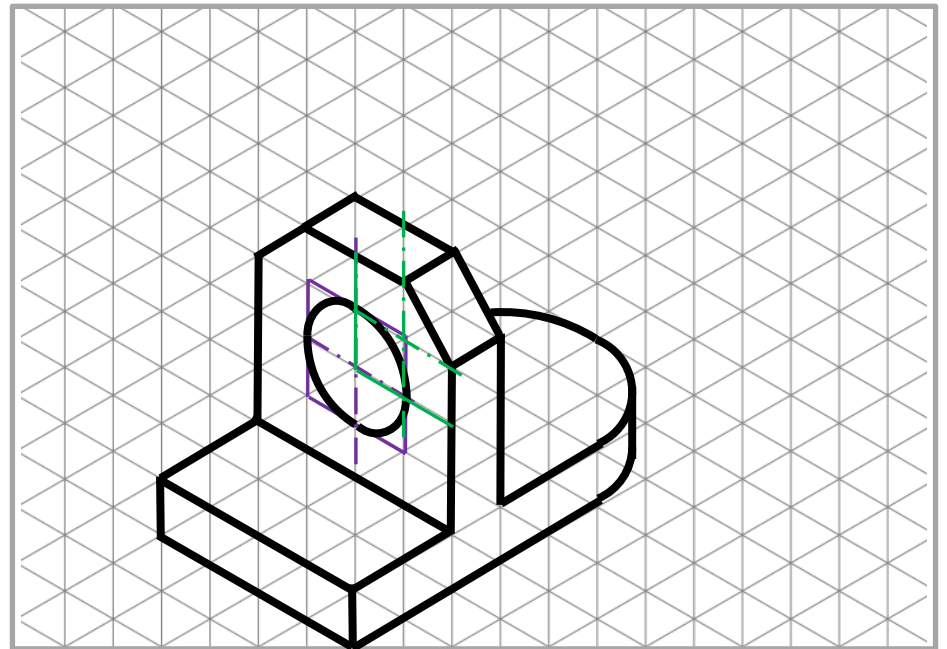
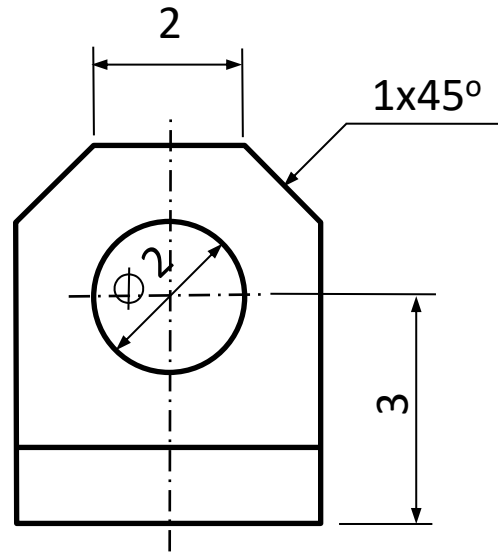
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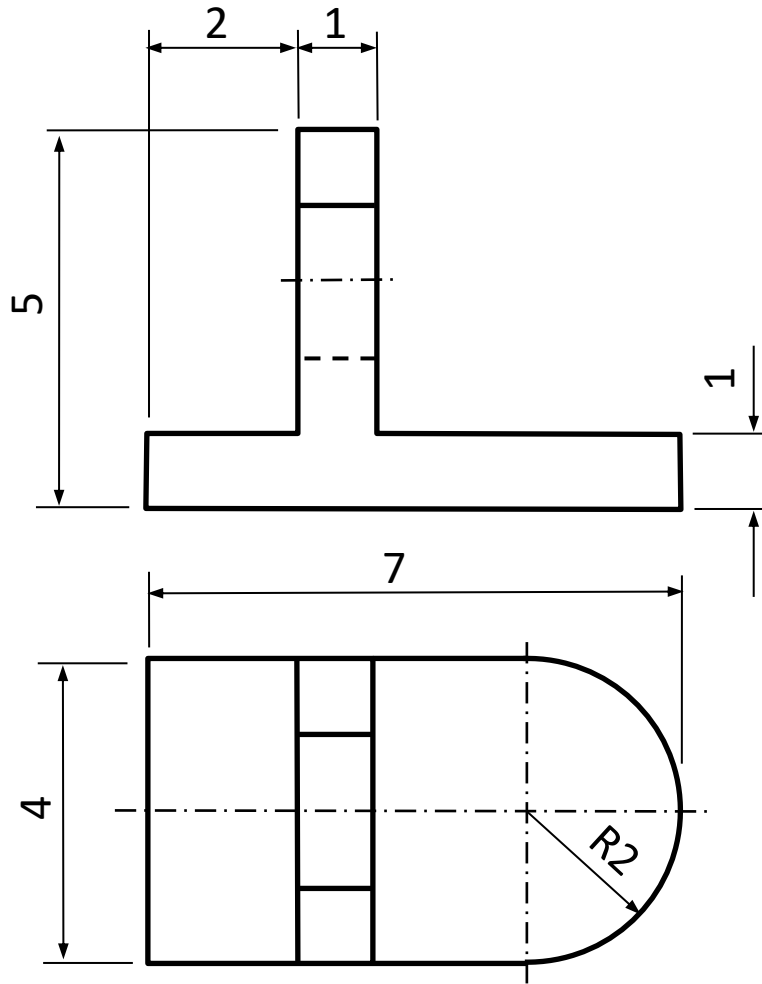
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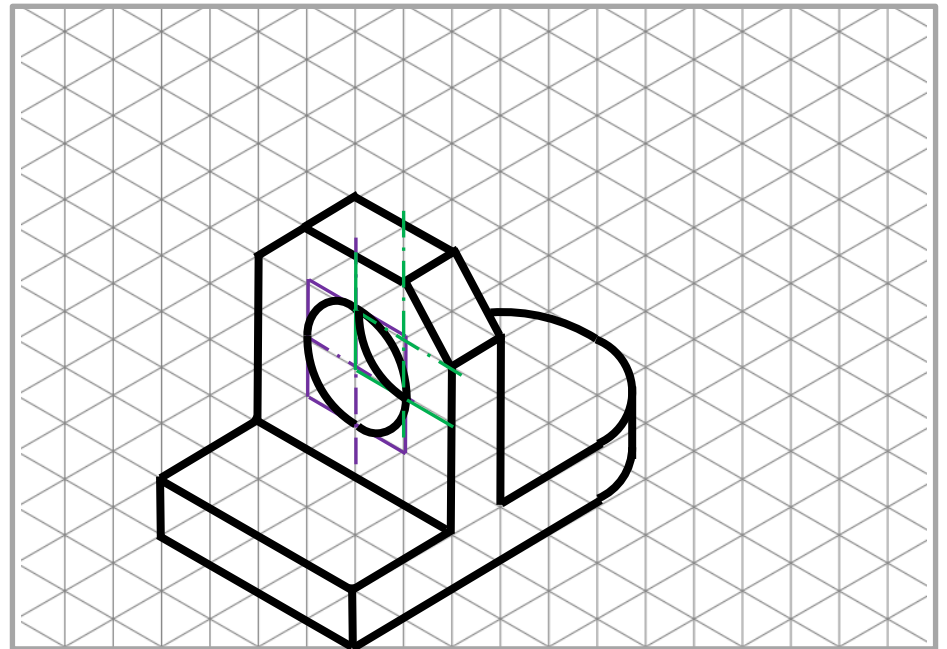
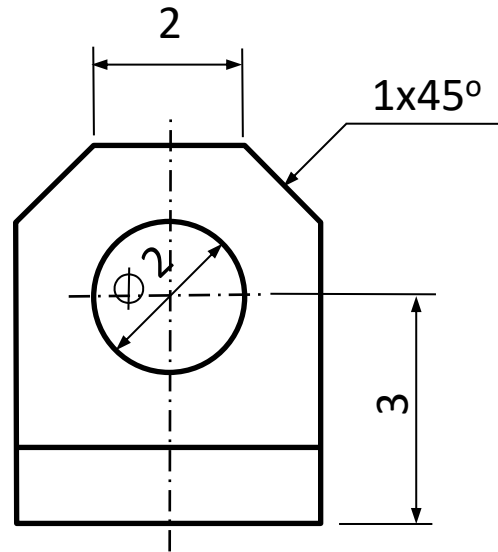
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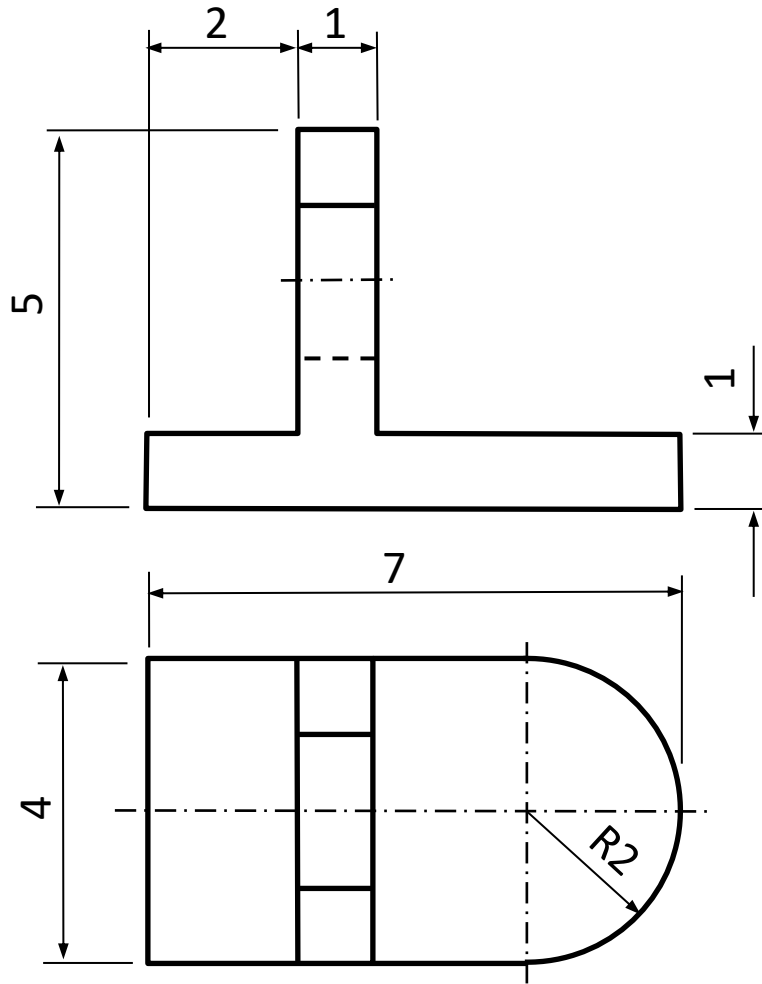
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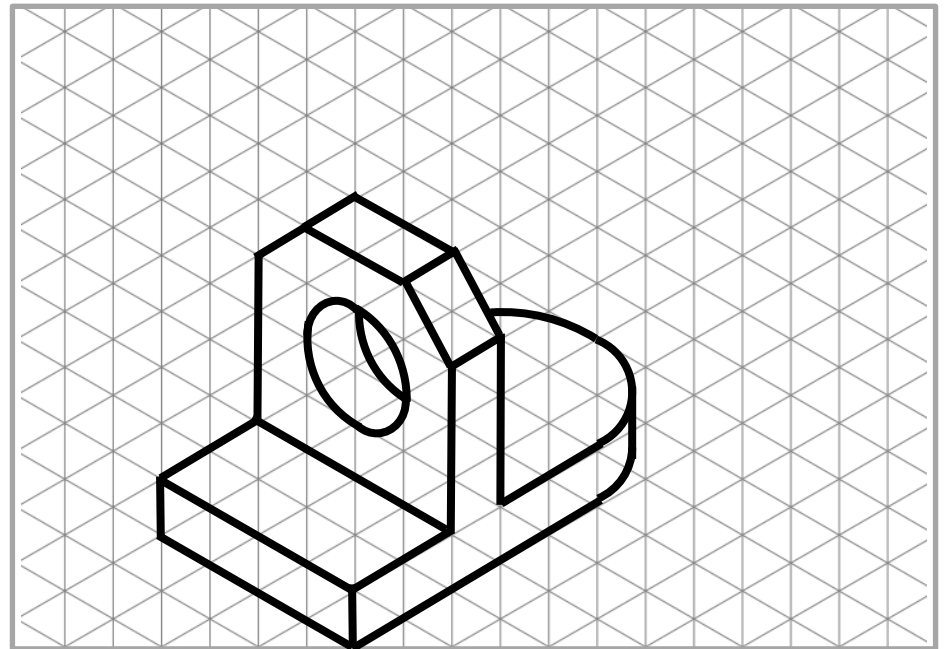
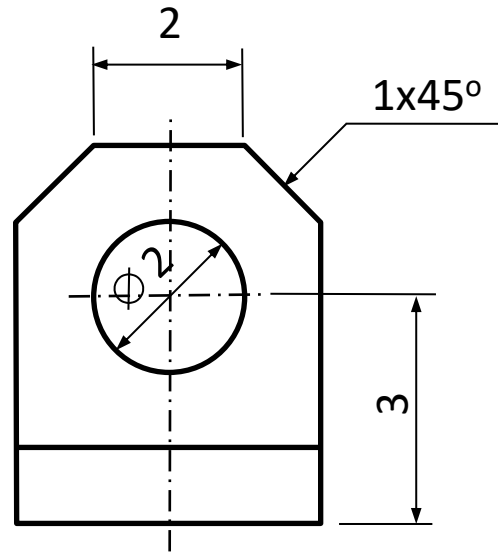
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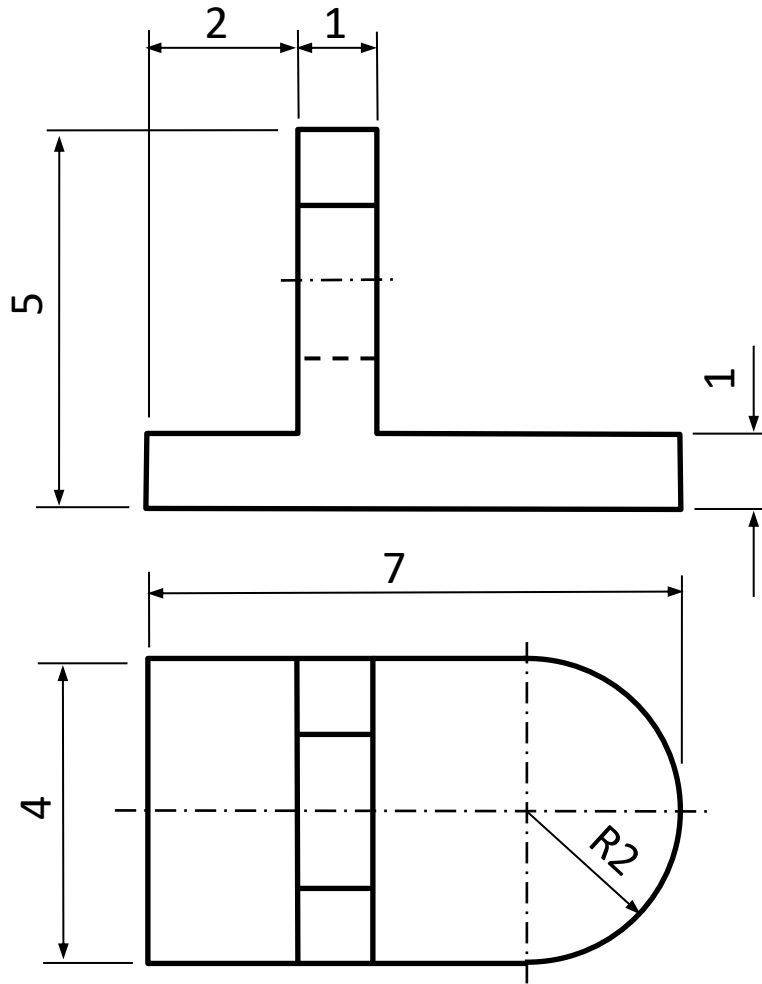
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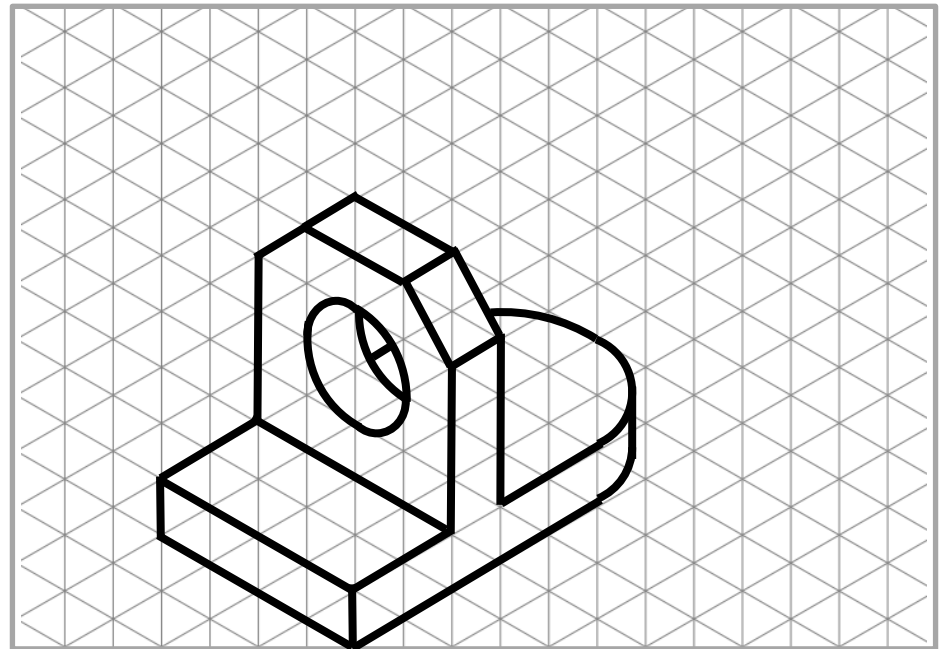
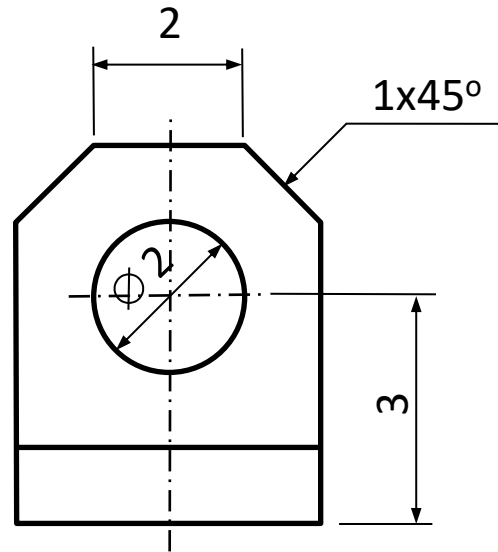
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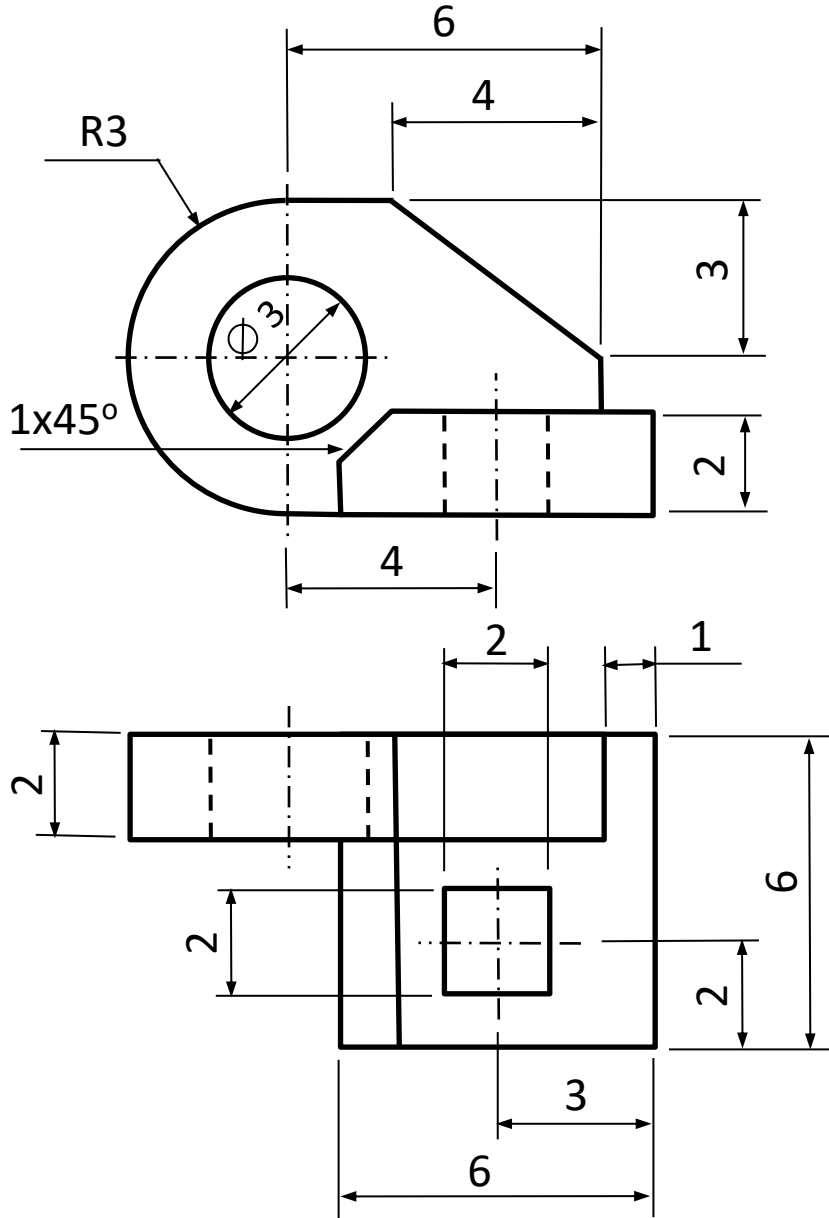
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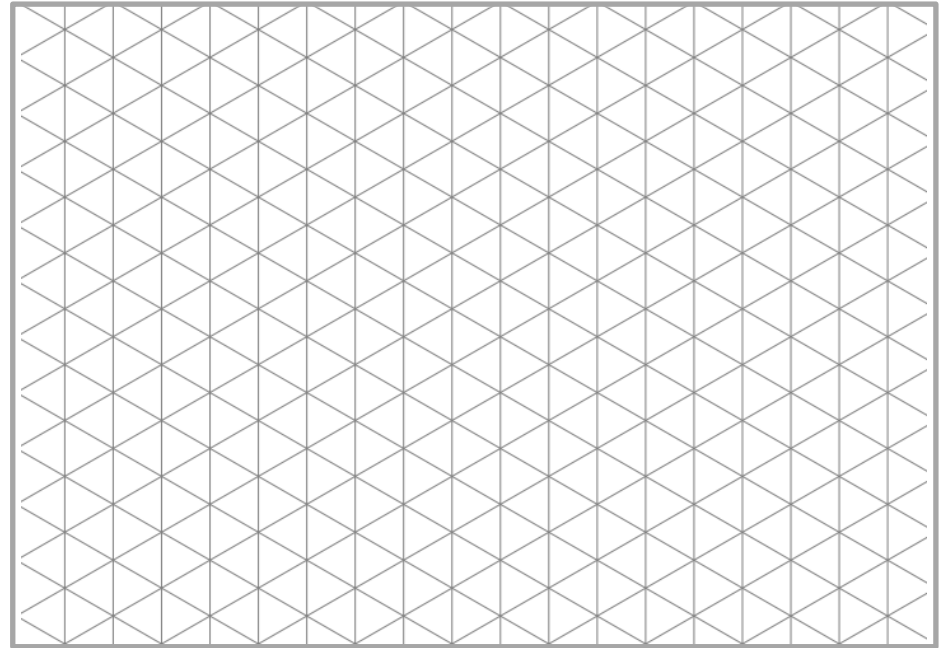
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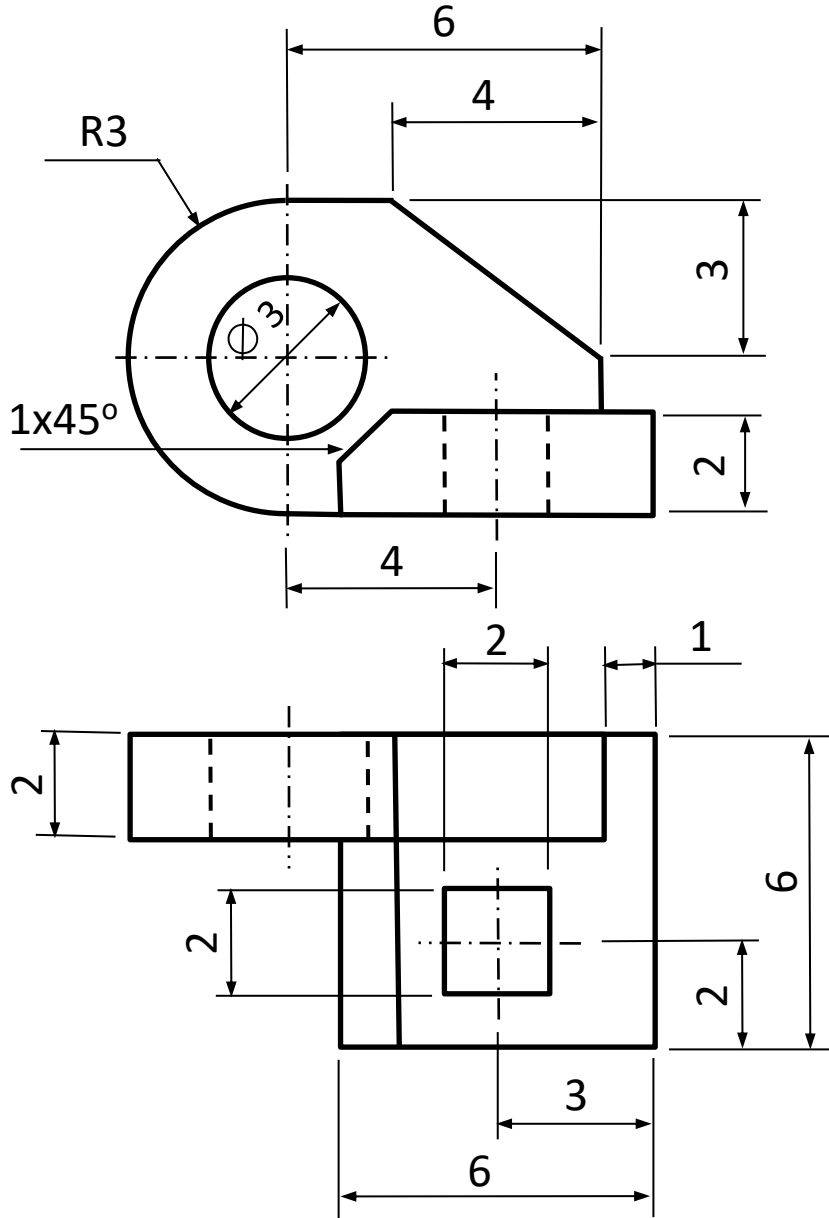
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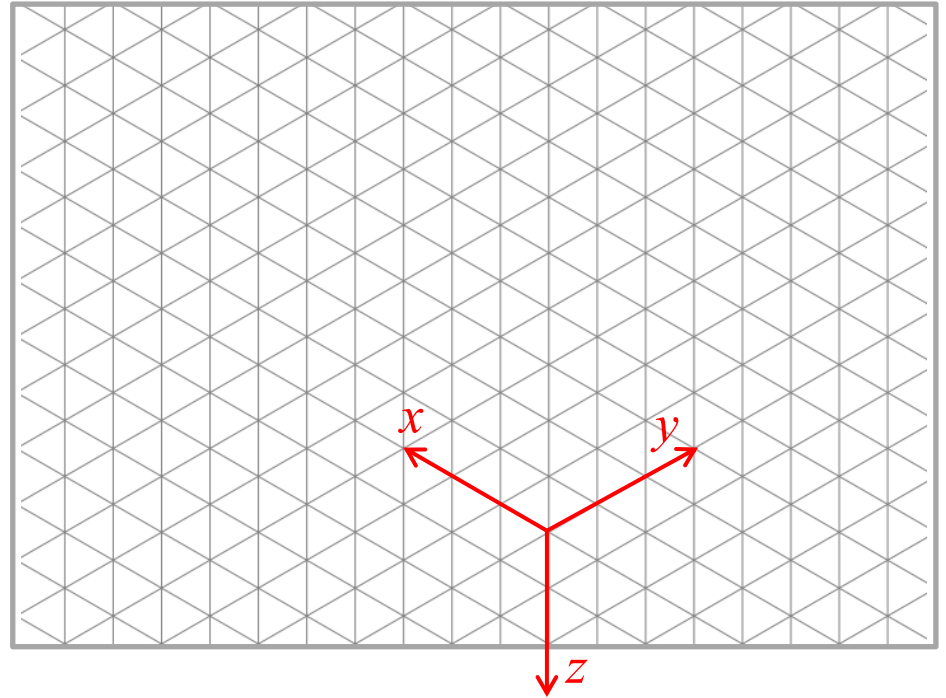
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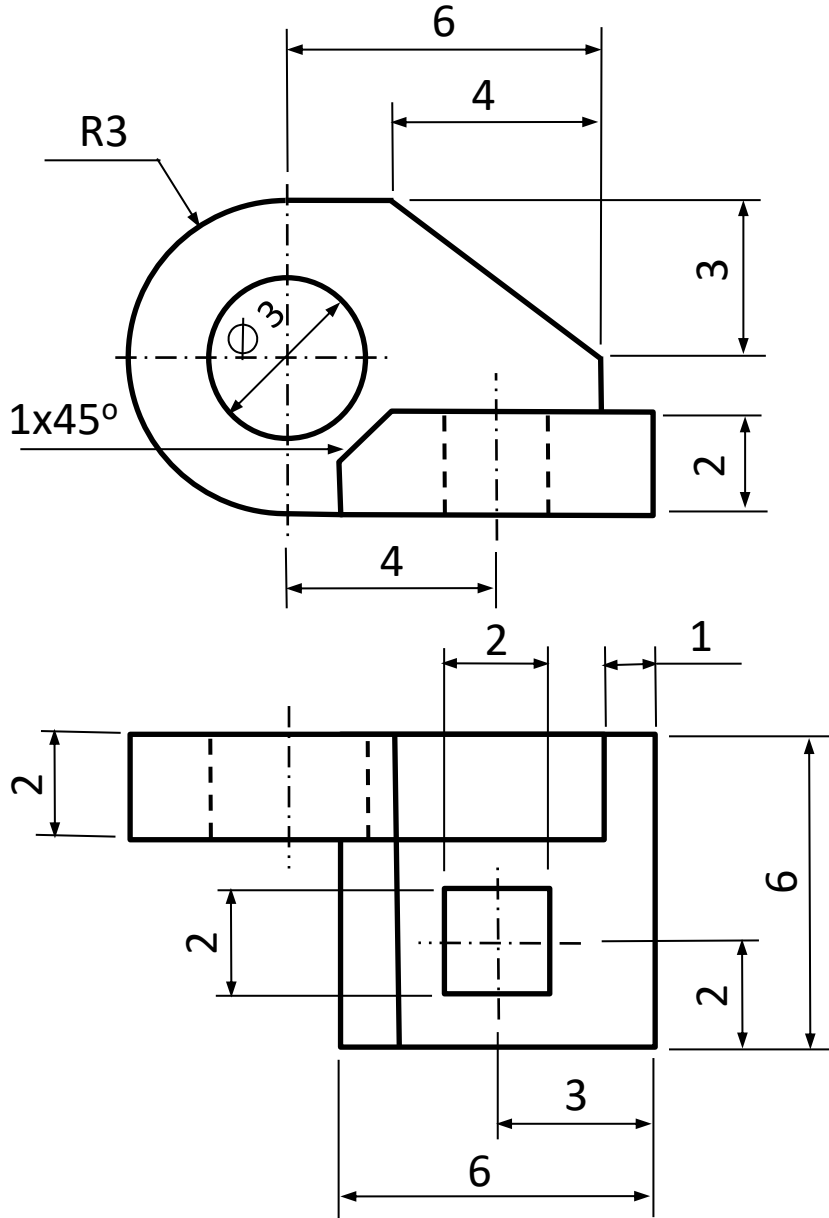
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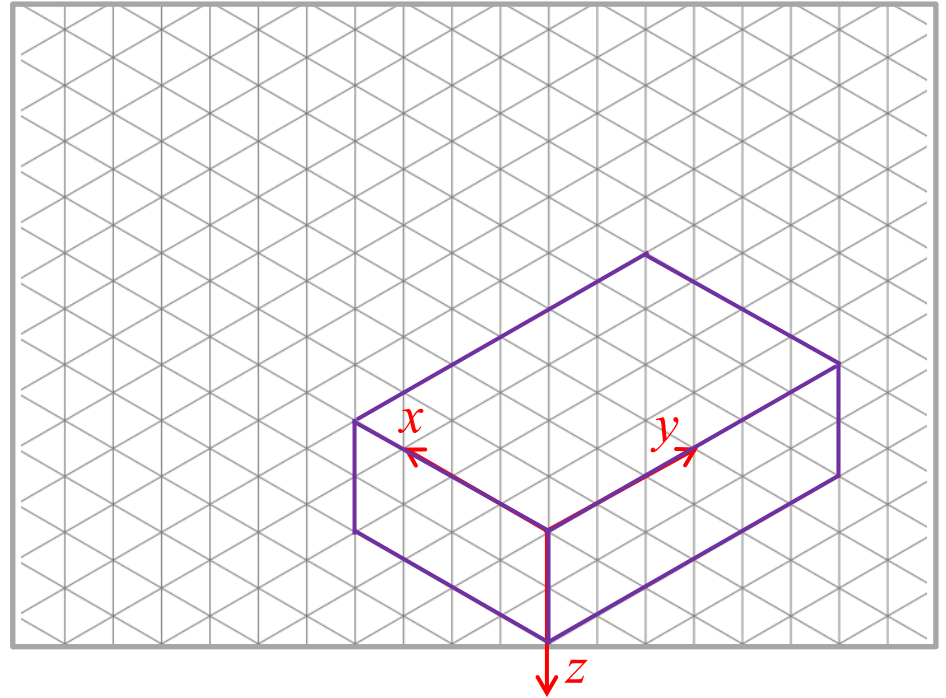
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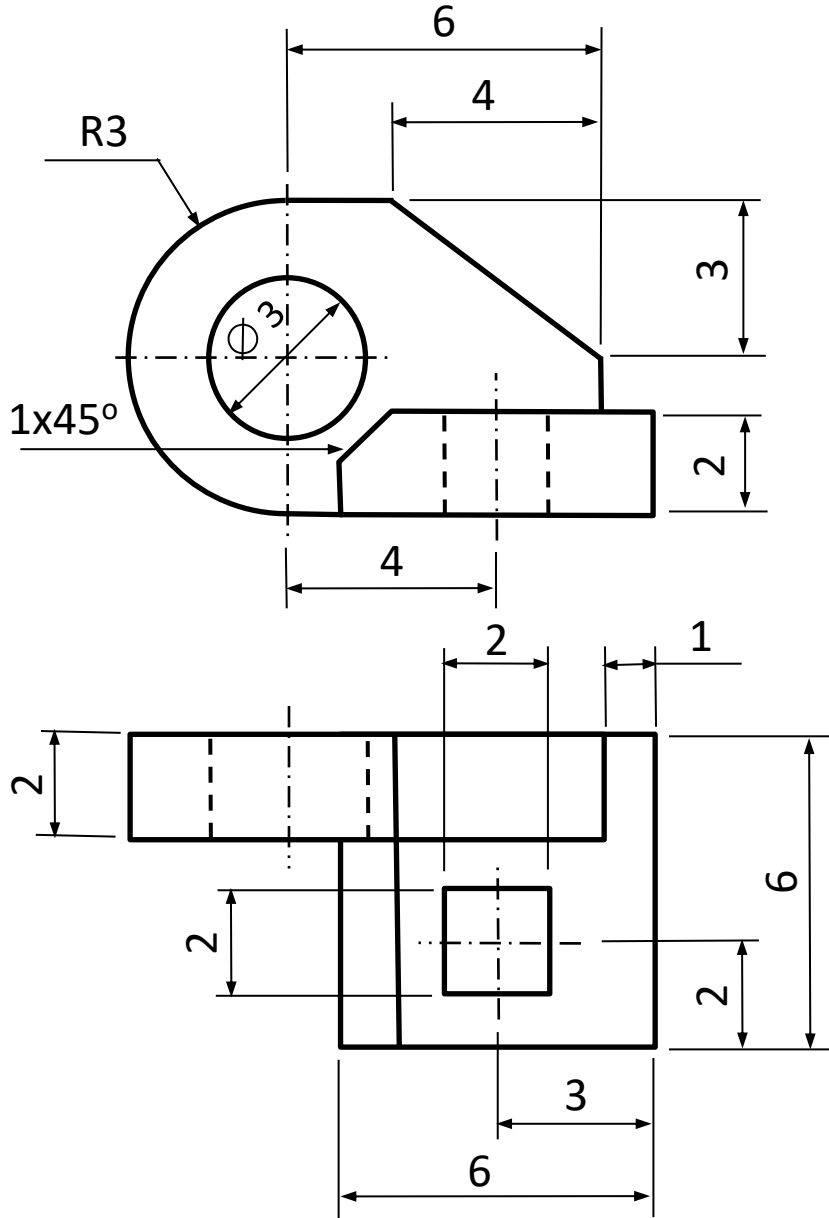
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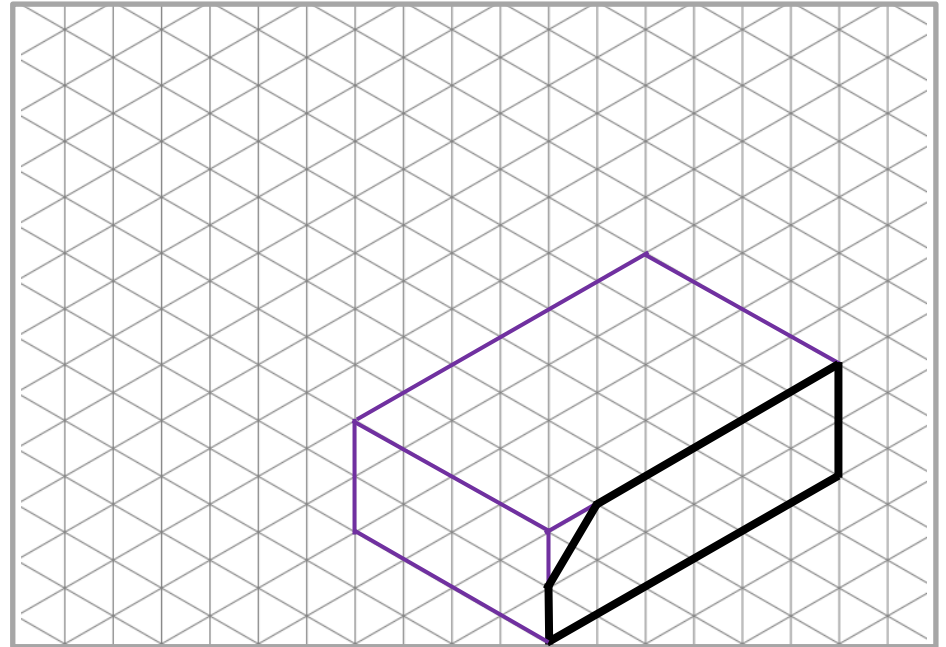
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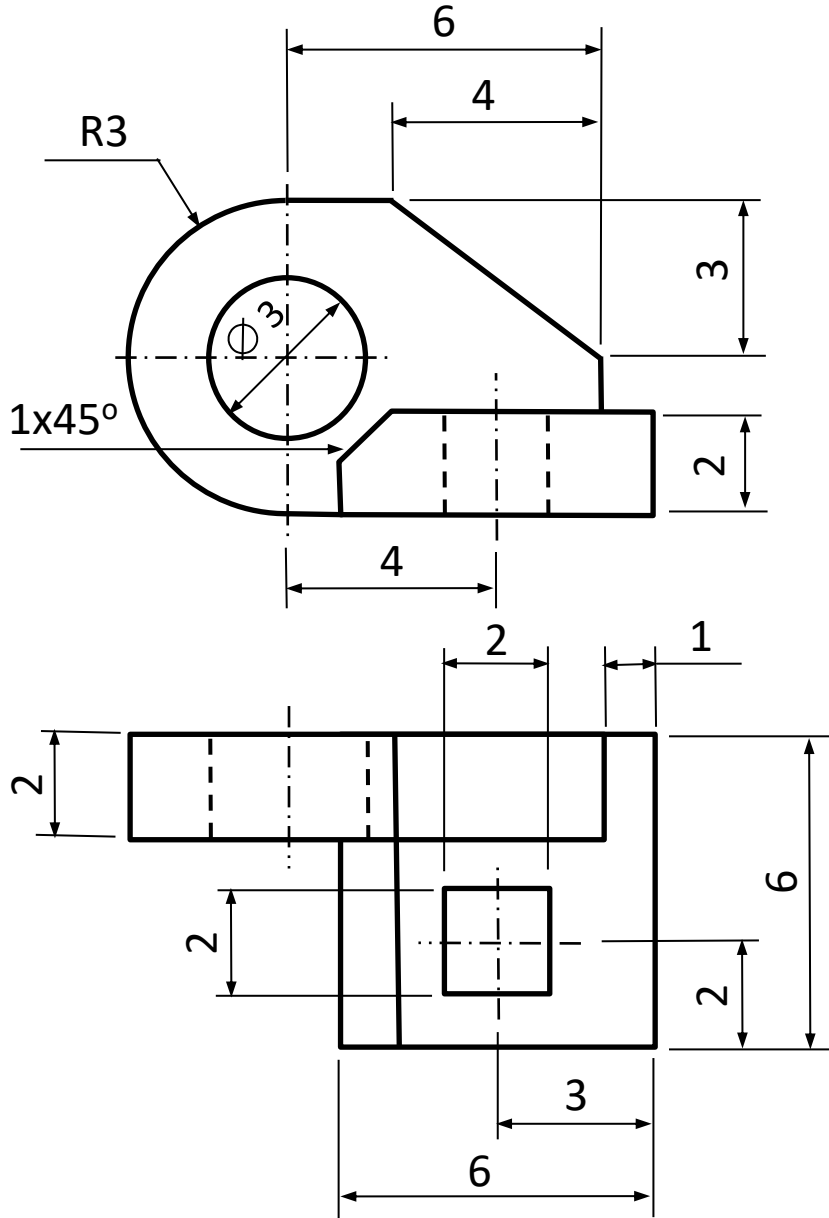
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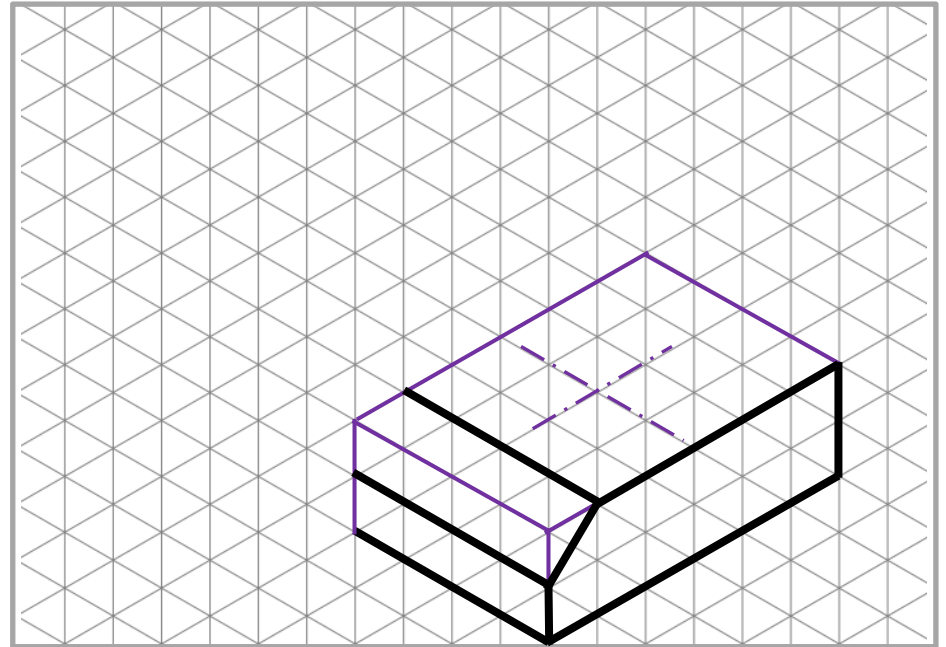
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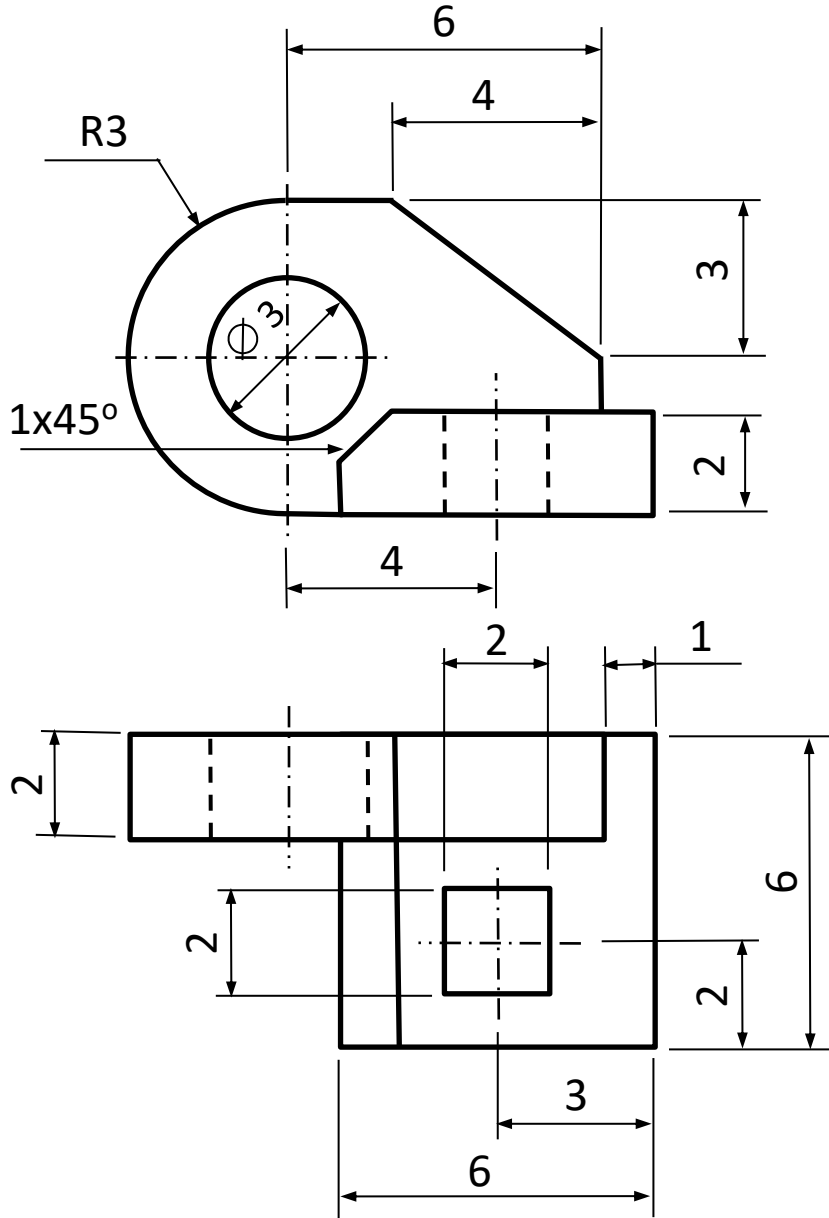
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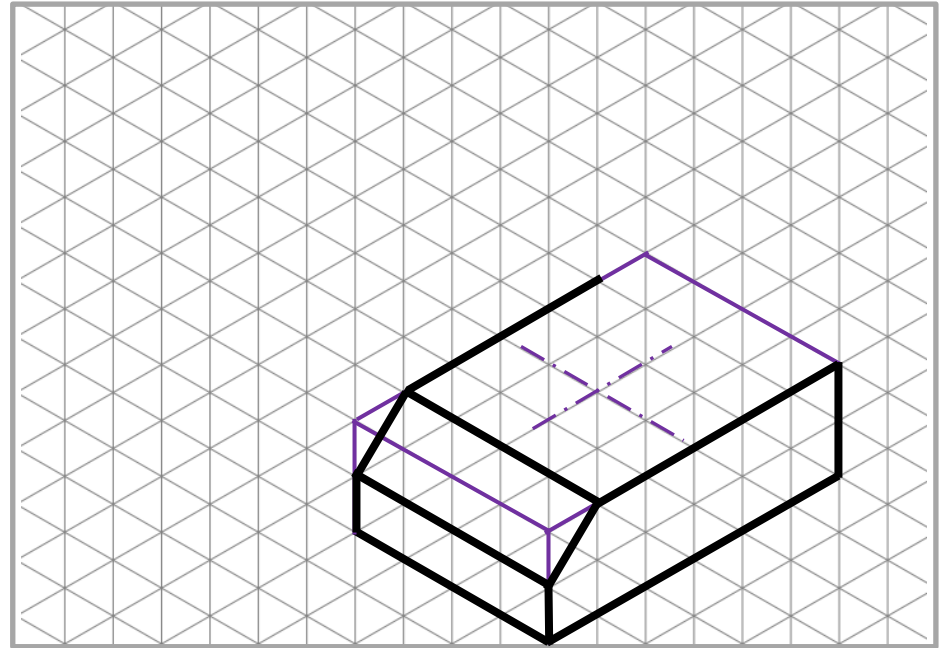
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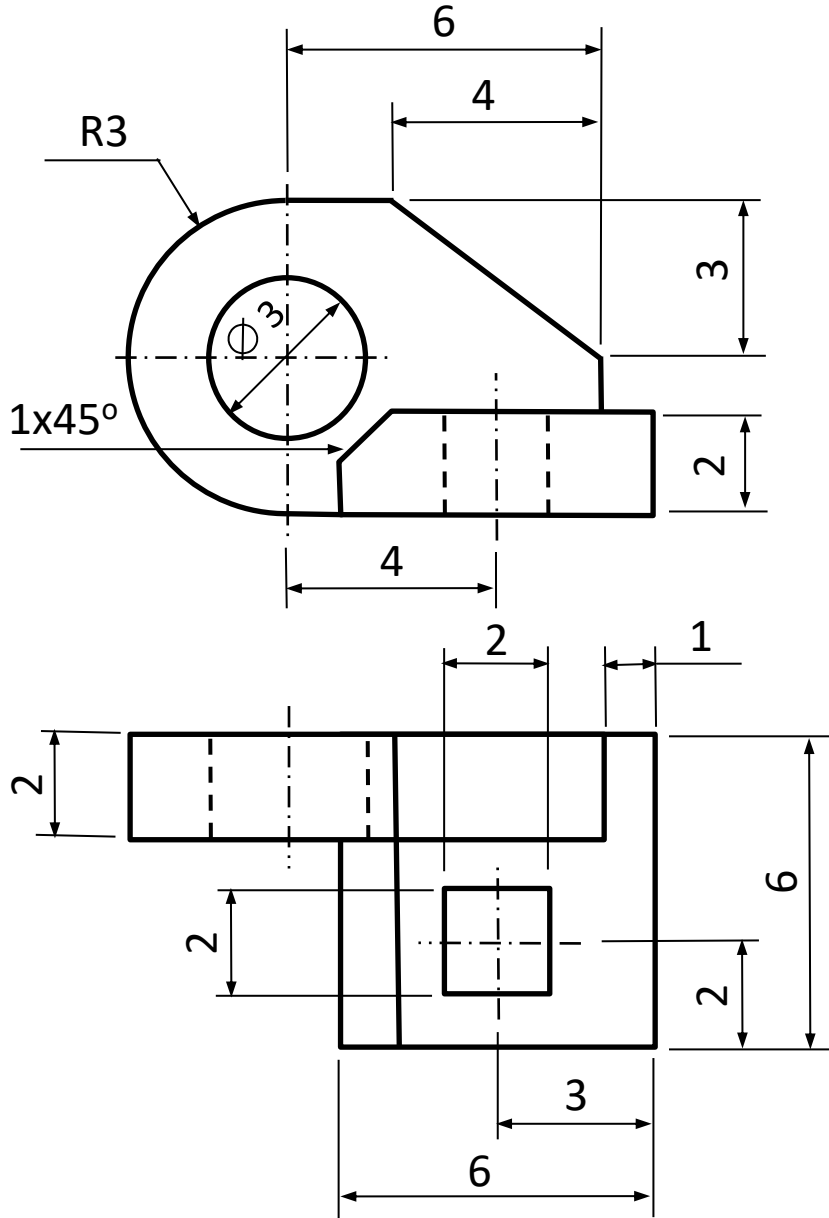
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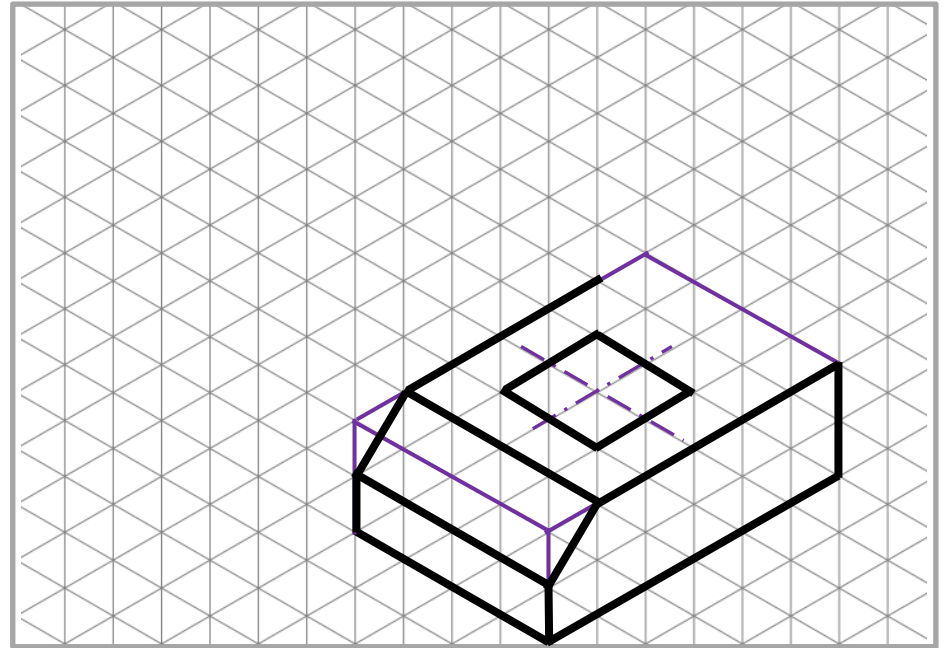
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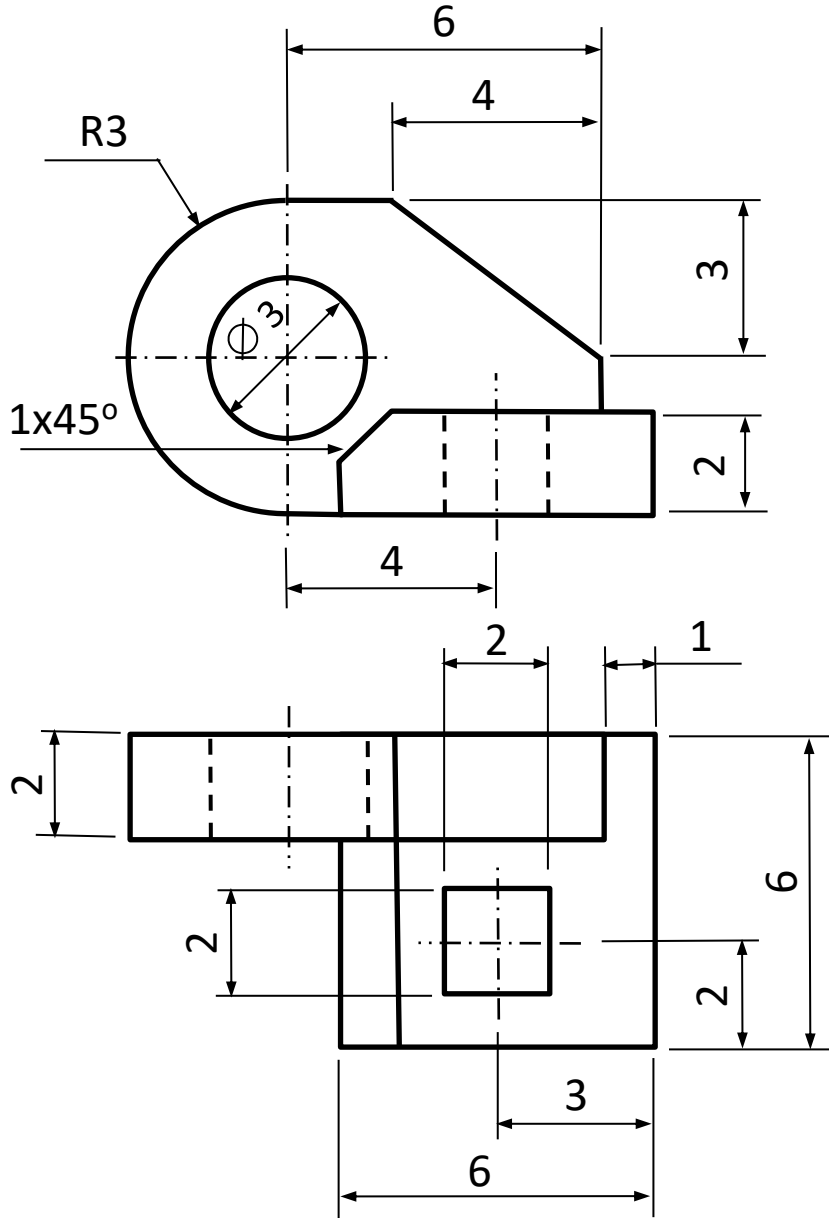
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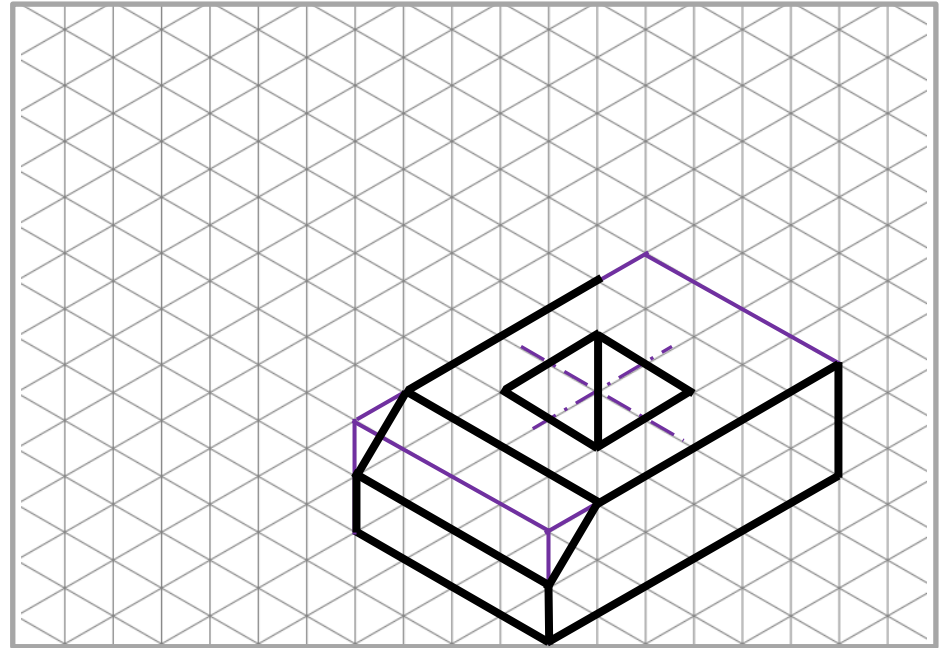
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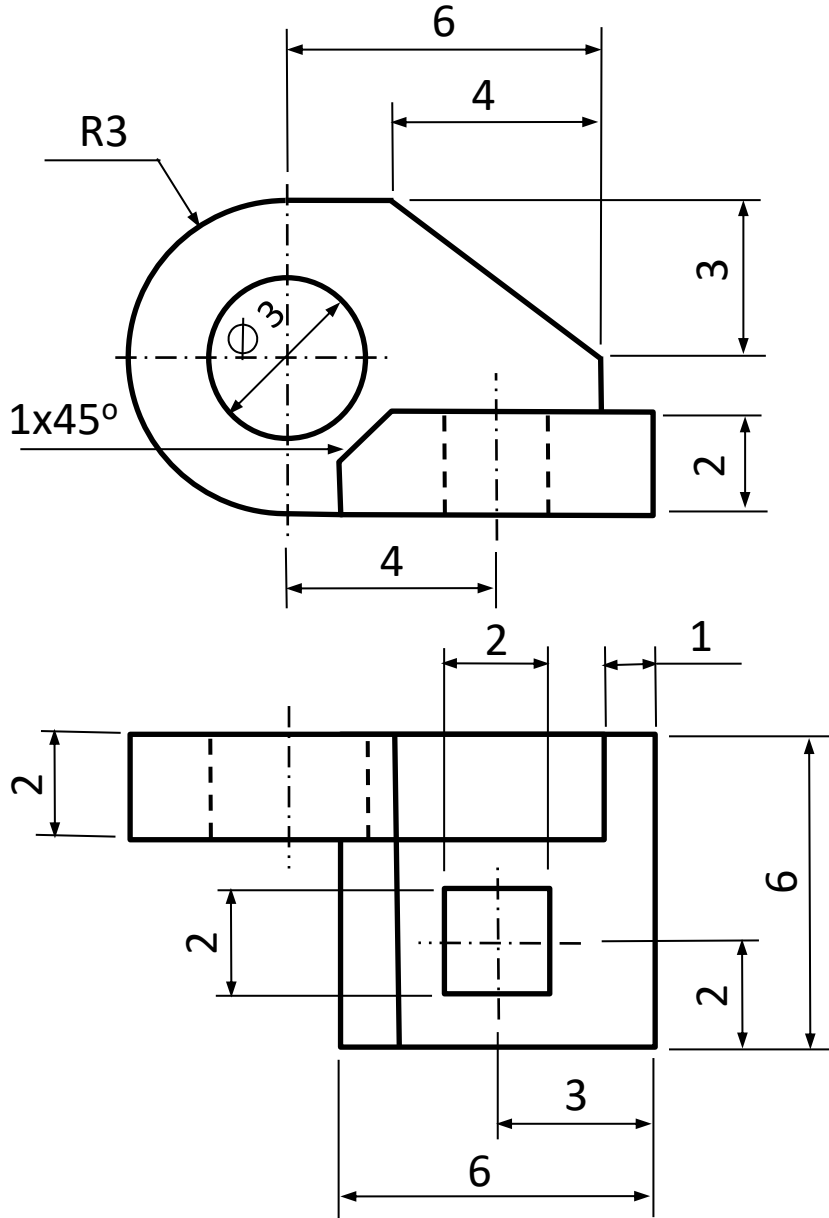
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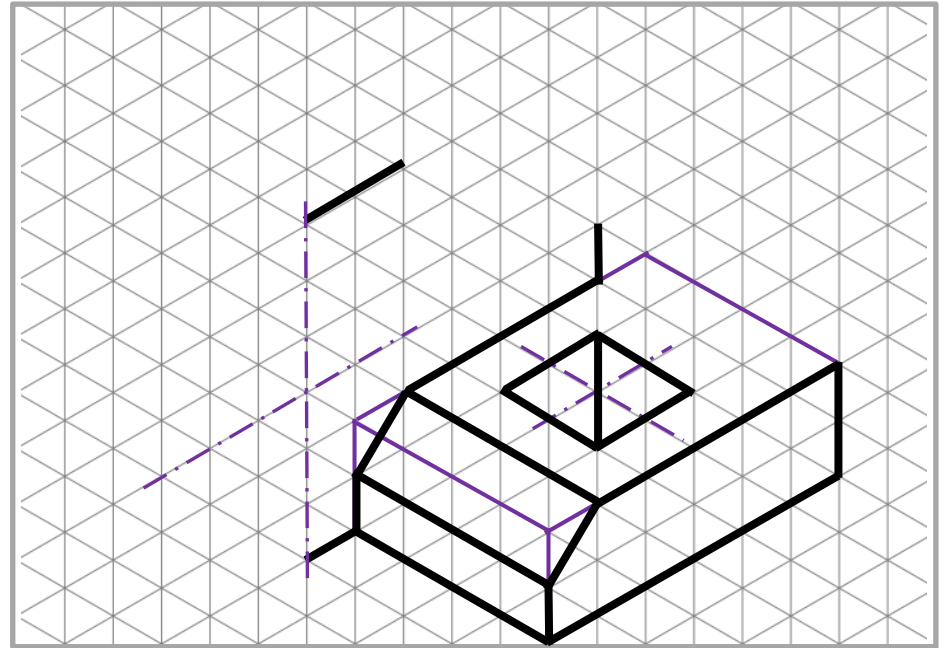
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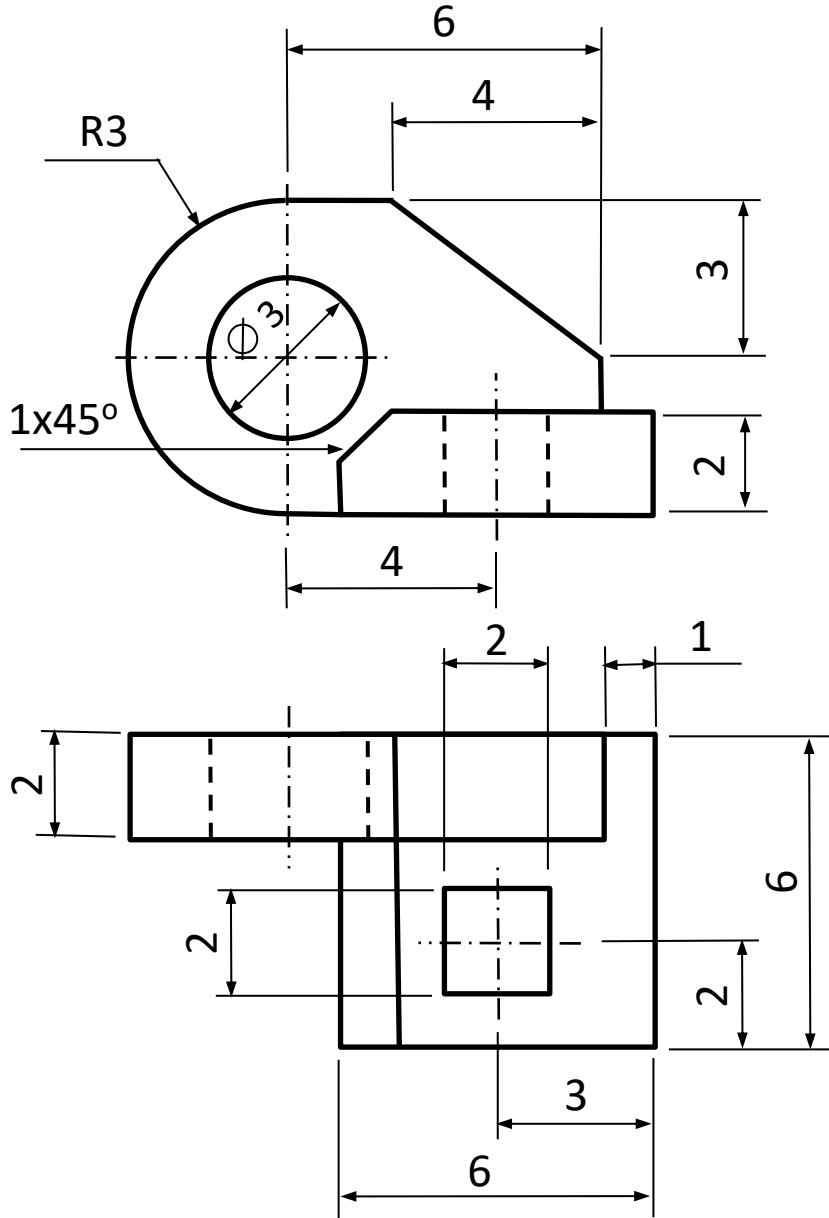
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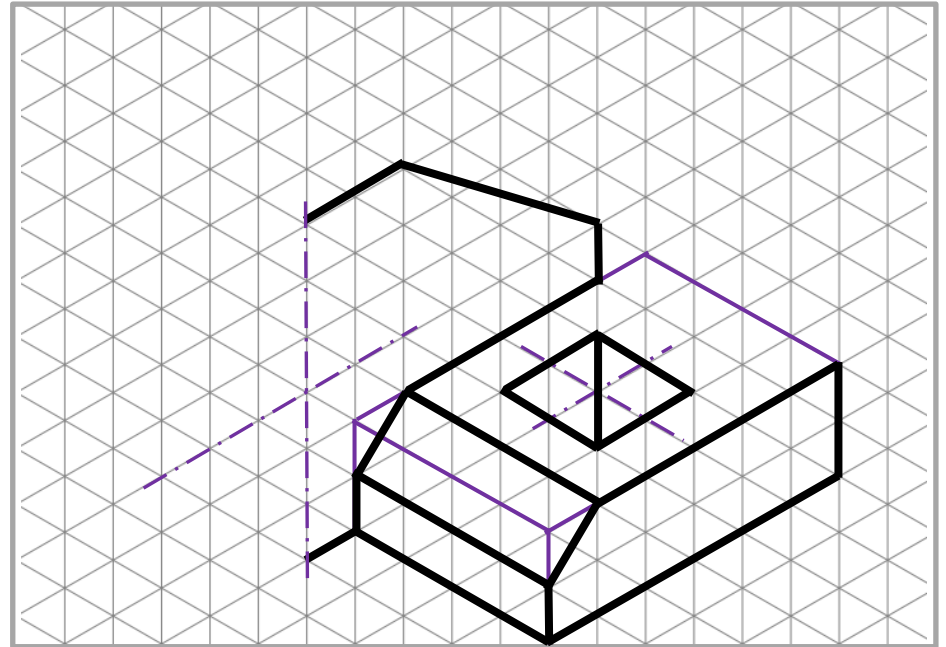
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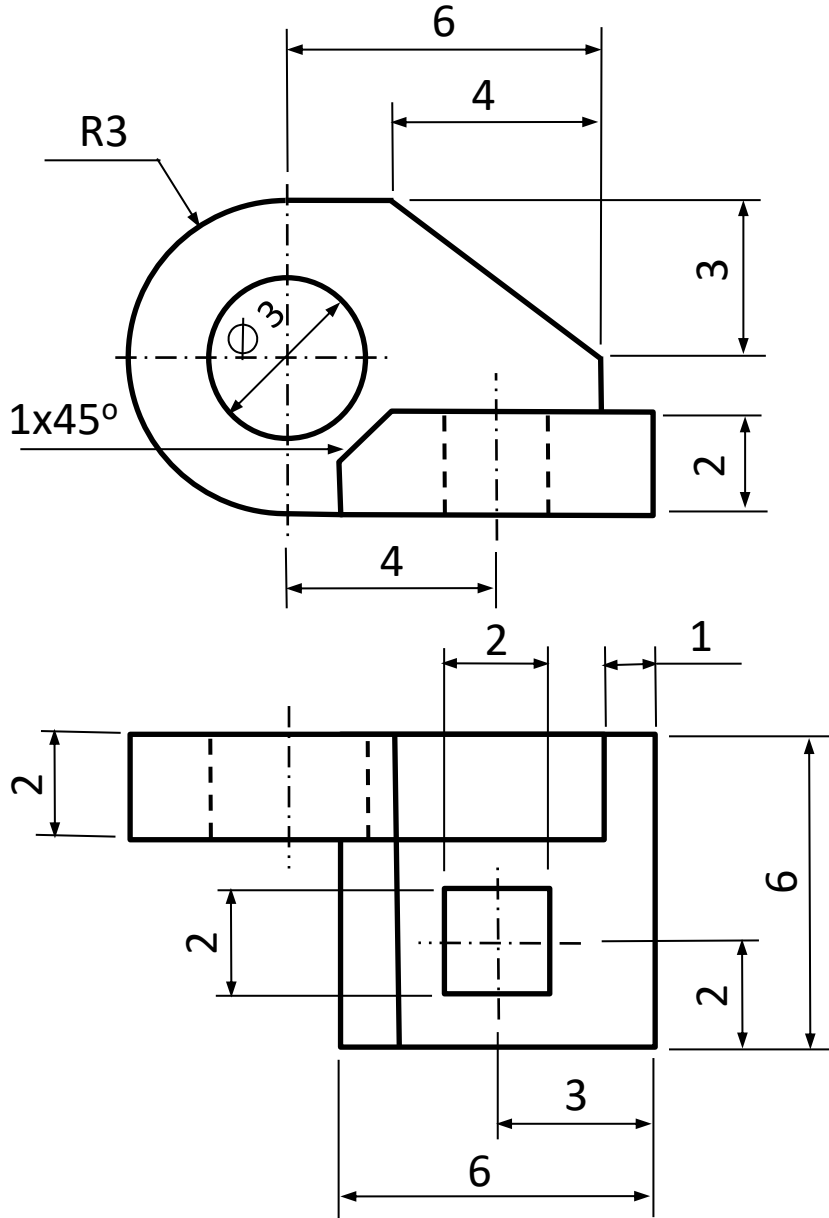
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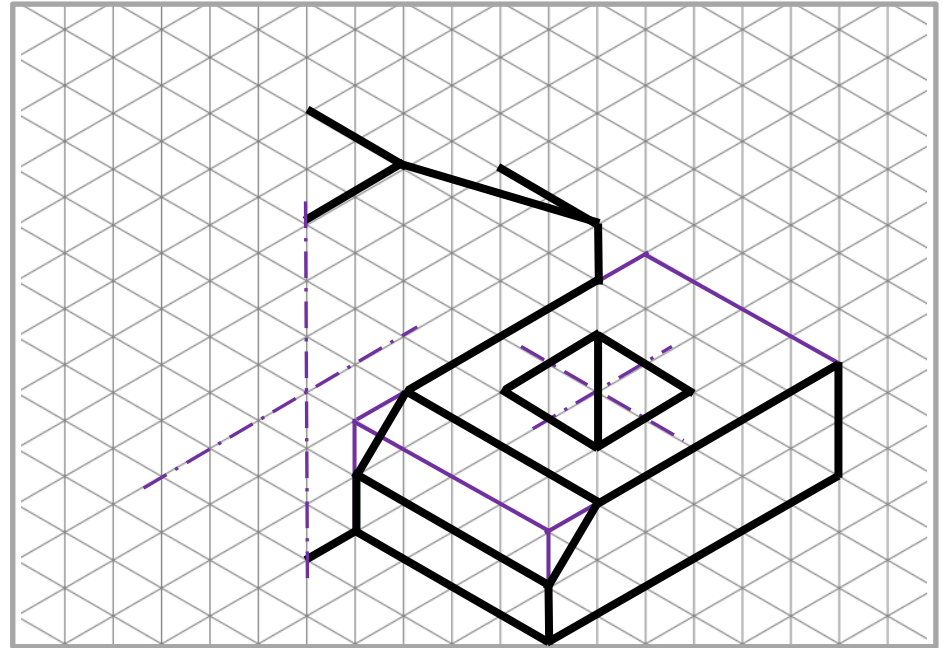
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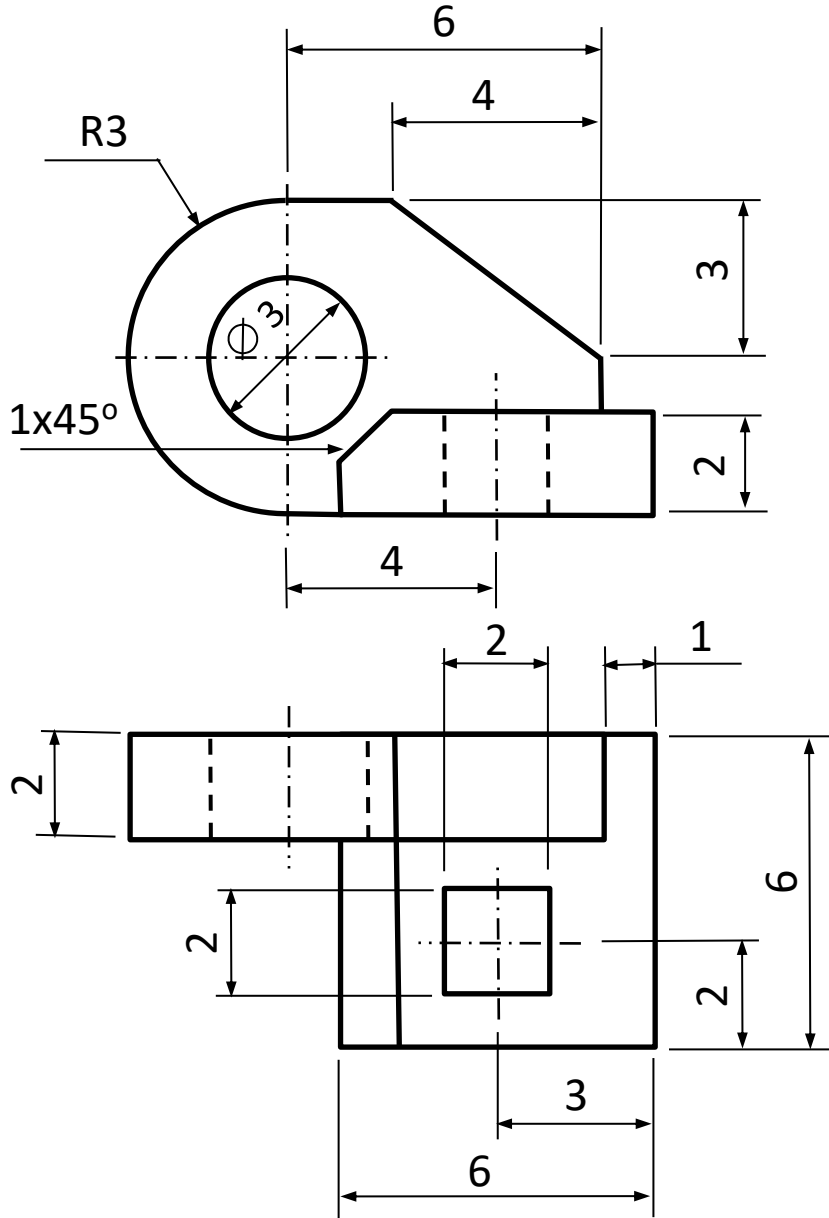
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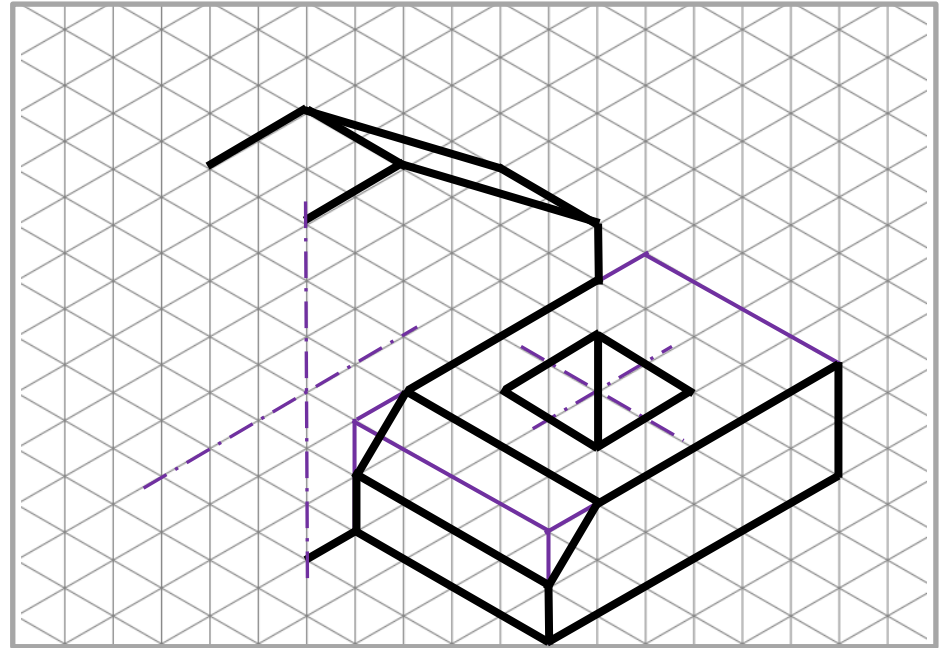
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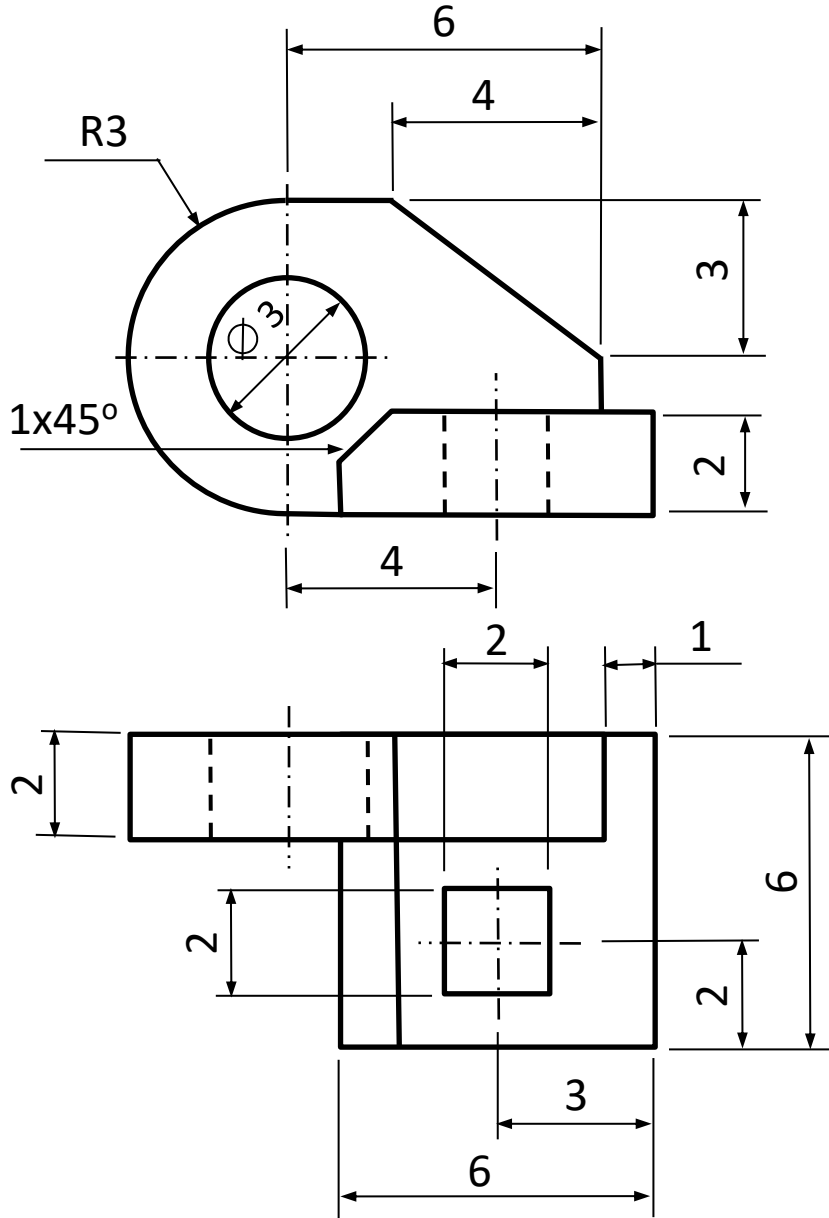
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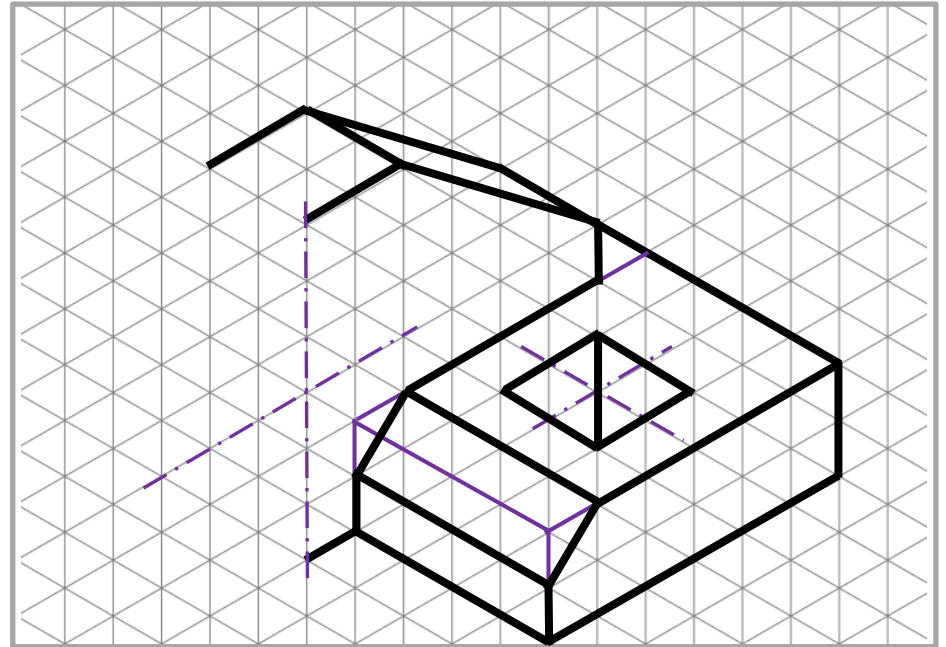
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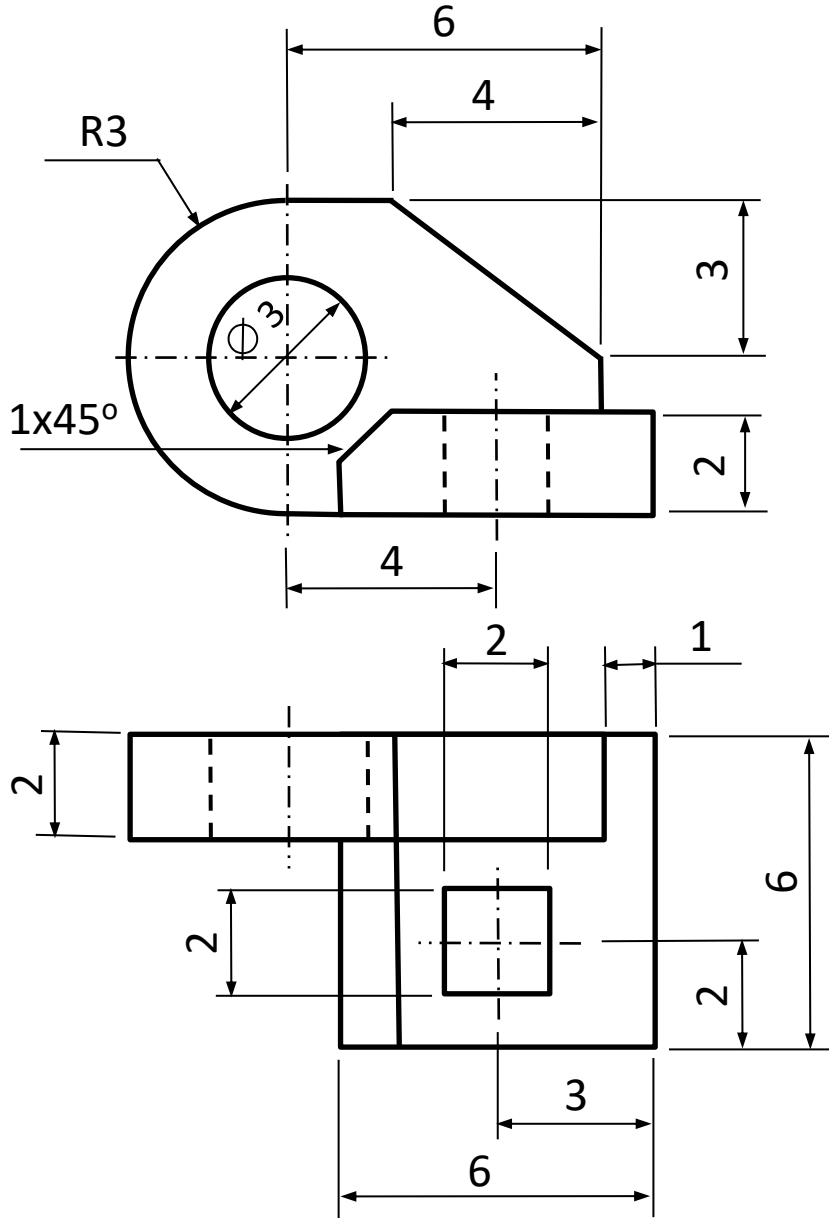
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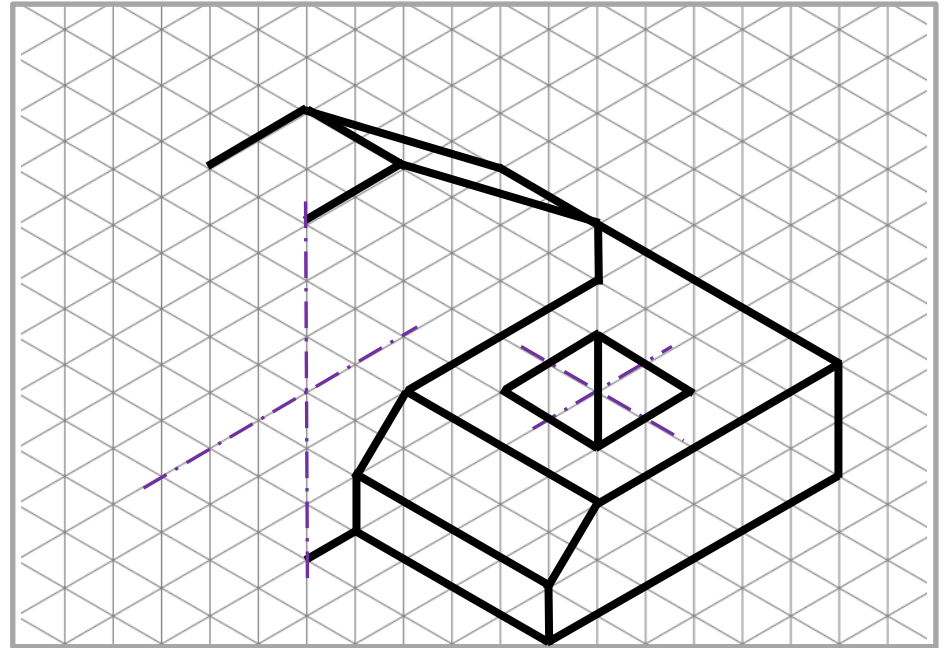
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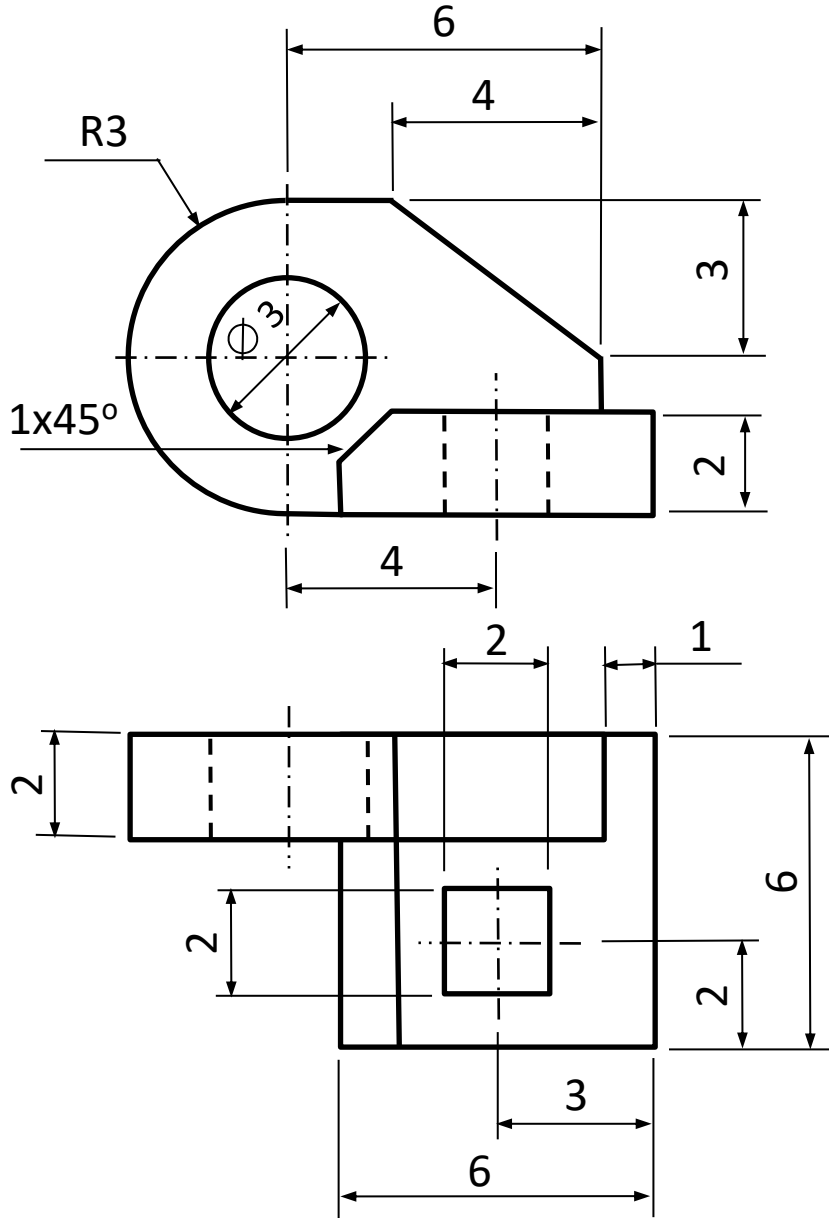
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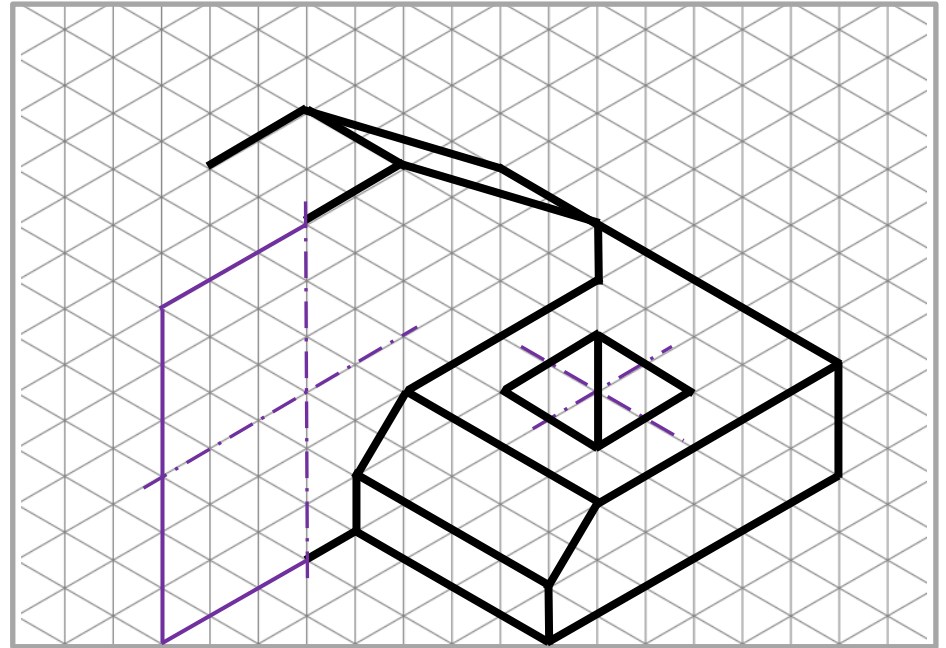
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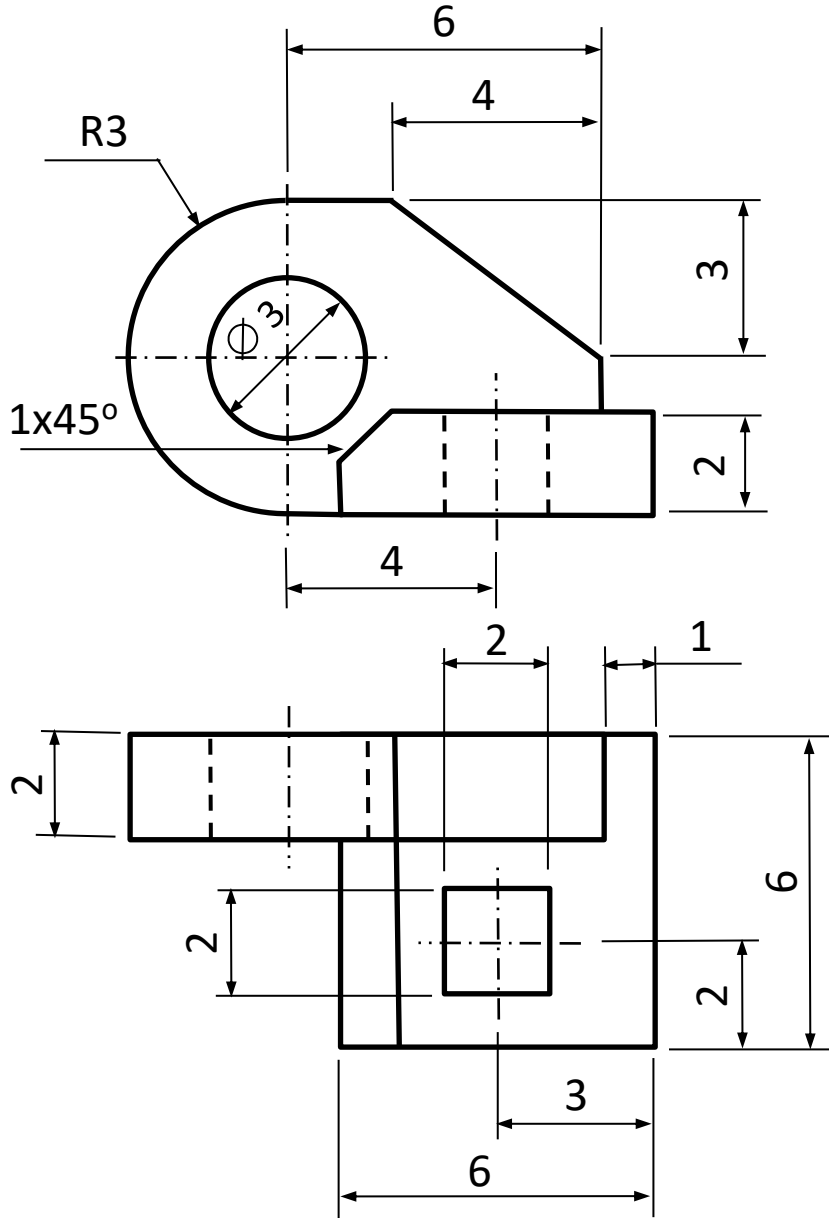
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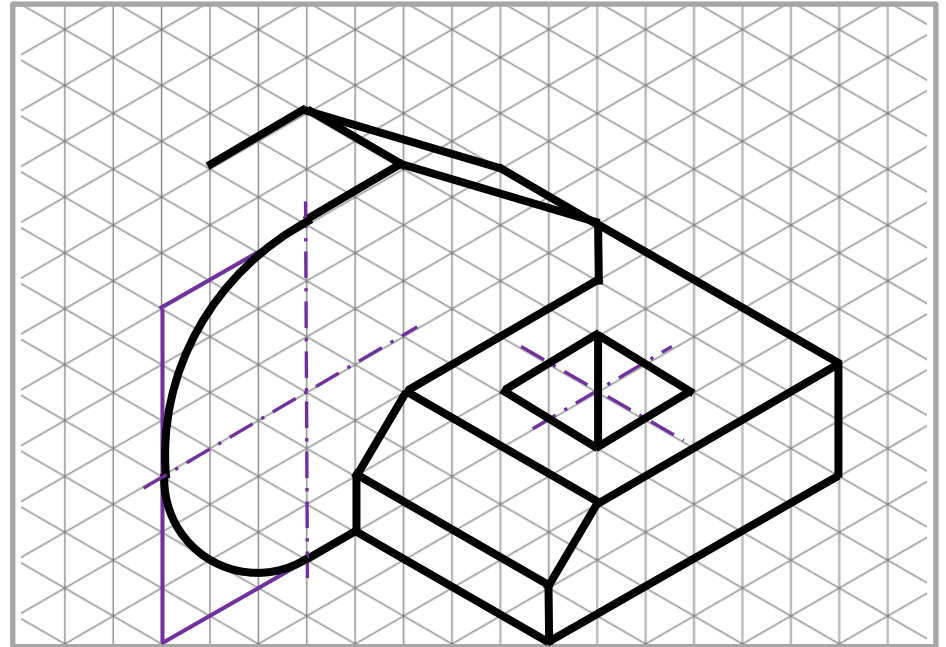
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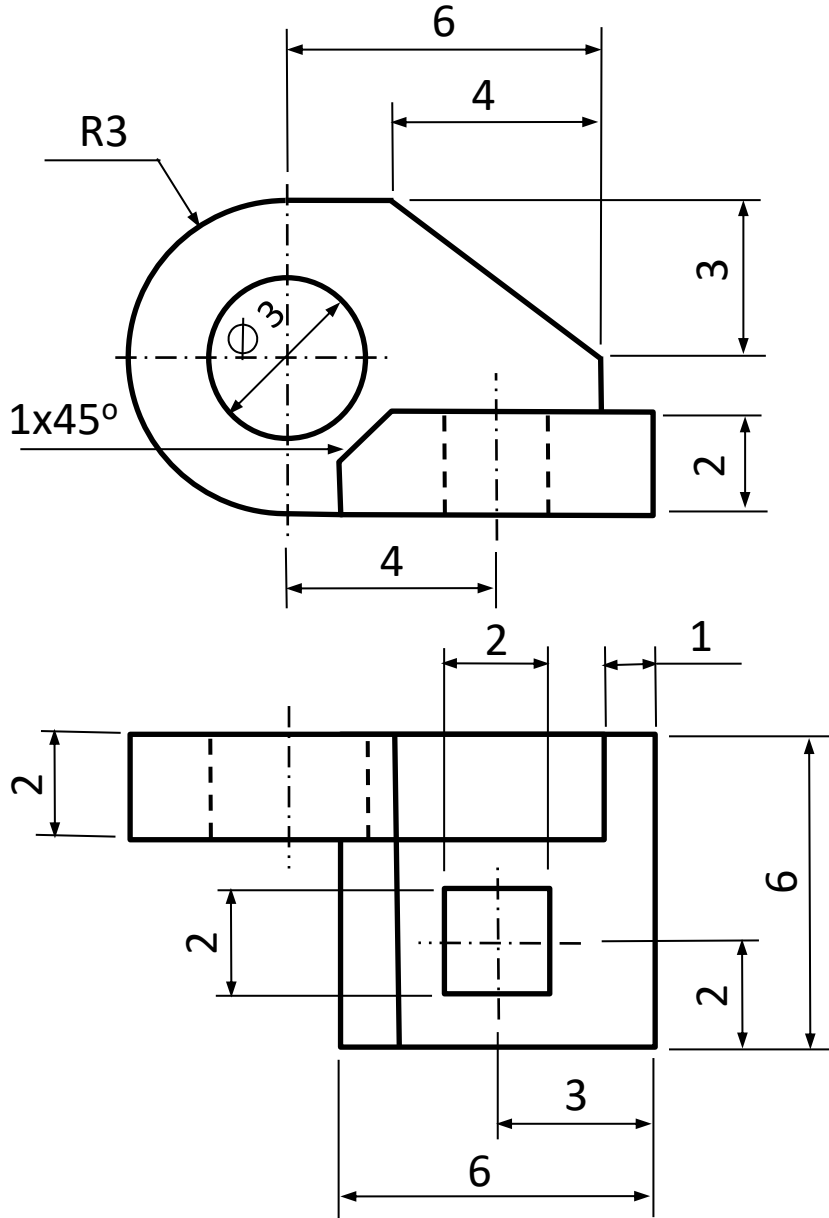
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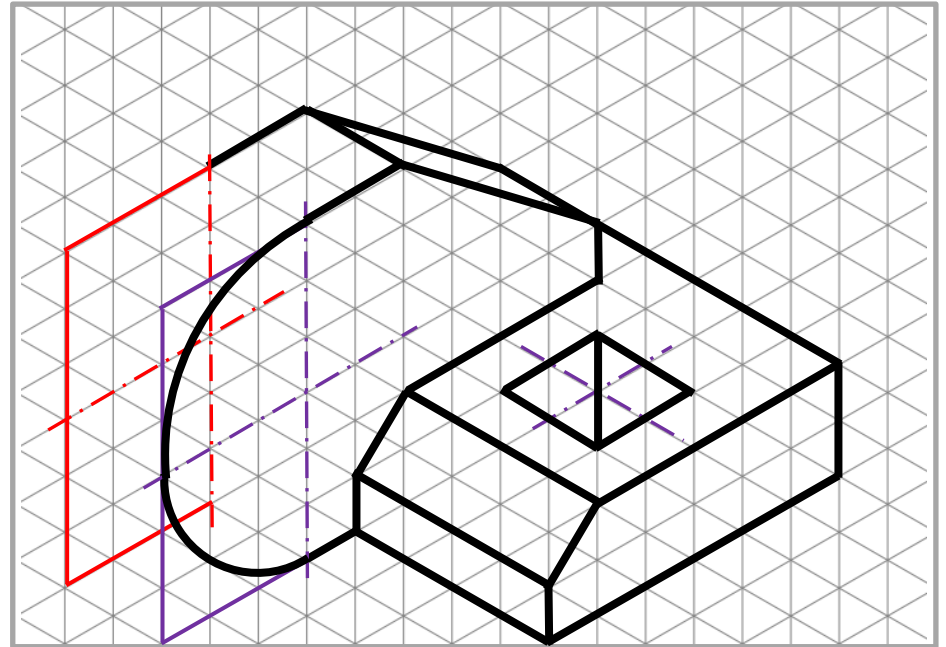
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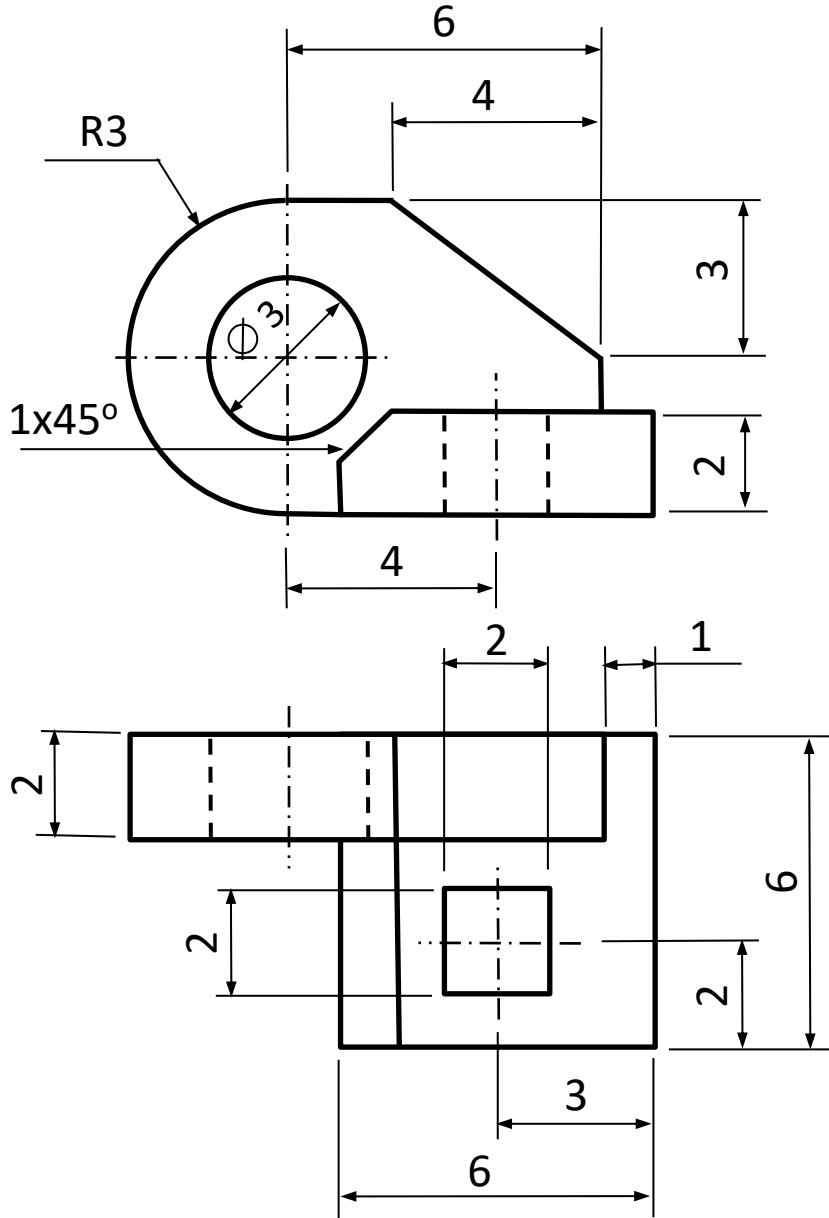
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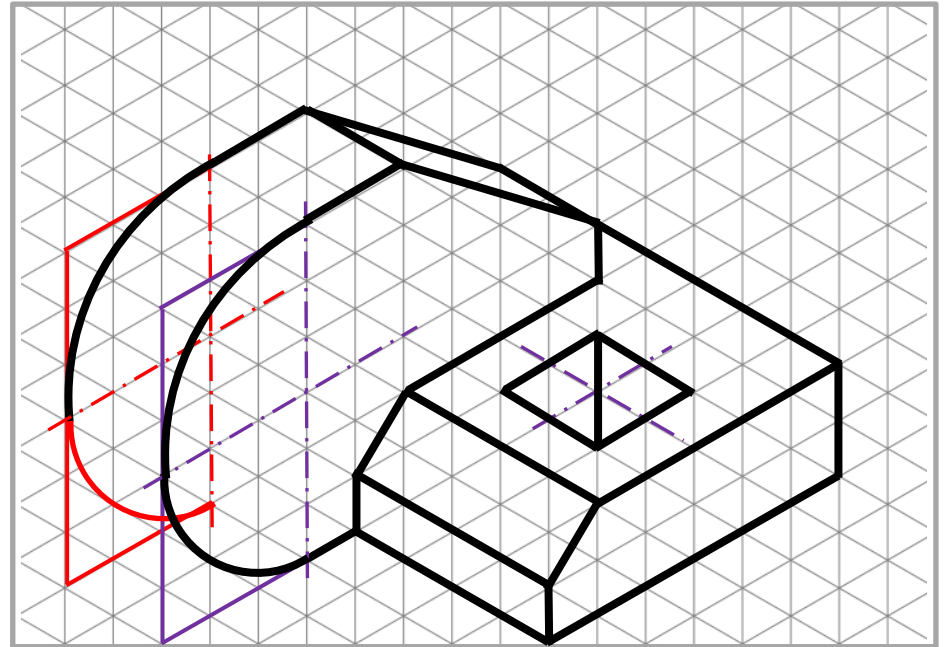
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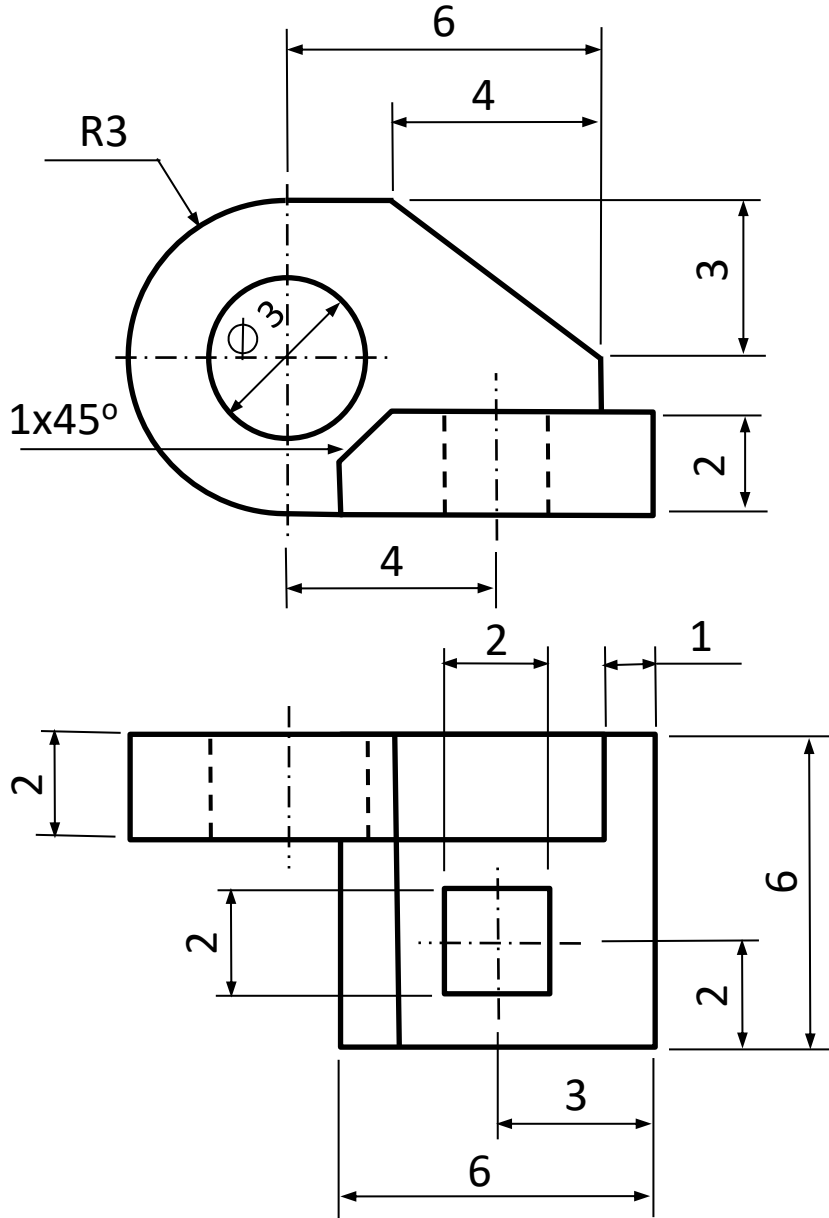
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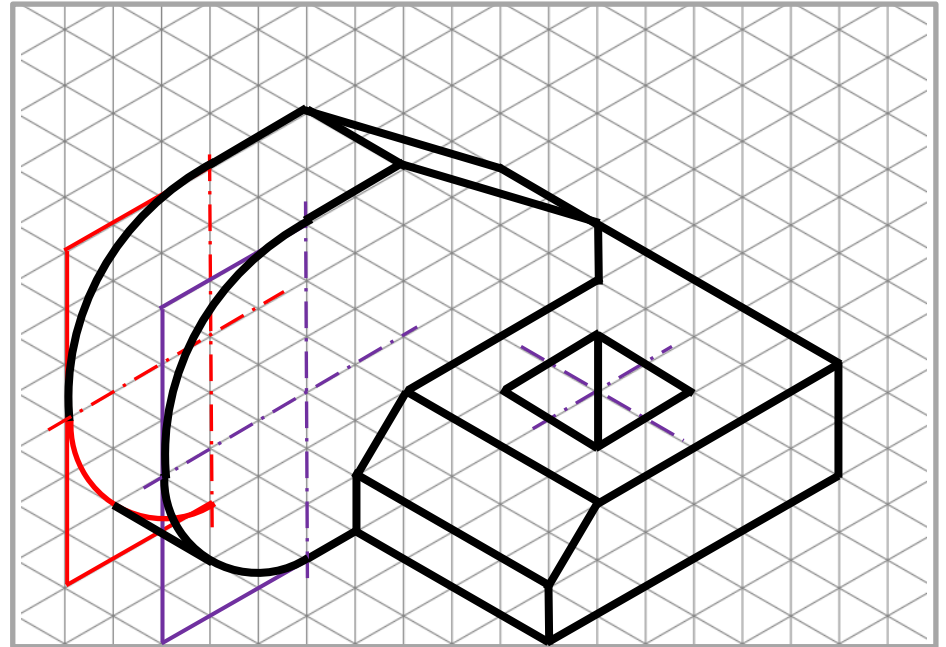
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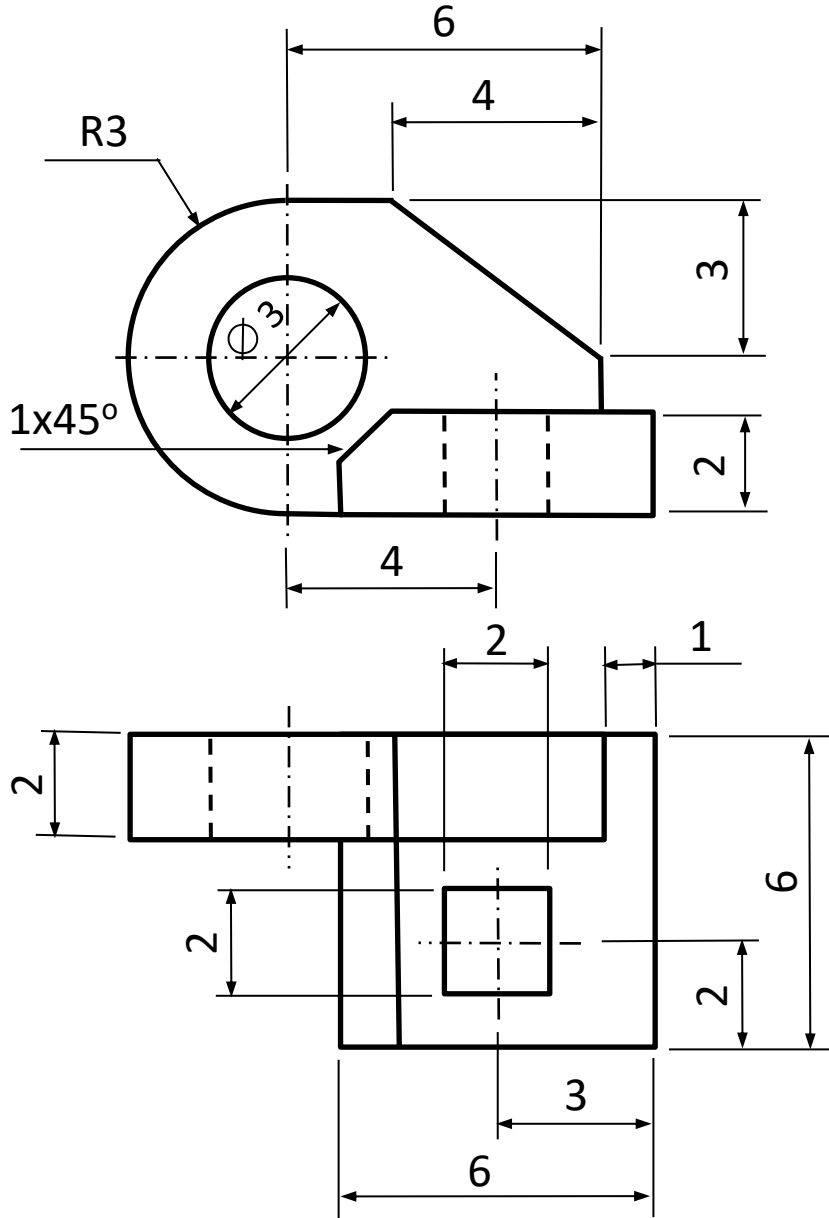
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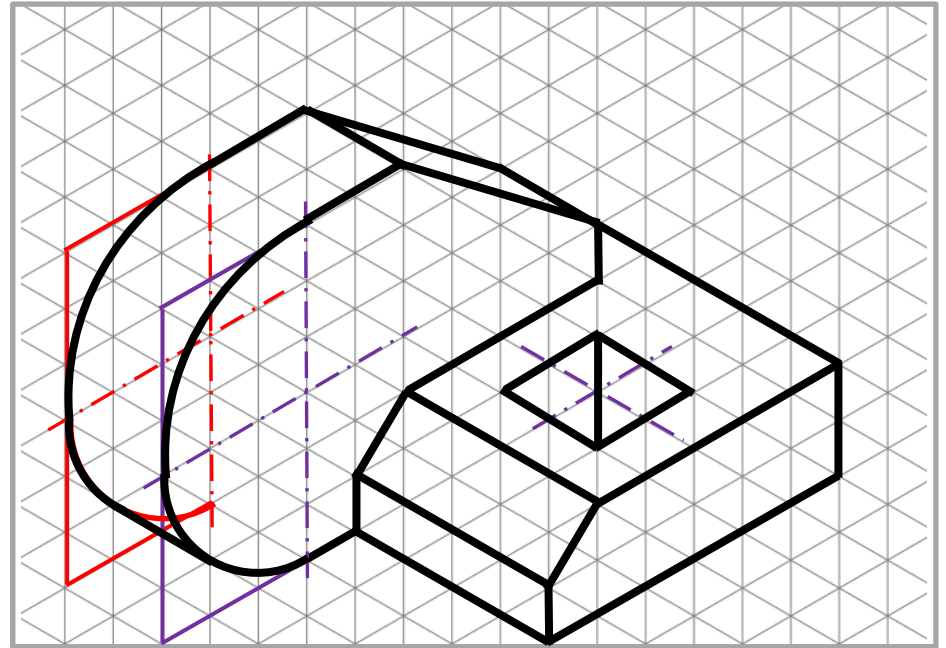
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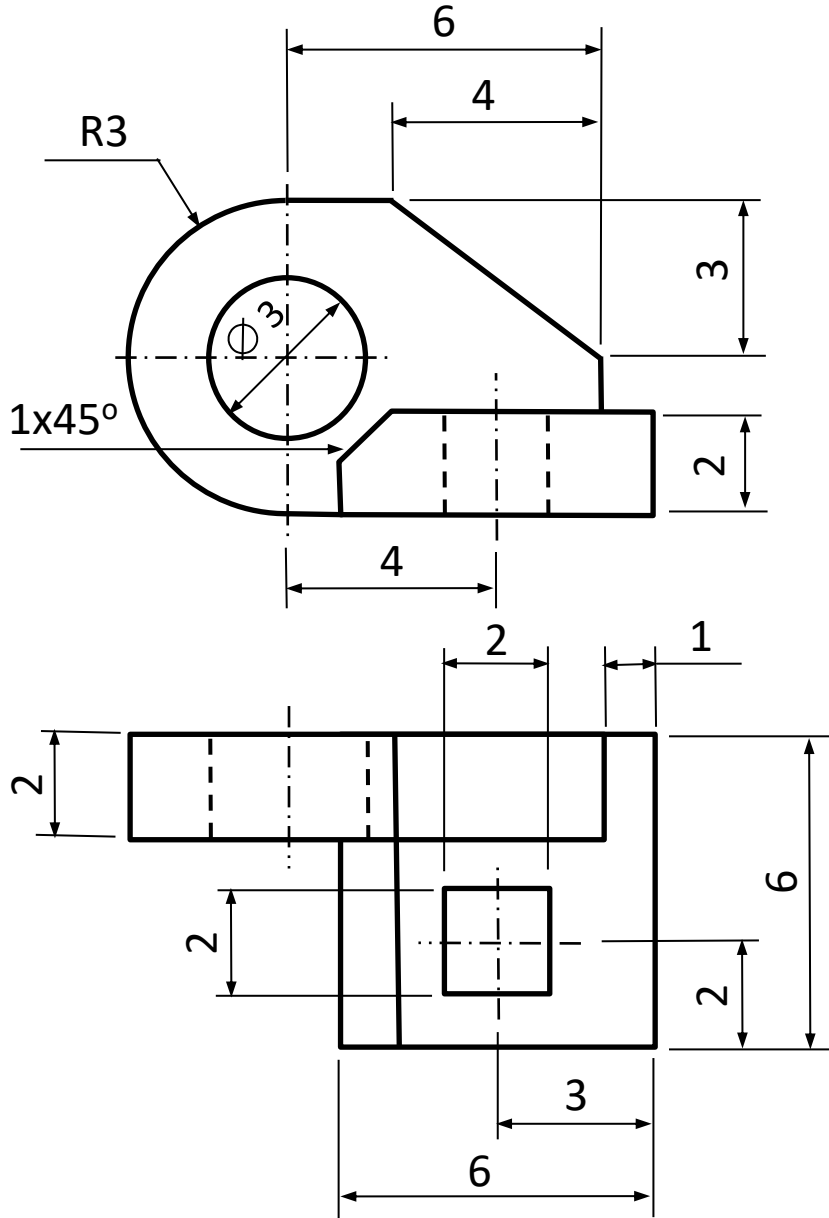
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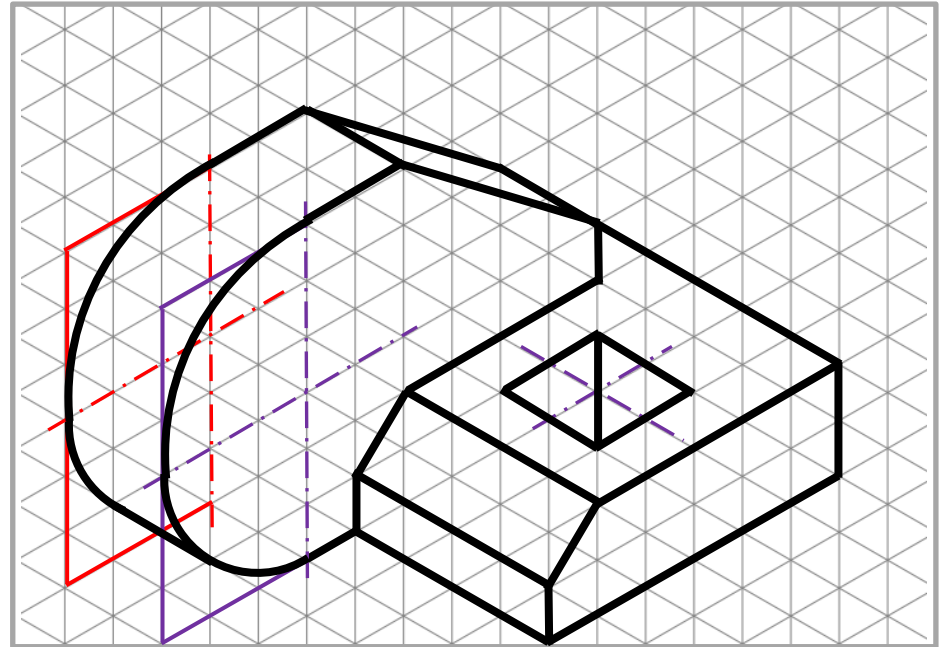
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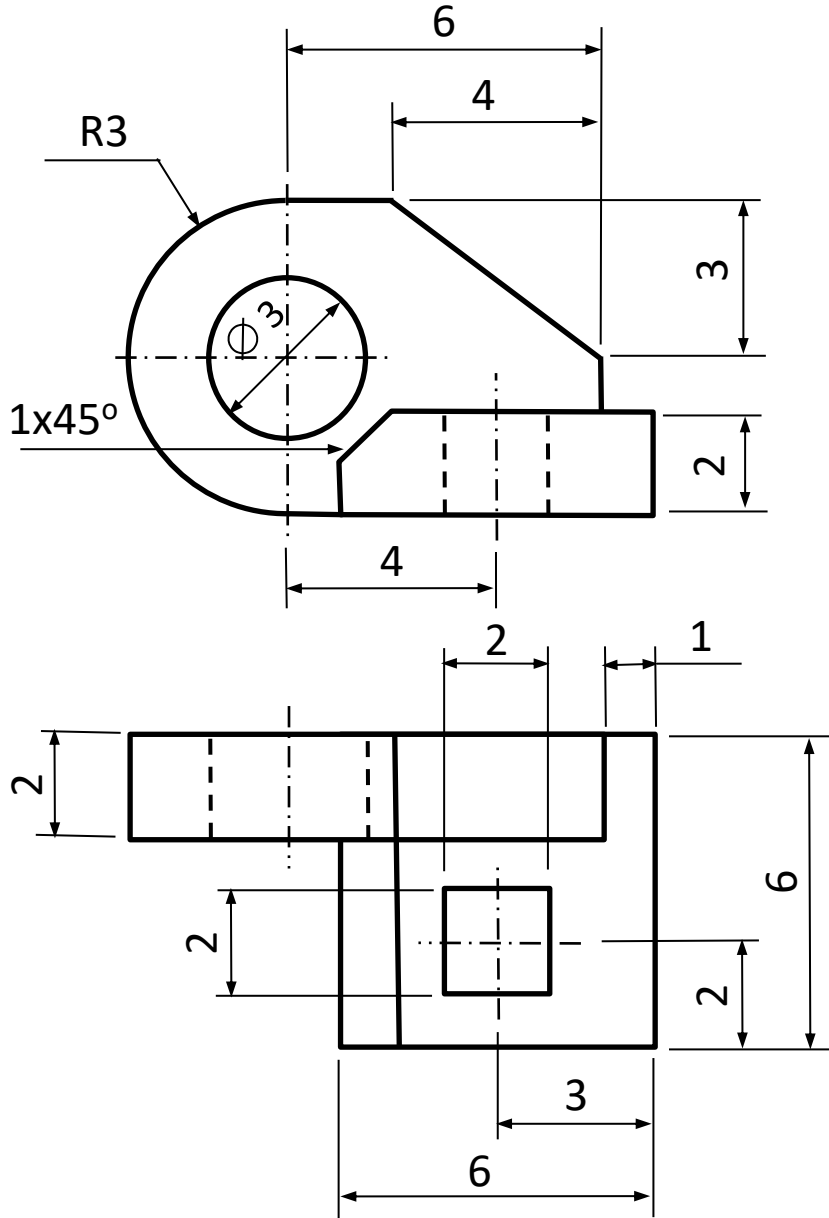
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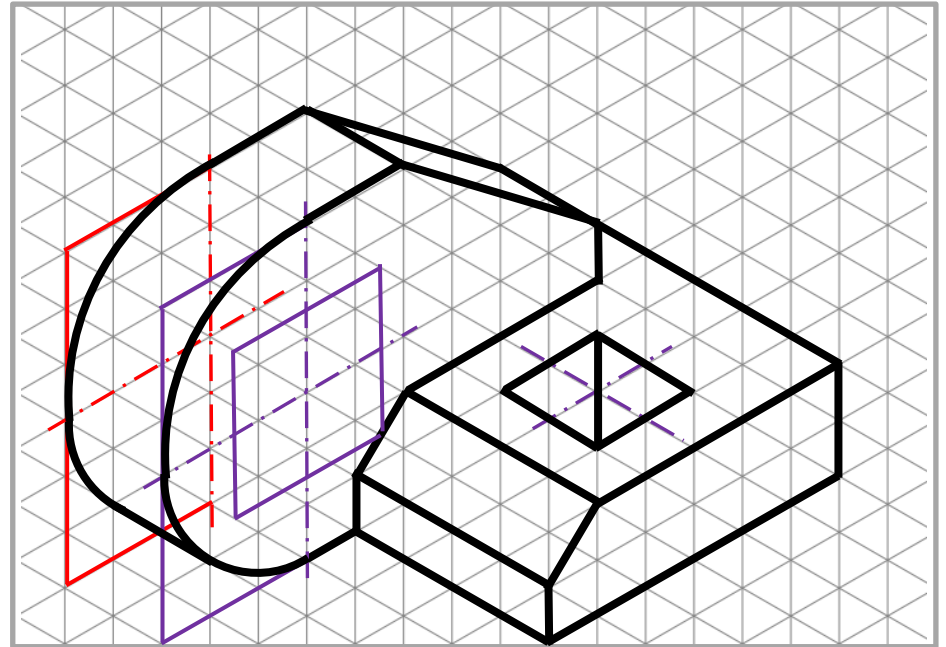
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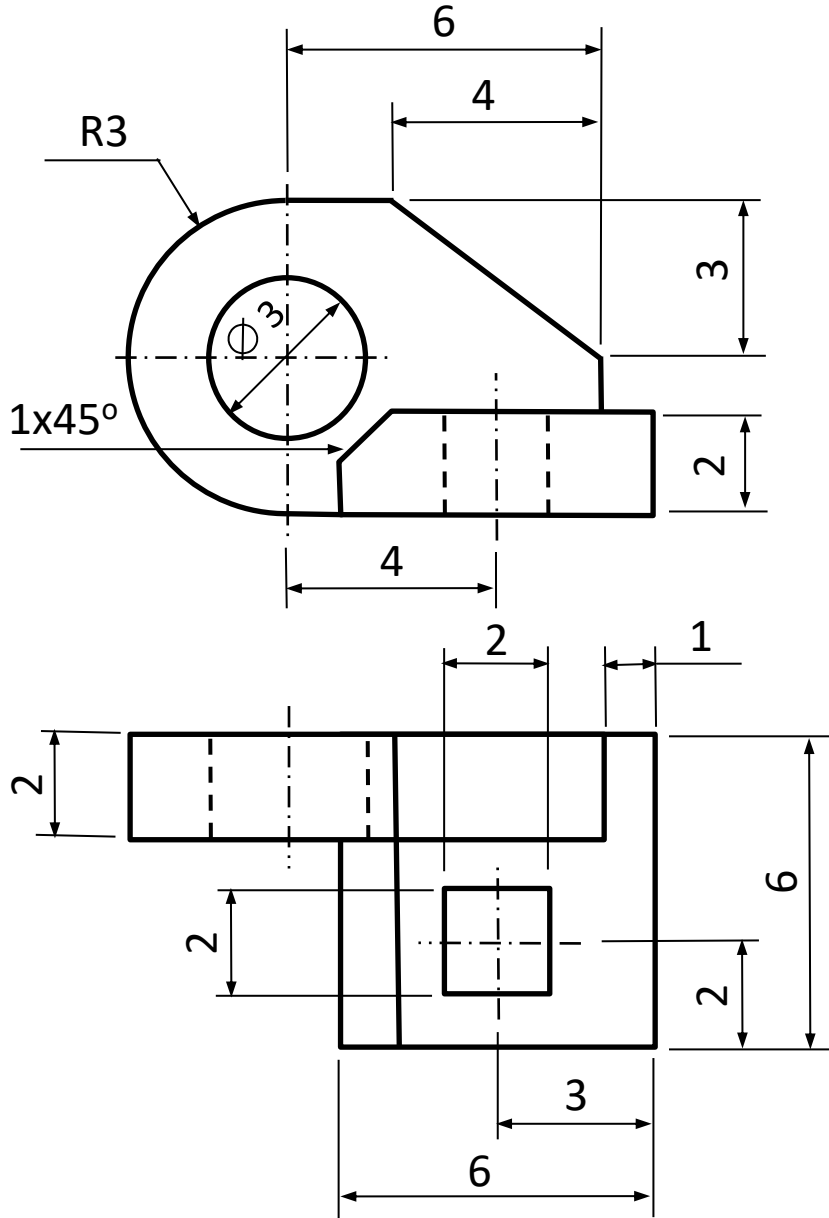
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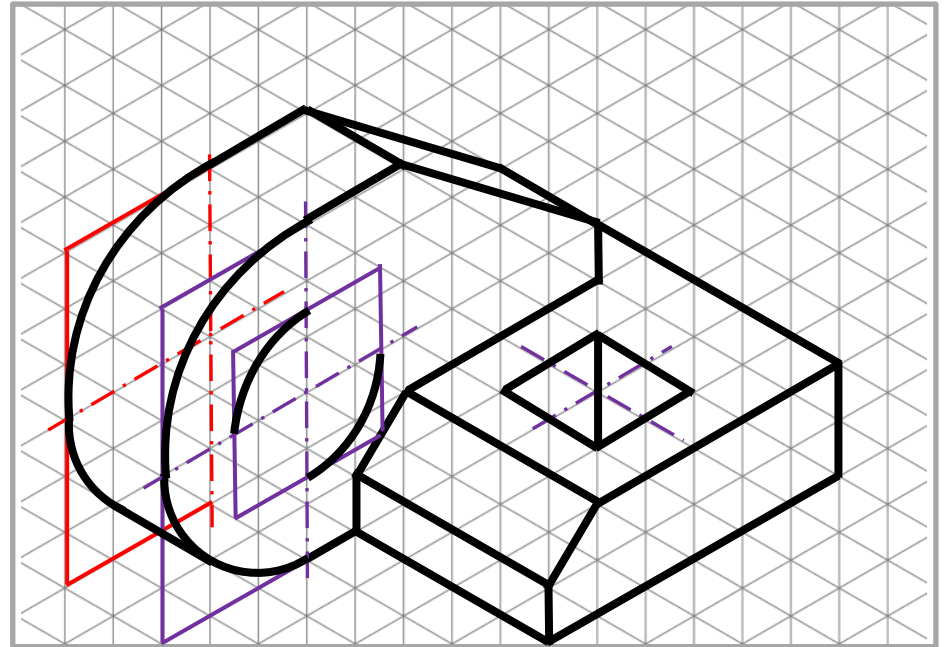
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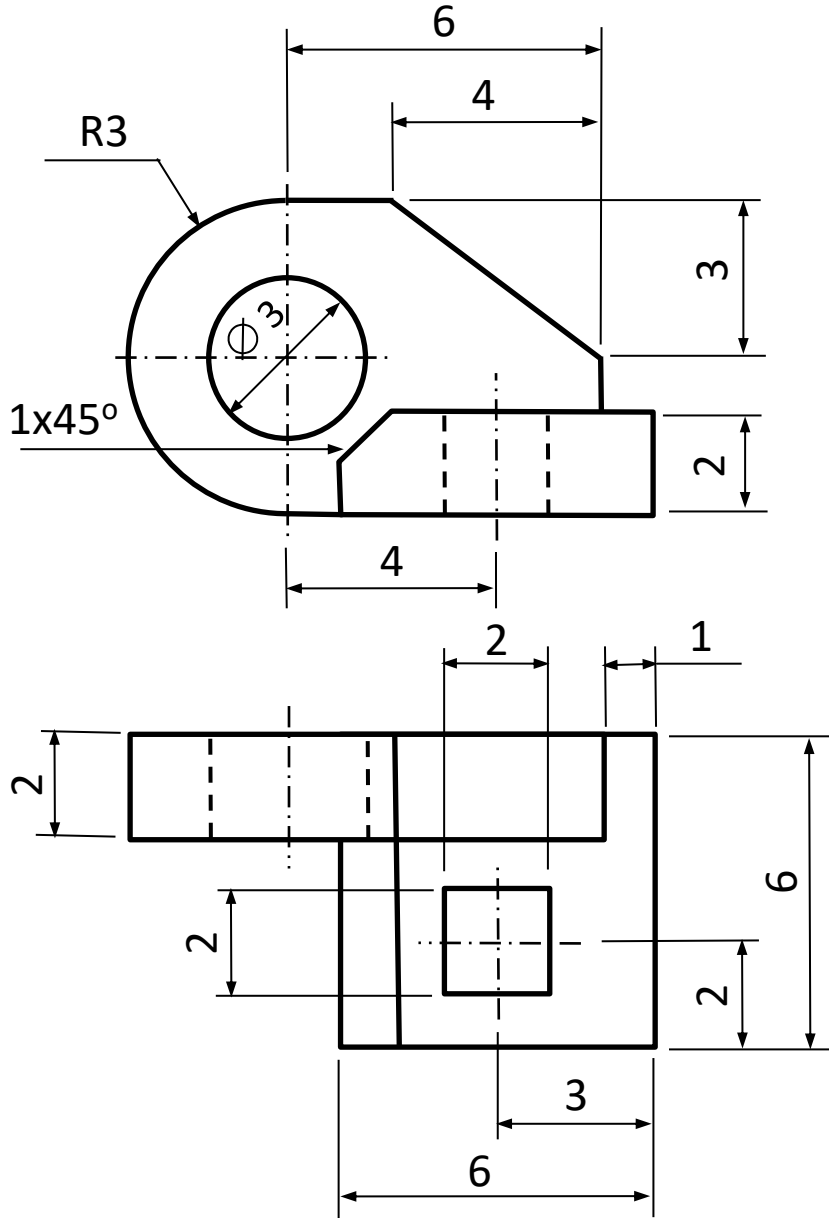
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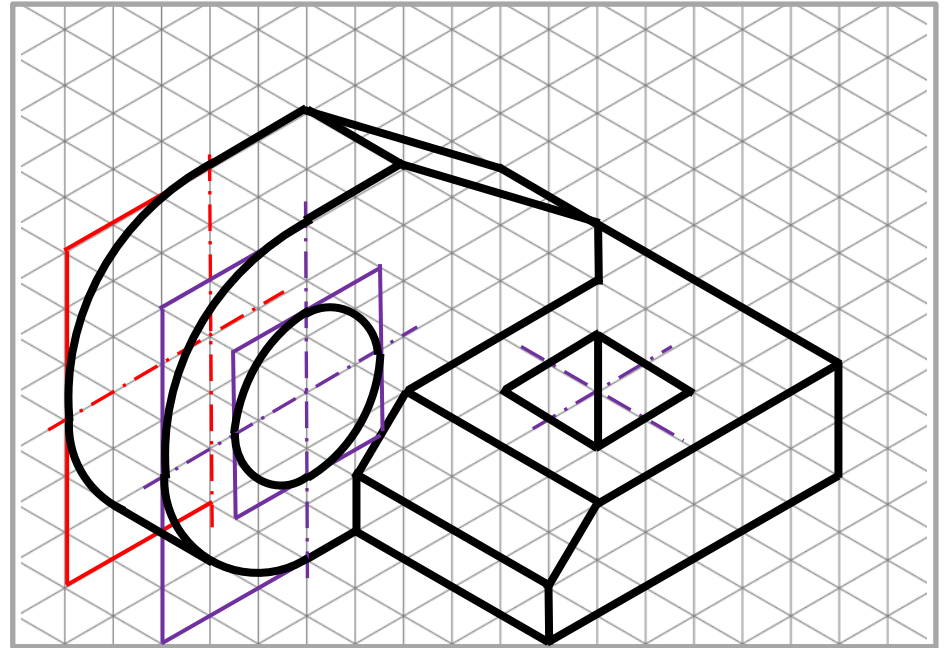
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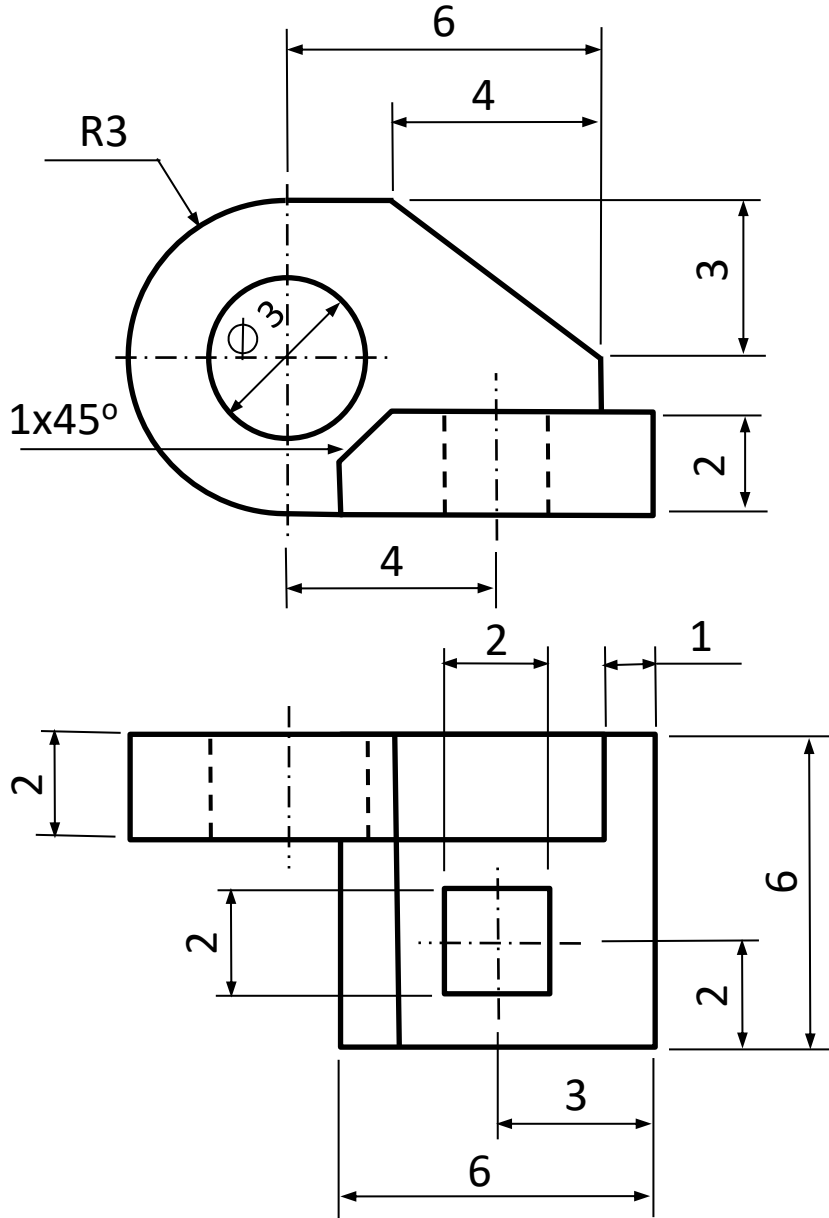
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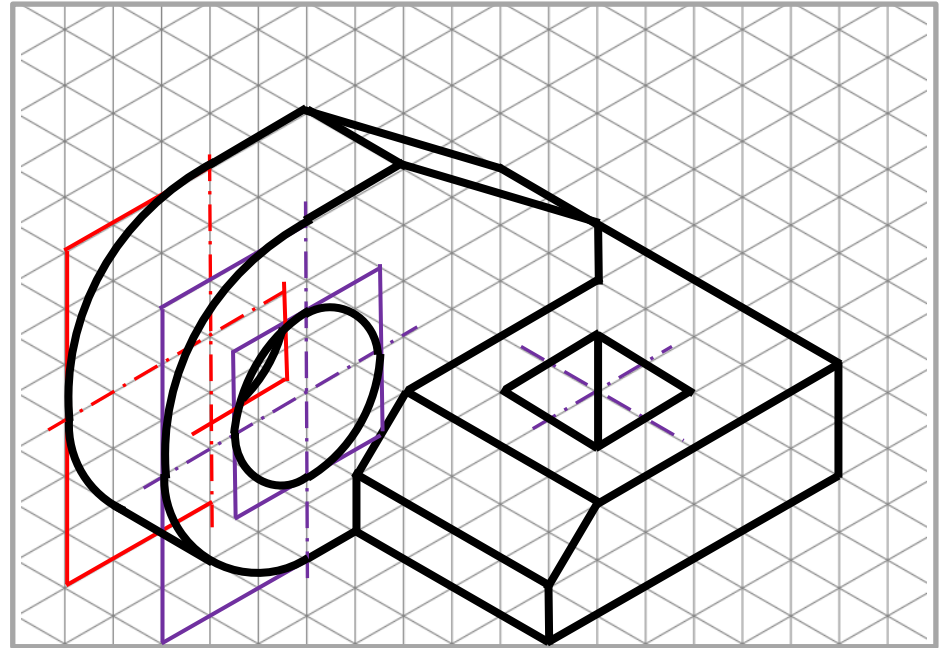
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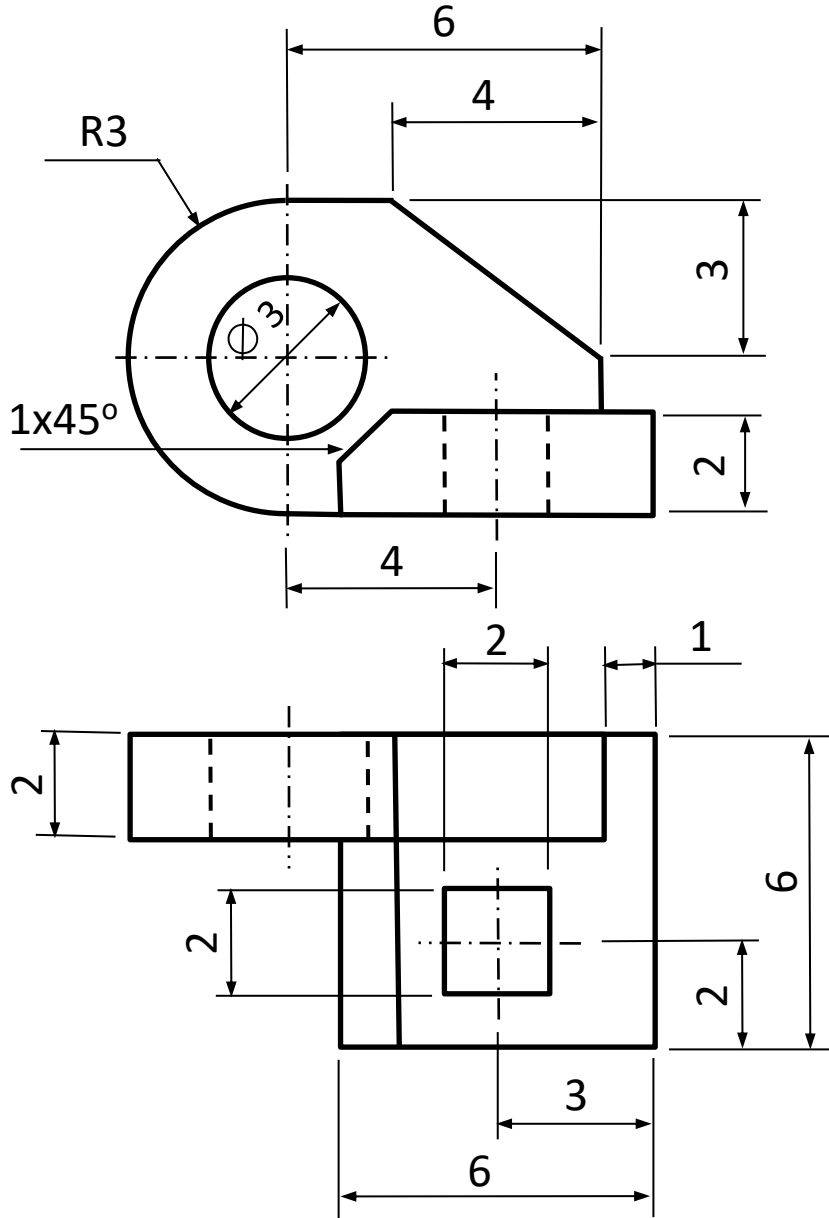
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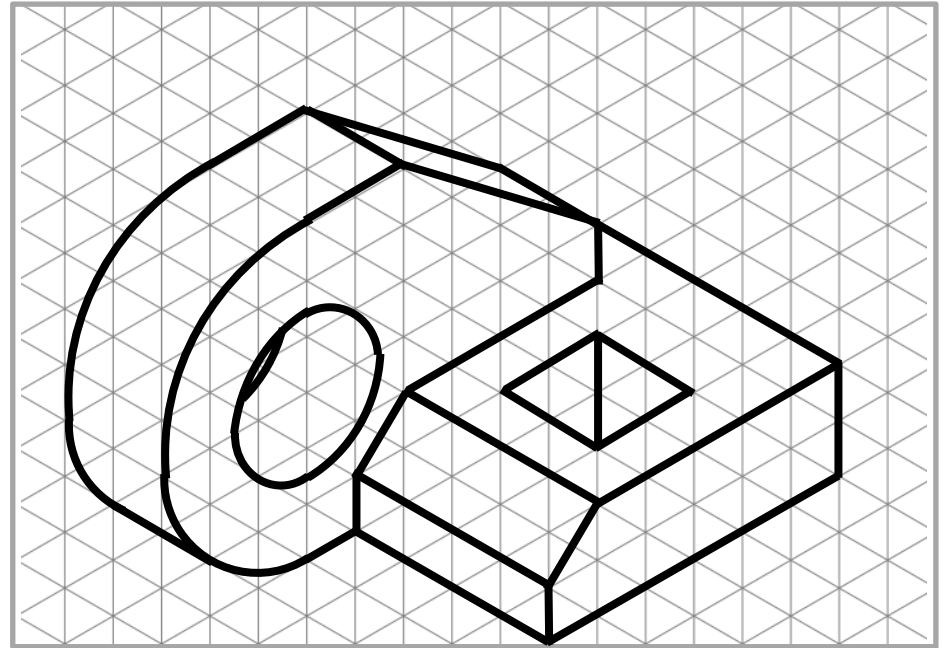
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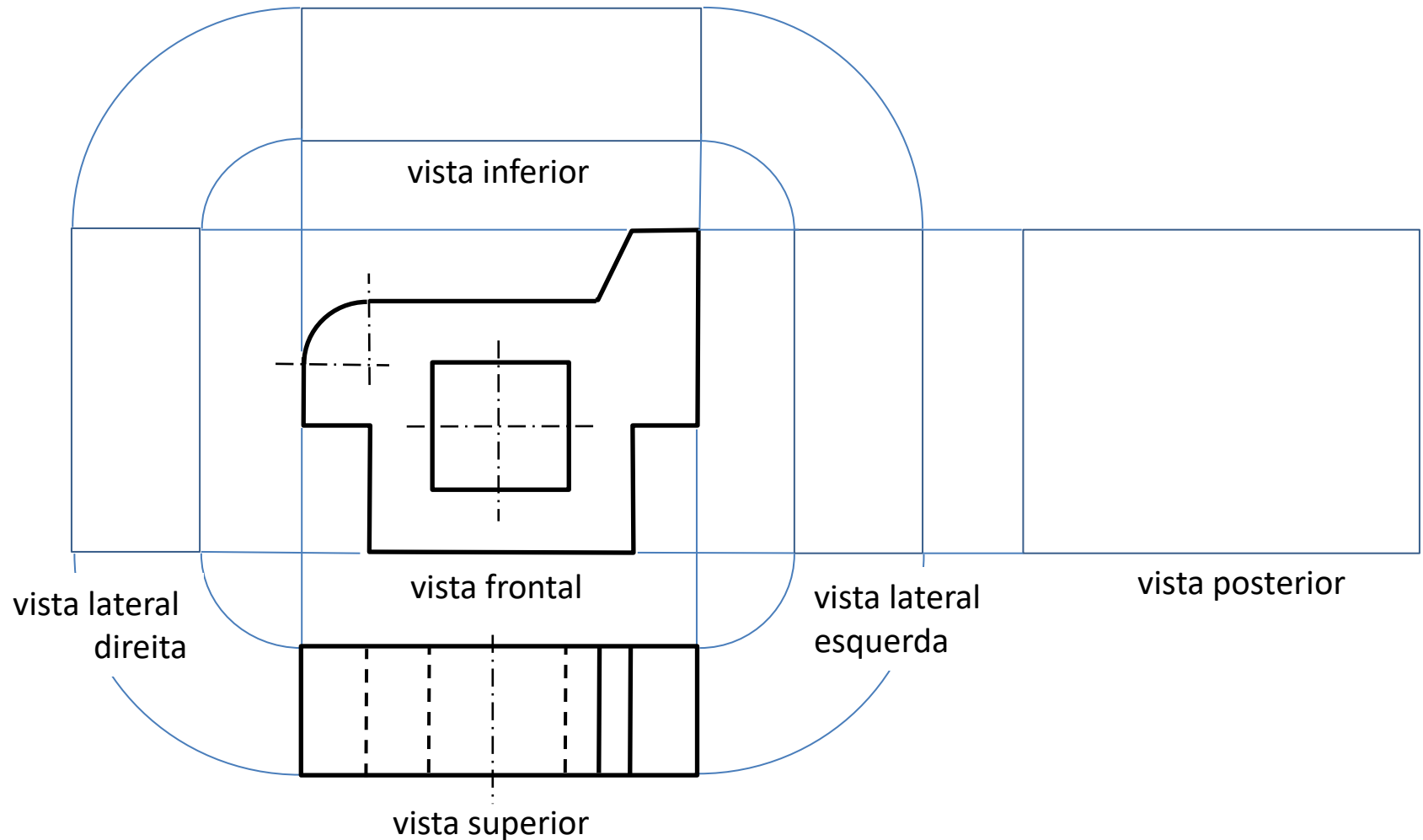
Vistas ortográficas



Perspectiva Isométrica - 7

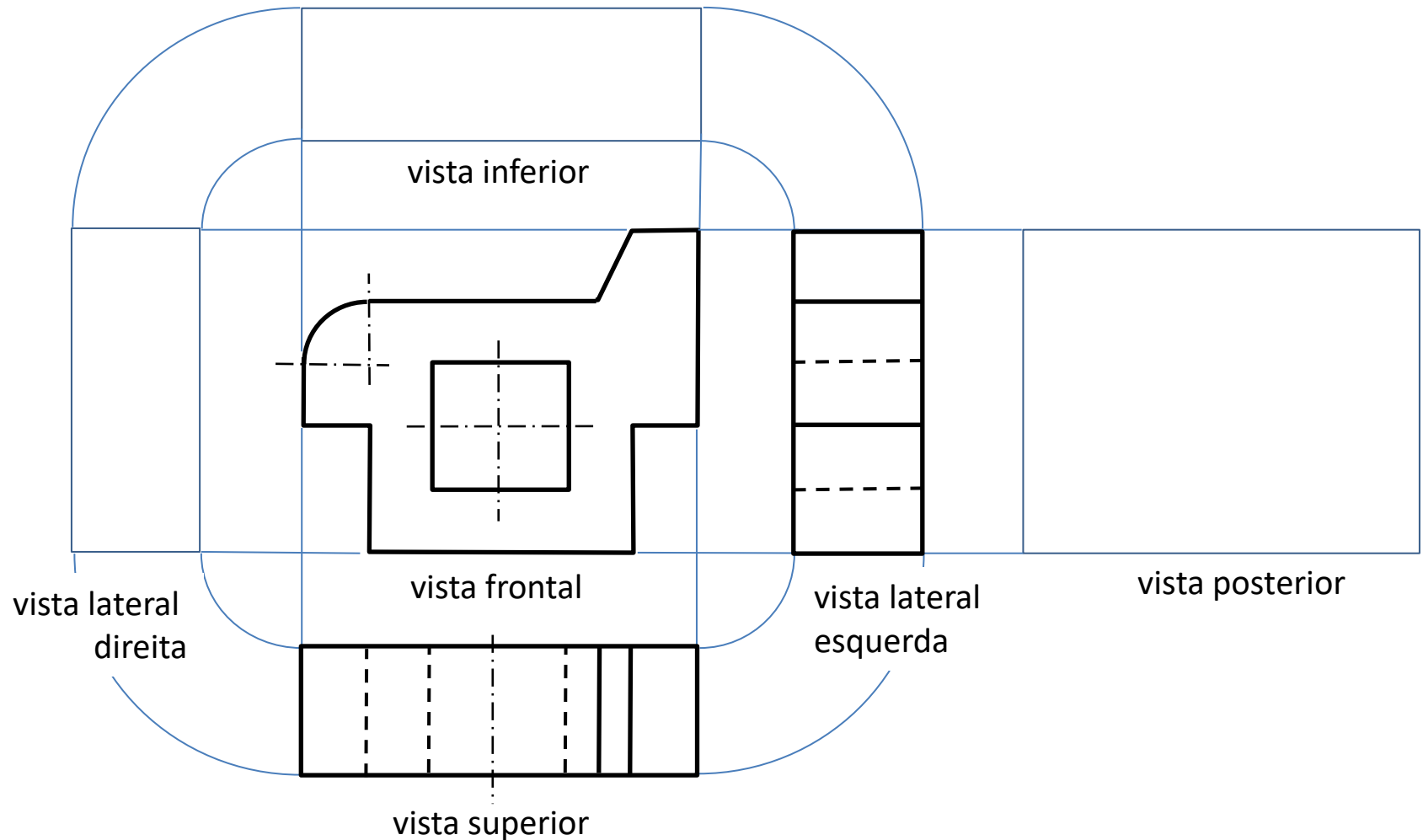


Desenhar vistas ortográficas no 1º diedro



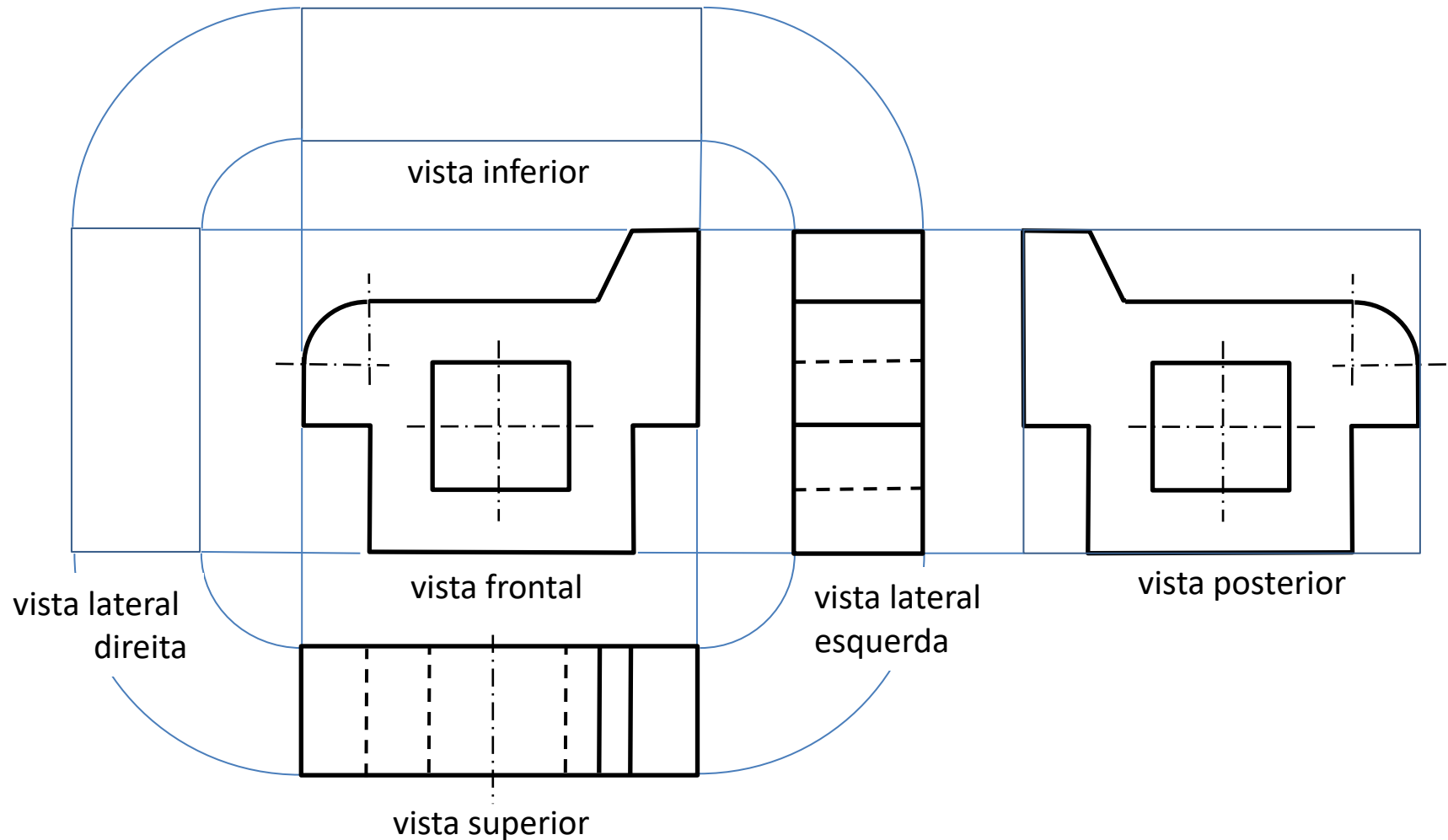
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Desenhar vistas ortográficas no 1º diedro



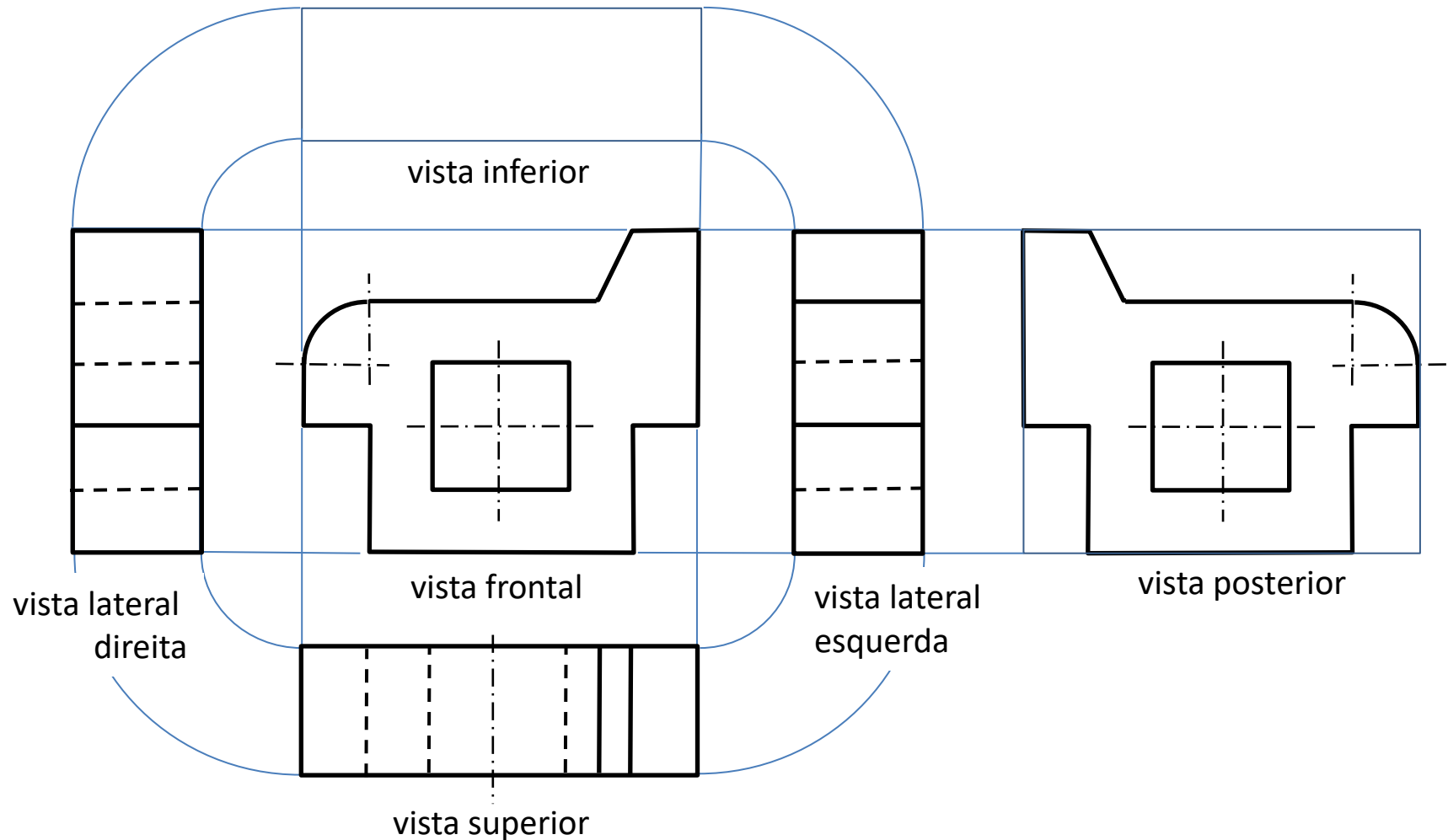
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Desenhar vistas ortográficas no 1º diedro



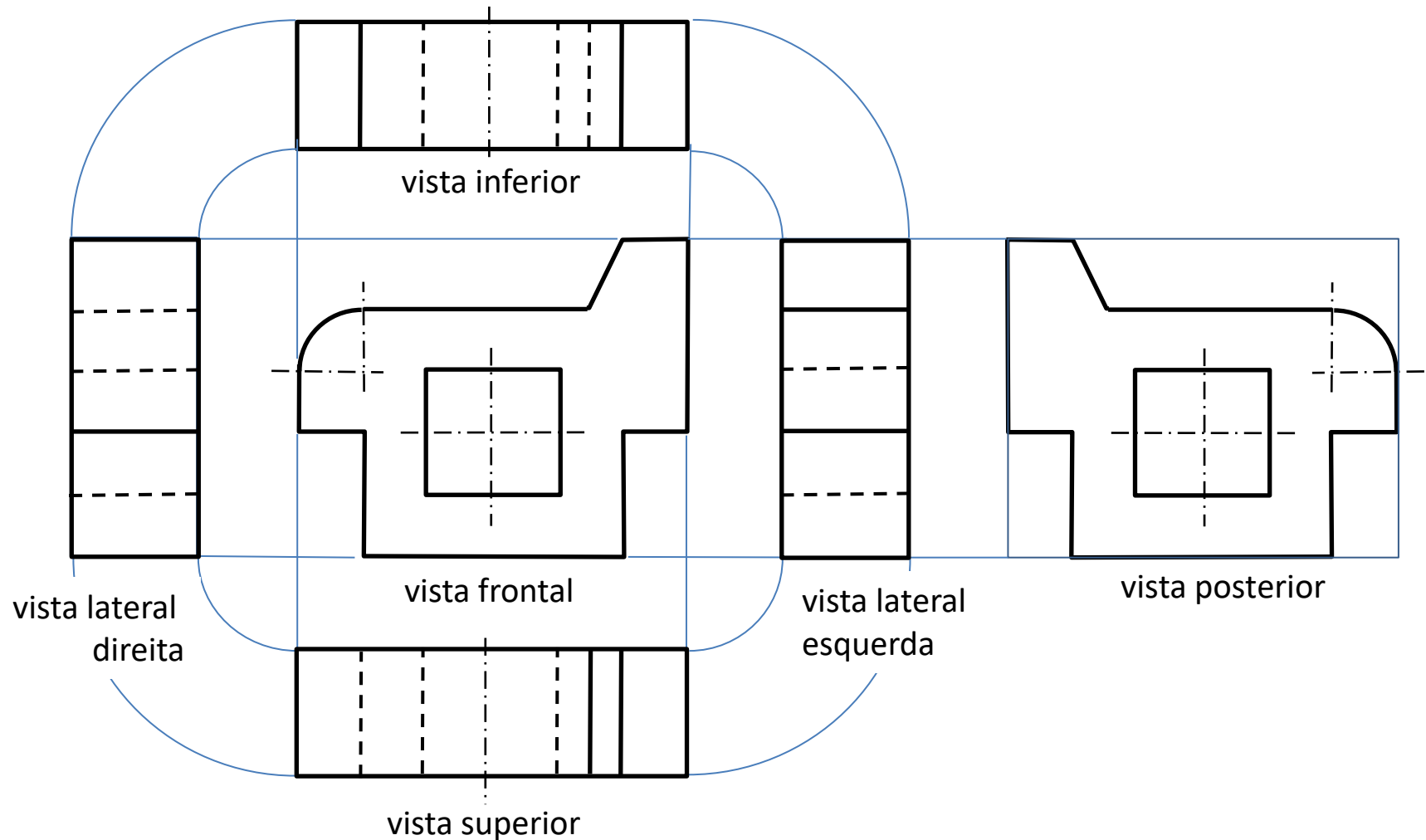
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Desenhar vistas ortográficas no 1º diedro



OBS.: O desenho deve obedecer: 1) posicionamento; 2) alinhamento; e 3) espaçamento.

Desenhar vistas ortográficas no 1º diedro



OBS.: O desenho deve obedecer: 1) posicionamento; 2) alinhamento; e 3) espaçamento.